

An OMG<sup>®</sup> Unified Architecture Framework<sup>®</sup> Domain Metamodel Publication



# Unified Architecture Framework (UAF) Domain Metamodel

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|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| <b>PREFACE .....</b>                                            | <b>1</b>  |
| <b>1. SCOPE .....</b>                                           | <b>3</b>  |
| 1.1 INTRODUCTION .....                                          | 3         |
| 1.2 UAF BACKGROUND .....                                        | 3         |
| 1.3 INTENDED USAGE .....                                        | 4         |
| 1.4 RELATED DOCUMENTS .....                                     | 4         |
| <b>2. CONFORMANCE .....</b>                                     | <b>5</b>  |
| <b>3. REFERENCES .....</b>                                      | <b>6</b>  |
| 3.1 NORMATIVE REFERENCES .....                                  | 6         |
| 3.2 OMG DOCUMENTS (NORMATIVE REFERENCES) .....                  | 6         |
| 3.3 OTHER NORMATIVE REFERENCES .....                            | 6         |
| 3.4 INFORMATIVE REFERENCES .....                                | 7         |
| <b>4. TERMS AND DEFINITIONS .....</b>                           | <b>8</b>  |
| <b>5. SYMBOLS .....</b>                                         | <b>9</b>  |
| <b>6. ADDITIONAL INFORMATION .....</b>                          | <b>11</b> |
| 6.1 CHANGES TO ADOPTED OMG SPECIFICATIONS .....                 | 11        |
| 6.2 LANGUAGE ARCHITECTURE .....                                 | 11        |
| 6.3 PHILOSOPHY .....                                            | 11        |
| 6.4 CORE PRINCIPLES .....                                       | 11        |
| <b>7. UAF GRID .....</b>                                        | <b>13</b> |
| 7.1 DESCRIPTIONS OF DOMAINS AND MODEL KINDS .....               | 15        |
| 7.2 DOMAIN INTERRELATIONSHIPS .....                             | 17        |
| <b>8. DOMAIN METAMODEL DIAGRAM LEGEND .....</b>                 | <b>17</b> |
| <b>9. DOMAIN METAMODEL DIAGRAMS .....</b>                       | <b>19</b> |
| 9.1 VIEW SPECIFICATIONS .....                                   | 19        |
| 9.1.1 <i>View Specifications::Metadata</i> .....                | 19        |
| 9.1.2 <i>View Specifications::Strategic</i> .....               | 20        |
| 9.1.3 <i>View Specifications::Operational</i> .....             | 28        |
| 9.1.4 <i>View Specifications::Services</i> .....                | 38        |
| 9.1.5 <i>View Specifications::Personnel</i> .....               | 47        |
| 9.1.6 <i>View Specifications::Resources</i> .....               | 62        |
| 9.1.7 <i>View Specifications::Security</i> .....                | 74        |
| 9.1.8 <i>View Specifications::Projects</i> .....                | 82        |
| 9.1.9 <i>View Specifications::Standards</i> .....               | 88        |
| 9.1.10 <i>View Specifications::Actual Resources</i> .....       | 91        |
| 9.1.11 <i>View Specifications::Dictionary</i> .....             | 94        |
| 9.1.12 <i>View Specifications::Summary &amp; Overview</i> ..... | 95        |
| 9.1.13 <i>View Specifications::Requirements</i> .....           | 96        |
| 9.1.14 <i>View Specifications::Information</i> .....            | 97        |
| 9.1.15 <i>View Specifications::Parameters</i> .....             | 98        |
| 9.1.16 <i>View Specifications::Other</i> .....                  | 101       |

|            |                                                      |            |
|------------|------------------------------------------------------|------------|
| <b>10.</b> | <b>DOMAIN METAMODEL (DMM) ELEMENTS .....</b>         | <b>105</b> |
| 10.1       | DOMAIN METAMODEL.....                                | 105        |
| 10.1.1     | <i>Domain MetaModel::Metadata.....</i>               | 105        |
| 10.1.2     | <i>Domain MetaModel::Strategic .....</i>             | 119        |
| 10.1.3     | <i>Domain MetaModel::Operational.....</i>            | 130        |
| 10.1.4     | <i>Domain MetaModel::Services.....</i>               | 144        |
| 10.1.5     | <i>Domain MetaModel::Personnel.....</i>              | 152        |
| 10.1.6     | <i>Domain MetaModel::Resources.....</i>              | 159        |
| 10.1.7     | <i>Domain MetaModel::Security.....</i>               | 177        |
| 10.1.8     | <i>Domain MetaModel::Projects.....</i>               | 192        |
| 10.1.9     | <i>Domain MetaModel::Standards.....</i>              | 199        |
| 10.1.10    | <i>Domain MetaModel::Actual Resources.....</i>       | 201        |
| 10.1.11    | <i>Domain MetaModel::Dictionary .....</i>            | 207        |
| 10.1.12    | <i>Domain MetaModel::Summary &amp; Overview.....</i> | 209        |
| 10.1.13    | <i>Domain MetaModel::Information.....</i>            | 213        |
| 10.1.14    | <i>Domain MetaModel::Parameters .....</i>            | 214        |

## TABLE OF FIGURES

|                                                                           |    |
|---------------------------------------------------------------------------|----|
| Figure 7:1- UAF Grid .....                                                | 13 |
| Figure 7:2 - Domain Interrelationships.....                               | 17 |
| Figure 7:3 - Legend of color codes for element types defined in UAF ..... | 18 |
| Figure 8:2 - Strategic Structure.....                                     | 22 |
| Figure 8:3 - Strategic Connectivity.....                                  | 23 |
| Figure 8:4 - Strategic States.....                                        | 24 |
| Figure 8:5 - Strategic Constraints .....                                  | 25 |
| Figure 8:6 - Strategic Roadmap: Deployment .....                          | 26 |
| Figure 8:7 - Strategic Roadmap: Phasing .....                             | 27 |
| Figure 8:8 - Strategic Traceability .....                                 | 28 |
| Figure 8:9 - Operational Taxonomy .....                                   | 29 |
| Figure 8:10 - Operational Structure .....                                 | 30 |
| Figure 8:11 - Operational Connectivity .....                              | 31 |
| Figure 8:12 - Operational Processes .....                                 | 32 |
| Figure 8:13 - Operational Processes BPMN Semantics.....                   | 34 |
| Figure 8:14 - Operational States .....                                    | 35 |
| Figure 8:15 - Operational Interaction Scenarios .....                     | 36 |
| Figure 8:16 - Operational Constraints .....                               | 37 |
| Figure 8:17 - Operational Traceability .....                              | 38 |
| Figure 8:18 - Services Taxonomy.....                                      | 39 |
| Figure 8:19 - Services Structure .....                                    | 40 |
| Figure 8:20 - Services Connectivity .....                                 | 41 |
| Figure 8:21 - Services Processes .....                                    | 42 |
| Figure 8:22 - Services Processes BPMN Semantics.....                      | 43 |
| Figure 8:23 - Services States .....                                       | 44 |
| Figure 8:24 - Services Interaction Scenarios .....                        | 45 |
| Figure 8:25 - Services Constraints.....                                   | 46 |
| Figure 8:26 - Services Roadmap.....                                       | 46 |
| Figure 8:27 - Services Traceability.....                                  | 47 |
| Figure 8:28 - Personnel Taxonomy .....                                    | 48 |
| Figure 8:29 - Personnel Structure .....                                   | 49 |
| Figure 8:30 - Personnel Connectivity .....                                | 50 |
| Figure 8:31 - Personnel Processes .....                                   | 51 |
| Figure 8:32 - Personnel States .....                                      | 52 |
| Figure 8:33 - Personnel Interaction Scenarios.....                        | 53 |
| Figure 8:34 - Personnel Constraints: Competence.....                      | 55 |
| Figure 8:35 - Personnel Constraints: Drivers.....                         | 56 |
| Figure 8:36 - Personnel Constraints: Performance .....                    | 57 |
| Figure 8:37 - Personnel Roadmap: Availability .....                       | 59 |
| Figure 8:38 - Personnel Roadmap: Evolution.....                           | 60 |
| Figure 8:39 - Personnel Roadmap: Forecast.....                            | 61 |
| Figure 8:40 - Personnel Traceability .....                                | 62 |
| Figure 8:41 - Resources Taxonomy .....                                    | 63 |
| Figure 8:42 - Resources Structure .....                                   | 64 |
| Figure 8:43 - Resources Connectivity .....                                | 65 |
| Figure 8:44 - Resources Processes .....                                   | 67 |
| Figure 8:45 - Resources Processes BPMN Semantics .....                    | 68 |
| Figure 8:46 - Resources States .....                                      | 69 |
| Figure 8:47 - Resources Interaction Scenarios .....                       | 70 |
| Figure 8:48 - Resources Constraints.....                                  | 71 |
| Figure 8:49 - Resources Roadmap: Evolution .....                          | 72 |

|                                                       |     |
|-------------------------------------------------------|-----|
| Figure 8:50 - Resources Roadmap: Forecast .....       | 73  |
| Figure 8:51 - Resources Traceability .....            | 74  |
| Figure 8:52 - Security Taxonomy .....                 | 75  |
| Figure 8:53 - Security Structure .....                | 76  |
| Figure 8:54 - Security Connectivity .....             | 77  |
| Figure 8:55 - Security Processes .....                | 79  |
| Figure 8:56 - Security Constraints .....              | 80  |
| Figure 8:57 - Security Traceability .....             | 82  |
| Figure 8:58 - Project Taxonomy .....                  | 83  |
| Figure 8:59 - Project Structure .....                 | 84  |
| Figure 8:60 - Project Connectivity .....              | 85  |
| Figure 8:61 - Project Processes .....                 | 86  |
| Figure 8:62 - Project Roadmap .....                   | 87  |
| Figure 8:63 - Project Traceability .....              | 88  |
| Figure 8:64 - Standards Taxonomy .....                | 89  |
| Figure 8:65 - Standards Structure .....               | 90  |
| Figure 8:66 - Standards Roadmap .....                 | 90  |
| Figure 8:67 - Standards Traceability .....            | 91  |
| Figure 8:68 - Actual Resources Structure .....        | 92  |
| Figure 8:69 - Actual Resources Connectivity .....     | 93  |
| Figure 8:70 - Actual Resources Traceability .....     | 94  |
| Figure 8:71 - Dictionary .....                        | 95  |
| Figure 8:72 - Summary & Overview .....                | 96  |
| Figure 8:73 - Requirements .....                      | 97  |
| Figure 8:74 - Information Model .....                 | 98  |
| Figure 8:75 - Parameters: Environment .....           | 99  |
| Figure 8:76 - Parameters: Measurements .....          | 100 |
| Figure 8:77 - BPMN .....                              | 101 |
| Figure 8:78 - IEPPV .....                             | 102 |
| Figure 8:79 - NIEM .....                              | 103 |
| Figure 9:1 - ArchitectureMetadata .....               | 105 |
| Figure 9:2 - InteractionScenarioGeneralization .....  | 106 |
| Figure 9:3 - Metadata .....                           | 106 |
| Figure 9:4 - ProcessGeneralization .....              | 107 |
| Figure 9:5 - PropertySetGeneralization .....          | 107 |
| Figure 9:6 - StateDescriptionGeneralization .....     | 108 |
| Figure 9:7 - EnvironmentProperty .....                | 108 |
| Figure 9:8 - Exchange .....                           | 109 |
| Figure 9:9 - ExchangeItem .....                       | 109 |
| Figure 9:10 - Resource .....                          | 110 |
| Figure 9:11 - ActivityPerformableUnderCondition ..... | 110 |
| Figure 9:12 - IsCapableToPerform .....                | 111 |
| Figure 9:13 - PerformsInContext .....                 | 111 |
| Figure 9:14 - Process .....                           | 112 |
| Figure 9:15 - Process .....                           | 112 |
| Figure 9:16 - ProcessEdge .....                       | 113 |
| Figure 9:17 - ProcessOperation .....                  | 113 |
| Figure 9:18 - ProcessParameter .....                  | 114 |
| Figure 9:19 - ProcessUsage .....                      | 114 |
| Figure 9:20 - StateDescription .....                  | 115 |
| Figure 9:21 - InteractionRole .....                   | 115 |
| Figure 9:22 - InteractionScenario .....               | 116 |
| Figure 9:23 - Information .....                       | 117 |
| Figure 9:24 - Rule .....                              | 117 |

|                                                    |     |
|----------------------------------------------------|-----|
| Figure 9:25 - ArchitecturalReference .....         | 118 |
| Figure 9:26 - Implements.....                      | 119 |
| Figure 9:27 - Capability .....                     | 120 |
| Figure 9:28 - CapabilityGeneralization .....       | 120 |
| Figure 9:29 - ActualEnduringTask .....             | 121 |
| Figure 9:30 - ActualEnterprisePhase .....          | 121 |
| Figure 9:31 - CapabilityRole .....                 | 122 |
| Figure 9:32 - EnduringTask.....                    | 122 |
| Figure 9:33 - EnterpriseGoal .....                 | 122 |
| Figure 9:34 - EnterprisePhase.....                 | 123 |
| Figure 9:35 - EnterpriseVision .....               | 123 |
| Figure 9:36 - StructuralPart .....                 | 124 |
| Figure 9:37 - TemporalPart .....                   | 124 |
| Figure 9:38 - VisionStatement.....                 | 125 |
| Figure 9:39 - WholeLifeEnterprise.....             | 125 |
| Figure 9:40 - CapabilityDependency .....           | 125 |
| Figure 9:41 - CapabilityRoleDependency .....       | 126 |
| Figure 9:42 - AchievedEffect .....                 | 126 |
| Figure 9:43 - Achiever.....                        | 127 |
| Figure 9:44 - DesiredEffect .....                  | 127 |
| Figure 9:45 - Desirer.....                         | 128 |
| Figure 9:46 - CapabilityForTask .....              | 128 |
| Figure 9:47 - CapableElement.....                  | 129 |
| Figure 9:48 - Exhibits .....                       | 129 |
| Figure 9:49 - MapsToCapability .....               | 130 |
| Figure 9:50 - OrganizationInEnterprise.....        | 130 |
| Figure 9:51 - ArbitraryConnector.....              | 131 |
| Figure 9:52 - ConceptItem.....                     | 131 |
| Figure 9:53 - HighLevelOperationalConcept .....    | 132 |
| Figure 9:54 - KnownResource.....                   | 132 |
| Figure 9:55 - OperationalAgent.....                | 133 |
| Figure 9:56 - OperationalArchitecture.....         | 133 |
| Figure 9:57 - OperationalMethod .....              | 134 |
| Figure 9:58 - OperationalParameter .....           | 134 |
| Figure 9:59 - OperationalPerformer .....           | 135 |
| Figure 9:60 - OperationalRole .....                | 135 |
| Figure 9:61 - ProblemDomain.....                   | 136 |
| Figure 9:62 - OperationalConnector .....           | 136 |
| Figure 9:63 - OperationalExchange .....            | 137 |
| Figure 9:64 - OperationalExchangeItem.....         | 137 |
| Figure 9:65 - OperationalInterface .....           | 138 |
| Figure 9:66 - OperationalPort.....                 | 138 |
| Figure 9:67 - OperationalSignal .....              | 139 |
| Figure 9:68 - OperationalActivity .....            | 139 |
| Figure 9:69 - OperationalActivityAction.....       | 140 |
| Figure 9:70 - OperationalActivityEdge .....        | 140 |
| Figure 9:71 - StandardOperationalActivity .....    | 141 |
| Figure 9:72 - OperationalStateDescription.....     | 141 |
| Figure 9:73 - OperationalInteractionScenario ..... | 142 |
| Figure 9:74 - OperationalMessage.....              | 142 |
| Figure 9:75 - InformationElement .....             | 143 |
| Figure 9:76 - OperationalConstraint.....           | 143 |
| Figure 9:77 - SubjectOfOperationalConstraint.....  | 143 |
| Figure 9:78 - ServiceSpecification .....           | 144 |

|                                                        |     |
|--------------------------------------------------------|-----|
| Figure 9:79 - ServiceSpecificationGeneralization ..... | 145 |
| Figure 9:80 - ServiceConnector .....                   | 145 |
| Figure 9:81 - ServiceMethod .....                      | 146 |
| Figure 9:82 - ServiceParameter .....                   | 146 |
| Figure 9:83 - ServiceSpecificationRole .....           | 147 |
| Figure 9:84 - ServiceInterface .....                   | 147 |
| Figure 9:85 - ServicePort .....                        | 148 |
| Figure 9:86 - ServiceFunction .....                    | 148 |
| Figure 9:87 - ServiceFunctionAction .....              | 149 |
| Figure 9:88 - ServiceFunctionEdge .....                | 149 |
| Figure 9:89 - ServiceStateDescription .....            | 150 |
| Figure 9:90 - ServiceInteractionScenario .....         | 150 |
| Figure 9:91 - ServiceMessage .....                     | 151 |
| Figure 9:92 - ServicePolicy .....                      | 151 |
| Figure 9:93 - Consumes .....                           | 152 |
| Figure 9:94 - Organization .....                       | 152 |
| Figure 9:95 - OrganizationalResource .....             | 153 |
| Figure 9:96 - Person .....                             | 153 |
| Figure 9:97 - Post .....                               | 154 |
| Figure 9:98 - Responsibility .....                     | 154 |
| Figure 9:99 - PostRole .....                           | 155 |
| Figure 9:100 - SubOrganization .....                   | 155 |
| Figure 9:101 - Command .....                           | 156 |
| Figure 9:102 - Control .....                           | 156 |
| Figure 9:103 - ResourceInteractionScenario .....       | 157 |
| Figure 9:104 - Competence .....                        | 157 |
| Figure 9:105 - CompetenceForRole .....                 | 158 |
| Figure 9:106 - RequiresCompetence .....                | 158 |
| Figure 9:107 - FillsPost .....                         | 159 |
| Figure 9:108 - CompetenceToConduct .....               | 159 |
| Figure 9:109 - CapabilityConfiguration .....           | 160 |
| Figure 9:110 - NaturalResource .....                   | 160 |
| Figure 9:111 - PhysicalResource .....                  | 160 |
| Figure 9:112 - ResourceArchitecture .....              | 161 |
| Figure 9:113 - ResourceArtifact .....                  | 161 |
| Figure 9:114 - ResourcePerformer .....                 | 162 |
| Figure 9:115 - Software .....                          | 162 |
| Figure 9:116 - System .....                            | 163 |
| Figure 9:117 - ResourceMethod .....                    | 163 |
| Figure 9:118 - ResourceParameter .....                 | 164 |
| Figure 9:119 - ResourcePort .....                      | 164 |
| Figure 9:120 - ResourceRole .....                      | 165 |
| Figure 9:121 - ResourceConnector .....                 | 166 |
| Figure 9:122 - ResourceExchange .....                  | 166 |
| Figure 9:123 - ResourceExchangeItem .....              | 167 |
| Figure 9:124 - ResourceInterface .....                 | 168 |
| Figure 9:125 - ResourceSignal .....                    | 168 |
| Figure 9:126 - Function .....                          | 169 |
| Figure 9:127 - FunctionAction .....                    | 169 |
| Figure 9:128 - FunctionEdge .....                      | 170 |
| Figure 9:129 - ResourceStateDescription .....          | 170 |
| Figure 9:130 - ResourceMessage .....                   | 171 |
| Figure 9:131 - DataElement .....                       | 171 |
| Figure 9:132 - DataRole .....                          | 172 |

|                                                  |     |
|--------------------------------------------------|-----|
| Figure 9:133 - ResourceConstraint.....           | 172 |
| Figure 9:134 - SubjectOfResourceConstraint.....  | 173 |
| Figure 9:135 - Forecast.....                     | 173 |
| Figure 9:136 - SubjectOfForecast.....            | 174 |
| Figure 9:137 - Technology .....                  | 174 |
| Figure 9:138 - VersionedElement.....             | 175 |
| Figure 9:139 - VersionOfConfiguration .....      | 175 |
| Figure 9:140 - VersionSuccession .....           | 176 |
| Figure 9:141 - WholeLifeConfiguration.....       | 176 |
| Figure 9:142 - ProtocolImplementation.....       | 177 |
| Figure 9:143 - Asset .....                       | 178 |
| Figure 9:144 - OperationalAsset.....             | 178 |
| Figure 9:145 - OperationalMitigation.....        | 179 |
| Figure 9:146 - ResourceAsset.....                | 179 |
| Figure 9:147 - ResourceMitigation.....           | 179 |
| Figure 9:148 - SecurityEnclave .....             | 180 |
| Figure 9:149 - AssetRole .....                   | 180 |
| Figure 9:150 - InformationRole.....              | 181 |
| Figure 9:151 - EnhancedSecurityControl .....     | 181 |
| Figure 9:152 - Enhances .....                    | 182 |
| Figure 9:153 - Protects .....                    | 182 |
| Figure 9:154 - ProtectsInContext.....            | 183 |
| Figure 9:155 - SecurityProcess .....             | 183 |
| Figure 9:156 - SecurityProcessAction .....       | 184 |
| Figure 9:157 - ActualRisk .....                  | 184 |
| Figure 9:158 - Caveat .....                      | 185 |
| Figure 9:159 - Risk .....                        | 185 |
| Figure 9:160 - SecurityAvailability .....        | 185 |
| Figure 9:161 - SecurityCategory .....            | 186 |
| Figure 9:162 - SecurityClassification .....      | 186 |
| Figure 9:163 - SecurityClassificationKind .....  | 187 |
| Figure 9:164 - SecurityConstraint .....          | 187 |
| Figure 9:165 - SecurityControl .....             | 188 |
| Figure 9:166 - SecurityControlFamily.....        | 188 |
| Figure 9:167 - SecurityIntegrity .....           | 189 |
| Figure 9:168 - SecurityMeasurement .....         | 189 |
| Figure 9:169 - SubjectOfSecurityConstraint ..... | 190 |
| Figure 9:170 - Affects.....                      | 190 |
| Figure 9:171 - AffectsInContext.....             | 190 |
| Figure 9:172 - Mitigates .....                   | 191 |
| Figure 9:173 - OwnsRisk.....                     | 191 |
| Figure 9:174 - OwnsRiskInContext.....            | 192 |
| Figure 9:175 - Project.....                      | 192 |
| Figure 9:176 - ProjectMilestone .....            | 193 |
| Figure 9:177 - ActualProjectMilestoneRole .....  | 193 |
| Figure 9:178 - ProjectMilestoneRole.....         | 194 |
| Figure 9:179 - ProjectStatus .....               | 194 |
| Figure 9:180 - ProjectTheme .....                | 195 |
| Figure 9:181 - StatusIndicators.....             | 195 |
| Figure 9:182 - MilestoneDependency.....          | 196 |
| Figure 9:183 - ProjectActivity .....             | 196 |
| Figure 9:184 - ProjectActivityAction .....       | 196 |
| Figure 9:185 - ProjectSequence.....              | 197 |
| Figure 9:186 - ActualProject .....               | 197 |

|                                                   |     |
|---------------------------------------------------|-----|
| Figure 9:187 - ActualProjectMilestone.....        | 198 |
| Figure 9:188 - ResponsibleFor .....               | 199 |
| Figure 9:189 - Protocol.....                      | 199 |
| Figure 9:190 - ProtocolStack.....                 | 200 |
| Figure 9:191 - Standard .....                     | 200 |
| Figure 9:192 - ProtocolLayer.....                 | 201 |
| Figure 9:193 - ActualOrganization.....            | 201 |
| Figure 9:194 - ActualOrganizationalResource ..... | 202 |
| Figure 9:195 - ActualPerson .....                 | 202 |
| Figure 9:196 - ActualPost.....                    | 203 |
| Figure 9:197 - ActualResource .....               | 203 |
| Figure 9:198 - ActualResourceRelationship.....    | 204 |
| Figure 9:199 - ActualResponsibility.....          | 204 |
| Figure 9:200 - ActualResponsibleResource.....     | 205 |
| Figure 9:201 - FieldedCapability .....            | 205 |
| Figure 9:202 - ActualService.....                 | 206 |
| Figure 9:203 - ProvidedServiceLevel .....         | 206 |
| Figure 9:204 - ProvidesCompetence.....            | 206 |
| Figure 9:205 - RequiredServiceLevel.....          | 207 |
| Figure 9:206 - Alias.....                         | 207 |
| Figure 9:207 - Definition .....                   | 208 |
| Figure 9:208 - SameAs .....                       | 208 |
| Figure 9:209 - ArchitecturalDescription .....     | 209 |
| Figure 9:210 - Architecture .....                 | 210 |
| Figure 9:211 - Concern.....                       | 211 |
| Figure 9:212 - Stakeholder .....                  | 211 |
| Figure 9:213 - UAFAElement .....                  | 212 |
| Figure 9:214 - View.....                          | 212 |
| Figure 9:215 - Viewpoint .....                    | 213 |
| Figure 9:216 - DataModel .....                    | 213 |
| Figure 9:217 - ActualCondition.....               | 214 |
| Figure 9:218 - ActualEnvironment.....             | 214 |
| Figure 9:219 - ActualLocation.....                | 215 |
| Figure 9:220 - ActualMeasurement .....            | 216 |
| Figure 9:221 - ActualMeasurementSet .....         | 216 |
| Figure 9:222 - ActualPropertySet.....             | 217 |
| Figure 9:223 - ActualState.....                   | 217 |
| Figure 9:224 - Condition .....                    | 218 |
| Figure 9:225 - Environment.....                   | 218 |
| Figure 9:226 - GeoPoliticalExtentType.....        | 219 |
| Figure 9:227 - ISO8601DateTime .....              | 219 |
| Figure 9:228 - Location .....                     | 220 |
| Figure 9:229 - LocationHolder .....               | 220 |
| Figure 9:230 - MeasurableElement.....             | 221 |
| Figure 9:231 - Measurement.....                   | 222 |
| Figure 9:232 - MeasurementSet.....                | 222 |
| Figure 9:233 - PropertySet.....                   | 223 |

## TABLE OF TABLES

|                                                                      |    |
|----------------------------------------------------------------------|----|
| Table 1:1 - Table of Related Documents .....                         | 4  |
| Table 5:1 - Description of acronyms used in this specification ..... | 9  |
| Table 7:1 - Definitions for the Domains .....                        | 15 |
| Table 7:2 - Definitions of the Model Kinds .....                     | 16 |

# Preface

## OMG

Founded in 1989, the Object Management Group, Inc. (OMG) is an open membership, not-for-profit computer industry standards consortium that produces and maintains computer industry specifications for interoperable, portable and reusable enterprise applications in distributed, heterogeneous environments. Membership includes Information Technology vendors, end users, government agencies and academia. OMG member companies write, adopt, and maintain its specifications following a mature, open process. OMG's specifications implement the Model Driven Architecture® (MDA®), maximizing ROI through a full-lifecycle approach to enterprise integration that covers multiple operating systems, programming languages, middleware and networking infrastructures, and software development environments. OMG's specifications include: UML® (Unified Modeling Language™); CORBA® (Common Object Request Broker Architecture); CWM™ (Common Warehouse Metamodel); and industry-specific standards for dozens of vertical markets. More information on the OMG is available at <https://www.omg.org/>.

## OMG Specifications

As noted, OMG specifications address middleware, modeling and vertical domain frameworks. All OMG Specifications are available from this URL: <https://www.omg.org/spec>

Specifications are organized by the following categories:

### Business Modeling Specifications<sup>[1]</sup>

- CORBA/IIOP
- Data Distribution Services
- Specialized CORBA IDL/Language Mapping Specifications
  
- UML, MOF, CWM, XMI
- UML Profile Specifications
  
- CORBAServices
- CORBAFacilities
- OMG Domain Specifications
- CORBA Embedded Intelligence Specifications
- CORBA Security Specifications

All of OMG's formal specifications may be downloaded without charge from our website. (Products implementing OMG specifications are available from individual suppliers.) Copies of specifications, available in PostScript and PDF format, may be obtained from the Specifications Catalog cited above or by contacting the Object Management Group, Inc. at: OMG Headquarters 109 Highland Avenue, Needham, MA 02494 USA Tel: +1- 781-444-0404 Fax: +1-781-444-0320 Email: [pubs@omg.org](mailto:pubs@omg.org)

Certain OMG specifications are also available as ISO standards. Please consult <http://www.iso.org>

## Typographical Conventions

The type styles shown below are used in this document to distinguish programming statements from ordinary English. However, these conventions are not used in tables or section headings where no distinction is necessary.

Times/Times New Roman - 10 pt.: Standard body text

**Helvetica/Arial - 10 pt. Bold** OMG Interface Definition Language (OMG IDL) and syntax elements.

**Courier - 10 pt. Bold** Programming language elements.

Helvetica/Arial - 10 pt: Exceptions

Terms that appear in *italics* are defined in the glossary. Italic text also represents the name of a document, specification, or other publication.

## Issues

All OMG specifications are subject to continuous review and improvement. As part of this process we encourage readers to report any ambiguities, inconsistencies, or inaccuracies they may find by completing the Issue Reporting Form listed on the main web page <https://www.omg.org>, under Documents, Report a Bug/Issue (<https://issues.omg.org/issues/create-new-issue>).

# 1 Scope

## 1.1 Introduction

There are four parts to this specification, two are normative and two informative. The normative parts are:

1. The UAF Domain Metamodel (DMM) (this document) that provides the definition of concepts, relationships and viewpoints for the framework. The UAF DMM is the basis for any implementation of UAF including non-UML/SysML implementations.
2. The UAF Profile (UAFP) (see document dtc/19-06-15) is a UML/SysML implementation of the UAF DMM

The informative parts are:

3. The UAF Traceability, Annex A (see document dtc/19-06-17), which details the mappings between the UAF and the various frameworks and languages that contribute to the UAF.
4. The UAF Example Model, Annex B (see document dtc/19-06-18), which illustrates a practical usage of UAF.

## 1.2 UAF Background

UAF evolved from the Unified Profile for DoDAF and MODAF (UPDM), version 2.1. UAF extends the scope of UPDM and generalizes it to make it applicable to commercial as well as military architectures. The intent of UAF is to provide a standard representation for describing enterprise architectures using a Model Based Systems Engineering (MBSE) approach.

The core concepts in the UAF are based upon the DoDAF 2.0.2 Domain Metamodel (DM2) and the MODAF ontological data exchange mechanism (MODEM), Security Views from Canada's Department of National Defense Architecture Framework (DNDAF) and the North Atlantic Treaty Organization (NATO) Architecture Framework (NAF) v 4.

UAF models describe a system<sup>1</sup> from a set of stakeholders' concerns such as security or information through a set of predefined viewpoints. Developed models can also reflect custom viewpoints or users can develop more formal extensions for new viewpoints.

The UAFP can be used to develop architectures compliant with:

- Department of Defense Architecture Framework (DoDAF) version 2.02
- Ministry of Defence Architecture Framework (MODAF) version 1.3
- North Atlantic Treaty Organization (NATO) Architecture Framework (NAF) version 3.1
- North Atlantic Treaty Organization (NATO) Architecture Framework (NAF) version 4

UAF v 1.1 supports the capability to:

- model architectures for a broad range of complex systems, which may include hardware, software, data, personnel, and facility elements,
- model consistent architectures for system-of-systems (SoS) down to lower levels of design and implementation,
- support the analysis, specification, design, and verification of complex systems; and
- improve the ability to exchange architecture information among related tools that are SysML based.

---

<sup>1</sup> The term system is used from: "Systems and software engineering -- Architecture description," [http://www.iso.org/iso/catalogue\\_detail.htm?csnumber=50508](http://www.iso.org/iso/catalogue_detail.htm?csnumber=50508)

## 1.3 Intended Usage

The UAF enables the modeling of strategic capabilities, operational scenarios, services, resources, personnel, security, projects, standards, measures and requirements; which supports best practices through, separation of concerns and abstractions. In addition, the UAF enables the modeling of related architecture concepts such as:

- System of Systems (SoS),
- information exchanges consistent with the National Information Exchange Model (NIEM),
- DoD's doctrine, organization, training material, leadership & education, personnel, and facilities (DOTMLPF)
- UK Ministry of Defence Lines of Development (DLOD) elements,
- Human Computer Interfaces (HCI).

Further, The UAF conforms to terms defined in the ISO/IEC/IEEE 42010 standard for architecture description, where the terms: architecture, architecture description (AD), architecture framework, architecture view, architecture viewpoint, concern, environment, model kind, stakeholder [ISO/IEC/IEEE 42010:2011] form correspondence rules specified as constraints on UAF.

## 1.4 Related Documents

The specification includes a metamodel and description as separate documents. Other appendices are also provided as separate documents. The table below provides a listing of these documents:

**Table 1:1 - Table of Related Documents**

|              |                                                                                                                                             |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| dtc/19-06-16 | The UAF Domain MetaModel (DMM)                                                                                                              |
| dtc/19-06-15 | The UAF Profile (UAFP)                                                                                                                      |
| dtc/19-06-17 | Appendix A that contains a separate traceability subsection from UAFP to each of the frameworks listed in Section 1.2 of this specification |
| dtc/19-06-18 | Appendix B: An example of how the language can be used to represent a UAFP architecture                                                     |
| dtc/19-06-19 | UAF XMI file                                                                                                                                |
| dtc/19-06-20 | UAF XMI Measurements library                                                                                                                |
| dtc/19-05-14 | Attachments                                                                                                                                 |

## 2 Conformance

UAF specifies four types of conformance.

*Type 1 Conformance: - UAF View specification conformance.* A tool demonstrating view specification conformance shall implement a version of all the view specifications defined in the UAF Grid, with the exception of the view specifications in the Metadata Domain. Optionally the tool vendor can implement other donor framework viewpoints, for instance DoDAF, MODAF or NAF based upon the mapping between them and UAF provided in Appendix A (dtc/19-06-17)

*Type 2 Conformance: - UAF Conceptual Syntax Conformance.* A tool demonstrating conceptual syntax conformance is consistent with the concepts, relationships and constraints defined in the UAF DMM (this document). UAF Conceptual Syntax Conformance implies Type 1 Conformance.

*Type 3 Conformance: - UAF Formal Syntax Conformance.* A tool demonstrating formal syntax conformance:

- enables instances of concrete UAFP stereotypes defined in the UAFP (dtc/19-06-15)
- complies with the constraints defined in the UAFP (dtc/19-06-15)
- complies with the SysML version 1.5 Concrete Syntax Conformance (formal/17-05-01)

UAF Formal Syntax Conformance implies Type 2 Conformance.

*Type 4 Conformance: - UAF Model interchange conformance.* A tool demonstrating model interchange conformance can import and export conformant XMI for all valid UAFP models. Model interchange conformance implies Type 3 Conformance.

## 3 References

### 3.1 Normative References

The following normative documents contain provisions which, through reference in this text, constitute provisions of this specification. For dated references, subsequent amendments to, or revisions of, any of these publications do not apply.

### 3.2 OMG Documents (Normative References)

- Unified Modeling Language (UML), 2.5.1, December 2017, <https://www.omg.org/spec/UML>
- Object Constraint Language (OCL), 2.4, February 2014, <https://www.omg.org/spec/OCL>
- System Modeling Language (SysML) ,1.5, May 2017, <https://www.omg.org/spec/SysML>
- Diagram Definition (DD), 1.1, June 2015, <https://www.omg.org/spec/DD>
- UML Profile for the National Information Exchange Model (NIEM UML), 3.0, April 2017, <https://www.omg.org/spec/NIEM-UML>
- Unified Profile for DoDAF and MODAF (UPDM), 2.1, August 2013, <https://www.omg.org/spec/UPDM>
- UML Profile for BPMN Processes, 1.0, July 2014, <https://www.omg.org/spec/BPMNProfile>
- Ontology Definition Metamodel (ODM), 1.1, September 2014, <https://www.omg.org/spec/ODM>
- Information Exchange Packaging Policy Vocabulary (IEPPV) 1.0, May 2015, <https://www.omg.org/spec/IEPPV>

### 3.3 Other Normative References

- Department of Defense Architecture Framework (DoDAF), Version 2.02, August 2010, <http://dodcio.defense.gov/Library/DoDArchitectureFramework.aspx>
- DM2 - DoDAF Meta-Model,
- The DM2 Conceptual Data Model, [http://dodcio.defense.gov/Library/DoDArchitectureFramework/dodaf20\\_conceptual.aspx](http://dodcio.defense.gov/Library/DoDArchitectureFramework/dodaf20_conceptual.aspx)
- DM2 Logical Data Model, [http://dodcio.defense.gov/Library/DoDArchitectureFramework/dodaf20\\_logical.aspx](http://dodcio.defense.gov/Library/DoDArchitectureFramework/dodaf20_logical.aspx)
- DM2 Formal Ontology. [http://dodcio.defense.gov/Library/DoDArchitectureFramework/dodaf20\\_ontology1.aspx](http://dodcio.defense.gov/Library/DoDArchitectureFramework/dodaf20_ontology1.aspx)
- Department National Defence and Canadian Forces (DND/ CF) Architecture Framework (DNDAF), Version 1.8.1, 25 January 2013
- International Defence Enterprise Architecture Specification for Exchange (IDEAS) Group, <http://www.ideasgroup.org/>
- IDEAS Foundation, <http://www.ideasgroup.org/foundation/>
- IDEAS Foundation v1.0 as XMI File (zipped), <http://www.ideasgroup.org/7Documents/>
- ISO/IEC/IEEE 42010:2011, Systems and software engineering – Architecture Description, [http://www.iso.org/iso/catalogue\\_detail.htm?csnumber=50508](http://www.iso.org/iso/catalogue_detail.htm?csnumber=50508)
- Ministry of Defence Architecture Framework (MODAF), <https://www.gov.uk/mod-architecture-framework>
- MODAF Ontological Data Exchange Mechanism (MODEM)
- [https://www.gov.uk/government/uploads/system/uploads/attachment\\_data/file/63980/20130117\\_MODAF\\_MODEM.pdf](https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/63980/20130117_MODAF_MODEM.pdf)
- NATO Architecture Framework (NAF),

- Version 3, NATO C3 BOARD (AC/322-D(2007)0048), <http://www.nhqc3s.nato.int/HomePage.asp> (no longer publicly available online as of 3 November 2015)
- NATO Architecture Framework v4.0 Documentation.

### 3.4 Informative References

- Business Process Model & Notation (BPMN), Version 2.0.2, January 2014 <https://www.omg.org/spec/BPMN>
- ISO 15704:2000, Industrial Automation Systems – “Requirements for Enterprise-Reference Architectures and Methodologies,” [http://www.iso.org/iso/home/store/catalogue\\_tc/catalogue\\_detail.htm?csnumber=28777](http://www.iso.org/iso/home/store/catalogue_tc/catalogue_detail.htm?csnumber=28777)
- ISO 8601:2004 Data elements and interchange formats – Information interchange – Representation of dates and times,  
[http://www.iso.org/iso/home/store/catalogue\\_ics/catalogue\\_detail\\_ics.htm?ics1=01&ics2=140&ics3=30&csnumber=40874](http://www.iso.org/iso/home/store/catalogue_ics/catalogue_detail_ics.htm?ics1=01&ics2=140&ics3=30&csnumber=40874)
- ISO/IEC 15288:2015, "Systems Engineering - Systems Life Cycle Processes,"  
[http://www.iso.org/iso/home/store/catalogue\\_tc/catalogue\\_detail.htm?csnumber=63711](http://www.iso.org/iso/home/store/catalogue_tc/catalogue_detail.htm?csnumber=63711)
- Object Management Group (OMG), Metamodel Extension Facility, Initial submission, ad/12-02-01,  
<https://www.omg.org/cgi-bin/doc?ad/12-02-01> (Requires OMG Member Access)
- OASIS SOA-RAF, Reference Architecture Foundation for Service Oriented Architecture Version 1.0, OASIS SOA Reference Model TC, 04 December 2012. <http://docs.oasis-open.org/soa-rm/soa-ra/v1.0/cs01/soa-ra-v1.0-cs01.pdf> (Authoritative)
- Object Management Group (OMG), Semantics of Business Vocabulary and Business Rules (SBVR), Version 1.3, May 2015, <https://www.omg.org/spec/SBVR>
- Business Motivation Model (BMM), Version 1.3, <http://www.omg.org/spec/BMM/1.3/>
- International Council On Systems Engineering (INCOSE), Systems Engineering Handbook V4, 2015,  
<http://www.incose.org/ProductsPublications/sehandbook>

## 4 Terms and Definitions

No new terms and definitions have been required to create this specification. All terms are available in the normative references or bibliographic citations for detailed explanation.

## 5 Symbols

For the purposes of this specification, the following List of symbols/abbreviations apply.

**Table 5:1 - Description of acronyms used in this specification**

|                    |                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------|
| AcV-* <sup>2</sup> | Acquisition View                                                                            |
| AD                 | Architecture Description                                                                    |
| AV-*               | All View                                                                                    |
| BMM                | Business Motivation Model                                                                   |
| BPMN               | Business Process Modeling Notation                                                          |
| C4ISR              | Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance |
| CaT                | Capability Team                                                                             |
| COI                | Communities of Interest                                                                     |
| CV-*               | Capability View                                                                             |
| DIV-*              | Data and Information Views                                                                  |
| DLOD               | Defence Lines of Development                                                                |
| DM2                | DoDAF Meta Model                                                                            |
| DMM                | Domain Meta Model                                                                           |
| DND AF             | Department National Defence and Canadian Forces (DND/ CF) Architecture Framework            |
| DoD                | United States Department of Defense                                                         |
| DoDAF              | Department of Defense Architecture Framework                                                |
| DOTMLP             | Doctrine, Organization, Training, Material, Leadership, Personnel, Facilities               |
| EIE                | Enterprise Information Environment                                                          |
| IDEAS              | International Defense Enterprise Architecture Specification for Exchange                    |
| IDEF               | Integrated DEFinition Methods                                                               |
| INCOSE             | International Council Of Systems Engineering                                                |
| JCIDS              | Joint Capabilities Integration and Development System                                       |
| MISIG              | Model Interchange Special Interest Group                                                    |
| MOD                | United Kingdom Ministry of Defence                                                          |
| MODAF              | Ministry of Defence Architecture Framework                                                  |
| MODEM              | MODAF Ontological Data Exchange Mechanism                                                   |
| NAF                | NATO Architecture Framework                                                                 |
| OASIS              | Organization for the Advancement of Structured Information Standards                        |
| OSLC               | Open Services for Lifecycle Collaboration                                                   |
| OV-*               | Operational View                                                                            |
| PES                | DoDAF Physical Exchange Specification                                                       |
| POC                | Proof of Concept                                                                            |
| PV-*               | Project View                                                                                |
| RDF                | Resource Description Framework                                                              |
| SoaML              | Service orientated architecture Modeling Language                                           |
| SoS                | System of Systems                                                                           |

---

<sup>2</sup> \* denotes a wildcard

|              |                                                                                                                |
|--------------|----------------------------------------------------------------------------------------------------------------|
| SOV-*        | Service Oriented View                                                                                          |
| StdV-*       | Standards View in DoDAF 2.02 compare TV-* in UAF                                                               |
| STV-*        | Strategic View                                                                                                 |
| SV-*         | System View                                                                                                    |
| SvcV-*       | Service View                                                                                                   |
| TEPID<br>OIL | Training, Equipment, Personnel, Information, Concepts and Doctrine, Organisation, Infrastructure,<br>Logistics |
| TOGAF        | The Open Group Architectural Framework©                                                                        |
| TPPU         | Task, Post, Process, and Use                                                                                   |
| TV-*         | Technical View                                                                                                 |
| UAF          | Unified Architecture Framework                                                                                 |
| UAFP         | Unified Architecture Framework Profile                                                                         |
| UPDM         | Unified Profile for DoDAF/MODAF                                                                                |

## 6 Additional Information

### 6.1 Changes to Adopted OMG Specifications

This specification completely replaces Unified Architecture Framework (UAF), version 1.0  
<https://www.omg.org/spec/UAF/About-UAF/>

### 6.2 Language Architecture

The UAF specification reuses a subset of UML 2.5.1 and SysML 1.5 and provides additional extensions needed to address requirements in the UPDM 3.0 RFP Mandatory Requirements. Those requirements form the basis for this specification. This specification documents the language architecture in terms of UML 2.5.1 and SysML 1.5 and specifies how to implement UAF. This clause explains design principles and how they are applied to define the UAF language architecture.

### 6.3 Philosophy

The UAF development uses a model-driven approach. A simple description of the work process is:

- A Domain Metamodel (DMM) uses UML Class models to represent individuals, types and tuples that aggregate the concepts defined in DoDAF, MODEM, NAF, DNDAAF and other frameworks.
- The aligned and renamed viewpoints from the various frameworks provide a common generic name for each viewpoint. It should be noted that the term viewpoint is in the context of ISO 42010 where a viewpoint is the specification of a view. The UAF viewpoints are mapped to the corresponding viewpoint in the relevant contributing framework. It is the viewpoints described in the DMM that provides the basis for the Unified Architecture Framework (UAF).
- The UAF provides an abstraction layer that separates the underlying UAF metamodel from the presentation layer. The results of this mapping are given in Appendix A (see document dtc/19-06-17 and an overview of the viewpoints in a grid format are given in this document).
- The intent of the UAF is to provide a Domain MetaModel usable by non-UML/SysML tool vendors who may wish to implement the UAF within their own tool and metalanguage.
- The Unified Architecture Framework Profile (UAFP) is the standard implementation of the UAF DMM. It was created by mapping the UAF concepts and relationships to corresponding stereotypes in the UAFP.
- The UAFP analysis and refactoring reflects language architecture, tool implementation, and reuse considerations.
- The specification is generated from the UML model used to describe the UAF DMM and UAFP. This approach allows the team to concentrate on architecture issues rather than documentation production. The UML tool automatically maintains consistency. The UML tool improves maintenance and enables traceability between the UAF and the UAFP where every stereotype is linkable to the UAF element using UML Abstraction relationship.

### 6.4 Core Principles

The fundamental design principles for UAF DMM are:

- : UAF is intended to satisfy the requirements of the UPDM 3.0 RFP Mandatory Requirements.
- The DMM was based upon an aggregation of concepts and relationships from the donor frameworks.
- The DMM was based upon a simplified version of the IDEAS ontology, see chapter 8.

- The DMM was expressed using UML class diagram notation.
- The UAF DMM reuses a number of concepts from the UML Metamodel, such as Statemachines, Activities and Interactions. The explicit relationship to these concepts enables the UAF DMM to reuse UML semantics instead of reinventing its own semantics.
- The UAF DMM reuses a number of concepts from BPMN, such as processes. The explicit relationship to these concepts enables the UAF DMM to reuse BPMN semantics instead of reinventing its own semantics.

## 7 UAF Grid

Due to the complexity of managing the multiple viewpoints with overlapping concerns and metamodels, the standard viewpoints are refactored as described in the donor frameworks into a more manageable format. This decision led to the development of the UAF grid which is described below.

The grid is a way of showing how the various viewpoints (known as *view specifications* in the rest of document) correspond to *domains* (horizontal rows) and the *model kinds* (the columns) that describe the view specification. The intent of the grid is not to be complete, but to capture the information that is present in the frameworks that contributes to the UAF, consequently, some gaps are evident.

|                             | Taxonomy Tx                          | Structure Sr                        | Connectivity Cn                | Processes Pr                | States St                | Interaction Scenarios Is                | Information <sup>c</sup> If                               | Parameters <sup>d</sup> Pm      | Constraints Ct                | Roadmap Rm                                             | Traceability Tr                |                           |
|-----------------------------|--------------------------------------|-------------------------------------|--------------------------------|-----------------------------|--------------------------|-----------------------------------------|-----------------------------------------------------------|---------------------------------|-------------------------------|--------------------------------------------------------|--------------------------------|---------------------------|
| Metadata <sup>a</sup><br>Md | Metadata Taxonomy Md-Tx <sup>f</sup> | Metadata Structure Md-Sr            | Metadata Connectivity Md-Cn    | Metadata Processes Md-Pr    | Metadata States Md-St    | -                                       | Conceptual Data Model,                                    | Environment <sup>d</sup> Pm-En  | Metadata Constraints Md-Ct    | Metadata Roadmap Md-Rm                                 | Metadata Traceability Md-Tr    |                           |
| Strategic St                | Strategic Taxonomy St-Tx             | Strategic Structure St-Sr           | Strategic Connectivity St-Cn   | -                           | Strategic States St-St   | -                                       |                                                           |                                 | Strategic Constraints St-Ct   | Strategic Deployment, St-Rm<br>Strategic Phasing St-Rm | Strategic Traceability St-Tr   |                           |
| Operational Op              | Operational Taxonomy Op-Tx           | Operational Structure Op-Sr         | Operational Connectivity Op-Cn | Operational Processes Op-Pr | Operational States Op-St | Operational Interaction Scenarios Op-Is |                                                           |                                 | Operational Constraints Op-Ct | -                                                      | Operational Traceability Op-Tr |                           |
| Services Sv                 | Service Taxonomy Sv-Tx               | Service Structure Sv-Sr             | Service Connectivity Sv-Cn     | Service Processes Sv-Pr     | Service States Sv-St     | Service Interaction Scenarios Sv-Is     |                                                           |                                 | Service Constraints Sv-Ct     | Service Roadmap Sv-Rm                                  | Service Traceability Sv-Tr     |                           |
| Personnel <sup>a</sup> Pe   | Personnel Taxonomy                   | Personnel Structure                 | Personnel Connectivity         | Personnel Processes         | Personnel States         | Personnel Interaction Scenarios         | Physical schema <sup>e</sup> real world results           | Measurements <sup>e</sup> Pm-Me | Competence, Drivers,          | Personnel Availability,                                | Personnel Traceability         |                           |
| Resource <sup>a</sup> Rs    | Resource Taxonomy Rs-Tx              | Resource Structure Rs-Sr            | Resource Connectivity Rs-Cn    | Resource Processes Rs-Pr    | Resource States Rs-St    | Resource Interaction Scenarios Rs-Is    |                                                           |                                 | Resource Constraints Rs-Ct    | Resource evolution, Resource forecast Rs-Rm            | Resource Traceability Rs-Tr    | Resources <sup>a</sup> Rs |
| Security <sup>a</sup> Sc    | Security Taxonomy Sc-Tx              | Security Structure Sc-Sr            | Security Connectivity Sc-Cn    | Security Processes Sc-Pr    | -                        | -                                       |                                                           |                                 | Security Constraints Sc-Ct    | -                                                      | Security Traceability Sc-Tr    | Security <sup>a</sup> Sc  |
| Project <sup>a</sup> Pj     | Project Taxonomy Pj-Tx               | Project Structure Pj-Sr             | Project Connectivity Pj-Cn     | Project Processes Pj-Pr     | -                        | -                                       |                                                           |                                 | -                             | Project Roadmap Pj-Rm                                  | Project Traceability Pj-Tr     | Projects <sup>a</sup> Pj  |
| Standard <sup>a</sup> Sd    | Standards Taxonomy Sd-Tx             | Standards Structure Sd-Sr           | -                              | -                           | -                        | -                                       |                                                           |                                 | -                             | Standards Roadmap Sd-Rm                                | Standards Traceability Sd-Tr   | Standards <sup>a</sup> Sd |
|                             | Actual Resources Structure Ar-Sr     | Actual Resources Connectivity Ar-Ch | Simulation <sup>b</sup>        |                             |                          |                                         | Parametric Execution <sup>b</sup> Evaluation <sup>b</sup> | -                               | -                             | Actual Resources <sup>a</sup> Ar                       |                                |                           |
| Dictionary Dc               |                                      |                                     |                                |                             |                          |                                         |                                                           |                                 |                               |                                                        |                                |                           |
| Summary & Overview Sm-Ov    |                                      |                                     |                                |                             |                          |                                         |                                                           |                                 |                               |                                                        |                                |                           |
| Requirements Req            |                                      |                                     |                                |                             |                          |                                         |                                                           |                                 |                               |                                                        |                                |                           |

Figure 7:1- UAF Grid

Notes related to suffixes in the grid:

- The view specifications in the Metadata Domain are not modeled as part of the UAF but are architectural artifacts that contribute to the success in defining and developing an architecture.
- To be able to evaluate architecture behavior and constraints (i.e., non-functional requirements) it is necessary to define actual instances of the architectural elements. The expectation is that tool vendors intending to implement the UAF have capabilities native to their tools to enable behavioral simulation and the evaluation of measures and constraints through parametric diagrams or a proprietary equivalent.
- The information model is a column across the domains and can be defined in any of its forms, i.e., Conceptual, Logical or Physical. The expectation is that most developers of the information model will use the Conceptual or Logical forms of the data model when using an abstract modeling tool.

- d. The parameters column captures the measures and environments across the architecture in all the different domains.
- e. The expectation is that the physical schema model would not be defined in the UAF. Any tool implementing the framework provides a means to import or link-to representations of the physical model.
- f. The Metadata Taxonomy view specification provides a means to extend the framework to other domains.

The detailed mapping between the view specifications of the UAF shown in the grid and the viewpoints from the donor frameworks is described in dtc\2019-06-17. A definition for each view specification in the grid is described in the following chapters.

## 7.1 Descriptions of Domains and Model Kinds

**Table 7:1 - Definitions for the Domains**

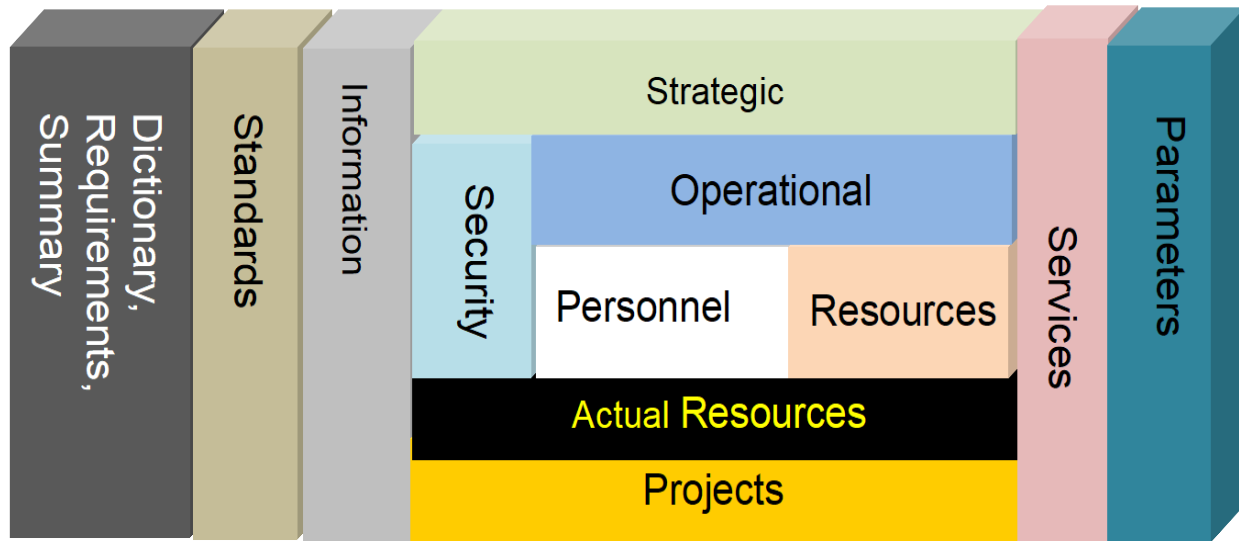
| Metadata         | Md | Identifies the metadata required to develop a suitable architecture that is fit for its purpose.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Strategic        | St | Capability management process. Describes the capability taxonomy, composition, dependencies and evolution.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Operational      | Op | Illustrates the Logical Architecture of the enterprise. Describes the requirements, operational behavior, structure, and exchanges required to support (exhibit) capabilities. Defines all operational elements in an implementation/solution independent manner.                                                                                                                                                                                                                                                                                                |
| Services         | Sv | The Service-Orientated View (SOV) is a description of services needed to directly support the operational domain as described in the Operational View. A service within MODAF is understood in its broadest sense, as a unit of work through which a provider provides a useful result to a consumer.<br>DoDAF: The Service Views within the Services Viewpoint describe the design for service-based solutions to support operational development processes (JCIDS) and Defense Acquisition System or capability development within the Joint Capability Areas. |
| Personnel        | Pr | Defines and explores organizational resource types. Shows the taxonomy of types of organizational resources as well as connections, interaction and growth over time.                                                                                                                                                                                                                                                                                                                                                                                            |
| Resources        | Rs | Captures a solution architecture consisting of resources, e.g., organizational, software, artifacts, capability configurations, and natural resources that implement the operational requirements. Further design of a resource is typically detailed in SysML or UML.                                                                                                                                                                                                                                                                                           |
| Security         | Sc | Security assets and security enclaves. Defines the hierarchy of security assets and asset owners, security constraints (policy, laws, and guidance) and details where they are located (security enclaves).                                                                                                                                                                                                                                                                                                                                                      |
| Projects         | Pj | Describes projects and project milestones, how those projects deliver capabilities, the organizations contributing to the projects and dependencies between projects.                                                                                                                                                                                                                                                                                                                                                                                            |
| Standards        | Sd | MODAF: Technical Standards Views are extended from the core DoDAF views to include non-technical standards such as operational doctrine, industry process standards, etc.<br>DoDAF: The Standards Views within the Standards Viewpoint are the set of rules governing the arrangement, interaction, and interdependence of solution parts or elements.                                                                                                                                                                                                           |
| Actual Resources | Ar | The analysis, e.g., evaluation of different alternatives, what-if, trade-offs, V&V on the actual resource configurations. Illustrates the expected or achieved actual resource configurations.                                                                                                                                                                                                                                                                                                                                                                   |

**Table 7:2 - Definitions of the Model Kinds**

| Taxonomy              | Tx | Presents all the elements as a standalone structure. Presents all the elements as a specialization hierarchy, provides a text definition for each one and references the source of the element                                 |
|-----------------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Structure             | Sr | Describes the definitions of the dependencies, connections, and relationships between the different elements.                                                                                                                  |
| Connectivity          | Cn | Describes the connections, relationships, and interactions between the different elements.                                                                                                                                     |
| Processes             | Pr | Captures activity based behavior and flows. It describes activities, their Inputs/Outputs, activity actions and flows between them.                                                                                            |
| States                | St | Captures state-based behavior of an element. It is a graphical representation of states of a structural element and how it responds to various events and actions.                                                             |
| Interaction Scenarios | Is | Expresses a time ordered examination of the exchanges as a result of a particular scenario. Provides a time-ordered examination of the exchanges between participating elements as a result of a particular scenario.          |
| Information           | If | Address the information perspective on operational, service, and resource architectures. Allows analysis of an architecture's information and data definition aspect, without consideration of implementation specific issues. |
| Constraints           | Ct | Details the measurements that set performance requirements constraining capabilities. Also defines the rules governing behavior and structure.                                                                                 |
| Roadmap               | Rm | Addresses how elements in the architecture change over time. Also, how at different points in time or different periods of time.                                                                                               |
| Traceability          | Tr | Describes the mapping between elements in the architecture. This can be between different viewpoints within domains as well as between domains. It can also be between structure and behaviors.                                |

## 7.2 Domain Interrelationships

Although the grid is the primary means of expressing the relationship between the Domains, Model Kinds and View Specifications, because of its two-dimensional nature it is not adequate to explain the abstract interrelationships that exist between the domains. The following diagram is an indication of how the domains are interrelated.



**Figure 7:2 - Domain Interrelationships**

Where a Domain is shown vertically the intent is to show that the Domain is a cross cutting concern that goes across the levels of abstraction in the architecture.

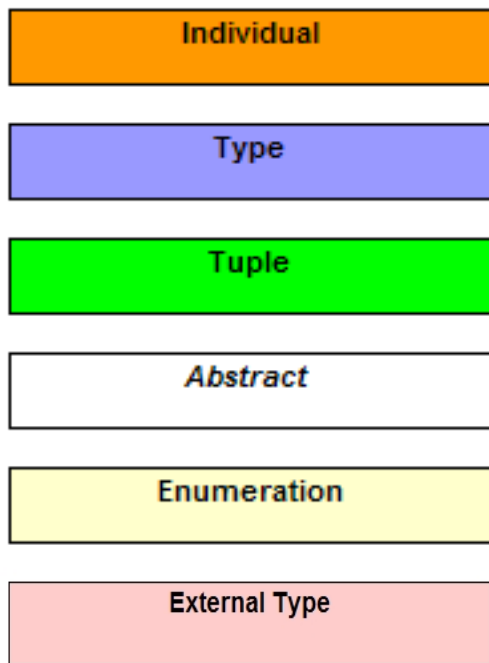
Where a Domain is shown horizontally the intent is to show that the Domain exists in a layer of abstraction between the Domains above and below it and there is an interrelationship with the Domains either side of it.

## 7.3 Domain Metamodel Diagram Legend

This Annex comprises of various diagrams that document the Domain Metamodel (DMM) that document the MoDAF 1.5 and MoDAF 1.2 integrated model. This model was used as a basis for creating the UPDM profile.

Note that the diagrams rely on color to aid the reader in understanding the model. Please refer to the legend below to understand the diagrams.

The following is the legend of element colors used in the DMM and what they denote.



**Figure 7:3 - Legend of color codes for element types defined in UAF**

The meaning of the element types in the UAF are based upon concepts put forth in the International Defence Enterprise Architecture Specification (IDEAS).

- An Individual denotes a single instance of an element.
- A Type denotes a set of Individuals.
- A Tuple denotes a relationship that exists between elements.
- An Abstract denotes that the element has no direct use but is a means of construction.
- An Enumeration is a complete, ordered listing of all the items in a collection.
- An External Type is an element that exists outside of the core DMM but is referenceable by elements in the DMM.

## 8 Domain Metamodel Diagrams

Note that the diagrams rely on color to aid the reader in understanding the model. Please refer to the legend in the various diagrams to understand the specific definitions.

### 8.1 View Specifications

This section documents each of the view specifications of UAF.

#### 8.1.1 View Specifications::Metadata

Stakeholders: Enterprise Architects, Technical Managers.

Concerns: architecture development process, architecture traceability, metamodel and its extensions, architecture versioning.

Definition: Identifies the metadata required to develop a suitable architecture that is fit for its purpose.

##### View Specifications::Metadata::Taxonomy

Stakeholders: Enterprise Architects, Technical Managers.

Concerns: metamodel and its extensions.

Definition: captures user defined metamodel extensions

Recommended Implementation: UML Profile Diagram, SysML Block Definition Diagram

##### View Specifications::Metadata::Structure

Stakeholders: Enterprise Architects, Technical Managers.

Concerns: domains, model kinds, and view specifications that are used to describe the architecture.

Definition: (i) lists predefined and custom domains, model kinds, and view specifications (ii) and identify the key stakeholders and their concerns.

Recommended Implementation: SysML Block Definition Diagram, SysML Package Diagram.

##### View Specifications::Metadata::Connectivity

Stakeholders: Enterprise Architects, people who want to understand relationships to related architectural descriptions, Technical Managers.

Concerns: high-level dependencies between architectural descriptions.

Definition: depicts and analyzes all relevant dependencies between architectural descriptions, e.g., reference architectures, as-is to to-be architectures.

Recommended Implementation: SysML Block Definition Diagram, SysML Package Diagram, matrix format.

##### View Specifications::Metadata::Processes

Stakeholders: Enterprise Architects, people who want to understand the architecture development process, Technical Managers.

Concerns: methodology used.

Definition: methodology used in developing the architecture.

Recommended Implementation: SysML Activity Diagram, text.

##### View Specifications::Metadata::States

Stakeholders: Enterprise Architects, people who want to understand the architecture governance, Technical Managers.

Concerns: architecture status.

Definition: captures version number and approval workflow of the architecture.

Recommended Implementation: SysML State Machine Diagram, state table, text.

### **View Specifications::Metadata::Constraints**

Stakeholders: Enterprise Architects, people who want to understand constraints for the architecture, Technical Managers.

Concerns: architectural constraints.

Definition: captures assumptions and constraints on the architecture.

Recommended Implementation: tabular format, text.

### **View Specifications::Metadata::Roadmap**

Stakeholders: Enterprise Architects, people who want to understand the architecture development plan, Technical Managers.

Concerns: architecture release schedule.

Definition: captures project timeline for the architecture.

Recommended Implementation: timeline, text.

### **View Specifications::Metadata::Traceability**

Stakeholders: Enterprise Architects, people who want to understand impact of change across the architecture supporting assets, Technical Managers.

Concerns: reuse of architectures.

Definition: shows references to asset libraries, legacy architectures, and external sources, e.g., documents.

Recommended Implementation: SysML Block Definition Diagram, SysML Package Diagram, tabular format.

## **8.1.2 View Specifications::Strategic**

Stakeholders: Capability Portfolio Managers.

Concerns: capability management process.

Definition: describe capability taxonomy, composition, dependencies and evolution.

### **View Specifications::Strategic::Taxonomy**

Contains the diagrams that document the Strategic Taxonomy Viewpoint.

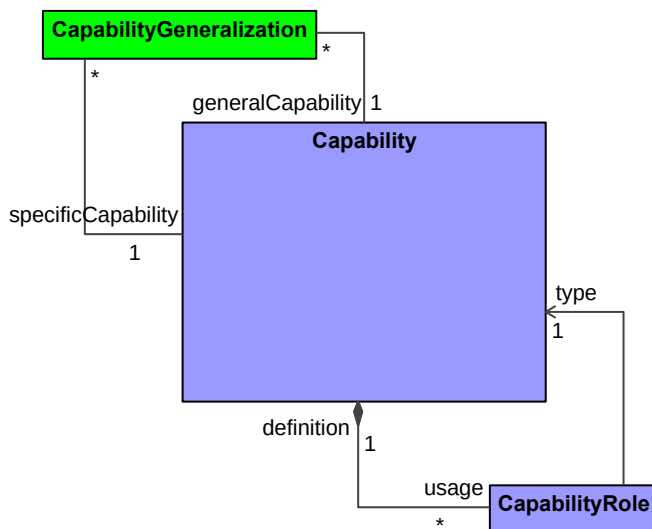
### **View Specifications::Strategic::Taxonomy::Strategic Taxonomy**

Stakeholders: PMs, Enterprise Architects, Executives.

Concerns: capability needs.

Definition: shows the taxonomy of capabilities.

Recommended Implementation: SysML Block Definition Diagram.



#### Elements

- [Capability](#)
- [CapabilityGeneralization](#)
- [CapabilityRole](#)

### View Specifications::Strategic::Structure

Contains the diagrams that document the Strategic Structure Viewpoint.

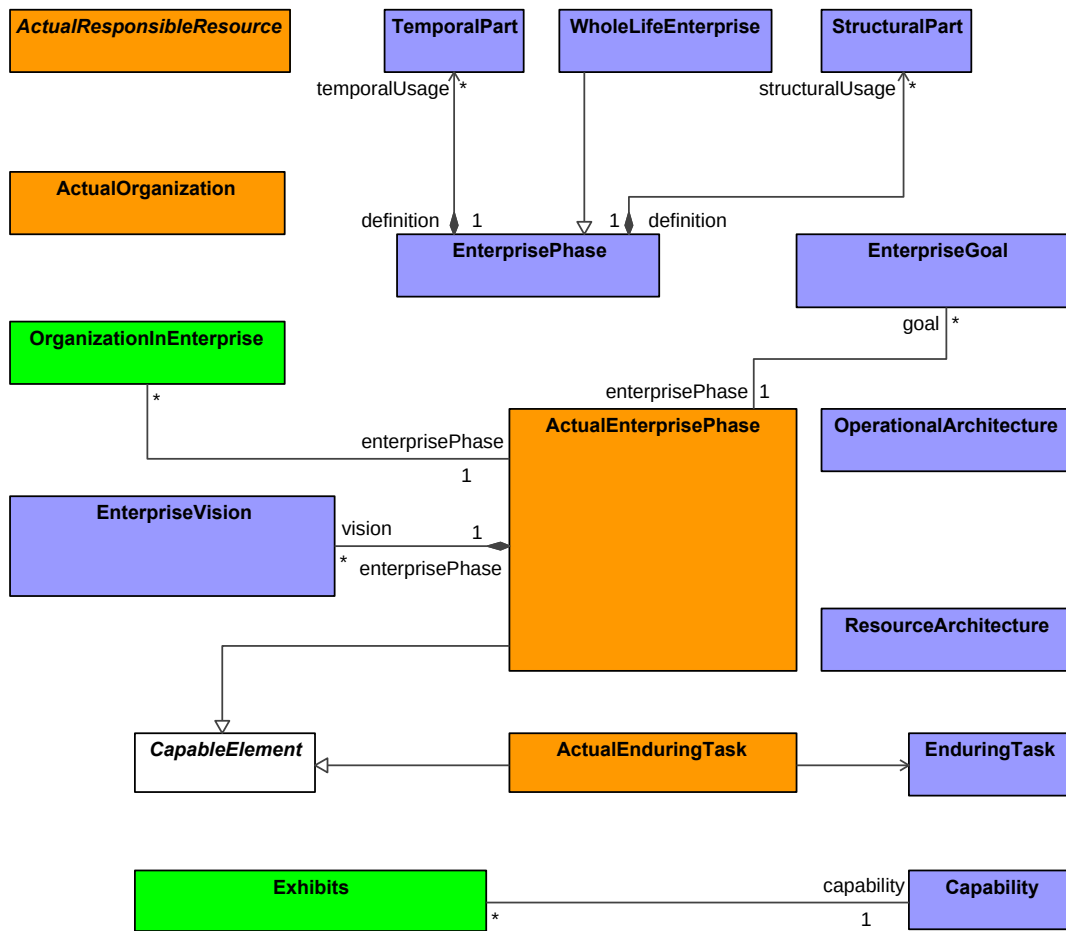
### View Specifications::Strategic::Structure::Strategic Structure

Stakeholders: PMs, Enterprise Architects, Executives.

Concerns: capability needs.

Definition: shows the relationship between EnterprisePhases and the Capabilities that are intended to be developed during the enterprise phases, and the organizations involved in the enterprise.

Recommended Implementation: SysML Block Definition Diagram.



**Figure 8:2 - Strategic Structure**

Elements

- [ActualEnduringTask](#)
- [ActualEnterprisePhase](#)
- [ActualOrganization](#)
- [ActualResponsibleResource](#)
- [Capability](#)
- [CapableElement](#)
- [EnduringTask](#)
- [EnterpriseGoal](#)
- [EnterprisePhase](#)
- [EnterpriseVision](#)
- [Exhibits](#)
- [OperationalArchitecture](#)
- [OrganizationInEnterprise](#)
- [ResourceArchitecture](#)
- [StructuralPart](#)
- [TemporalPart](#)
- [WholeLifeEnterprise](#)

## View Specifications::Strategic::Connectivity

Contains the diagrams that document the Strategic Connectivity Viewpoint.

### View Specifications::Strategic::Connectivity::Strategic Connectivity

Stakeholders: PMs, Executives, Enterprise Architects.

Concerns: capability dependencies.

Definition: describes the dependencies between planned capabilities.

Recommended Implementation: SysML Block Definition Diagram. SysML Internal Block Diagram.

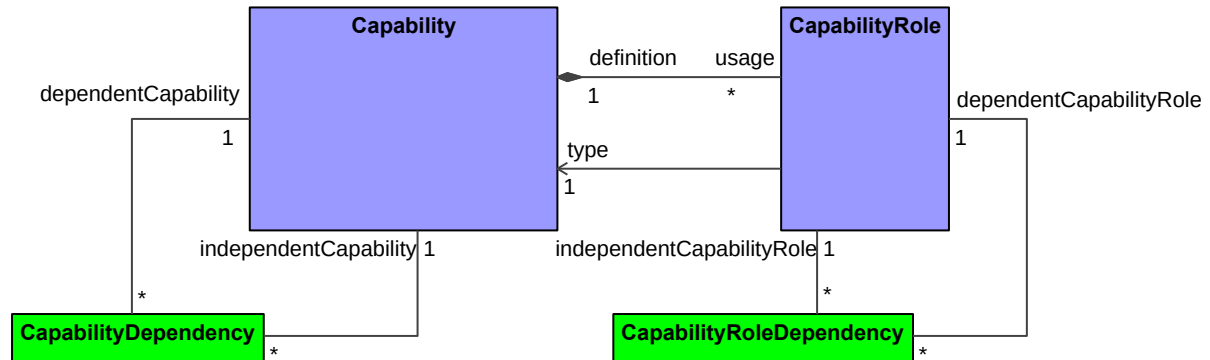


Figure 8:3 - Strategic Connectivity

Elements

- [Capability](#)
- [CapabilityDependency](#)
- [CapabilityRole](#)
- [CapabilityRoleDependency](#)

## View Specifications::Strategic::States

Contains the diagrams that document the Strategic States Viewpoint.

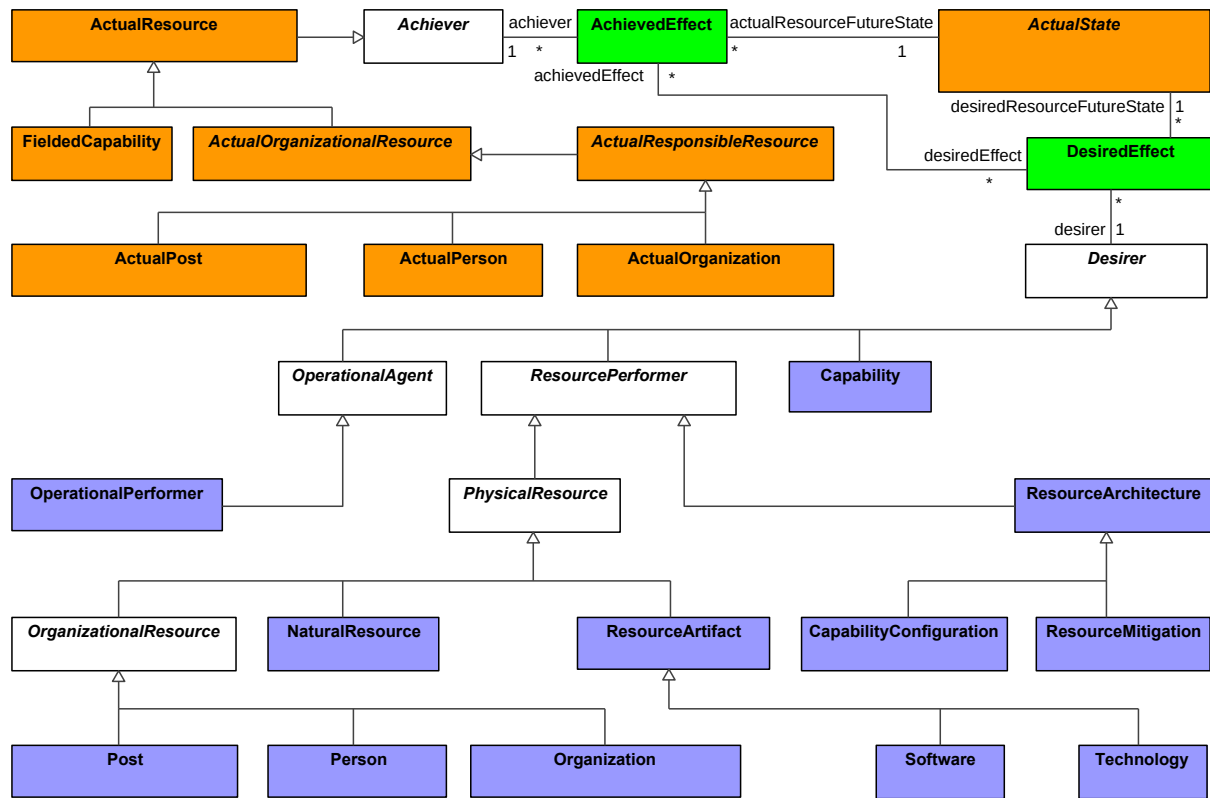
### View Specifications::Strategic::States::Strategic States

Stakeholders: PMs, Enterprise Architects.

Concerns: effects that the implementation(s) of capabilities are expected to deliver.

Definition: captures the relationships between capability(ies) and desired effect(s) that implementation(s) of capability(ies) should achieve.

Recommended Implementation: SysML Block Definition Diagram.



**Figure 8:4 - Strategic States**

Elements

- [AchievedEffect](#)
- [Achiever](#)
- [ActualOrganization](#)
- [ActualOrganizationalResource](#)
- [ActualPerson](#)
- [ActualPost](#)
- [ActualResource](#)
- [ActualResponsibleResource](#)
- [ActualState](#)
- [Capability](#)
- [CapabilityConfiguration](#)
- [DesiredEffect](#)
- [Desirer](#)
- [FieldedCapability](#)
- [NaturalResource](#)
- [OperationalAgent](#)
- [OperationalPerformer](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourceArchitecture](#)

- [ResourceArtifact](#)
- [ResourceMitigation](#)
- [ResourcePerformer](#)
- [Software](#)
- [Technology](#)

## View Specifications::Strategic::Constraints

Contains the diagrams that document the Strategic Constraints Viewpoint.

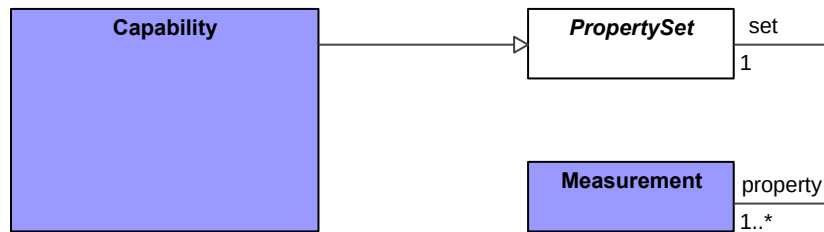
### View Specifications::Strategic::Constraints::Strategic Constraints

Stakeholders: PMs, Enterprise Architects.

Concerns: capability constraints.

Definition: details the measurements that set performance requirements constraining capabilities.

Recommended Implementation: tabular format, SysML Block Definition Diagram.



**Figure 8:5 - Strategic Constraints**

Elements

- [Capability](#)
- [Measurement](#)
- [PropertySet](#)

## View Specifications::Strategic::Roadmap

Contains the diagrams that document the Strategic Roadmap Viewpoint.

### View Specifications::Strategic::Roadmap::Deployment

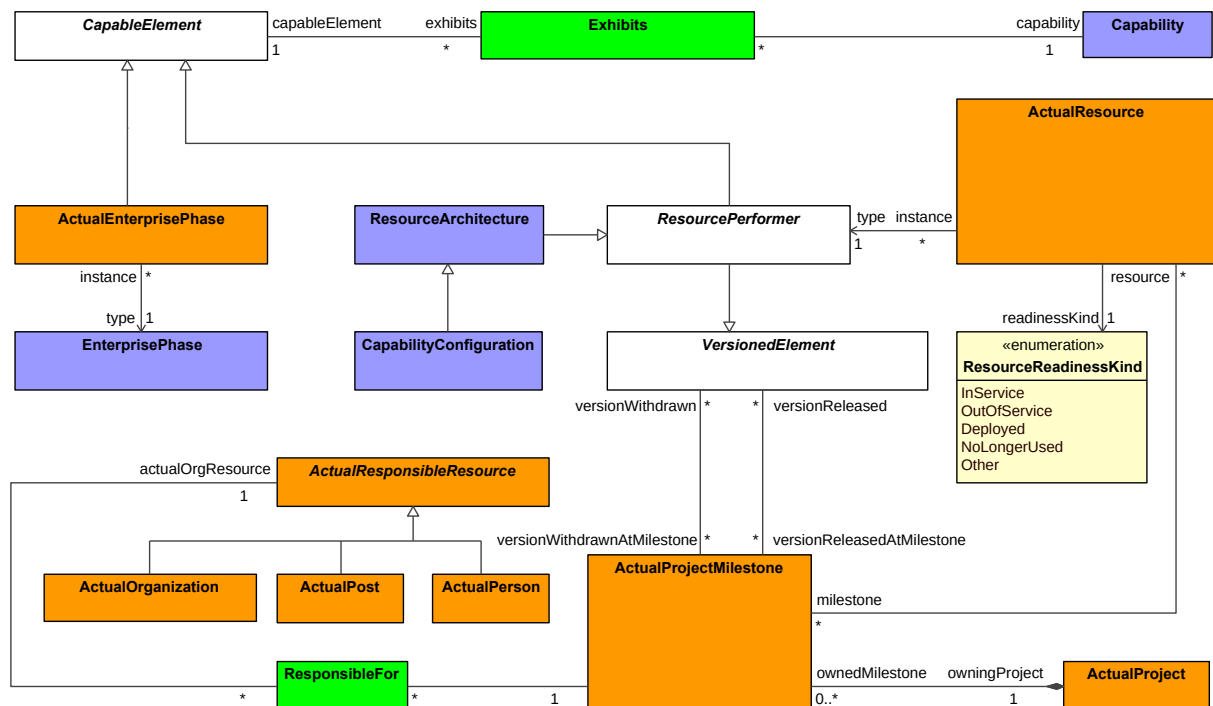
#### View Specifications::Strategic::Roadmap::Deployment::Strategic Roadmap: Deployment

Stakeholders: PMs, Executives, Enterprise Architects.

Concerns: capability deployment to organizations over time.

Definition: addresses the deployment of capability(ies) to actual organizations over time.

Recommended Implementation: timeline, tabular format, SysML Block Definition Diagram.



**Figure 8:6 - Strategic Roadmap: Deployment**

## Elements

- [ActualEnterprisePhase](#)
- [ActualOrganization](#)
- [ActualPerson](#)
- [ActualPost](#)
- [ActualProject](#)
- [ActualProjectMilestone](#)
- [ActualResource](#)
- [ActualResponsibleResource](#)
- [Capability](#)
- [CapabilityConfiguration](#)
- [CapableElement](#)
- [EnterprisePhase](#)
- [Exhibits](#)
- [ResourceArchitecture](#)
- [ResourcePerformer](#)
- [ResponsibleFor](#)
- [VersionedElement](#)

**View Specifications::Strategic::Roadmap::Phasing**

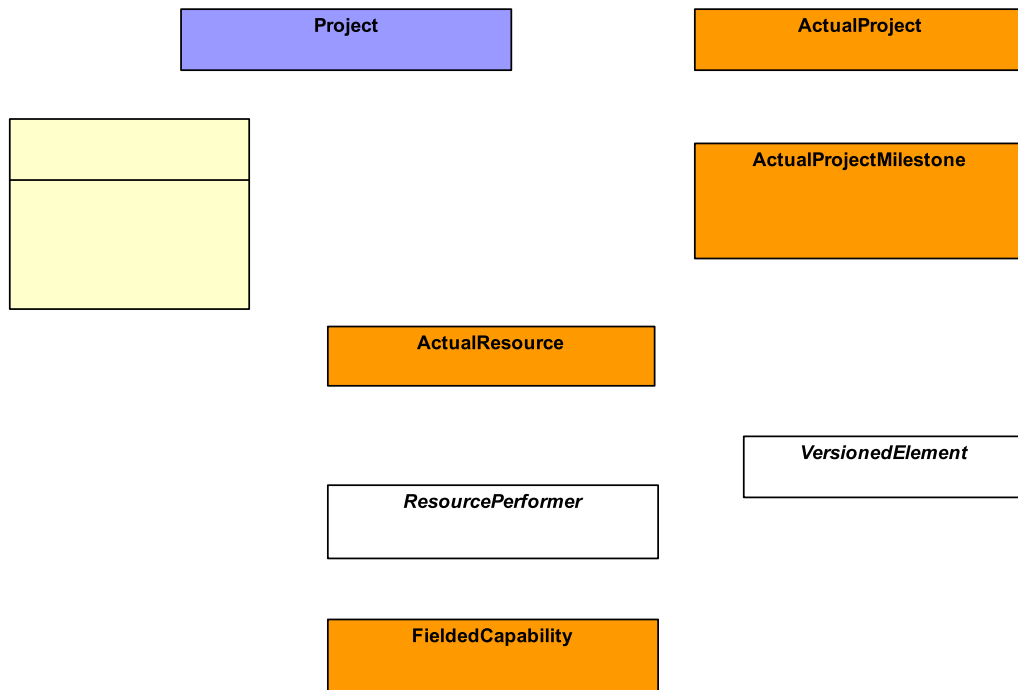
**View Specifications::Strategic::Roadmap::Phasing::Strategic Roadmap: Phasing**

Stakeholders: PMs, Executives, Enterprise Architects.

Concerns: capability(ies) achievement over time.

Definition: the planned achievement of capability(ies) at different points in time or during specific periods of time.

Recommended Implementation: timeline, tabular format, SysML Block Definition Diagram.



**Figure 8:7 - Strategic Roadmap: Phasing**

Elements

- [ActualProject](#)
- [ActualProjectMilestone](#)
- [ActualResource](#)
- [Capability](#)
- [CapabilityConfiguration](#)
- [CapableElement](#)
- [Exhibits](#)
- [FieldedCapability](#)
- [Project](#)
- [ResourceArchitecture](#)
- [ResourcePerformer](#)
- [VersionedElement](#)

## **View Specifications::Strategic::Traceability**

Contains the diagrams that document the Strategic Traceability Viewpoint.

### **View Specifications::Strategic::Traceability::Strategic Traceability**

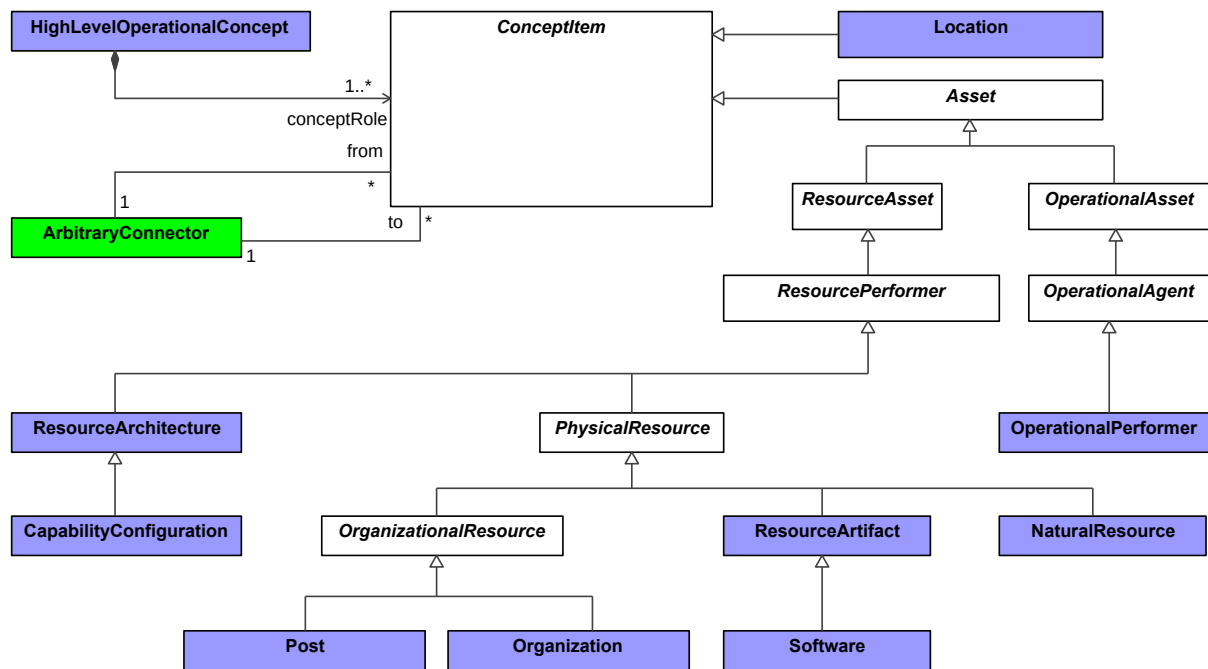
Stakeholders: PMs, Enterprise Architects, Business Architects.

Concerns: traceability between capabilities and operational activities.

Definition: describes the mapping between the capabilities required by an Enterprise and the supporting operational activities.

Recommended Implementation: matrix format, SysML Block Definition Diagram.





**Figure 8:9 - Operational Taxonomy**

Elements

- [ArbitraryConnector](#)
- [Asset](#)
- [CapabilityConfiguration](#)
- [ConceptItem](#)
- [HighLevelOperationalConcept](#)
- [Location](#)
- [NaturalResource](#)
- [OperationalAgent](#)
- [OperationalAsset](#)
- [OperationalPerformer](#)
- [Organization](#)
- [OrganizationalResource](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourceAsset](#)
- [ResourcePerformer](#)
- [Software](#)

## View Specifications::Operational::Structure

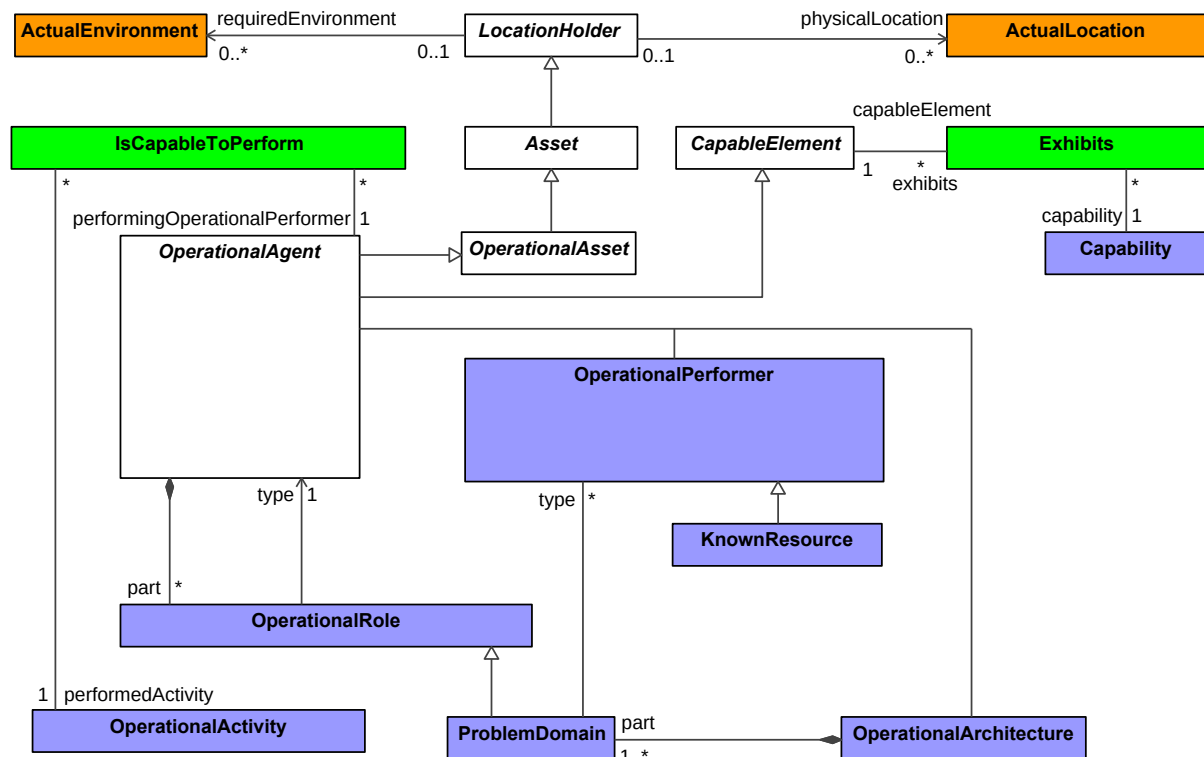
Contains the diagrams that document the Operational Structure Viewpoint.

### View Specifications::Operational::Structure::Operational Structure

Stakeholders: Business Architects, Systems Engineers, Enterprise Architects, Owners responsible for Operational Agents.

Concerns: identifies the operational exchange requirements between nodes.

Definition: defines operational architecture and exchange requirements necessary to support a specific set of Capability(ies).  
Recommended Implementation: SysML Block Definition Diagram, SysML Internal Block Diagram.



**Figure 8:10 - Operational Structure**

Elements

- [ActualEnvironment](#)
- [ActualLocation](#)
- [Asset](#)
- [Capability](#)
- [CapableElement](#)
- [Exhibits](#)
- [IsCapableToPerform](#)
- [KnownResource](#)
- [LocationHolder](#)
- [OperationalActivity](#)
- [OperationalAgent](#)
- [OperationalArchitecture](#)
- [OperationalAsset](#)
- [OperationalPerformer](#)
- [OperationalRole](#)
- [ProblemDomain](#)

## View Specifications::Operational::Connectivity

Contains the diagrams that document the Operational Connectivity Viewpoint.

## View Specifications::Operational::Connectivity::Operational Connectivity

Stakeholders: Systems Engineers, Architects, Solution Providers.

Concerns: capture the interfaces between OperationalPerformers.

Definition: summarizes logical exchanges between OperationalPerformers of information, systems, personnel, energy etc. and the logical activities that produce and consume them. Measurements can optionally be included.

Recommended Implementation: SysML Internal Block Diagram, tabular format.

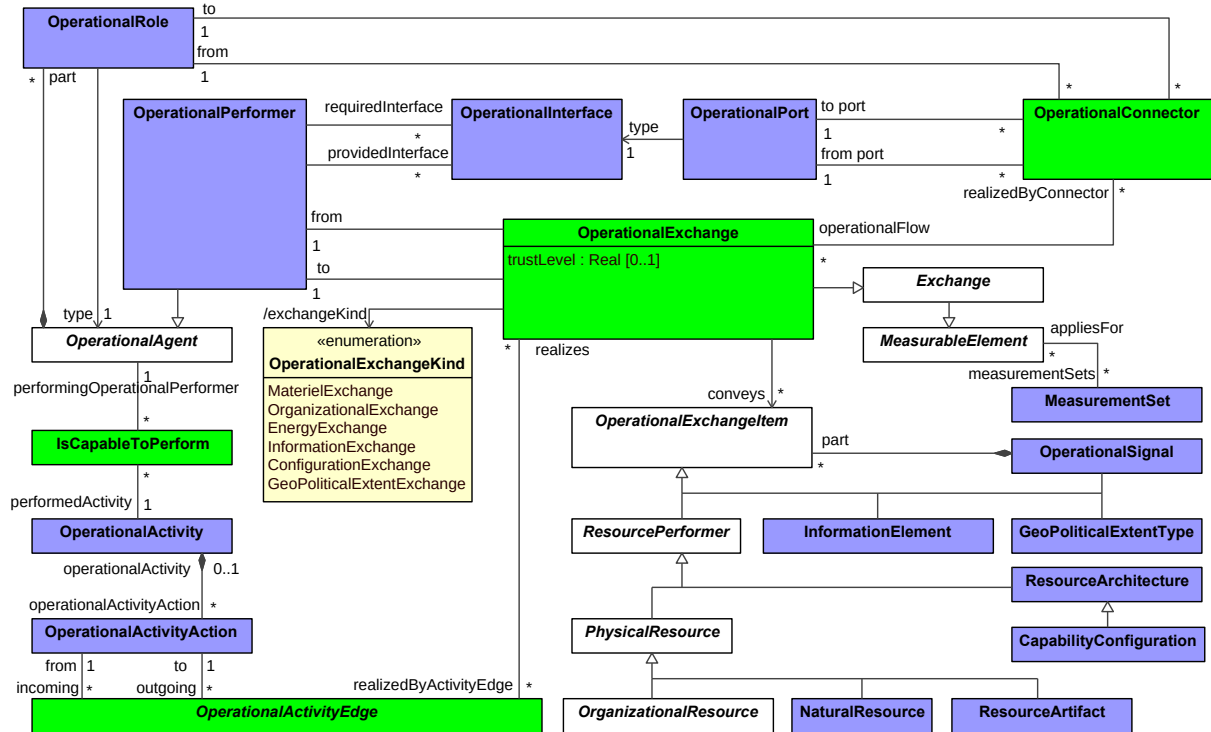


Figure 8:11 - Operational Connectivity

Elements

- [CapabilityConfiguration](#)
- [Exchange](#)
- [GeoPoliticalExtentType](#)
- [InformationElement](#)
- [IsCapableToPerform](#)
- [MeasurableElement](#)
- [MeasurementSet](#)
- [NaturalResource](#)
- [OperationalActivity](#)
- [OperationalActivityAction](#)
- [OperationalActivityEdge](#)
- [OperationalAgent](#)
- [OperationalConnector](#)
- [OperationalExchange](#)
- [OperationalExchangeItem](#)
- [OperationalInterface](#)
- [OperationalPerformer](#)
- [OperationalPort](#)



## Elements

- [ActivityPerformableUnderCondition](#)
- [ActualCondition](#)
- [ActualMeasurementSet](#)
- [ActualService](#)
- [IsCapableToPerform](#)
- [MeasurableElement](#)
- [OperationalActivity](#)
- [OperationalActivityAction](#)
- [OperationalActivityEdge](#)
- [OperationalAgent](#)
- [OperationalExchange](#)
- [OperationalExchangeItem](#)
- [OperationalMethod](#)
- [OperationalParameter](#)
- [OperationalPerformer](#)
- [OperationalRole](#)
- [PerformsInContext](#)
- [Process](#)
- [ProcessEdge](#)
- [ProcessOperation](#)
- [ProcessParameter](#)
- [ProcessUsage](#)
- [RequiredServiceLevel](#)
- [ServiceSpecification](#)
- [StandardOperationalActivity](#)
- UML2.5Metamodel::Activity
- UML2.5Metamodel::ActivityEdge
- UML2.5Metamodel::CallBehaviorAction
- UML2.5Metamodel::Operation
- UML2.5Metamodel::Parameter

## **View Specifications::Operational::Processes::Operational Processes BPMN Semantics**

Stakeholders: Business Architect, Enterprise Architects.

Concerns: captures activity based behavior and flows using BPMN notation.

Definition: describes the BPMN processes that are normally conducted in the course of achieving business goals that support a capability. It describes operational activities, their Inputs/Outputs, operational activity actions and flows between them using BPMN notation.

Recommended Implementation: BPMN Process Diagram.



- [OperationalAgent](#)
- [OperationalExchange](#)
- [OperationalExchangeItem](#)
- [OperationalMethod](#)
- [OperationalParameter](#)
- [OperationalPerformer](#)
- [OperationalRole](#)
- [PerformsInContext](#)
- [Process](#)
- [ProcessEdge](#)
- [ProcessUsage](#)
- [RequiredServiceLevel](#)
- [ServiceSpecification](#)
- [StandardOperationalActivity](#)

## View Specifications::Operational::States

Contains the diagrams that document the Operational States Viewpoint.

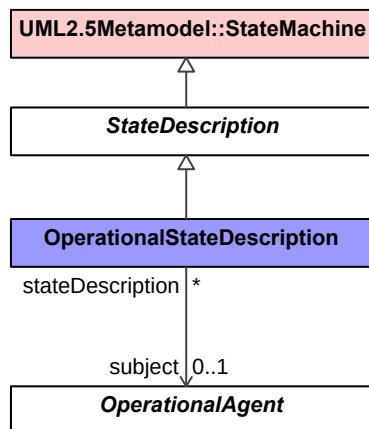
### View Specifications::Operational::States::Operational States

Stakeholders: Systems Engineers, Software Engineers.

Concerns: capture state-based behavior of an operational OperationalPerformer.

Definition: it is a graphical representation of states of an operational OperationalPerformer and how that operational OperationalPerformer responds to various events and actions.

Recommended Implementation: SysML State Machine Diagram.



**Figure 8:14 - Operational States**

Elements

- [OperationalAgent](#)
- [OperationalStateDescription](#)
- [StateDescription](#)
- UML2.5Metamodel::StateMachine

## View Specifications::Operational::Interaction Scenarios

Contains the diagrams that document the Operational Interaction Scenarios Viewpoint.



## View Specifications::Operational::Constraints

Contains the diagrams that document the Operational Constraints Viewpoint.

### View Specifications::Operational::Constraints::Operational Constraints

Stakeholders: Systems Engineers, Architects, Program Sponsors

Concerns: define operational limitations, constraints and performance parameters for the enterprise.

Definition: specifies traditional textual operational or business rules that are constraints on the way that business is done in the enterprise. The addition of SysML parametrics provides a computational means of defining operational constraints across the enterprise or within a specific operational context.

Recommended Implementation: tabular format, SysML Block Definition Diagram, SysML Parametric Diagram.

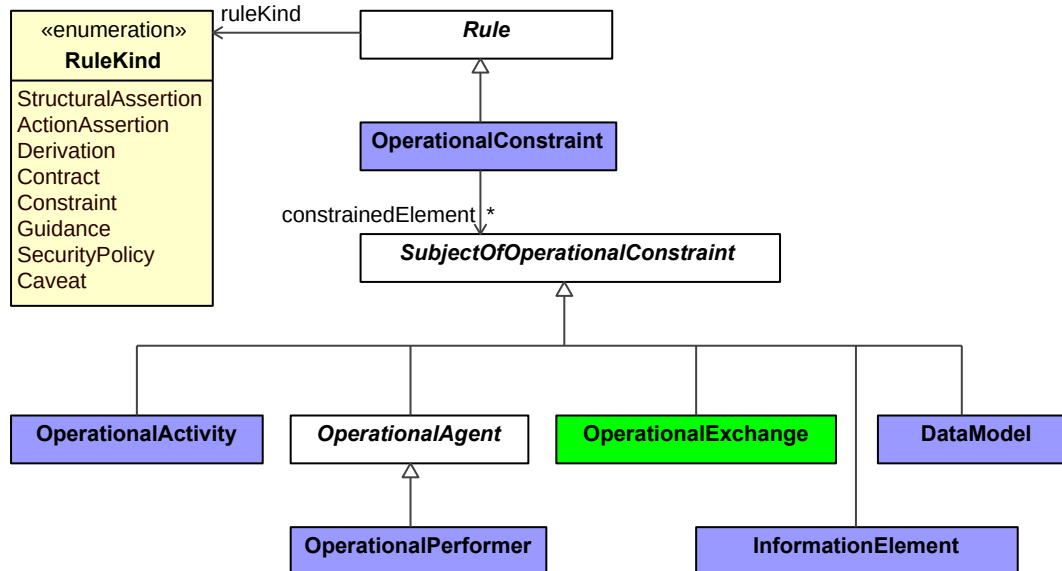


Figure 8:16 - Operational Constraints

Elements

- [DataModel](#)
- [InformationElement](#)
- [OperationalActivity](#)
- [OperationalAgent](#)
- [OperationalConstraint](#)
- [OperationalExchange](#)
- [OperationalPerformer](#)
- [Rule](#)
- [SubjectOfOperationalConstraint](#)

## View Specifications::Operational::Traceability

Contains the diagrams that document the Operational Traceability Viewpoint.

### View Specifications::Operational::Traceability::Operational Traceability

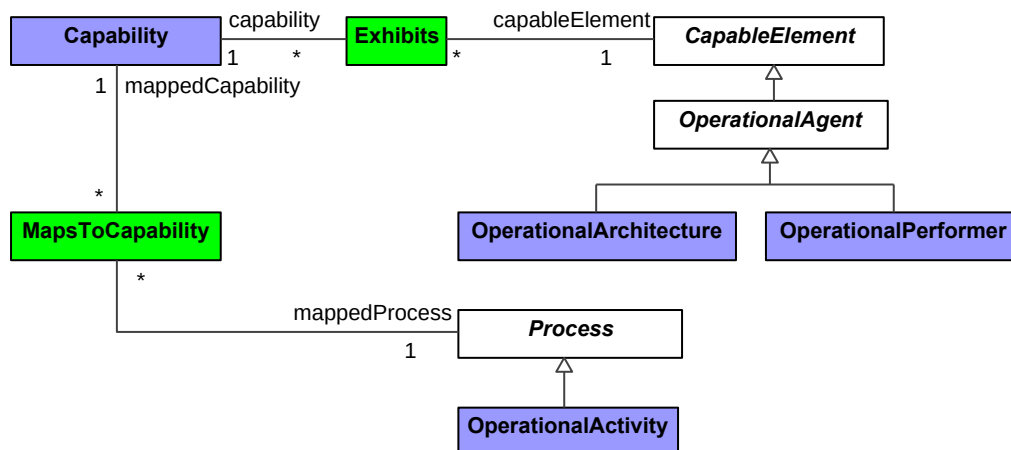
Stakeholders: PMs, Enterprise Architects, Business Architects.

Concerns: traceability between capabilities and operational activities and capabilities and operational agents.

Definition: describes the mapping between the capabilities required by an Enterprise and the supporting operational

activities and operational agents.

Recommended Implementation: matrix format, SysML Block Definition Diagram.



**Figure 8:17 - Operational Traceability**

Elements

- [Capability](#)
- [CapableElement](#)
- [Exhibits](#)
- [MapsToCapability](#)
- [OperationalActivity](#)
- [OperationalAgent](#)
- [OperationalArchitecture](#)
- [OperationalPerformer](#)
- [Process](#)

## 8.1.4 View Specifications::Services

Stakeholders: Enterprise Architects, Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: specifications of services required to exhibit a Capability.

Definition: shows Service Specifications and required and provided service levels of these specifications required to exhibit a Capability or to support an Operational Activity.

### View Specifications::Services::Taxonomy

Contains the diagrams that document the Services Taxonomy Viewpoint.

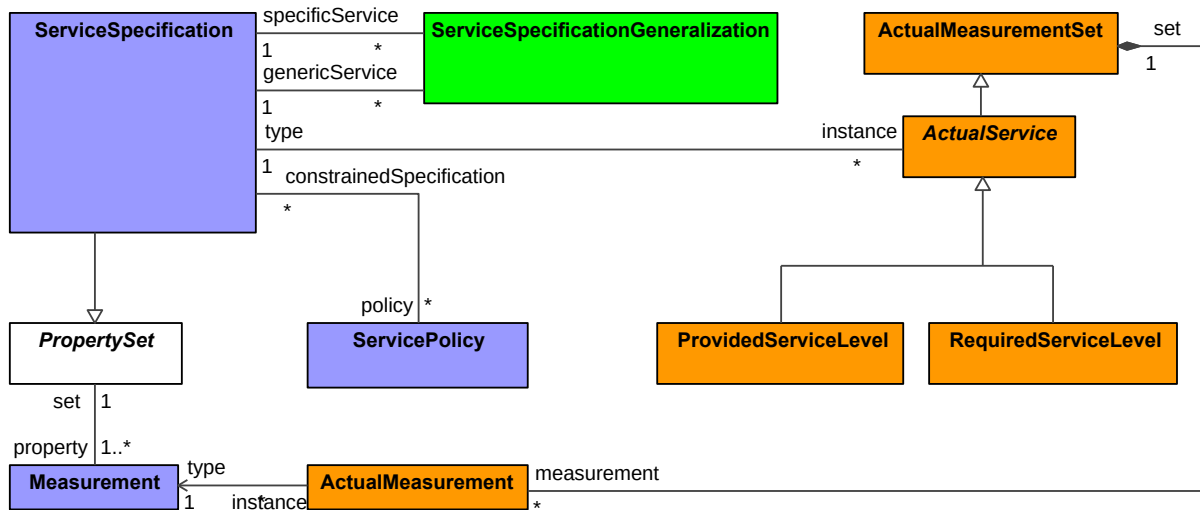
### View Specifications::Services::Taxonomy::Services Taxonomy

Stakeholders: Enterprise Architects, Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: service specification types and required and provided service levels of these types.

Definition: shows the taxonomy of types of services and the level of service that they are expected to provide or are required to meet through the display of ActualMeasurements associated with the Provided and Required Service Level.

Recommended Implementation: SysML Block Definition Diagram.



**Figure 8:18 - Services Taxonomy**

Elements

- [ActualMeasurement](#)
- [ActualMeasurementSet](#)
- [ActualService](#)
- [Measurement](#)
- [PropertySet](#)
- [ProvidedServiceLevel](#)
- [RequiredServiceLevel](#)
- [ServicePolicy](#)
- [ServiceSpecification](#)
- [ServiceSpecificationGeneralization](#)

## View Specifications::Services::Structure

Contains the diagrams that document the Services Structure Viewpoint.

### View Specifications::Services::Structure::Services Structure

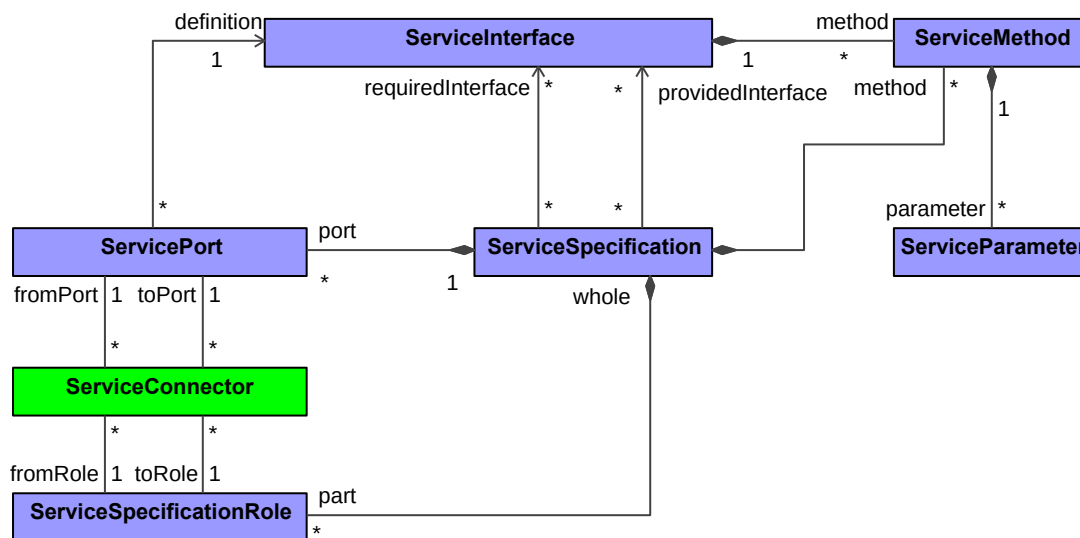
Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: combination of services required to exhibit a capability.

Definition: shows the composition of services and how services are combined into a higher level service required to exhibit a capability or support an operational activity.

Recommended Implementation: SysML Block Definition Diagram, SysML Internal Block Diagram.





**Figure 8:20 - Services Connectivity**

Elements

- [ServiceConnector](#)
- [ServiceInterface](#)
- [ServiceMethod](#)
- [ServiceParameter](#)
- [ServicePort](#)
- [ServiceSpecification](#)
- [ServiceSpecificationRole](#)

## View Specifications::Services::Processes

Contains the diagrams that document the Services Processes Viewpoint.

### View Specifications::Services::Processes::Services Processes

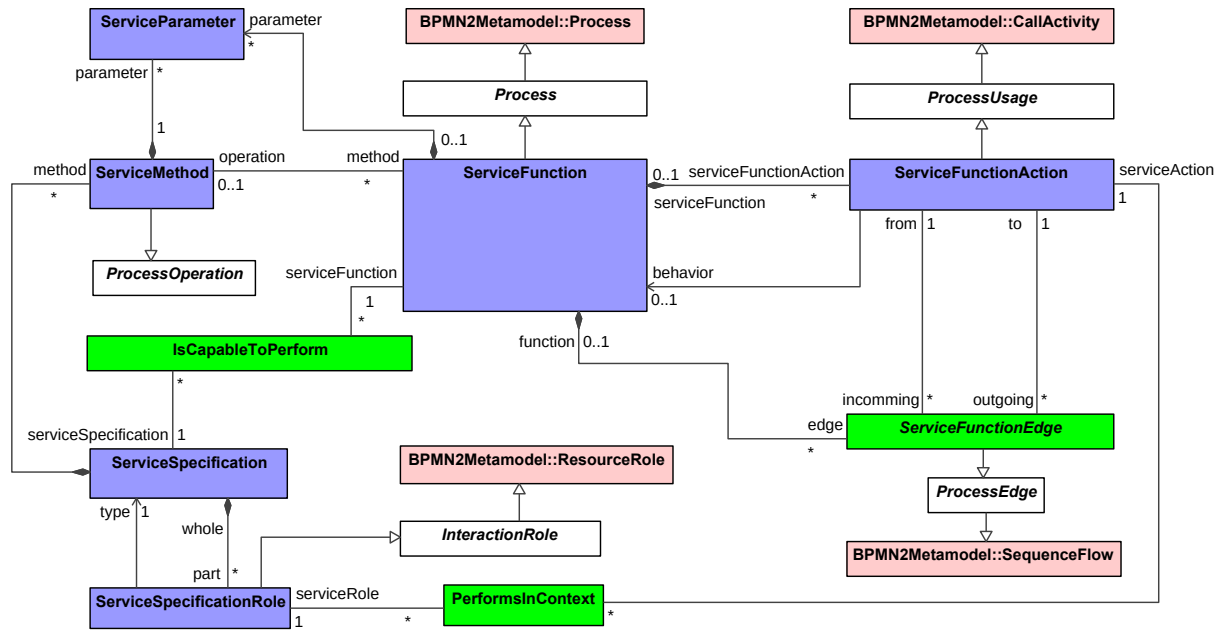
Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: the behavior of a service in terms of the operational activities it is expected to support.

Definition: provides detailed information regarding the allocation of service functions to service specifications, and data flows between service functions.

Recommended Implementation: SysML Activity Diagram, SysML Block Definition Diagram.





**Figure 8:22 - Services Processes BPMN Semantics**

Elements

- BPMN2Metamodel::CallActivity
- BPMN2Metamodel::Process
- BPMN2Metamodel::ResourceRole
- BPMN2Metamodel::SequenceFlow
- [InteractionRole](#)
- [IsCapableToPerform](#)
- [PerformsInContext](#)
- [Process](#)
- [ProcessEdge](#)
- [ProcessOperation](#)
- [ProcessUsage](#)
- [ServiceFunction](#)
- [ServiceFunctionAction](#)
- [ServiceFunctionEdge](#)
- [ServiceMethod](#)
- [ServiceParameter](#)
- [ServiceSpecification](#)
- [ServiceSpecificationRole](#)

## View Specifications::Services::States

Contains the diagrams that document the Services States Viewpoint.

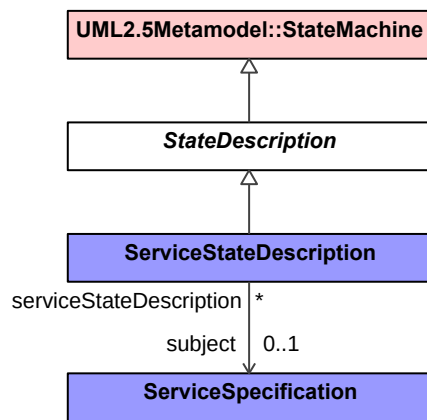
### View Specifications::Services::States::Services States

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: the behavior of a service specification in terms of states and events causing transitions between states.

Definition: specifies the possible states a service specification may have, and the possible transitions between those states.

Recommended Implementation: SysML State Machine Diagram.



**Figure 8:23 - Services States**

Elements

- [ServiceSpecification](#)
- [ServiceStateDescription](#)
- [StateDescription](#)
- UML2.5Metamodel::StateMachine

## View Specifications::Services::Interaction Scenarios

Contains the diagrams that document the Services Interaction Scenarios Viewpoint.

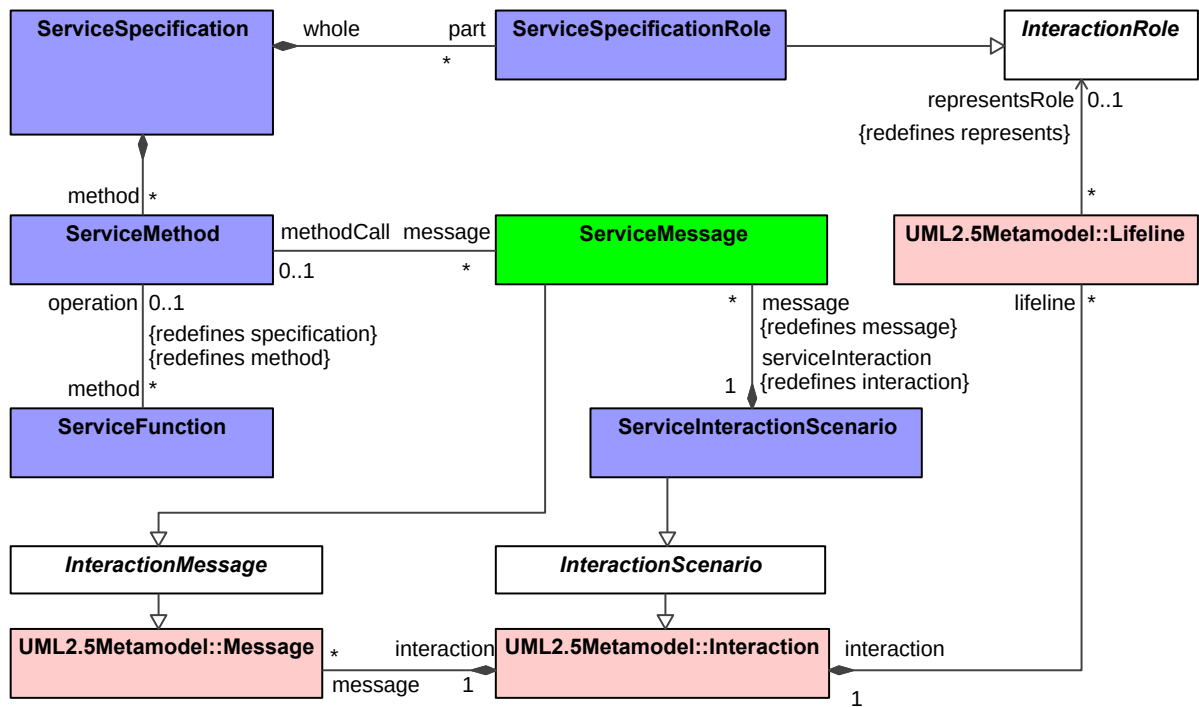
### View Specifications::Services::Interaction Scenarios::Services Interaction Scenarios

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: the behavior of a service specification in terms of expected time-ordered examination of the interactions between service roles.

Definition: specifies how a service roles interact with each other, service providers and consumers, and the sequence and dependencies of those interactions.

Recommended Implementation: SysML Sequence Diagram.



**Figure 8:24 - Services Interaction Scenarios**

Elements

- [InteractionMessage](#)
- [InteractionRole](#)
- [InteractionScenario](#)
- [ServiceFunction](#)
- [ServiceInteractionScenario](#)
- [ServiceMessage](#)
- [ServiceMethod](#)
- [ServiceSpecification](#)
- [ServiceSpecificationRole](#)
- UML2.5Metamodel::Interaction
- UML2.5Metamodel::Lifeline
- UML2.5Metamodel::Message

## View Specifications::Services::Constraints

Contains the diagrams that document the Services Constraints Viewpoint.

### View Specifications::Services::Constraints::Services Constraints

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: service policies that apply to implementations of service specifications.

Definition: specifies traditional textual service policies that are constraints on the way that service specifications are implemented within resources. The addition of SysML parametrics provide a computational means of defining service policies across the enterprise or within a specific service configuration.

Recommended Implementation: tabular format, SysML Parametric Diagram.



**Figure 8:25 - Services Constraints**

Elements

- [Rule](#)
- [ServicePolicy](#)
- [ServiceSpecification](#)

## View Specifications::Services::Roadmap

Contains the diagrams that document the Services Roadmap Viewpoint.

### View Specifications::Services::Roadmap::Services Roadmap

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: service specification changes over time.

Definition: provides an overview of how a service specification changes over time. It shows the combination of several service specifications mapped against a timeline.

Recommended Implementation: timeline, SysML Block Definition Diagram, SysML Internal Block Diagram.



**Figure 8:26 - Services Roadmap**

Elements

- [ActualProject](#)
- [ActualProjectMilestone](#)
- [MilestoneDependency](#)
- [ServiceSpecification](#)

- [VersionedElement](#)
- [VersionOfConfiguration](#)
- [VersionSuccession](#)
- [WholeLifeConfiguration](#)

## View Specifications::Services::Traceability

Contains the diagrams that document the Services Traceability Viewpoint.

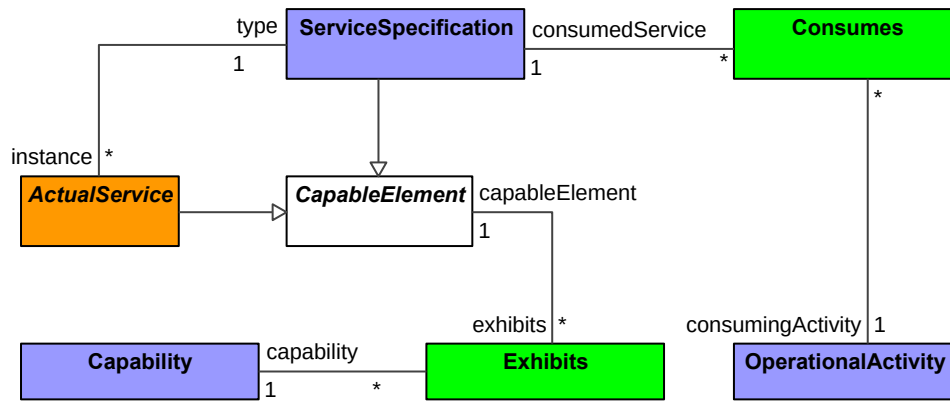
### View Specifications::Services::Traceability::Services Traceability

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: traceability between operational activities and service specifications that support them.

Definition: depicts the mapping of service specifications to operational activities and how service specifications contribute to the achievement of a capability.

Recommended Implementation: tabular or matrix format.



**Figure 8:27 - Services Traceability**

Elements

- [ActualService](#)
- [Capability](#)
- [CapableElement](#)
- [Consumes](#)
- [Exhibits](#)
- [OperationalActivity](#)
- [ServiceSpecification](#)

## 8.1.5 View Specifications::Personnel

Stakeholders: Human resources, Solution Providers, PMs.

Concerns: human factors.

Definition: aims to clarify the role of Human Factors (HF) when creating architectures in order to facilitate both Human Factors Integration (HFI) and systems engineering (SE).

### View Specifications::Personnel::Taxonomy

Contains the diagrams that document the Personnel Taxonomy Viewpoint.

## View Specifications::Personnel::Taxonomy::Personnel Taxonomy

Stakeholders: Human resources, Solution Providers, PMs.

Concerns: organizational resource types.

Definition: shows the taxonomy of types of organizational resources.

Recommended Implementation: SysML Block Definition Diagram.

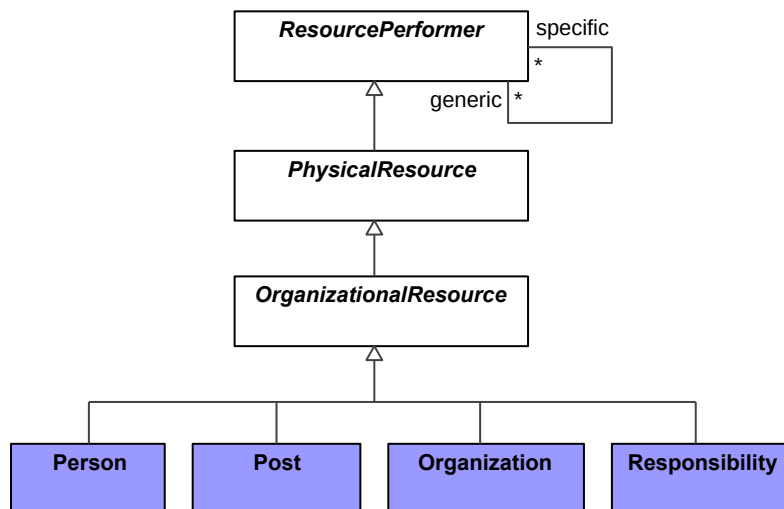


Figure 8:28 - Personnel Taxonomy

Elements

- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourcePerformer](#)
- [Responsibility](#)

## View Specifications::Personnel::Structure

Contains the diagrams that document the Personnel Structure Viewpoint.

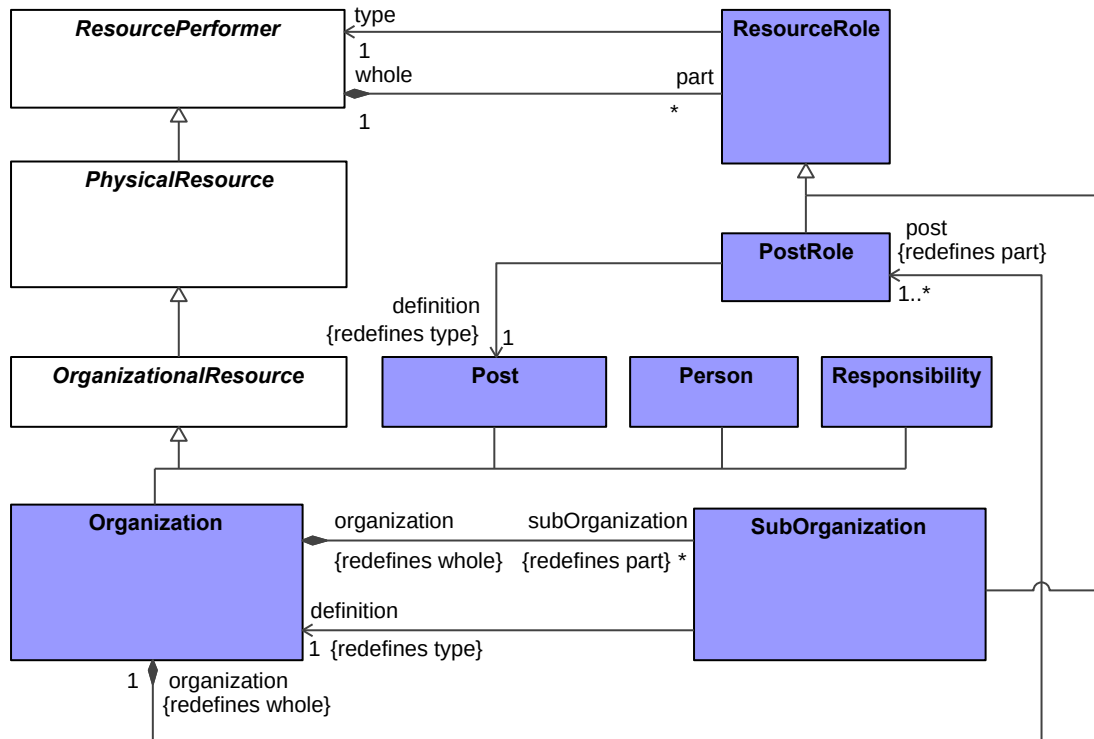
## View Specifications::Personnel::Structure::Personnel Structure

Stakeholders: Human resources, Solution Providers, PMs.

Concerns: typical organizational structure used to support a capability(ies).

Definition: shows organizational structures and possible interactions between organizational resources.

Recommended Implementation: SysML Block Definition Diagram, SysML Internal Block Diagram.



**Figure 8:29 - Personnel Structure**

Elements

- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [PostRole](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [Responsibility](#)
- [SubOrganization](#)

## View Specifications::Personnel::Connectivity

Contains the diagrams that document the Personnel Connectivity Viewpoint.

### View Specifications::Personnel::Connectivity::Personnel Connectivity

Stakeholders: Solution providers.

Concerns: interaction of organizational resources.

Definition: captures the possible interactions between organizational resources, including command and control relationships. Interactions typically illustrate the fundamental roles and management responsibilities.

Recommended Implementation: tabular format.



## View Specifications::Personnel::Processes

Contains the diagrams that document the Personnel Processes Viewpoint.

### View Specifications::Personnel::Processes::Personnel Processes

Stakeholders: Systems engineers, Solution providers.

Concerns: functions that have to be carried out by organizational resources.

Definition: specifies organizational resource functions in relation to resource definitions.

Recommended Implementation: SysML Activity Diagram, SysML Block Definition Diagram, BPMN Process Diagram as described in the Resources Processes section.

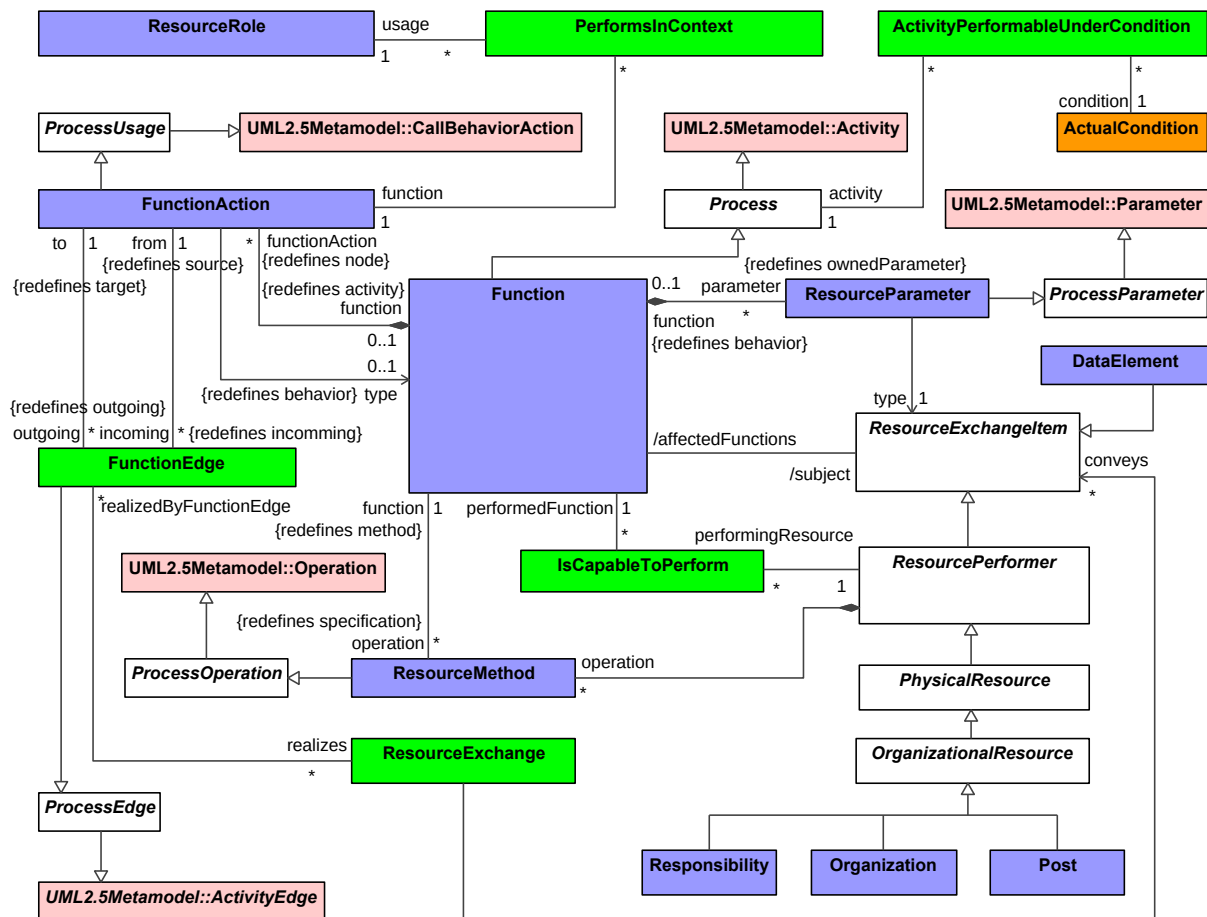


Figure 8:31 - Personnel Processes

Elements

- [ActivityPerformableUnderCondition](#)
- [ActualCondition](#)
- [DataElement](#)
- [Function](#)
- [FunctionAction](#)
- [FunctionEdge](#)
- [IsCapableToPerform](#)
- [Organization](#)
- [OrganizationalResource](#)

- [PerformsInContext](#)
- [PhysicalResource](#)
- [Post](#)
- [Process](#)
- [ProcessEdge](#)
- [ProcessOperation](#)
- [ProcessParameter](#)
- [ProcessUsage](#)
- [ResourceExchange](#)
- [ResourceExchangeItem](#)
- [ResourceMethod](#)
- [ResourceParameter](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [Responsibility](#)
- UML2.5Metamodel::Activity
- UML2.5Metamodel::ActivityEdge
- UML2.5Metamodel::CallBehaviorAction
- UML2.5Metamodel::Operation
- UML2.5Metamodel::Parameter

## View Specifications::Personnel::States

Contains the diagrams that document the Personnel States Viewpoint.

### View Specifications::Personnel::States::Personnel States

Stakeholders: Systems Engineers, Software Engineers.

Concerns: capture state-based behavior of an organizational resource.

Definition: it is a graphical representation of states of an organizational resource and how that organizational resource responds to various events and actions.

Recommended Implementation: SysML State Machine Diagram.

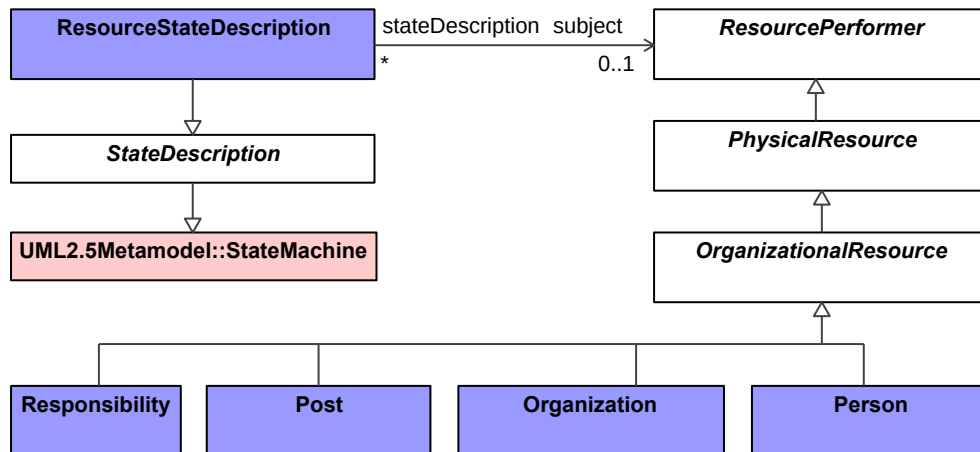


Figure 8:32 - Personnel States

- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourcePerformer](#)
- [ResourceStateDescription](#)
- [Responsibility](#)
- [StateDescription](#)
- UML2.5Metamodel::StateMachine

Contains the diagrams that document the Personnel Interaction Scenarios Viewpoint.

Stakeholders: Software Engineers, Systems Engineers.  
Concerns: interactions between organizational resources (roles).  
Definition: provides a time-ordered examination of the interactions between organizational resources.  
Recommended Implementation: SysML Sequence Diagram, BPMN Collaboration Diagram.



## Elements

- [DataElement](#)
- [Function](#)
- [InteractionMessage](#)
- [InteractionRole](#)
- [InteractionScenario](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [Process](#)
- [ResourceExchange](#)
- [ResourceExchangeItem](#)
- [ResourceInteractionScenario](#)
- [ResourceMessage](#)
- [ResourceMethod](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [Responsibility](#)
- UML2.5Metamodel::Interaction
- UML2.5Metamodel::Lifeline
- UML2.5Metamodel::Message

## View Specifications::Personnel::Constraints

Contains the diagrams that document the Personnel Constraints Viewpoint.

## View Specifications::Personnel::Constraints::Personnel Constraints: Competence

Stakeholders: Systems engineers, Solution providers.

Concerns: allocation of competencies to actual posts.

Definition: specifies requirements for actual organizational resources – by linking competencies and actual posts.

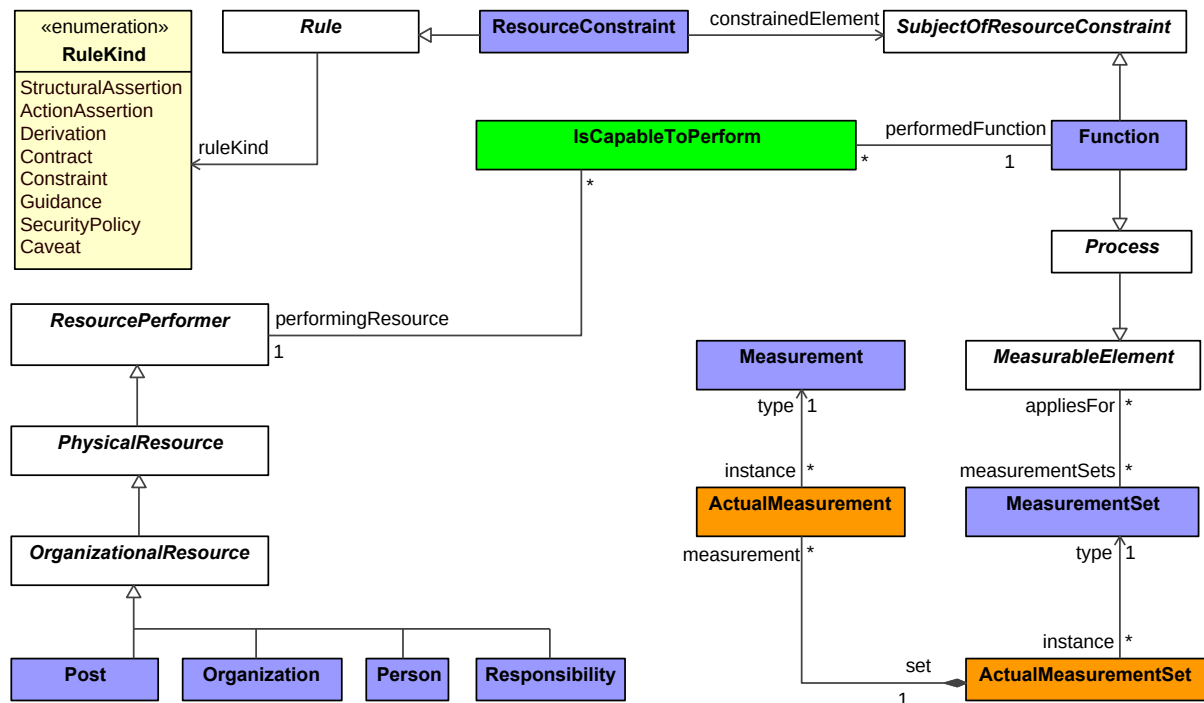
Recommended Implementation: SysML Block Definition Diagram.

**Figure 8:34 - Personnel Constraints: Competence**

Elements

- [ActualOrganization](#)
- [ActualOrganizationalResource](#)
- [ActualPerson](#)
- [ActualPost](#)
- [ActualResponsibility](#)
- [ActualResponsibleResource](#)
- [Competence](#)
- [CompetenceForRole](#)
- [CompetenceToConduct](#)
- [Function](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ProvidesCompetence](#)
- [RequiresCompetence](#)
- [ResourcePerformerfor](#)

performance predictions at the level of actual persons and actual organizations. It creates a bridge between static architectural definitions and behavior predictions through executable models.  
Recommended Implementation: tabular format, SysML Parametric Diagram, SysML Block Definition Diagram.



**Figure 8:35 - Personnel Constraints: Drivers**

Elements

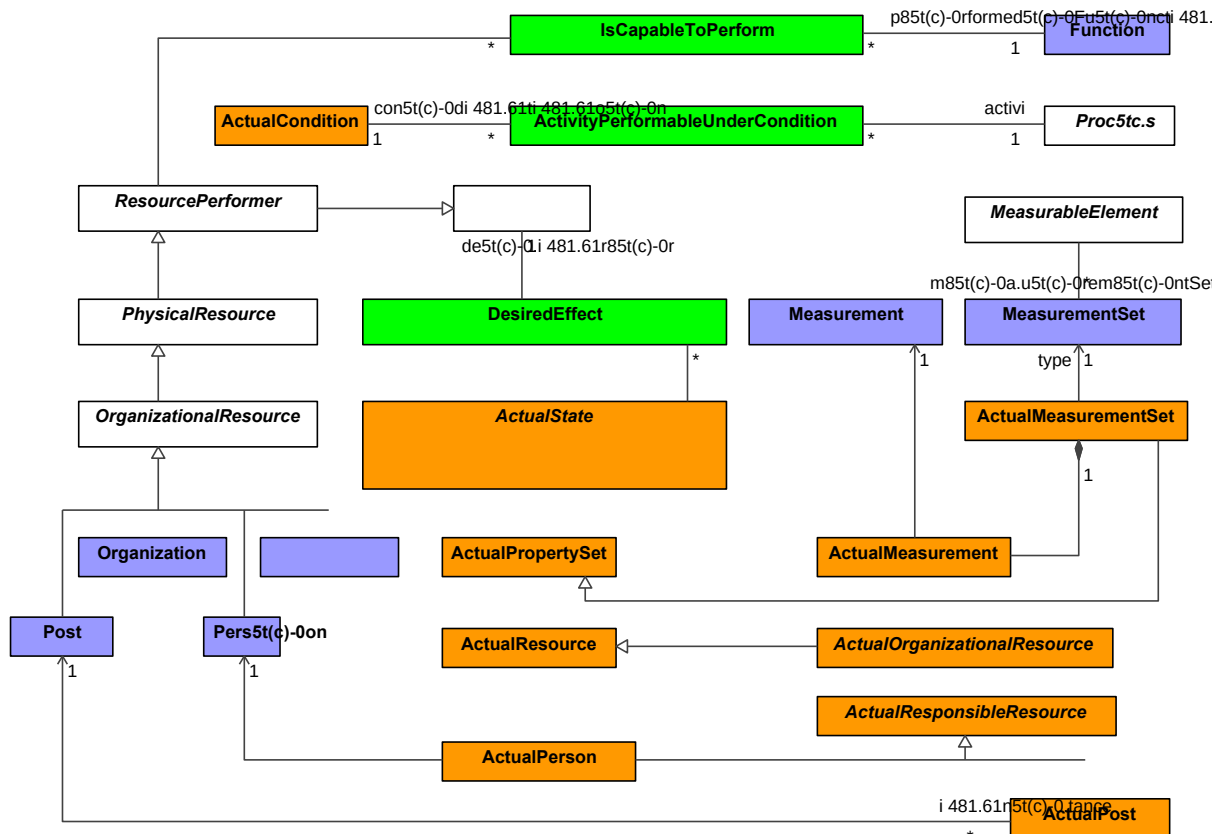
- [ActualMeasurement](#)
- [ActualMeasurementSet](#)
- [Function](#)
- [IsCapableToPerform](#)
- [MeasurableElement](#)
- [Measurement](#)
- [MeasurementSet](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [Process](#)
- [ResourceConstraint](#)
- [ResourcePerformer](#)
- [Responsibility](#)
- [Rule](#)
- [SubjectOfResourceConstraint](#)

## View Specifications::Personnel::Constraints::Personnel Constraints: Performance

Stakeholders: Human resources, solution providers.

Concerns: how well an actual organizational resource matches the needs of the actual organization.

**Definition:** provides a repository for human-related measures (i.e., quality objectives and performance criteria (HFI values)), targets and competences.



## Elements

- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
  - [Post](#)
  - [Process](#)
  - [ResourcePerformer](#)
  - [Responsibility](#)

## **View Specifications::Personnel::Roadmap**

Contains the diagrams that document the Personnel Roadmap Viewpoint.

### **View Specifications::Personnel::Roadmap::Personnel Roadmap: Availability**

Stakeholders: Human Resources, Training, Logisticians, Solution Providers.

Concerns: the staffing and training of resources.

Definition: defines the requirements and functions to ensure that actual persons with the right competencies, and in the right numbers, are available to fulfill actual posts.

Recommended Implementation: Timeline, SysML Block Definition Diagram.

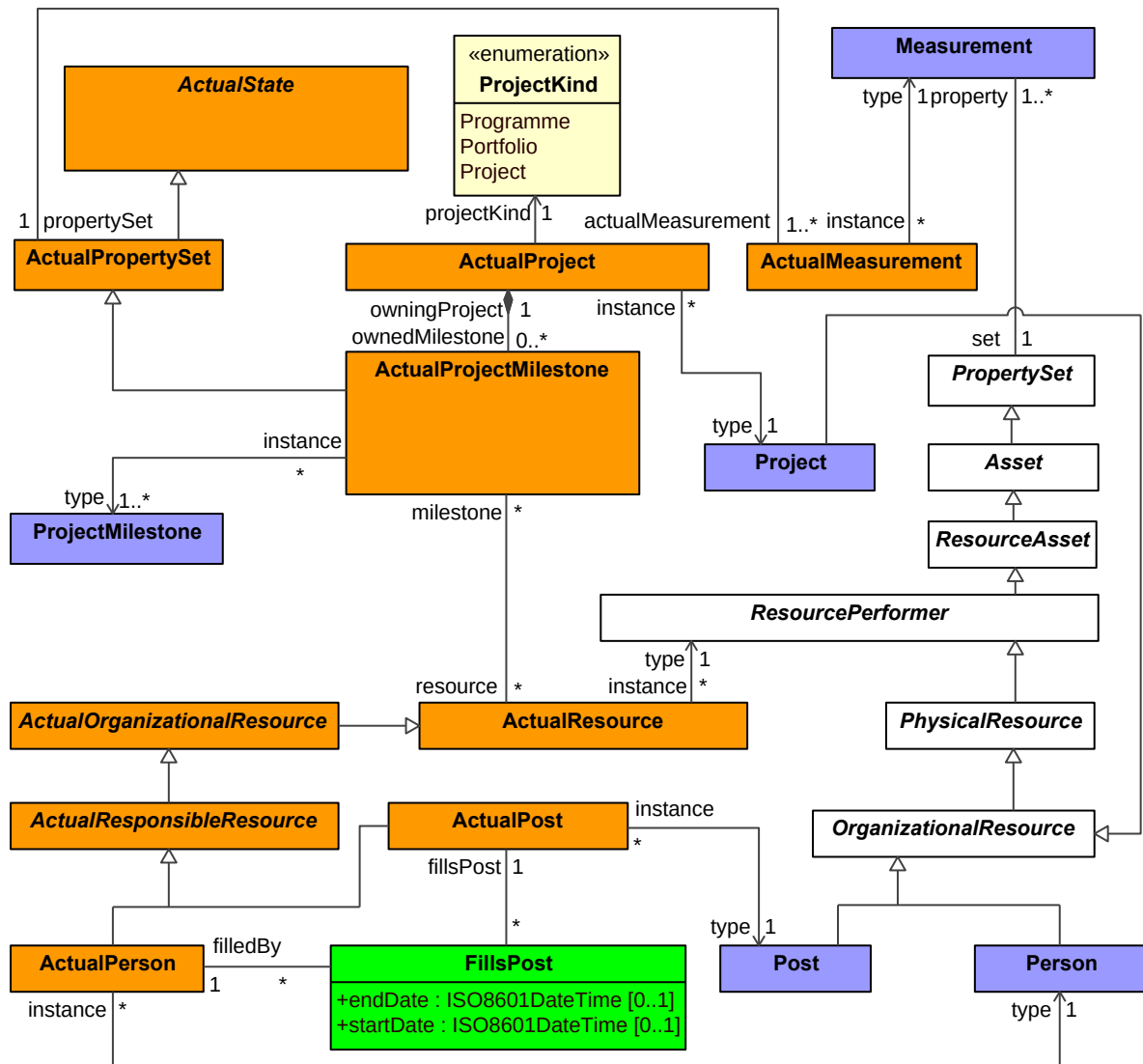


Figure 8:37 - Personnel Roadmap: Availability

Elements

- [ActualMeasurement](#)
- [ActualOrganizationalResource](#)
- [ActualPerson](#)
- [ActualPost](#)
- [ActualProject](#)
- [ActualProjectMilestone](#)
- [ActualPropertySet](#)
- [ActualResource](#)
- [ActualResponsibleResource](#)
- [ActualState](#)
- [Asset](#)
- [FillsPost](#)
- [Measurement](#)

- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [Project](#)
- [ProjectMilestone](#)
- [PropertySet](#)
- [ResourceAsset](#)
- [ResourcePerformer](#)

## View Specifications::Personnel::Roadmap::Personnel Roadmap: Evolution

Stakeholders: Human resources, Solution Providers.

Concerns: organizational structure changes over time.

Definition: provides an overview of how an organizational structure changes over time. It shows the structure of several organizational structures mapped against a timeline.

Recommended Implementation: timeline, SysML Block Definition Diagram, SysML Internal Block Diagram.

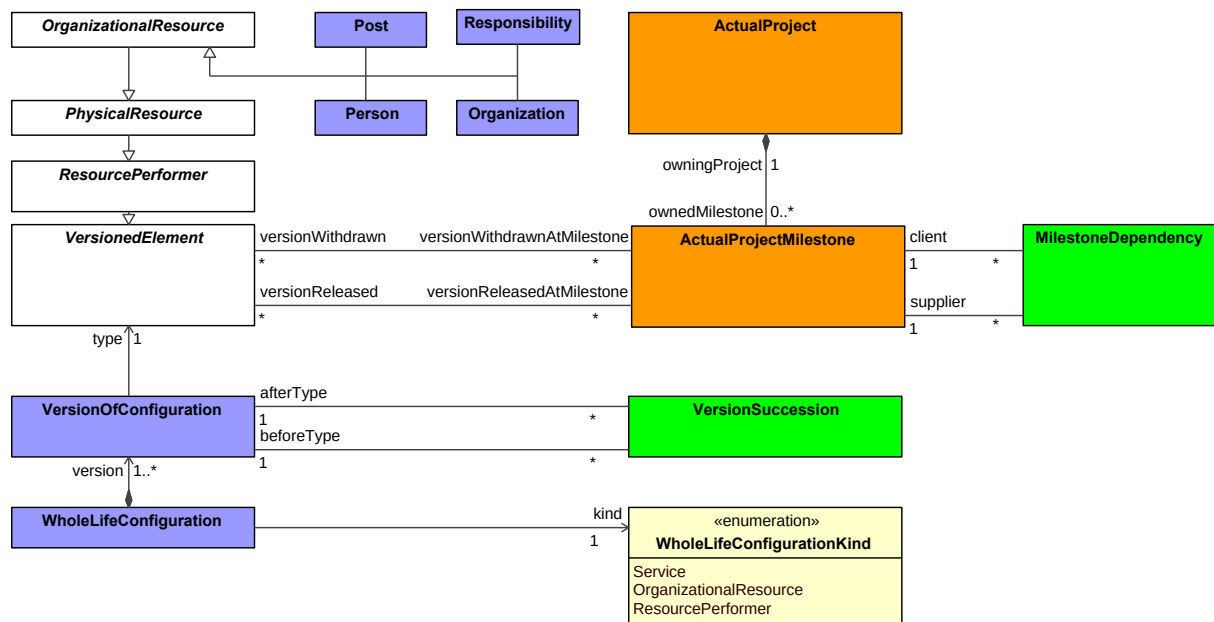


Figure 8:38 - Personnel Roadmap: Evolution

Elements

- [ActualProject](#)
- [ActualProjectMilestone](#)
- [MilestoneDependency](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourcePerformer](#)
- [Responsibility](#)
- [VersionedElement](#)
- [VersionOfConfiguration](#)

- [VersionSuccession](#)
- [WholeLifeConfiguration](#)

## View Specifications::Personnel::Roadmap::Personnel Roadmap: Forecast

Stakeholders: Human resources, Logisticians, Solution Providers.

Concerns: competencies and skills forecast.

Definition: defines the underlying current and expected supporting competencies and skills of organizational resources.

Recommended Implementation: timeline, tabular format, SysML Block Definition Diagram.

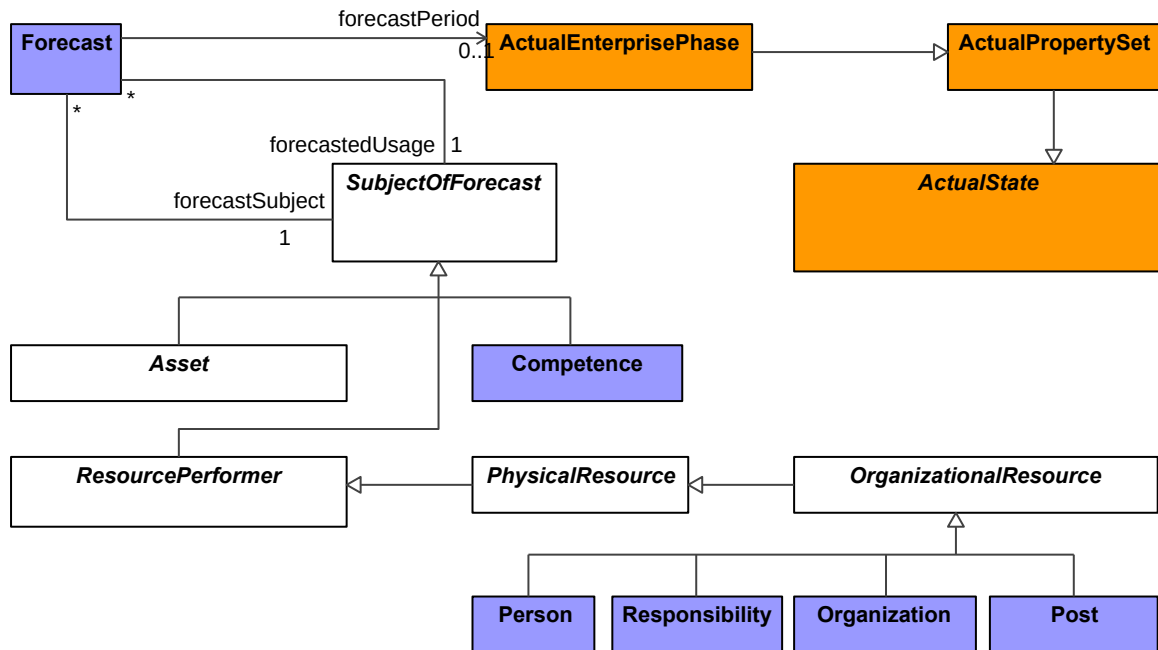


Figure 8:39 - Personnel Roadmap: Forecast

Elements

- [ActualEnterprisePhase](#)
- [ActualPropertySet](#)
- [ActualState](#)
- [Asset](#)
- [Competence](#)
- [Forecast](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourcePerformer](#)
- [Responsibility](#)
- [SubjectOfForecast](#)

## View Specifications::Personnel::Traceability

Contains the diagrams that document the Personnel Traceability Viewpoint.

## View Specifications::Personnel::Traceability::Personnel Traceability

Stakeholders: Systems Engineers, Enterprise Architects, Solution Providers, Business Architects.

Concerns: traceability between operational activities and functions that implements them.

Definition: depicts the mapping of functions (performed by organizational resources) to operational activities and thus identifies the transformation of an operational need into a purposeful function performed by an organizational resource or solution.

Recommended Implementation: Matrix format, SysML Block Definition Diagram.

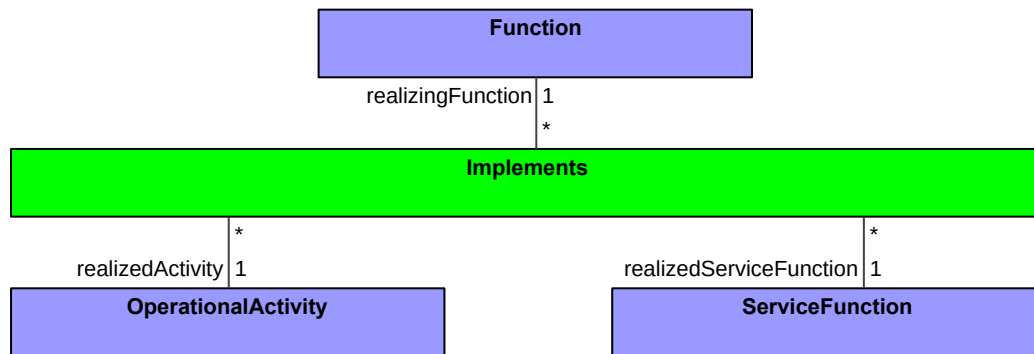


Figure 8:40 - Personnel Traceability

Elements

- [Function](#)
- [Implements](#)
- [OperationalActivity](#)
- [ServiceFunction](#)

## 8.1.6 View Specifications::Resources

Stakeholders: Systems Engineers, Resource Owners, Implementers, Solution Providers, IT Architects.

Concerns: definition of solution architectures to implement operational requirements.

Definition: captures a solution architecture consisting of resources, e.g., organizational, software, artifacts, capability configurations, natural resources that implement the operational requirements. Further design of a resource is typically detailed in SysML or UML.

## View Specifications::Resources::Taxonomy

Contains the diagrams that document the Resources Taxonomy Viewpoint.

## View Specifications::Resources::Taxonomy::Resources Taxonomy

Stakeholders: Solution Providers, Systems Engineers, IT Architects, Implementers.

Concerns: resource types.

Definition: shows the taxonomy of types of resources.

Recommended Implementation: SysML Block Definition Diagram.

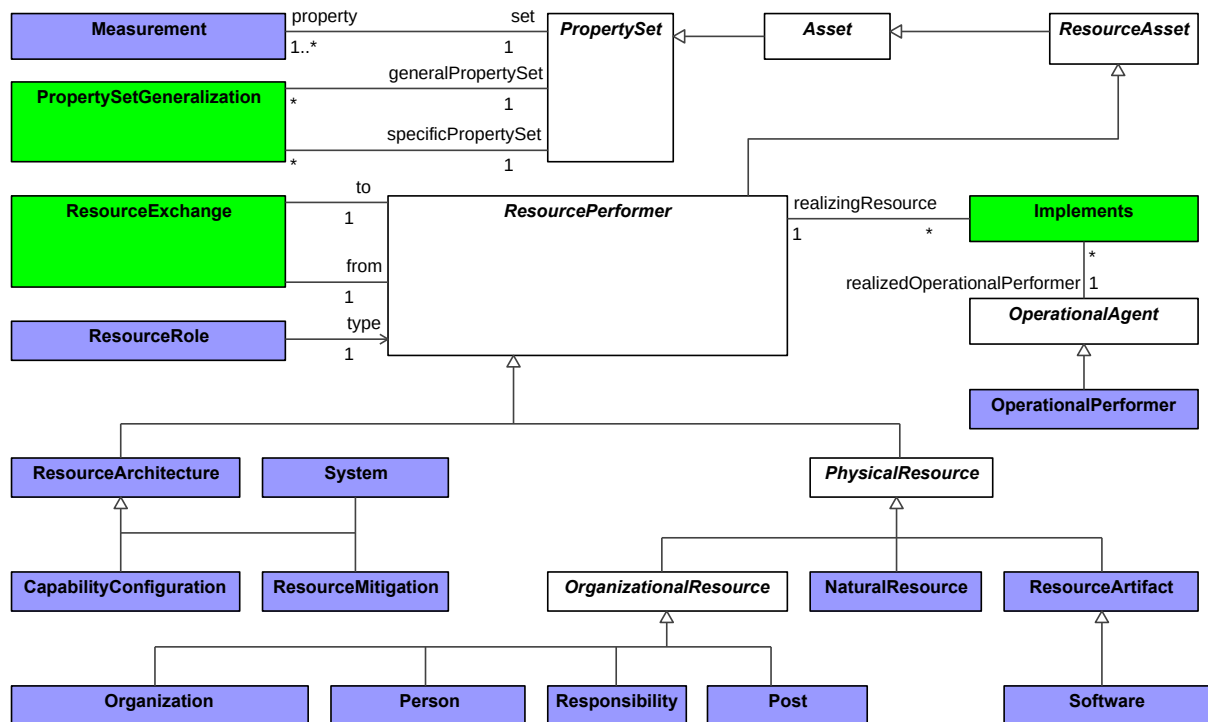


Figure 8:41 - Resources Taxonomy

Elements

- [Asset](#)
- [CapabilityConfiguration](#)
- [Implements](#)
- [Measurement](#)
- [NaturalResource](#)
- [OperationalAgent](#)
- [OperationalPerformer](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [PropertySet](#)
- [PropertySetGeneralization](#)
- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourceAsset](#)
- [ResourceExchange](#)
- [ResourceMitigation](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [Responsibility](#)
- [Software](#)
- [System](#)

## View Specifications::Resources::Structure

Contains the diagrams that document the Resources Structure Viewpoint.

### View Specifications::Resources::Structure::Resources Structure

Stakeholders: Systems Engineers, Resource Owners, Implementers, Solution Providers.

Concerns: reference the resource structure, connectors and interfaces in a specific context.

Definition: defines the physical resources, e.g., capability configuration(s)/system(s) and interactions necessary to implement a specific set of OperationalPerformer(s). Can be used to represent communications networks and pathways that link communications resources and provides details regarding their configuration.

Recommended Implementation: SysML Internal Block Diagram, SysML Block Definition Diagram.

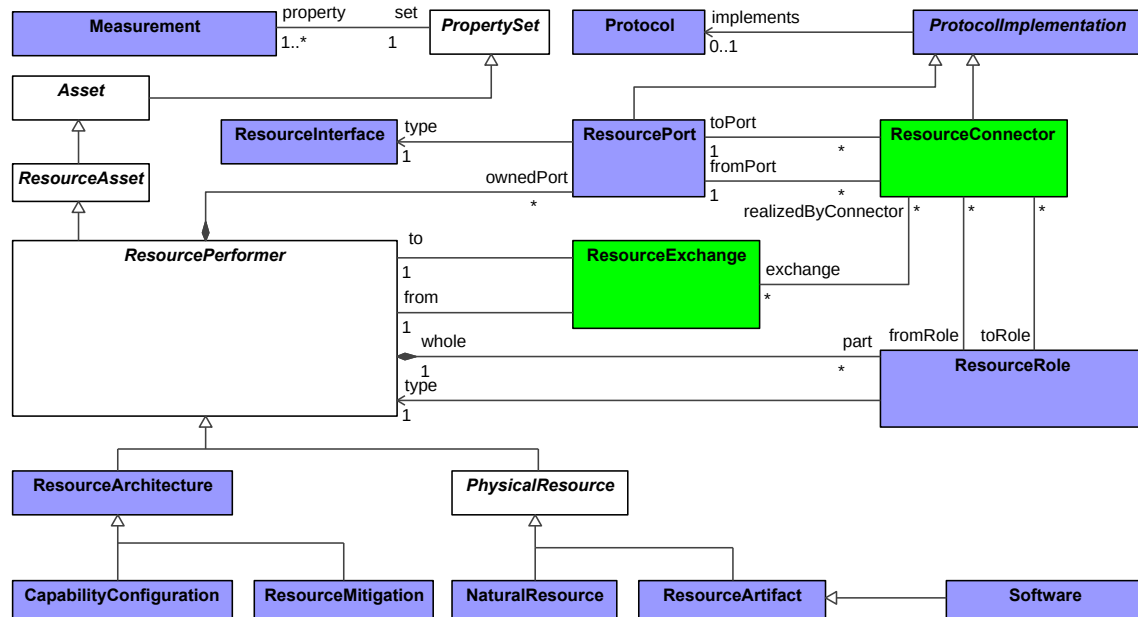


Figure 8:42 - Resources Structure

Elements

- [Asset](#)
- [CapabilityConfiguration](#)
- [Measurement](#)
- [NaturalResource](#)
- [PhysicalResource](#)
- [PropertySet](#)
- [Protocol](#)
- [ProtocolImplementation](#)
- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourceAsset](#)
- [ResourceConnector](#)
- [ResourceExchange](#)
- [ResourceInterface](#)
- [ResourceMitigation](#)
- [ResourcePerformer](#)
- [ResourcePort](#)



- [MeasurableElement](#)
- [Measurement](#)
- [MeasurementSet](#)
- [NaturalResource](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [Process](#)
- [PropertySet](#)
- [Resource](#)
- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourceConnector](#)
- [ResourceExchange](#)
- [ResourceExchangeItem](#)
- [ResourceInterface](#)
- [ResourceMitigation](#)
- [ResourcePerformer](#)
- [ResourcePort](#)
- [ResourceRole](#)
- [ResourceSignal](#)
- [Software](#)
- [Technology](#)

## **View Specifications::Resources::Processes**

Contains the diagrams that document the Resources Processes Viewpoint.

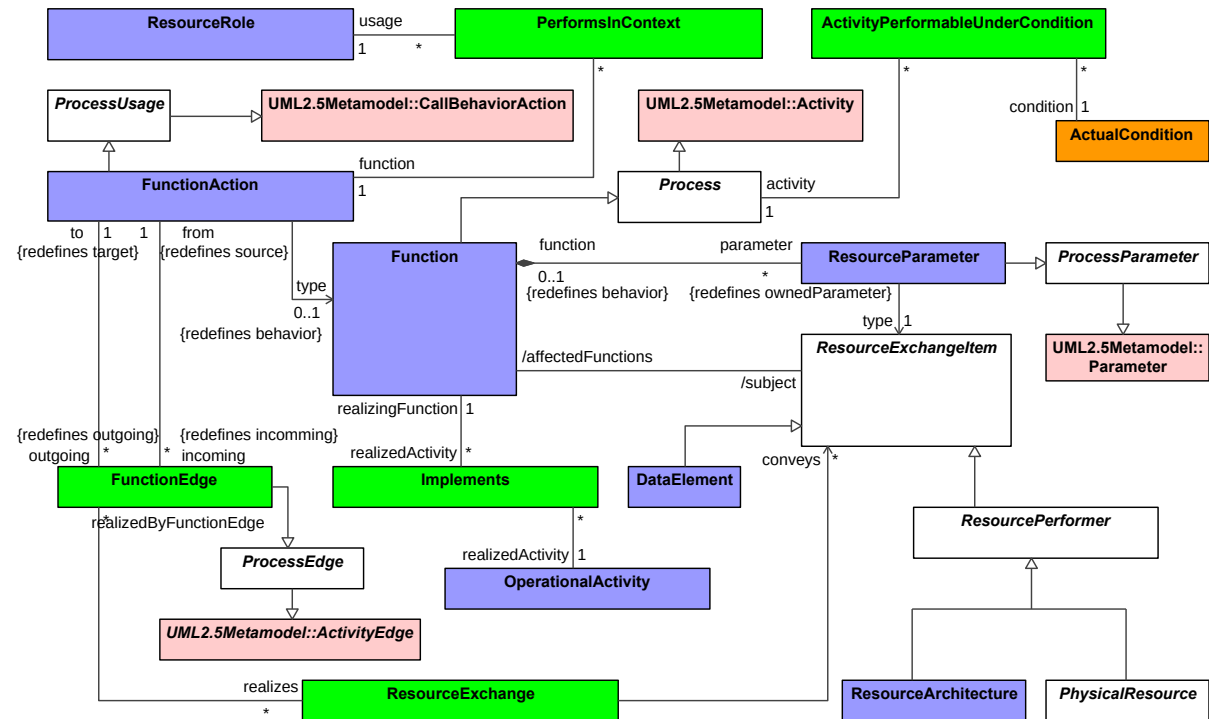
## **View Specifications::Resources::Processes::Resources Processes**

Stakeholders: Solution Providers, Systems Engineers, IT Architects.

Concerns: captures activity based behavior and flows.

Definition: describes the functions that are normally conducted in the course of implementing operational activity(ies) in support of capability(ies). It describes the functions, their Inputs/Outputs, function actions and flows between them.

Recommended Implementation: SysML Activity Diagram, SysML Block Definition Diagram.



**Figure 8:44 - Resources Processes**

Elements

- [ActivityPerformableUnderCondition](#)
- [ActualCondition](#)
- [DataElement](#)
- [Function](#)
- [FunctionAction](#)
- [FunctionEdge](#)
- [Implements](#)
- [OperationalActivity](#)
- [PerformsInContext](#)
- [PhysicalResource](#)
- [Process](#)
- [ProcessEdge](#)
- [ProcessParameter](#)
- [ProcessUsage](#)
- [ResourceArchitecture](#)
- [ResourceExchange](#)
- [ResourceExchangeItem](#)
- [ResourceParameter](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- UML2.5Metamodel::Activity
- UML2.5Metamodel::ActivityEdge
- UML2.5Metamodel::CallBehaviorAction
- UML2.5Metamodel::Parameter

Stakeholders: Solution Providers, IT Architects.  
Concerns: captures activity based behavior and flows using BPMN.  
Definition: describes the functions that are normally conducted in the course of implementing operational activity(ies) in support of capability(ies). It describes the functions, their Inputs/Outputs, function actions and flows between them using BPMN.  
Recommended Implementation: BPMN Process Diagram.



- [ActivityPerformableUnderCondition](#)
- [AssetRole](#)
- BPMN2Metamodel::BPMNMessage
- BPMN2Metamodel::CallActivity
- BPMN2Metamodel::MessageFlow
- BPMN2Metamodel::Process
- BPMN2Metamodel::ResourceRole
- BPMN2Metamodel::SequenceFlow
- [DataElement](#)

- [Exchange](#)
- [ExchangeItem](#)
- [Function](#)
- [FunctionAction](#)
- [FunctionEdge](#)
- [Implements](#)
- [OperationalActivity](#)
- [PerformsInContext](#)
- [PhysicalResource](#)
- [Process](#)
- [ProcessEdge](#)
- [ProcessUsage](#)
- [ResourceArchitecture](#)
- [ResourceExchange](#)
- [ResourceExchangeItem](#)
- [ResourceParameter](#)
- [ResourcePerformer](#)
- [ResourceRole](#)

## View Specifications::Resources::States

Contains the diagrams that document the Resources States Viewpoint.

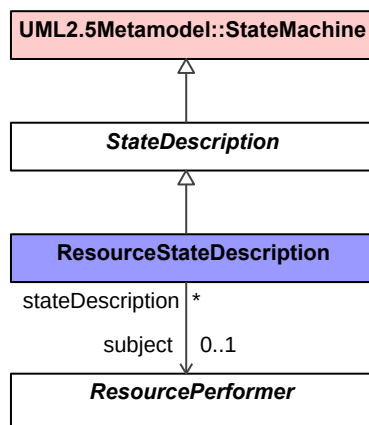
### View Specifications::Resources::States::Resources States

Stakeholders: Systems Engineers, Software Engineers.

Concerns: capture state-based behavior of a resource.

Definition: it is a graphical representation of states of a resource and how that resource responds to various events and actions.

Recommended Implementation: SysML State Machine Diagram.



**Figure 8:46 - Resources States**

Elements

- [ResourcePerformer](#)
- [ResourceStateDescription](#)
- [StateDescription](#)
- UML2.5Metamodel::StateMachine

Contains the diagrams that document the Resources Interaction Scenarios Viewpoint.

Stakeholders: Software Engineers, Systems Engineers.

**Definition:** provides a time-ordered examination of the interactions between resources.

### Recommended Implementation: SysML Sequence Diagram.



- [Function](#)
- [InteractionMessage](#)
- [InteractionRole](#)
- [InteractionScenario](#)
- [ResourceExchange](#)
- [ResourceInteractionScenario](#)
- [ResourceMessage](#)
- [ResourceMethod](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- UML2.5Metamodel::Interaction
- UML2.5Metamodel::Lifeline
- UML2.5Metamodel::Message

Contains the diagrams that document the Resources Constraints Viewpoint.





## View Specifications::Resources::Roadmap::Resources Roadmap: Forecast

Stakeholders: Solution Providers, Systems Engineers, IT Architects.

Concerns: technology forecast.

Definition: defines the underlying current and expected supporting technologies. Expected supporting technologies are those that can be reasonably forecast given the current state of technology, and expected improvements / trends.

Recommended Implementation: timeline, tabular format, SysML Block Definition Diagram.

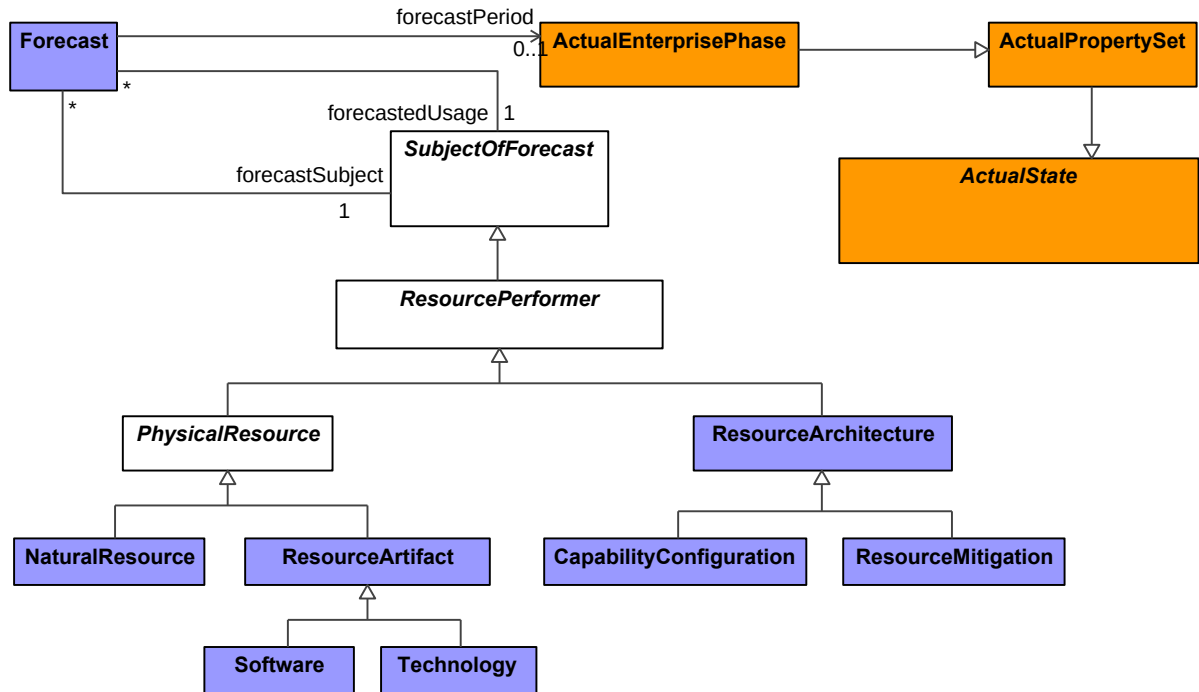


Figure 8:50 - Resources Roadmap: Forecast

Elements

- [ActualEnterprisePhase](#)
- [ActualPropertySet](#)
- [ActualState](#)
- [CapabilityConfiguration](#)
- [Forecast](#)
- [NaturalResource](#)
- [PhysicalResource](#)
- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourceMitigation](#)
- [ResourcePerformer](#)
- [Software](#)
- [SubjectOfForecast](#)
- [Technology](#)

## View Specifications::Resources::Traceability

Contains the diagrams that document the Resources Traceability Viewpoint.

## View Specifications::Resources::Traceability::Resources Traceability

Stakeholders: Systems Engineers, Enterprise Architects, Solution Providers, Business Architects.

Concerns: traceability between operational activities and functions that implements them.

Definition: depicts the mapping of functions to operational activities and thus identifies the transformation of an operational need into a purposeful function performed by a resource or solution.

Recommended Implementation: Matrix format, SysML Block Definition Diagram.

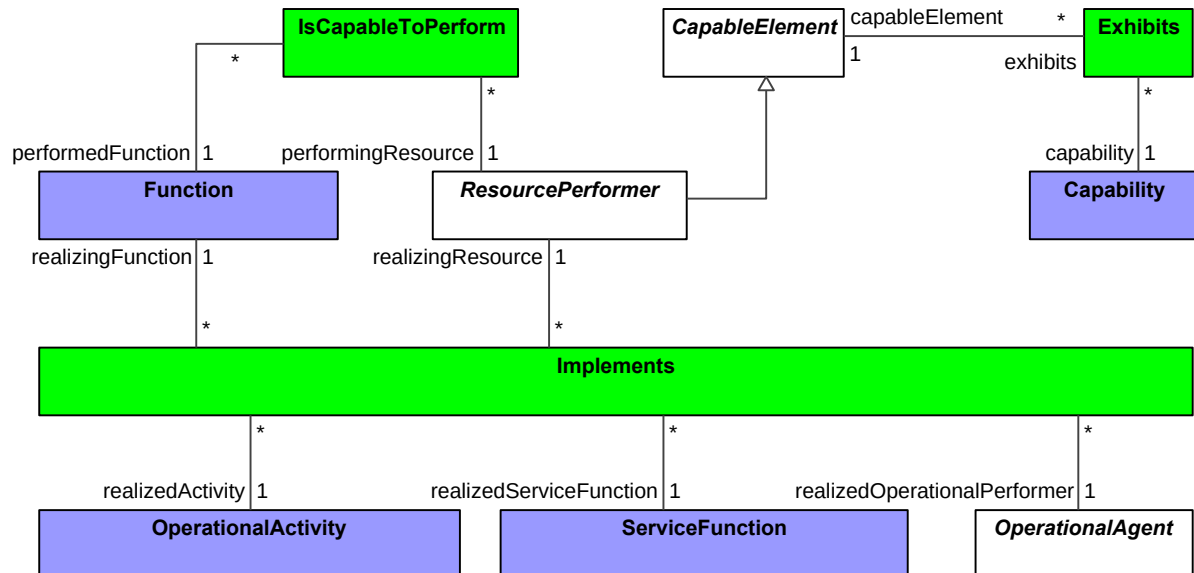


Figure 8:51 - Resources Traceability

Elements

- [Capability](#)
- [CapableElement](#)
- [Exhibits](#)
- [Function](#)
- [Implements](#)
- [IsCapableToPerform](#)
- [OperationalActivity](#)
- [OperationalAgent](#)
- [ResourcePerformer](#)
- [ServiceFunction](#)

### 8.1.7 View Specifications::Security

Stakeholders: Security Architects, Security Engineers. Systems Engineers, Operational Architects.

Concerns: addresses the security constraints and information assurance attributes that exist on exchanges between resources and OperationalPerformers

Definition: illustrates the security assets, security constraints, security controls, families, and measures required to address specific security concerns.

## View Specifications::Security::Taxonomy

Contains the diagrams that document the Security Taxonomy Viewpoint.

## View Specifications::Security::Taxonomy::Security Taxonomy

Stakeholders: Security Architects, Security Engineers.

Concerns: Security assets and security enclaves.

Definition: Defines the hierarchy of security assets and asset owners that are available to implement security, security constraints (policy, guidance, laws and regulations) and details where they are located (security enclaves).

Recommended Implementation: tabular format, SysML Block Definition Diagram.

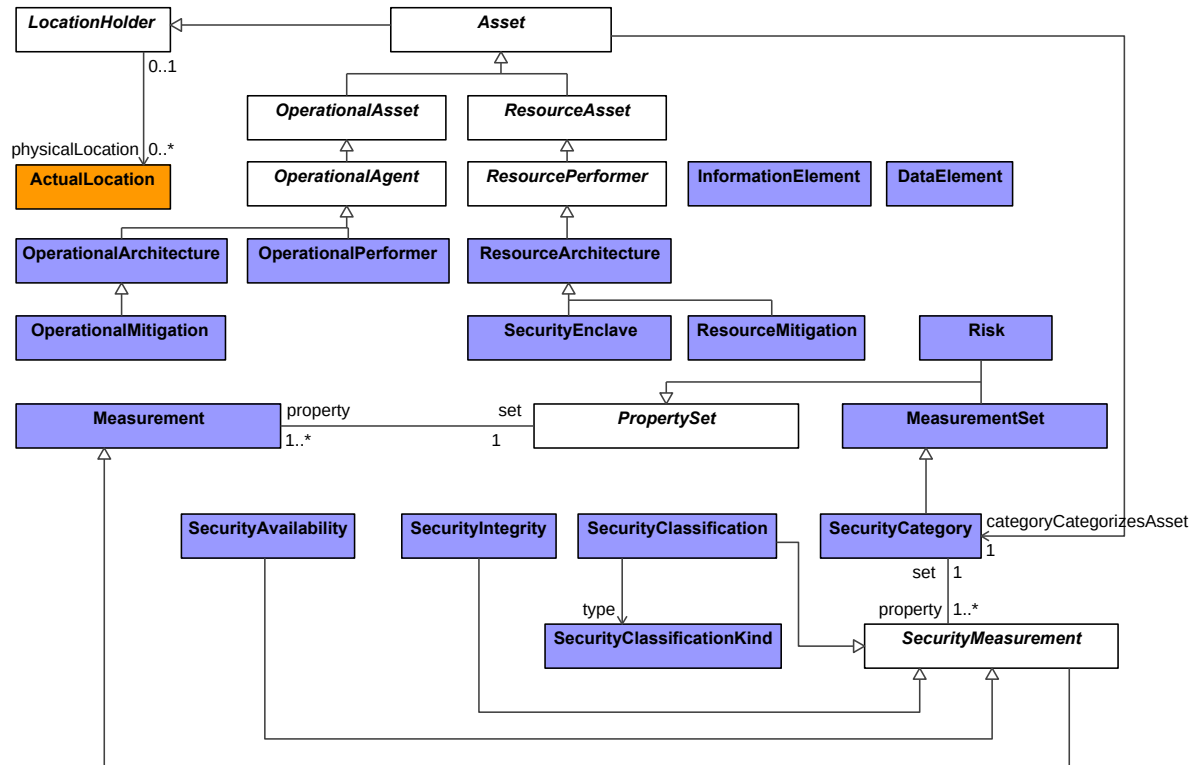


Figure 8:52 - Security Taxonomy

Elements

- [ActualLocation](#)
- [Asset](#)
- [DataElement](#)
- [InformationElement](#)
- [LocationHolder](#)
- [Measurement](#)
- [MeasurementSet](#)
- [OperationalAgent](#)
- [OperationalArchitecture](#)
- [OperationalMitigation](#)
- [OperationalPerformer](#)
- [PropertySet](#)
- [ResourceArchitecture](#)
- [ResourceAsset](#)
- [ResourceMitigation](#)
- [ResourcePerformer](#)

- [Risk](#)
- [SecurityAvailability](#)
- [SecurityCategory](#)
- [SecurityClassification](#)
- [SecurityClassificationKind](#)
- [SecurityEnclave](#)
- [SecurityIntegrity](#)
- [SecurityMeasurement](#)

## View Specifications::Security::Structure

Contains the diagrams that document the Security Structure Viewpoint.

### View Specifications::Security::Structure::Security Structure

Stakeholders: Security Architects, Security Engineers.

Concerns: The structure of security information and where it is used at the operational and resource level.

Definition: Captures the allocation of assets (operational and resource, information and data) across the security enclaves, shows applicable security controls necessary to protect organizations, systems and information during processing, while in storage (bdd), and during transmission (flows on an ibd). This view also captures Asset Aggregation and allocates the usage of the aggregated information at a location through the use of the SecurityProperty.

Recommended Implementation: SysML Internal Block Diagram, SysML Block Definition Diagram.

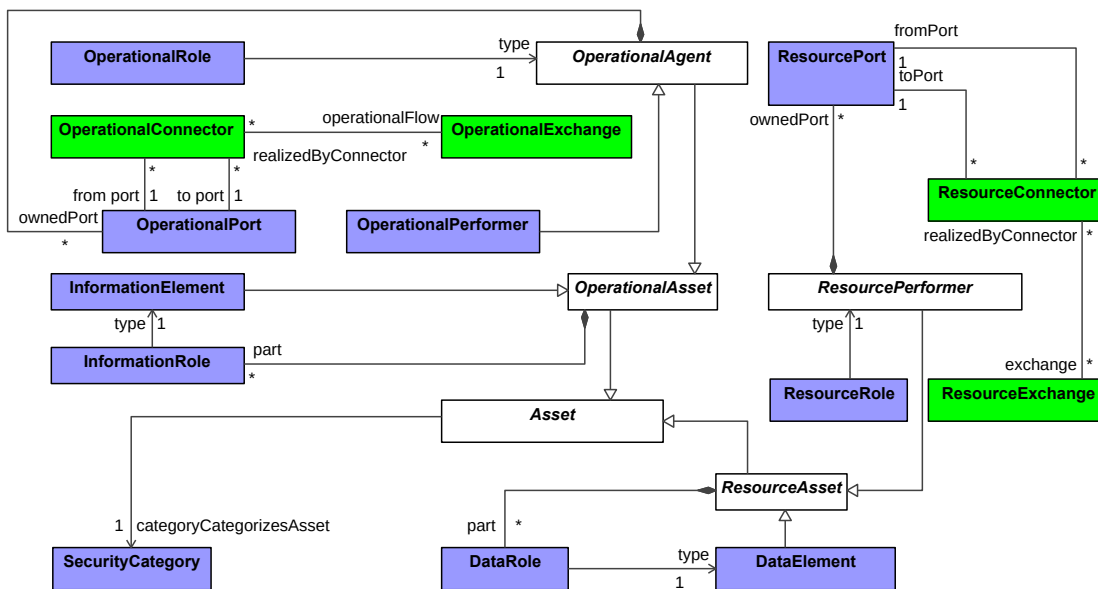


Figure 8:53 - Security Structure

Elements

- [Asset](#)
- [DataElement](#)
- [DataRole](#)
- [InformationElement](#)
- [InformationRole](#)
- [OperationalAgent](#)
- [OperationalAsset](#)
- [OperationalConnector](#)
- [OperationalExchange](#)

- [OperationalPerformer](#)
- [OperationalPort](#)
- [OperationalRole](#)
- [ResourceAsset](#)
- [ResourceConnector](#)
- [ResourceExchange](#)
- [ResourcePerformer](#)
- [ResourcePort](#)
- [ResourceRole](#)
- [SecurityCategory](#)

## View Specifications::Security::Connectivity

Contains the diagrams that document the Security Connectivity Viewpoint.

### View Specifications::Security::Connectivity::Security Connectivity

Stakeholders: Security Architects, Security Engineers.

Concerns: Addresses the security constraints and information assurance attributes that exist on exchanges across resources and across performers.

Definition: Lists security exchanges across security assets; the applicable security controls; and the security enclaves that house the producers and consumers of the exchanges. Measurements can optionally be included.

Recommended Implementation: SysML Internal Block Diagram, tabular format.

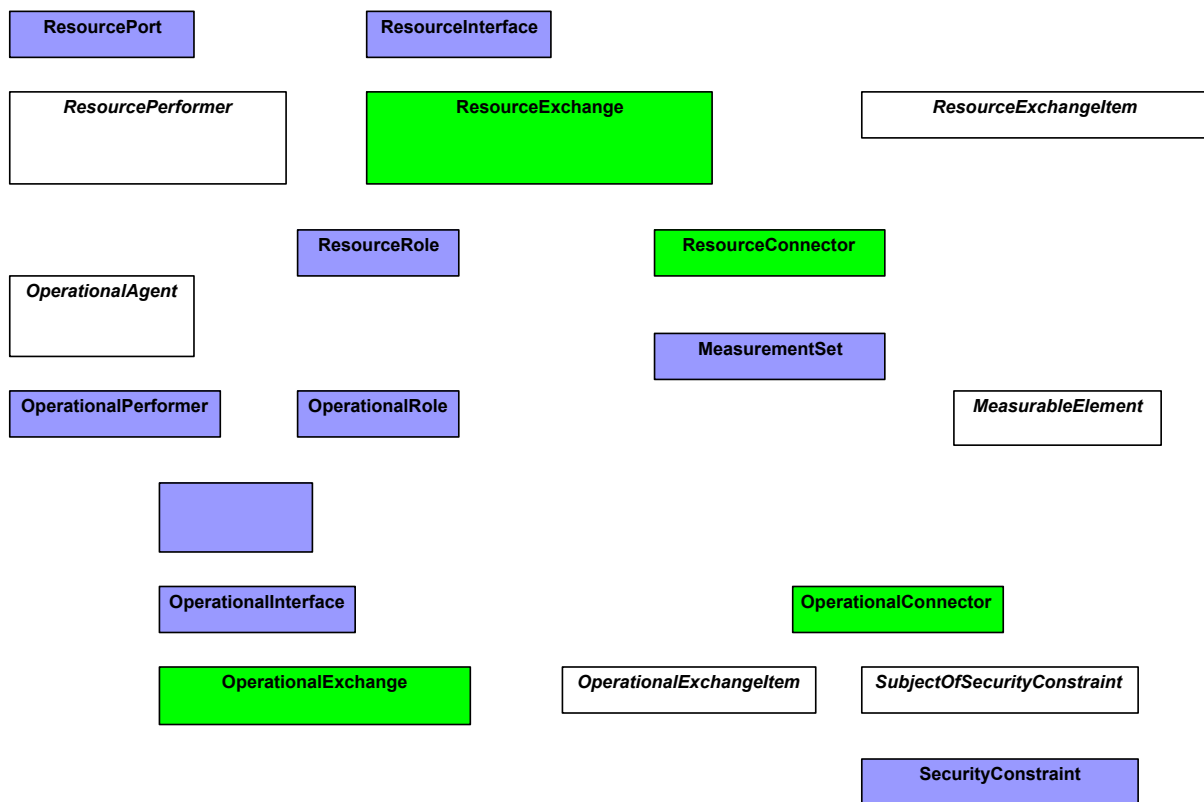


Figure 8:54 - Security Connectivity

## Elements

- [Caveat](#)
- [MeasurableElement](#)
- [MeasurementSet](#)
- [OperationalAgent](#)
- [OperationalConnector](#)
- [OperationalExchange](#)
- [OperationalExchangeItem](#)
- [OperationalInterface](#)
- [OperationalPerformer](#)
- [OperationalPort](#)
- [OperationalRole](#)
- [ResourceConnector](#)
- [ResourceExchange](#)
- [ResourceExchangeItem](#)
- [ResourceInterface](#)
- [ResourcePerformer](#)
- [ResourcePort](#)
- [ResourceRole](#)
- [SecurityConstraint](#)
- [SubjectOfSecurityConstraint](#)

## View Specifications::Security::Processes

Contains the diagrams that document the Security Processes Viewpoint.

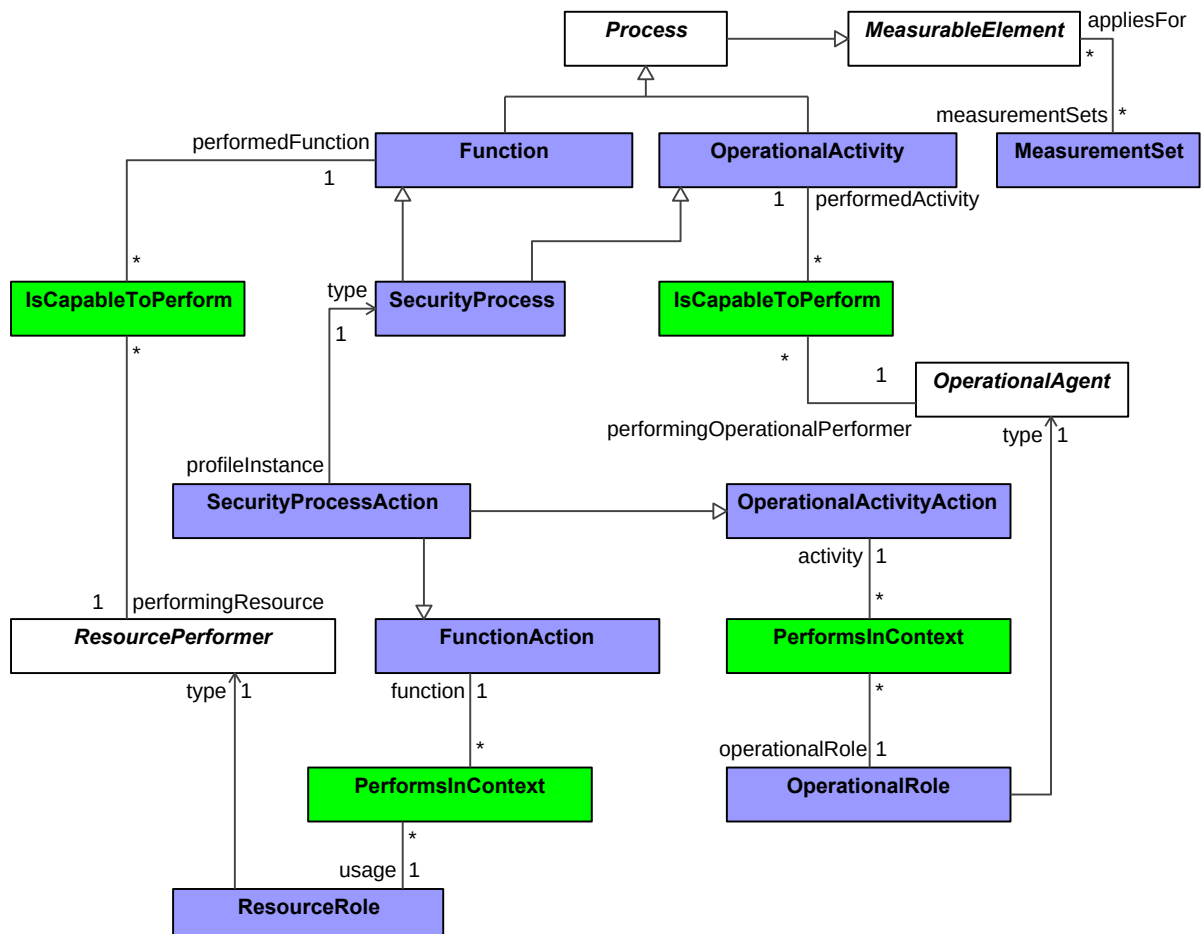
## View Specifications::Security::Processes::Security Processes

Stakeholders: Security Architects, Security Engineers.

Concerns: The specification of the Security Control families, security controls, and measures required to address a specific security baseline.

Definition: Provides a set of Security Controls and any possible enhancements as applicable to assets. The activity diagram describes operational or resource level processes that apply (operational level) or implement (resource level) security controls/enhancements to assets located in enclaves and across enclaves. This Security Process view can be instantiated either as a variant of an activity/flow diagram or as a hierarchical work breakdown structure.

Recommended Implementation: SysML Activity Diagram, SysML Block Definition Diagram, BPMN Process Diagram as described in Operational Processes and Resources Processes sections.



**Figure 8:55 - Security Processes**

Elements

- [Function](#)
- [FunctionAction](#)
- [IsCapableToPerform](#)
- [MeasurableElement](#)
- [MeasurementSet](#)
- [OperationalActivity](#)
- [OperationalActivityAction](#)
- [OperationalAgent](#)
- [OperationalRole](#)
- [PerformsInContext](#)
- [Process](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [SecurityProcess](#)
- [SecurityProcessAction](#)

## View Specifications::Security::Constraints

Contains the diagrams that document the Security Constraints Viewpoint.

## View Specifications::Security::Constraints::Security Constraints

Stakeholders: Security Architects, Security Engineers, Risk Analysts.

Concerns: (i) Security-related policy, guidance, laws and regulations as applicable to assets, (ii) threats, vulnerabilities, and risk assessments as applicable to assets.

Definition: (i) Specifies textual rules/non-functional requirements that are security constraints on resources, information and data (e.g. security-related in the form of rules (e.g. access control policy)). A common way of representing access control policy is through the use of XACML (eXtensible Access Control Markup Language), it is expected that implementations of UAF allow users to link security constraints to external files represented in XACML. (ii) Identifies risks, specifies risk likelihood, impact, asset criticality, other measurements and enables risk assessment.

Recommended Implementation: tabular or Matrix format, SysML Block Definition Diagram, SysML Parametric Diagram, or OCL.

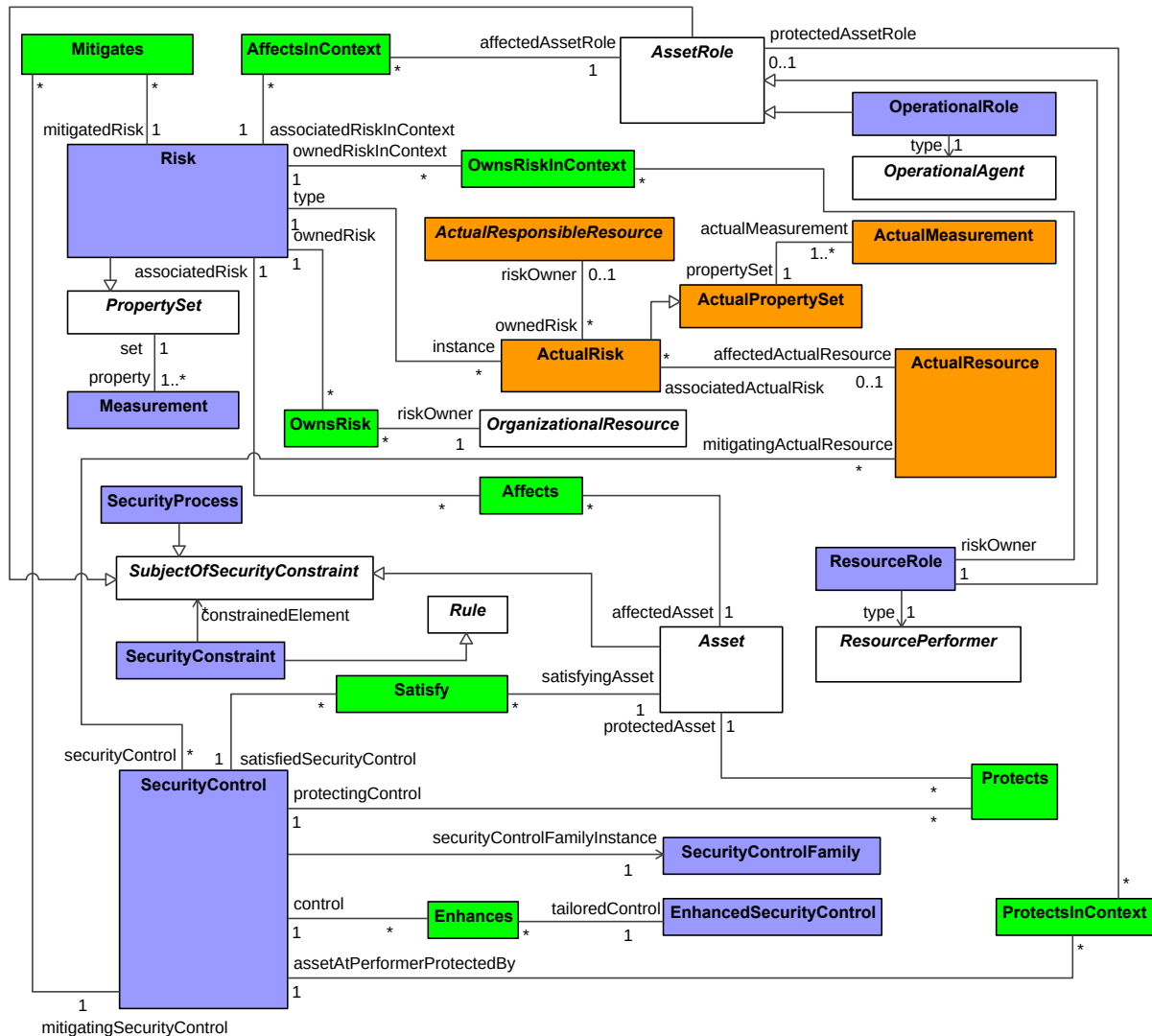


Figure 8:56 - Security Constraints

Elements

- [ActualMeasurement](#)
- [ActualPropertySet](#)
- [ActualResource](#)

- [ActualResponsibleResource](#)
- [ActualRisk](#)
- [Affects](#)
- [AffectsInContext](#)
- [Asset](#)
- [AssetRole](#)
- [EnhancedSecurityControl](#)
- [Enhances](#)
- [Measurement](#)
- [Mitigates](#)
- [OperationalAgent](#)
- [OperationalRole](#)
- [OrganizationalResource](#)
- [OwnsRisk](#)
- [OwnsRiskInContext](#)
- [PropertySet](#)
- [Protects](#)
- [ProtectsInContext](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [Risk](#)
- [Rule](#)
- Satisfy
- [SecurityConstraint](#)
- [SecurityControl](#)
- [SecurityControlFamily](#)
- [SecurityProcess](#)
- [SubjectOfSecurityConstraint](#)

## **View Specifications::Security::Traceability**

Contains the diagrams that document the Security Traceability Viewpoint.

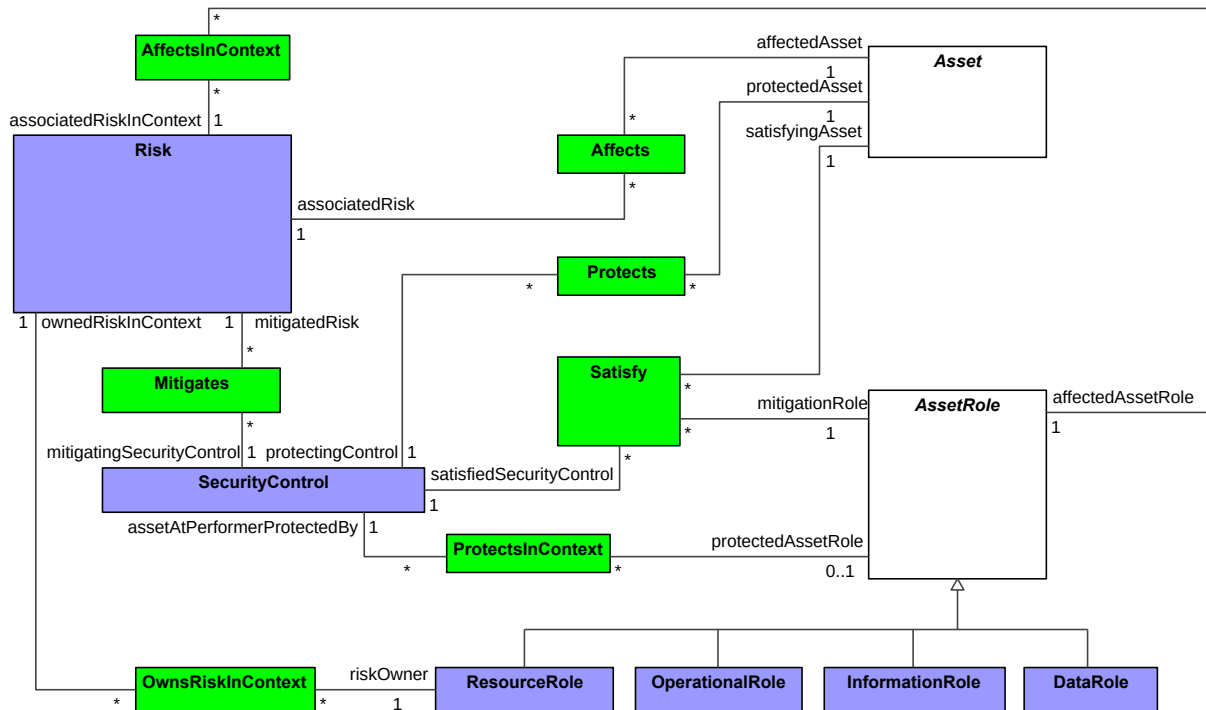
### **View Specifications::Security::Traceability::Security Traceability**

Stakeholders: Security Architects, Security Engineers, Risk Analysts.

Concerns: traceability between risk and risk owner, risk mitigations, and affected asset roles.

Definition: depicts the mapping of a risk to each of the following: risk owner, risk mitigations, and affected asset roles.

Recommended Implementation: Matrix format, SysML Block Definition Diagram.



**Figure 8:57 - Security Traceability**

#### Elements

- [Affects](#)
- [AffectsInContext](#)
- [Asset](#)
- [AssetRole](#)
- [DataRole](#)
- [InformationRole](#)
- [Mitigates](#)
- [OperationalRole](#)
- [OwnsRiskInContext](#)
- [Protects](#)
- [ProtectsInContext](#)
- [ResourceRole](#)
- [Risk](#)
- [Satisfy](#)
- [SecurityControl](#)

### 8.1.8 View Specifications::Projects

Stakeholders: PMs, Project Portfolio Managers, Enterprise Architects.

Concerns: project portfolio, projects and project milestones.

Definition: describes projects and project milestones, how those projects deliver capabilities, the organizations contributing to the projects and dependencies between projects.

#### View Specifications::Projects::Taxonomy

Contains the diagrams that document the Project Taxonomy Viewpoint.

## View Specifications::Projects::Taxonomy::Project Taxonomy

Stakeholders: PMs, Project Portfolio Managers, Enterprise Architects. Concerns: types of projects and project milestones. Definition: shows the taxonomy of types of projects and project milestones. Recommended Implementation: SysML Block Definition Diagram.

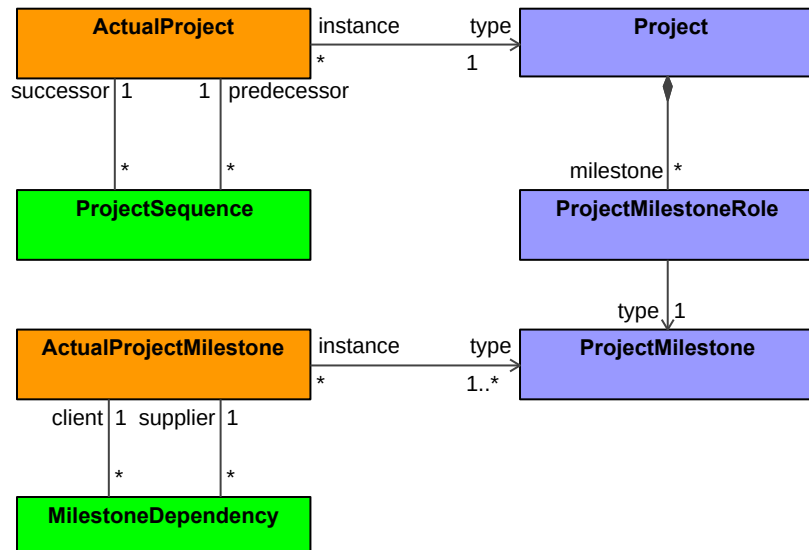


Figure 8:58 - Project Taxonomy

### Elements

- [ActualProject](#)
- [ActualProjectMilestone](#)
- [MilestoneDependency](#)
- [Project](#)
- [ProjectMilestone](#)
- [ProjectMilestoneRole](#)
- [ProjectSequence](#)

## View Specifications::Projects::Structure

Contains the diagrams that document the Project Structure Viewpoint.

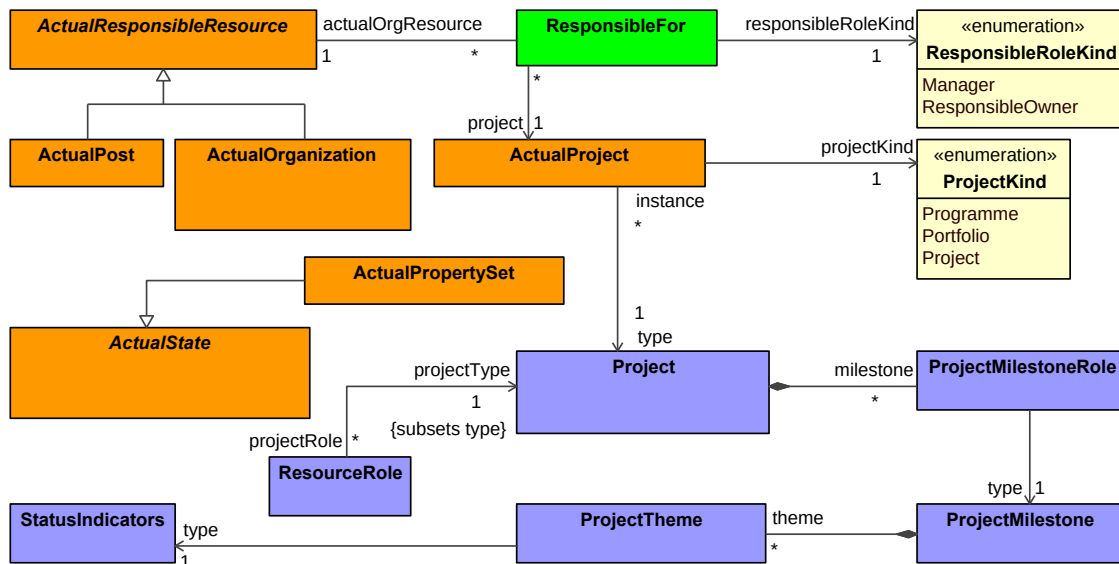
### View Specifications::Projects::Structure::Project Structure

Stakeholders: PMs.

Concerns: relationships between types of projects and project milestones.

Definition: provides a template for an actual project(s) road map(s) to be implemented.

Recommended Implementation: SysML Block Definition Diagram.



**Figure 8:59 - Project Structure**

Elements

- [ActualOrganization](#)
- [ActualPost](#)
- [ActualProject](#)
- [ActualPropertySet](#)
- [ActualResponsibleResource](#)
- [ActualState](#)
- [Project](#)
- [ProjectMilestone](#)
- [ProjectMilestoneRole](#)
- [ProjectTheme](#)
- [ResourceRole](#)
- [ResponsibleFor](#)
- [StatusIndicators](#)

## View Specifications::Projects::Connectivity

Contains the diagrams that document the Project Connectivity Viewpoint.

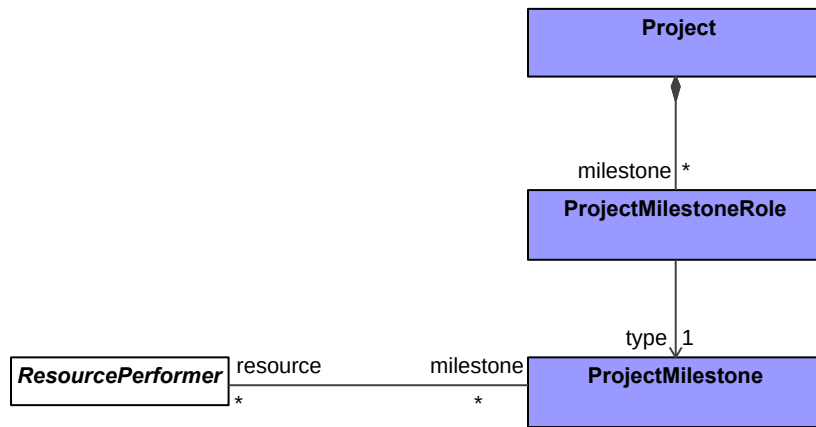
## View Specifications::Projects::Connectivity::Project Connectivity

Stakeholders: PMs.

Concerns: relationships between projects and project milestones.

Definition: shows how projects and project milestones are related in sequence.

Recommended Implementation: SysML Block Definition Diagram.



**Figure 8:60 - Project Connectivity**

Elements

- [Project](#)
- [ProjectMilestone](#)
- [ProjectMilestoneRole](#)
- [ResourcePerformer](#)

## View Specifications::Projects::Processes

Contains the diagrams that document the Project Processes Viewpoint.

### View Specifications::Projects::Processes::Project Processes

Stakeholders: PMs.

Concerns: captures project tasks (ProjectActivities) and flows between them.

Definition: describes the ProjectActivities that are normally conducted in the course of projects to support capability(ies) and implement resources. It describes the ProjectActivities, their Inputs/Outputs, ProjectActivityActions and flows between them.

Recommended Implementation: SysML Activity Diagram, SysML Block Definition Diagram, BPMN Process Diagram as described in Resources Processes section.



## View Specifications::Projects::Roadmap::Project Roadmap

Stakeholders: PMs, Capability Owners, Solution Providers, Enterprise Architects.

Concerns: the product portfolio management; a planning of capability delivery.

Definition: provides a timeline perspective on programs or projects

Recommended Implementation: timeline, tabular format, SysML Block Definition Diagram.

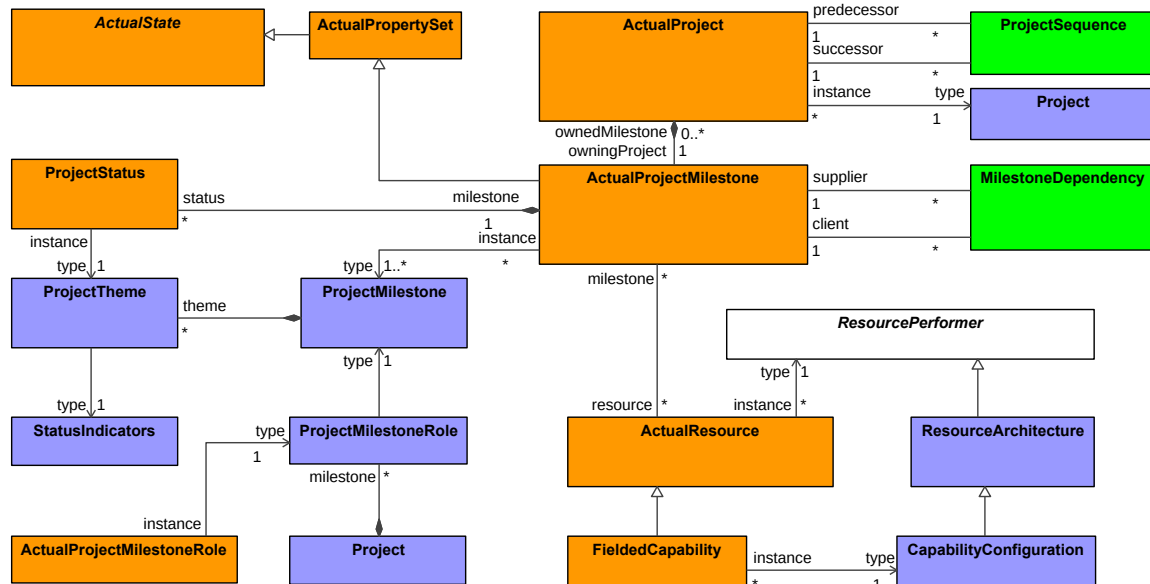


Figure 8:62 - Project Roadmap

Elements

- [ActualProject](#)
- [ActualProjectMilestone](#)
- [ActualProjectMilestoneRole](#)
- [ActualPropertySet](#)
- [ActualResource](#)
- [ActualState](#)
- [CapabilityConfiguration](#)
- [FieldedCapability](#)
- [MilestoneDependency](#)
- [Project](#)
- [ProjectMilestone](#)
- [ProjectMilestoneRole](#)
- [ProjectSequence](#)
- [ProjectStatus](#)
- [ProjectTheme](#)
- [ResourceArchitecture](#)
- [ResourcePerformer](#)
- [StatusIndicators](#)

## View Specifications::Projects::Traceability

Contains the diagrams that document the Project Traceability Viewpoint.

## View Specifications::Projects::Traceability::Project Traceability

Stakeholders: PMs, Project Portfolio Managers, Enterprise Architects.

Concerns: traceability between capabilities and projects that deliver them.

Definition: depicts the mapping of projects to capabilities and thus identifies the transformation of a capability(ies) into a purposeful implementation via projects.

Recommended Implementation: Matrix format, SysML Block Definition Diagram.

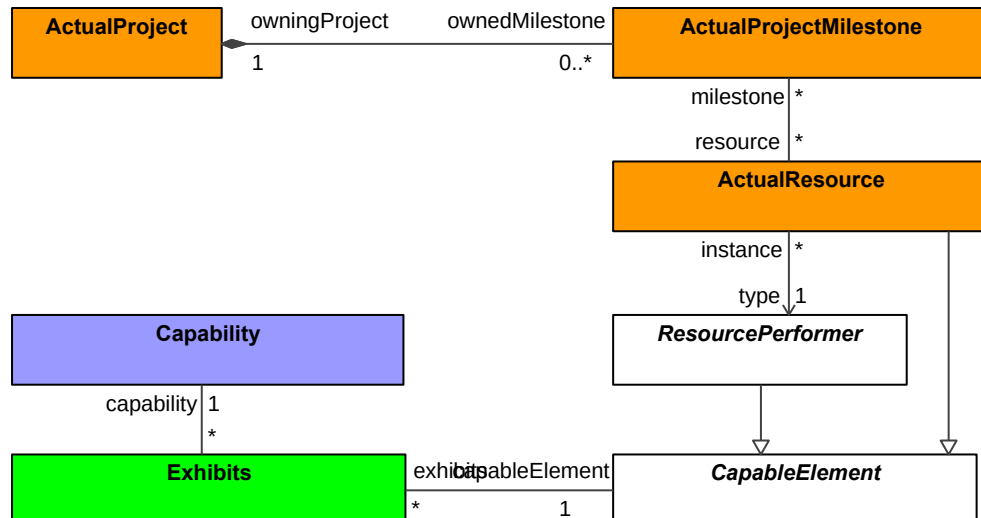


Figure 8:63 - Project Traceability

Elements

- [ActualProject](#)
- [ActualProjectMilestone](#)
- [ActualResource](#)
- [Capability](#)
- [CapableElement](#)
- [Exhibits](#)
- [ResourcePerformer](#)

## 8.1.9 View Specifications::Standards

Stakeholders: Solution Providers, Systems Engineers, Software Engineers, Systems Architects, Business Architects.

Concerns: technical and non-technical Standards applicable to the architecture.

Definition: shows the technical, operational, and business Standards applicable to the architecture. Defines the underlying current and expected Standards.

## View Specifications::Standards::Taxonomy

Contains the diagrams that document the Standards Taxonomy Viewpoint.

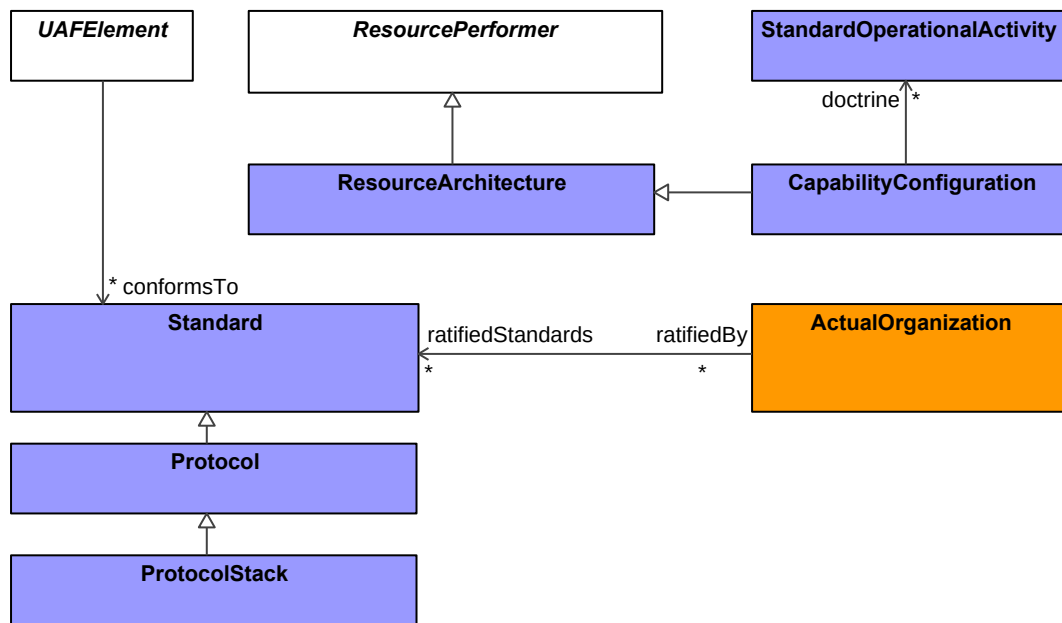
## View Specifications::Standards::Taxonomy::Standards Taxonomy

Stakeholders: Solution Providers, Systems Engineers, Software Engineers, Systems Architects, Business Architects.

Concerns: technical and non-technical standards, guidance and policy applicable to the architecture.

Definition: shows the taxonomy of types of technical, operational, and business standards, guidance and policy applicable to the architecture.

Recommended Implementation: SysML Block Definition Diagram.



**Figure 8:64 - Standards Taxonomy**

Elements

- [ActualOrganization](#)
- [CapabilityConfiguration](#)
- [Protocol](#)
- [ProtocolStack](#)
- [ResourceArchitecture](#)
- [ResourcePerformer](#)
- [Standard](#)
- [StandardOperationalActivity](#)
- [UAFElement](#)

## View Specifications::Standards::Structure

Contains the diagrams that document the Standards Structure Viewpoint.

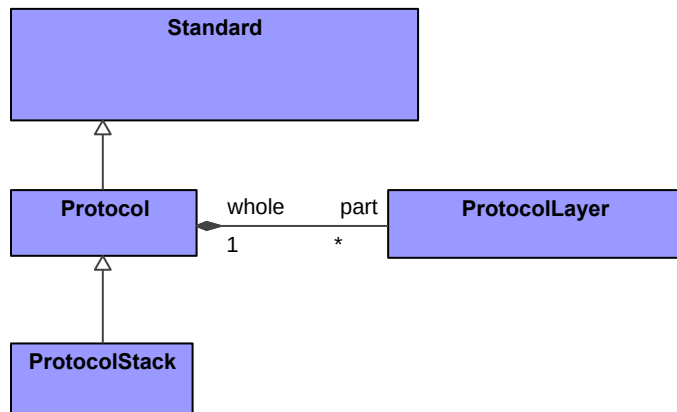
### View Specifications::Standards::Structure::Standards Structure

Stakeholders: Solution Providers, Systems Engineers, Software Engineers, Systems Architects.

Concerns: the specification of the protocol stack used in the architecture.

Definition: shows the composition of standards required to achieve the architecture's objectives.

Recommended Implementation: SysML Internal Block Diagram.



**Figure 8:65 - Standards Structure**

Elements

- [Protocol](#)
- [ProtocolLayer](#)
- [ProtocolStack](#)
- [Standard](#)

## View Specifications::Standards::Roadmap

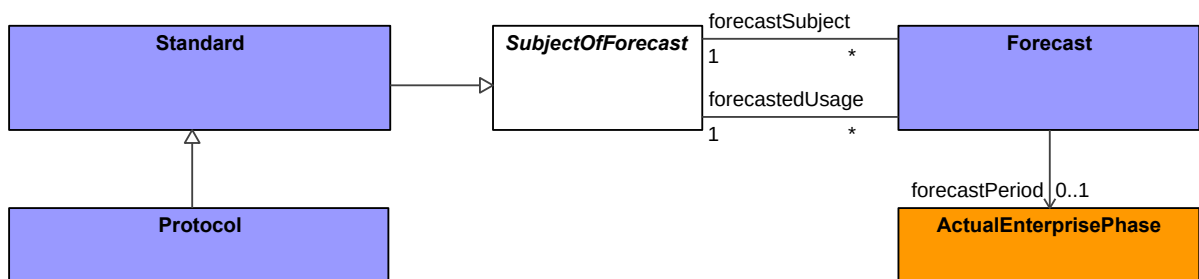
Contains the diagrams that document the Standards Roadmap Viewpoint.

### View Specifications::Standards::Roadmap::Standards Roadmap

Stakeholders: Solution Providers, Systems Engineers, Systems Architects, Software Engineers, Business Architects.  
 Concerns: expected changes in technology-related standards and conventions, operational standards, or business standards and conventions.

Definition: defines the underlying current and expected standards. Expected standards are those that can be reasonably forecast given the current state of technology, and expected improvements / trends.

Recommended Implementation: timeline, tabular format, SysML Block Definition Diagram.



**Figure 8:66 - Standards Roadmap**

Elements

- [ActualEnterprisePhase](#)
- [Forecast](#)
- [Protocol](#)
- [Standard](#)
- [SubjectOfForecast](#)

## View Specifications::Standards::Traceability

Contains the diagrams that document the Standards Traceability Viewpoint.

### View Specifications::Standards::Traceability::Standards Traceability

Stakeholders: Solution Providers, Systems Engineers, Software Engineers, Systems Architects, Business Architects.

Concerns: standards that need to be taken in account to ensure the interoperability of the implementation of architectural elements.

Definition: shows the applicability of standards to specific elements in the architecture.

Recommended Implementation: tabular format, matrix format, SysML Block Definition Diagram.

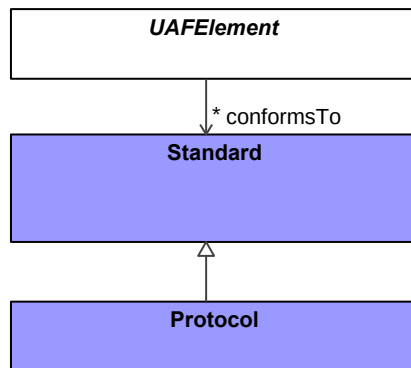


Figure 8:67 - Standards Traceability

Elements

- [Protocol](#)
- [Standard](#)
- [UAElement](#)

## 8.1.10 View Specifications::Actual Resources

Stakeholders: Solution Providers, Systems Engineers, Business Architects, Human Resources.

Concerns: the analysis, e.g., evaluation of different alternatives, what-if, trade-offs, V&V on the actual resource configurations.

Definition: illustrates the expected or achieved actual resource configurations and actual relationships between them.

### View Specifications::Actual Resources::Structure

Contains the diagrams that document the Actual Resources Structure Viewpoint.

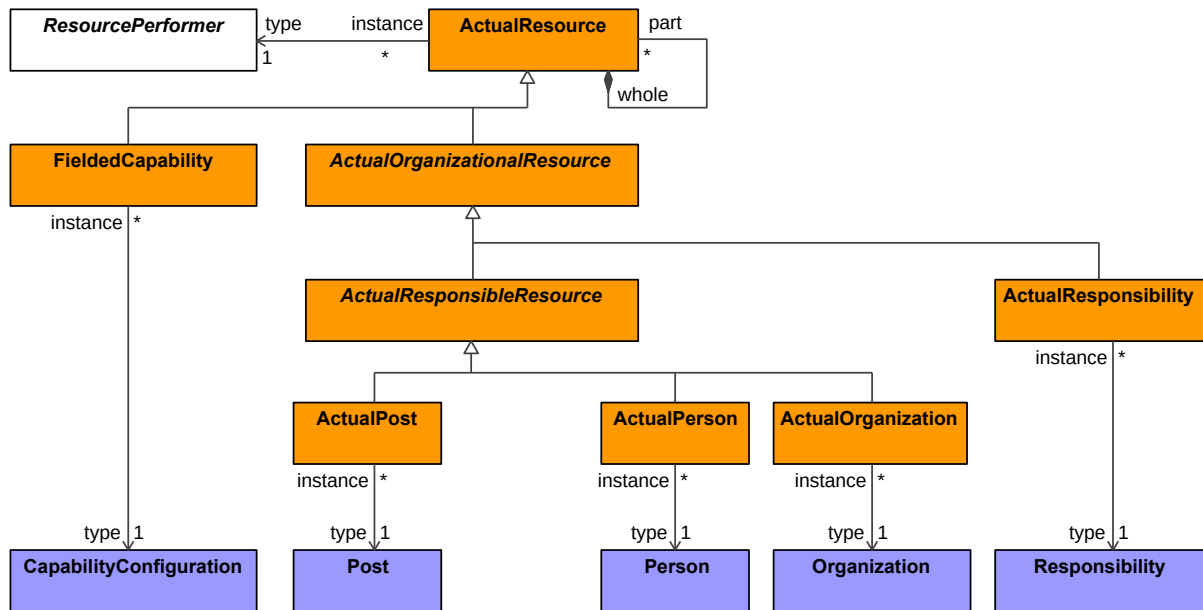
#### View Specifications::Actual Resources::Structure::Actual Resources Structure

Stakeholders: Solution Providers, Systems Engineers, Business Architects.

Concerns: the analysis, e.g., evaluation of different alternatives, what-if, trade-offs, V&V on the actual resource configurations as it provides a means to capture different solution architectures. The detailed analysis (trade-off, what-if etc.) is carried out using the Resource Constraints view.

Definition: illustrates the expected or achieved actual resource configurations required to meet an operational need.

Recommended Implementation: SysML Block Definition Diagram.



**Figure 8:68 - Actual Resources Structure**

Elements

- [ActualOrganization](#)
- [ActualOrganizationalResource](#)
- [ActualPerson](#)
- [ActualPost](#)
- [ActualResource](#)
- [ActualResponsibility](#)
- [ActualResponsibleResource](#)
- [CapabilityConfiguration](#)
- [FieldedCapability](#)
- [Organization](#)
- [Person](#)
- [Post](#)
- [ResourcePerformer](#)
- [Responsibility](#)

## View Specifications::Actual Resources::Connectivity

Contains the diagrams that document the Actual Resources Connectivity Viewpoint.

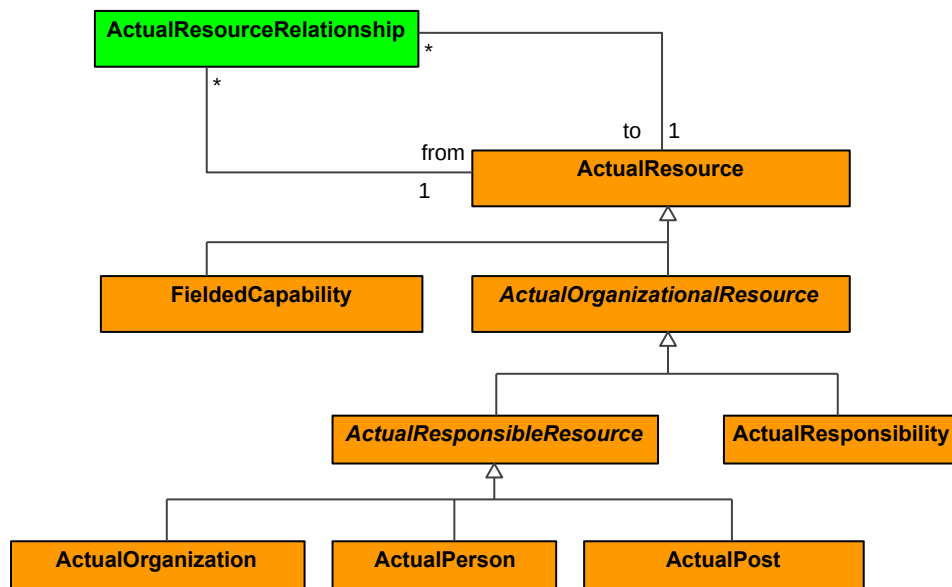
### View Specifications::Actual Resources::Connectivity::Actual Resources Connectivity

Stakeholders: Solution Providers, Systems Engineers, Business Architects.

Concerns: the communication of actual resource.

Definition: illustrates the actual resource configurations and actual relationships between them.

Recommended Implementation: tabular format, SysML Block Definition Diagram, SysML Internal Block Diagram, SysML Sequence Diagram.



**Figure 8:69 - Actual Resources Connectivity**

Elements

- [ActualOrganization](#)
- [ActualOrganizationalResource](#)
- [ActualPerson](#)
- [ActualPost](#)
- [ActualResource](#)
- [ActualResourceRelationship](#)
- [ActualResponsibility](#)
- [ActualResponsibleResource](#)
- [FieldedCapability](#)

## View Specifications::Actual Resources::Traceability

Contains the diagrams that document the Actual Resources Traceability Viewpoint.

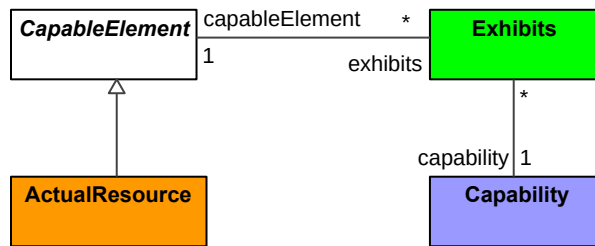
### View Specifications::Actual Resources::Traceability::Actual Resources Traceability

Stakeholders: Systems Engineers, Enterprise Architects, Solution Providers, Business Architects.

Concerns: traceability between operational activities and functions that implements them.

Definition: depicts the mapping of functions to operational activities and thus identifies the transformation of an operational need into a purposeful function performed by a resource or solution.

Recommended Implementation: Matrix format, SysML Block Definition Diagram.



**Figure 8:70 - Actual Resources Traceability**

Elements

- [ActualResource](#)
- [Capability](#)
- [CapableElement](#)
- [Exhibits](#)

### 8.1.11 View Specifications::Dictionary

Stakeholders: Architects, users of the architecture, Capability Owners, Systems Engineers, Solution Providers.

Concerns: Definitions for all the elements in the architecture, libraries of environments and measurements.

Definition: Presents all the elements used in an architecture. Can be used specifically to capture:

- a. Elements and relationships that are involved in defining the environments applicable to capability, operational concept or set of systems.
- b. Measurable properties that can be used to support analysis such as KPIs, MoEs, TPIs etc.

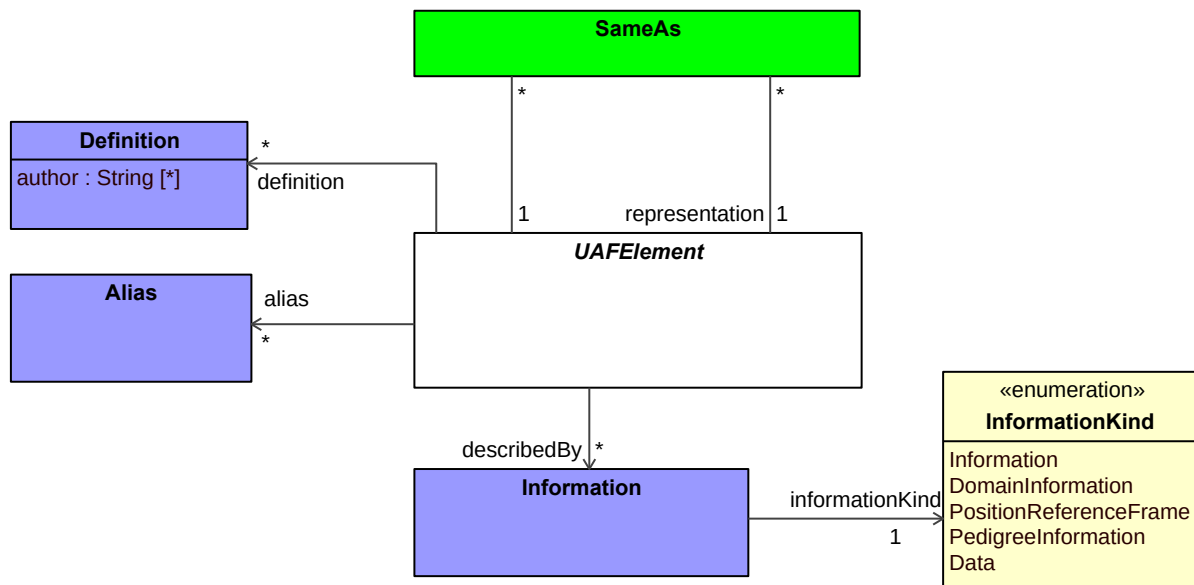
#### View Specifications::Dictionary::Dictionary

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: provides a central reference for a given architecture's data and metadata. It enables the set of architecture description to stand alone, with minimal reference to outside resources.

Definition: contains definitions of terms used in the given architecture. It consists of textual definitions in the form of a glossary, their taxonomies, and their metadata (i.e., data about architecture data), including metadata for any custom-tailored views. Architects should use standard terms where possible (i.e., terms from existing, approved dictionaries, glossaries, and lexicons).

Recommended Implementation: text, table format.



**Figure 8:71 - Dictionary**

Elements

- [Alias](#)
- [Definition](#)
- [Information](#)
- [SameAs](#)
- [UAFElement](#)

### 8.1.12 View Specifications::Summary & Overview

Stakeholders: Executives, PMs, Enterprise Architects.

Concerns: executive-level summary information in a consistent form.

Definition: provides executive-level summary information in a consistent form that allows quick reference and comparison between architectural descriptions. Includes assumptions, constraints, and limitations that may affect high-level decisions relating to an architecture-based work program.

#### View Specifications::Summary & Overview::Summary & Overview

Stakeholders: Decision makers, Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: quick overview of an architecture description and summary of analysis. In the initial phases of architecture development, it serves as a planning guide. Upon completion of an architecture, it provides a summary of findings, and any conducted analysis.

Definition: provides executive-level summary information in a consistent form that allows quick reference and comparison among architectures. The Summary and Overview includes assumptions, constraints, and limitations that may affect high-level decision processes involving the architecture.

Recommended Implementation: text, free form diagram, table format.



## View Specifications::Requirements::Requirements

Stakeholders: Requirement Engineers, Solution Providers, Systems Engineers, Software Engineers, Systems Architects, Business Architects.

Concerns: provides a central reference for a set of stakeholder needs expressed as requirements, their relationship (via traceability) to more detailed requirements and the solution described by the architecture that will meet those requirements.

Definition: used to represent requirements, their properties, and relationships (trace, verify, satisfy, refine) between each other and to UAF architectural elements.

Recommended Implementation: SysML Requirement Diagram, tabular format, matrix format.

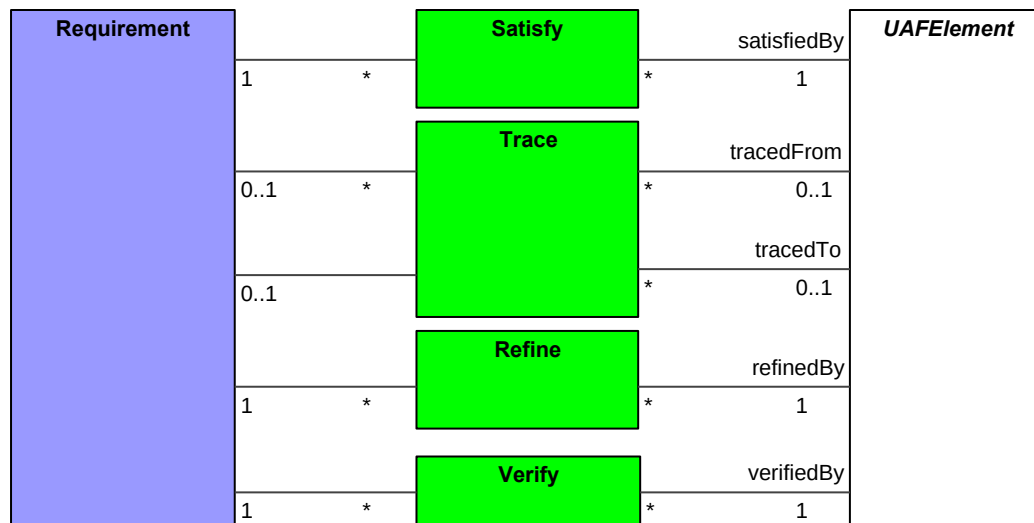


Figure 8:73 - Requirements

Elements

- Refine
- Requirement
- Satisfy
- Trace
- [UAFElement](#)
- Verify

## 8.1.14 View Specifications::Information

Stakeholders: Data Modelers, Software Engineers, Systems Engineers

Concerns: address the information perspective on operational, service, and resource architectures.

Definition: allows analysis of an architecture's information and data definition aspect, without consideration of implementation specific issues.

Recommended Implementation: SysML Block Definition Diagram.

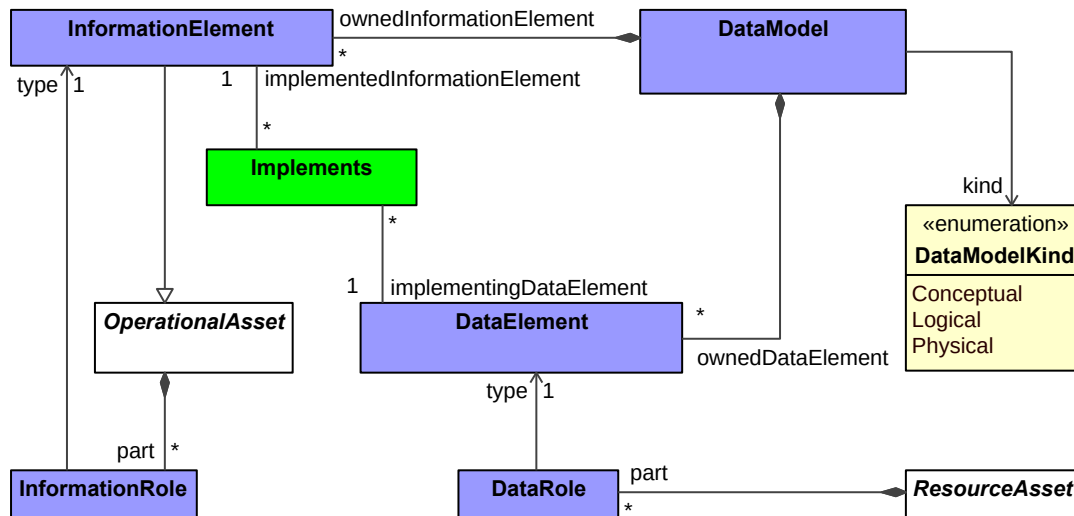
### View Specifications::Information::Information Model

Stakeholders: Data Modelers, Software Engineers, Systems Engineers

Concerns: address the information perspective on operational, service, and resource architectures.

Definition: allows analysis of an architecture's information and data definition aspect, without consideration of implementation specific issues.

Recommended Implementation: SysML Block Definition Diagram.



**Figure 8:74 - Information Model**

Elements

- [DataElement](#)
- [DataModel](#)
- [DataRole](#)
- [Implements](#)
- [InformationElement](#)
- [InformationRole](#)
- [OperationalAsset](#)
- [ResourceAsset](#)

### 8.1.15 View Specifications::Parameters

Stakeholders: Capability owners, Systems Engineers, Solution Providers.

Concerns: identifies measurable properties that can be used to support engineering analysis and environment for the Capabilities

Definition: Shows the measurable properties of something in the physical world and elements and relationships that are involved in defining the environments applicable to capability, operational concept or set of systems.

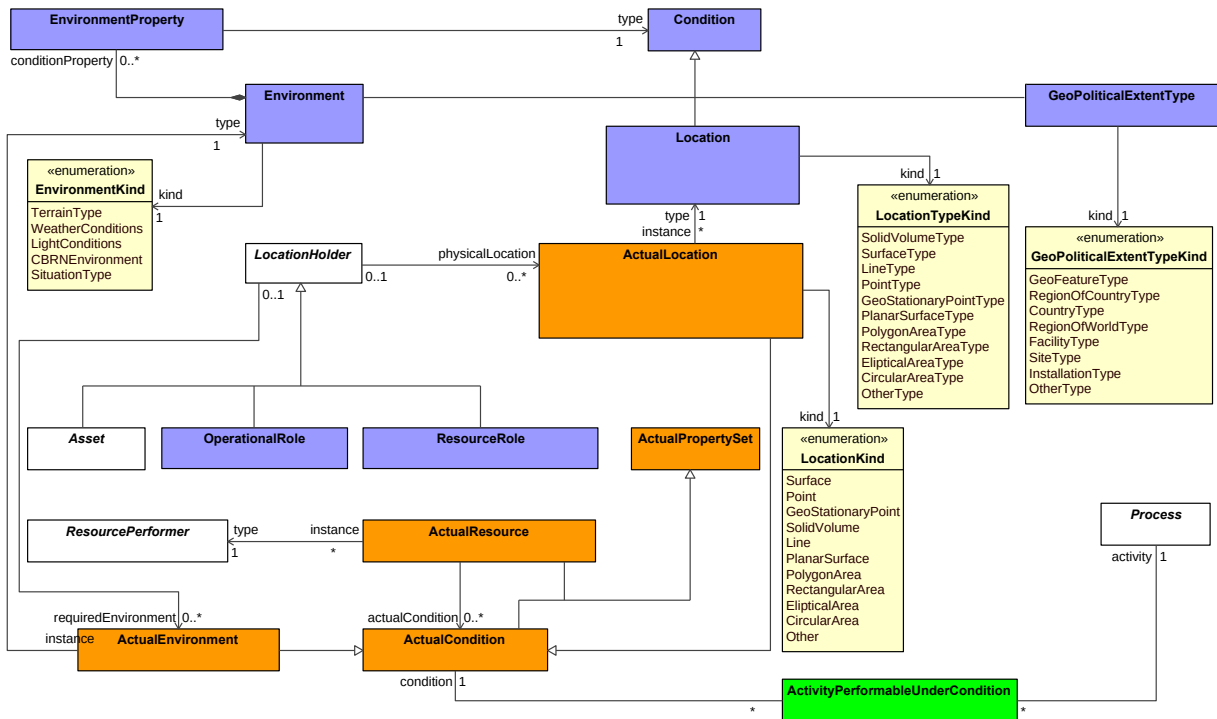
#### View Specifications::Parameters::Parameters: Environment

Stakeholders: Capability owners, Systems Engineers, Solution Providers.

Concerns: defines the environment for the capabilities.

Definition: shows the elements and relationships that are involved in defining the environments applicable to capability, operational concept or set of systems.

Recommended Implementation: SysML Block Definition Diagram.



**Figure 8:75 - Parameters: Environment**

Elements

- [ActivityPerformableUnderCondition](#)
- [ActualCondition](#)
- [ActualEnvironment](#)
- [ActualLocation](#)
- [ActualPropertySet](#)
- [ActualResource](#)
- [Asset](#)
- [Condition](#)
- [Environment](#)
- [EnvironmentProperty](#)
- [GeoPoliticalExtentType](#)
- [Location](#)
- [LocationHolder](#)
- [OperationalRole](#)
- [Process](#)
- [ResourcePerformer](#)
- [ResourceRole](#)

## View Specifications::Parameters::Parameters: Measurements

Stakeholders: Capability owners, Systems Engineers, Solution Providers.

Concerns: identifies measurable properties that can be used to support analysis such as KPIs, MoEs, TPIs etc.

Definition: Shows the measurable properties of something in the physical world, expressed in amounts of a unit of measure that can be associated with any element in the architecture.

Recommended Implementation: SysML Block Definition Diagram.

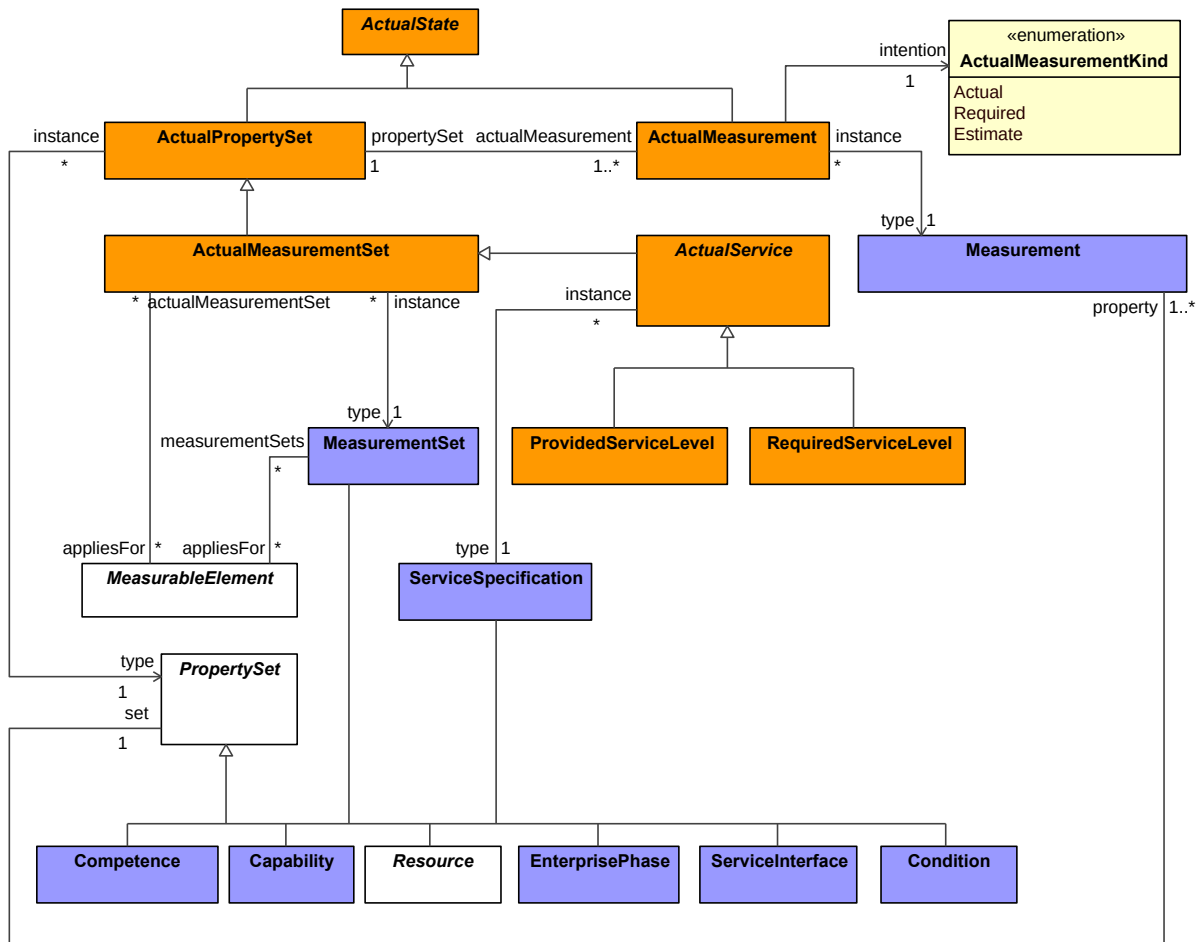


Figure 8:76 - Parameters: Measurements

Elements

- [ActualMeasurement](#)
- [ActualMeasurementSet](#)
- [ActualPropertySet](#)
- [ActualService](#)
- [ActualState](#)
- [Capability](#)
- [Competence](#)
- [Condition](#)
- [EnterprisePhase](#)
- [MeasurableElement](#)
- [Measurement](#)
- [MeasurementSet](#)
- [PropertySet](#)
- [ProvidedServiceLevel](#)
- [RequiredServiceLevel](#)
- [Resource](#)
- [ServiceInterface](#)
- [ServiceSpecification](#)

## 8.1.16 View Specifications::Other

Contains the diagrams that document the use of BPMN, NIEM, IEPPV in the context of UAF.

### View Specifications::Other::BPMN

Stakeholders: Business Architects, Enterprise Architects

Concerns: captures activity based behavior and flows.

Definition: describes the activities that are normally conducted in the course of achieving business goals that support a capability. It describes operational activities, their Inputs/Outputs, operational activity actions and flows between them using BPMN.

Recommended Implementation: BPMN Process Diagram.

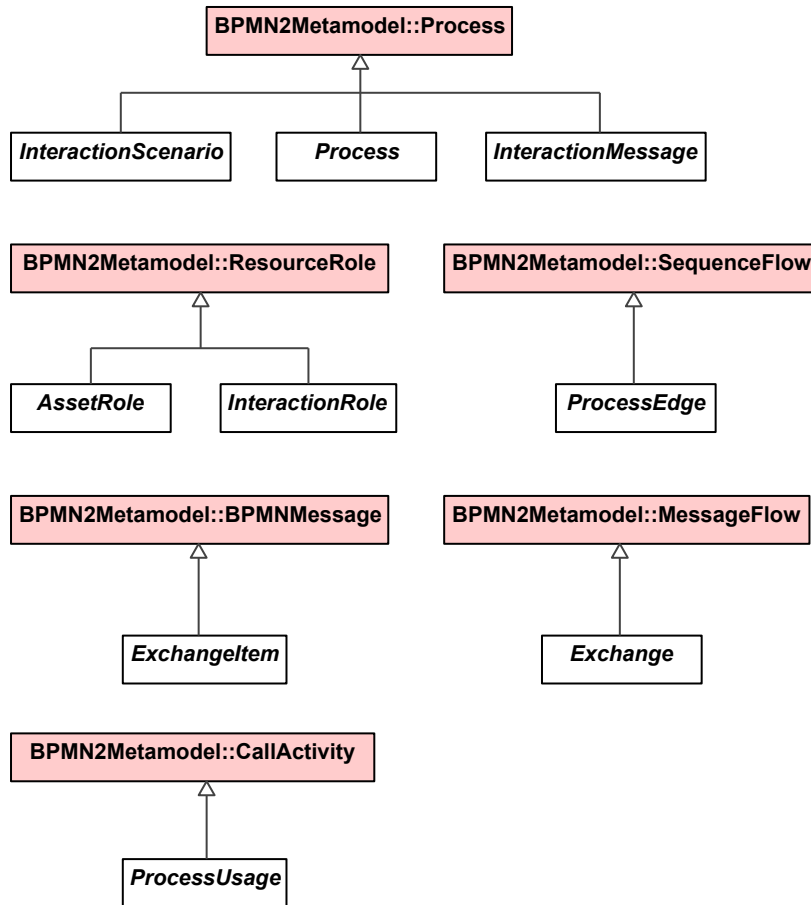


Figure 8:77 - BPMN

Elements

- [AssetRole](#)
- BPMN2Metamodel::BPMNMessage
- BPMN2Metamodel::CallActivity
- BPMN2Metamodel::MessageFlow
- BPMN2Metamodel::Process
- BPMN2Metamodel::ResourceRole
- BPMN2Metamodel::SequenceFlow
- [Exchange](#)

- [ExchangeItem](#)
- [InteractionMessage](#)
- [InteractionRole](#)
- [InteractionScenario](#)
- [Process](#)
- [ProcessEdge](#)
- [ProcessUsage](#)

## View Specifications::Other::IEPPV

Stakeholders: Data Modelers, Solution Providers, Systems Engineers, Software Engineers, Systems Architects, Business Architects, information architects.

Concerns: information exchanges, information interfaces, information interoperability, information sharing and safeguarding.

Definition: UAFP supports information modeling and traceability to IEPPV model elements using the IEPPV-defined elements: Message, SemanticElement, and FilteredSemanticElement, used to represent data, properties/attributes, structure, format, and relationships. The IEPPV profile enables the specification of the policies, rules and constraints governing the packaging (assembly, transformation, marking, redaction) of data elements conforming to information sharing and safeguarding requirements. The IEPPV profile also governs the processing (parsing, transformation, and marshalling) received information and data element.

Recommended Implementation: UML Class Diagram, SysML Block Diagram.

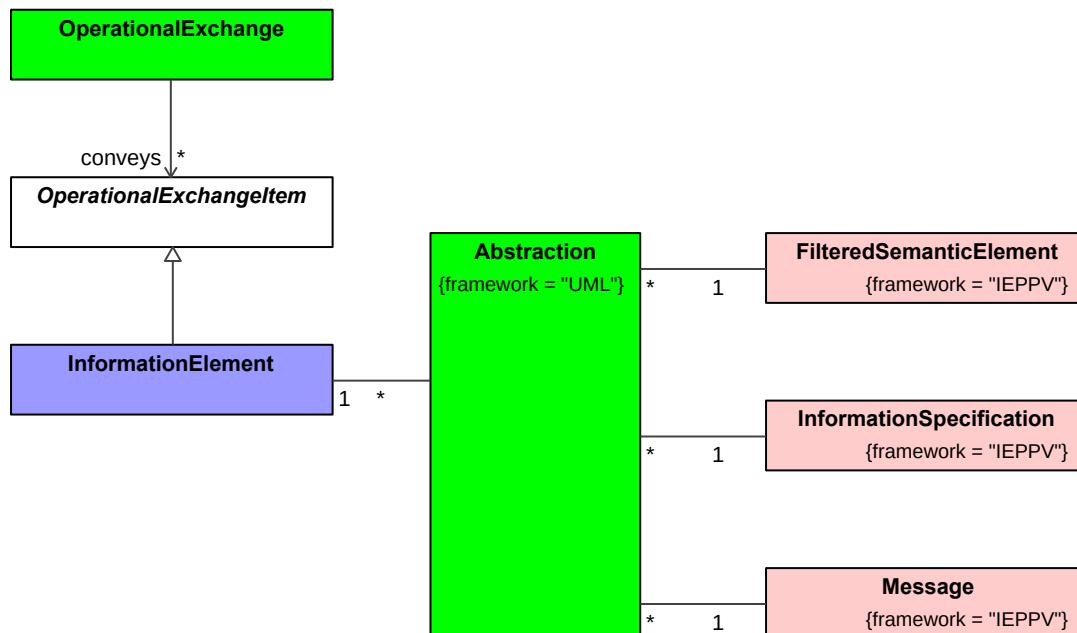


Figure 8:78 - IEPPV

Elements

- Abstraction
- FilteredSemanticElement
- [InformationElement](#)
- InformationSpecification
- Message
- [OperationalExchange](#)
- [OperationalExchangeItem](#)

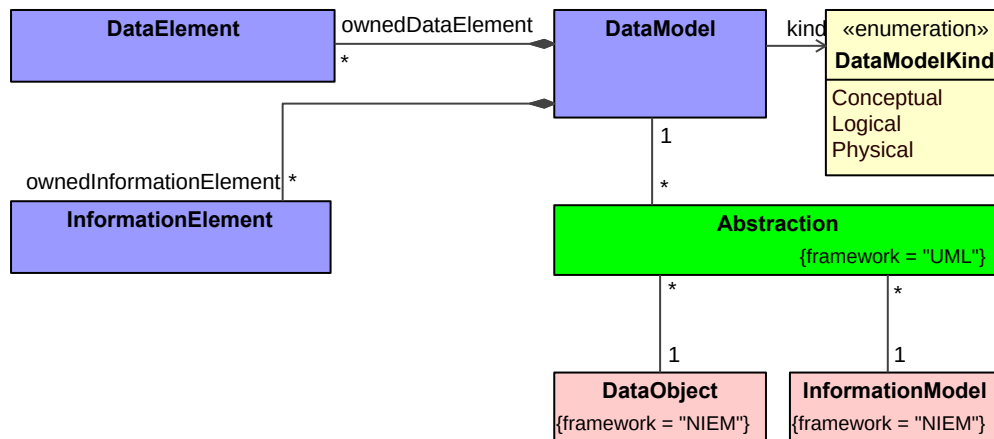
## View Specifications::Other::NIEM

Stakeholders: Data Modelers, Solution Providers, Systems Engineers, Software Engineers, Systems Architects, Business Architects.

Concerns: information exchanges, information interoperability, data schema.

Definition: A specification representing the structure, semantics, and relationships of data objects that satisfy an information exchange requirement. Used for organizing and packaging Model Package Descriptions (MPDs) and Information Exchange Package Documentation (IEPD) as defined by the National Information Exchange Model (NIEM). An IEPD is a type of MPD. The NIEM MPD defines an Enterprise Information Exchange Model (EIEM) as an MPD that contains NIEM-conforming schemas that define and declare data components to be consistently reused in the IEPDs of an enterprise. An EIEM is a collection of schemas organized into a collection of subset schemas and one or more extension schemas. An information sharing enterprise creates and maintains an EIEM.

Recommended Implementation: UML Class Diagram, SysML Block Diagram.



**Figure 8:79 - NIEM**

Elements

- Abstraction
- [DataElement](#)
- [DataModel](#)
- DataObject
- [InformationElement](#)
- InformationModel

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## 9 Domain Metamodel (DMM) Elements

### 9.1 Domain MetaModel

This package contains the elements of the DMM.

#### 9.1.1 Domain MetaModel::Metadata

Enterprise Architects, people who want to discover the architecture, Technical Managers.

Captures meta-data relevant to the entire architecture

Provide information pertinent to the entire architecture. Present supporting information rather than architectural models.

#### Domain MetaModel::Metadata::Taxonomy

##### ArchitectureMetadata

Taxonomy

No

[Metadata](#)

Description

Information associated with an ArchitecturalDescription, that supplements the standard set of tags used to summarize the Architecture. It states things like what methodology was used, notation, etc.

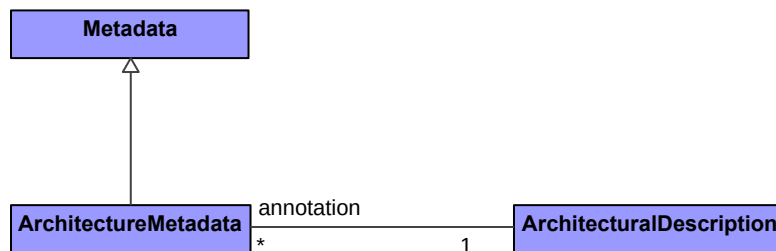


Figure 9:1 - ArchitectureMetadata

#### InteractionScenarioGeneralization

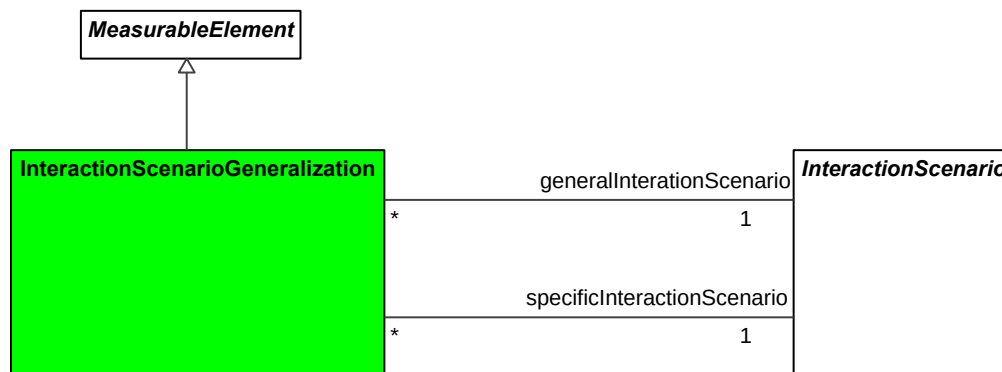
Taxonomy

No

UML2.5Metamodel::Generalization, [MeasurableElement](#)

Description

An InteractionScenarioGeneralization is a taxonomic relationship between a more general InteractionScenario and a more specific InteractionScenario.



**Figure 9:2 - InteractionScenarioGeneralization**

## Metadata

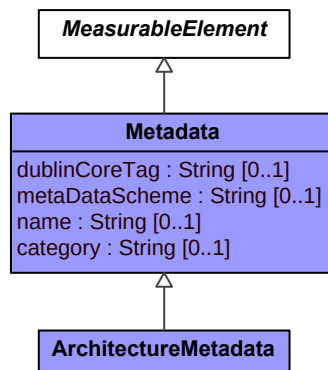
Taxonomy

No

[MeasurableElement](#)

## Description

A comment that can be applied to any element in the architecture. The attributes associated with this element details the relationship between the element and its related dublinCoreElement, metaDataScheme, category and name. This allows the element to be referenced using the Semantic Web.



**Figure 9:3 - Metadata**

## Attributes

category : String[0..1]

Defines the category of a Metadata element example:  
http://purl.org/dc/terms/abstract.

dublinCoreTag : String[0..1]

A metadata category that is a DublinCore tag.

metaDataScheme : String[0..1]

A representation scheme that defines a set of Metadata.

name : String[0..1]

The name of the Metadata.

## ProcessGeneralization

Taxonomy

No

UML2.5Metamodel::Generalization, [MeasurableElement](#)

Description

A ProcessGeneralization is a taxonomic relationship between a more general Process and a more specific Process.

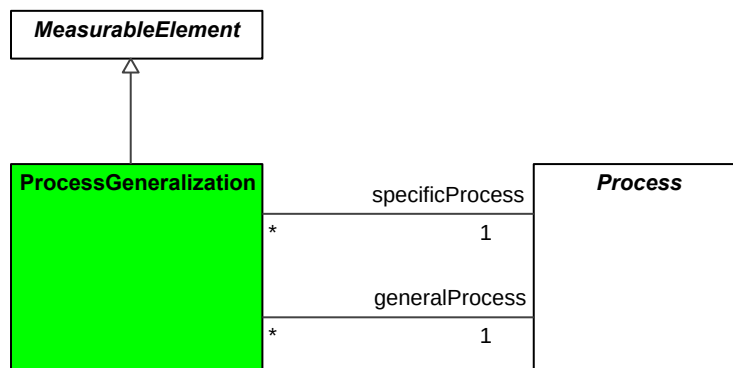


Figure 9:4 - ProcessGeneralization

## PropertySetGeneralization

Taxonomy

No

UML2.5Metamodel::Generalization, [MeasurableElement](#)

Description

A PropertySetGeneralization is a taxonomic relationship between a more general PropertySet and a more specific PropertySet.

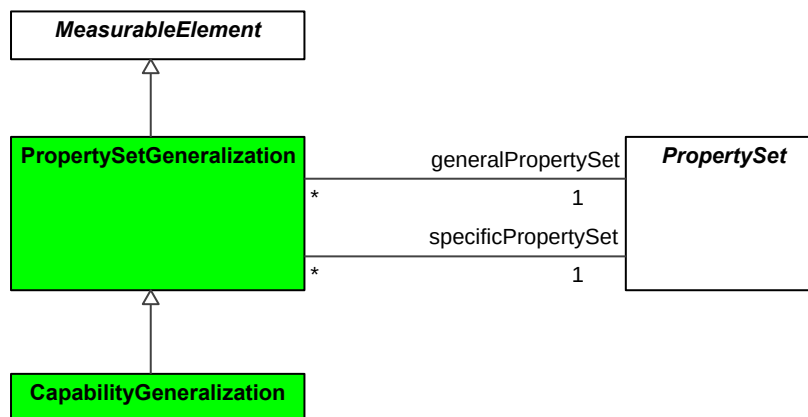


Figure 9:5 - PropertySetGeneralization

## StateDescriptionGeneralization

Taxonomy

No

UML2.5Metamodel::Generalization, [MeasurableElement](#)

Description

A StateDescriptionGeneralization is a taxonomic relationship between a more general StateDescription and a more specific StateDescription.

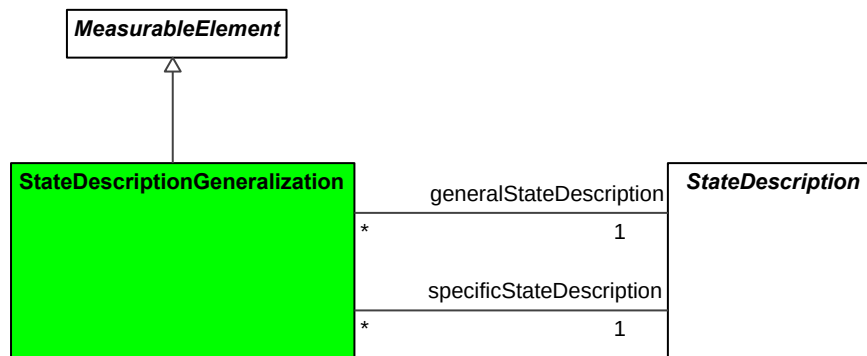


Figure 9:6 - StateDescriptionGeneralization

## Domain MetaModel::Metadata::Structure

### EnvironmentProperty

Structure

No

[MeasurableElement](#)

Description

A property of an Environment that is typed by a Condition. The kinds of Condition that can be represented are Location, GeoPoliticalExtentType and Environment.

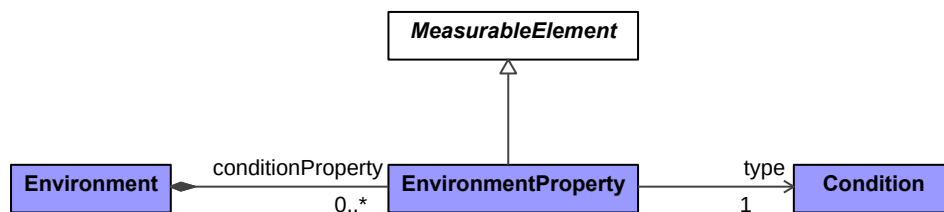


Figure 9:7 - EnvironmentProperty

## Domain MetaModel::Metadata::Connectivity

### Exchange

Connectivity

Yes

[MeasurableElement](#), BPMN2Metamodel::MessageFlow, [SubjectOfSecurityConstraint](#)

Description

Abstract tuple, grouping OperationalExchanges and ResourceExchanges that exchange Resources.

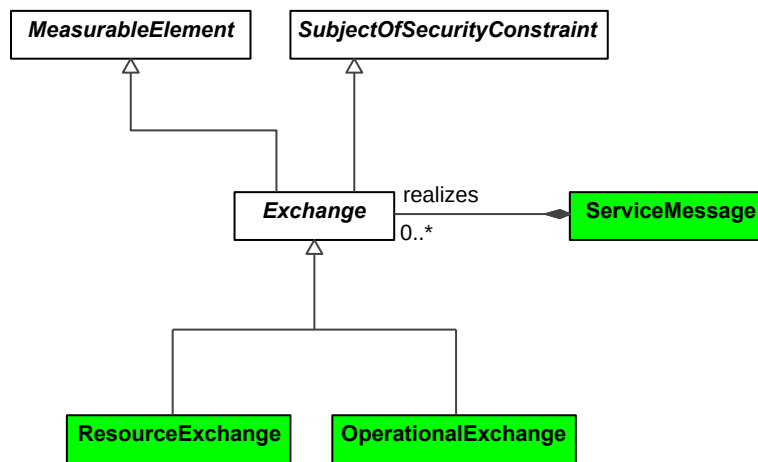


Figure 9:8 - Exchange

## Exchangeltem

Connectivity

Yes

BPMN2Metamodel::BPMNMessage

Description

An abstract grouping for elements that defines the types of elements that can be exchanged between Assets and conveyed by an Exchange.

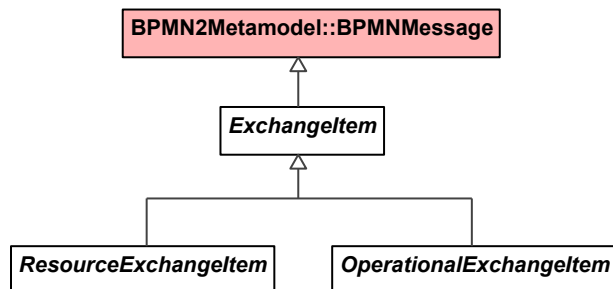


Figure 9:9 - Exchangeltem

## Resource

Connectivity

Yes

[PropertySet](#)

Description

Abstract type grouping all elements that can be conveyed by an Exchange.

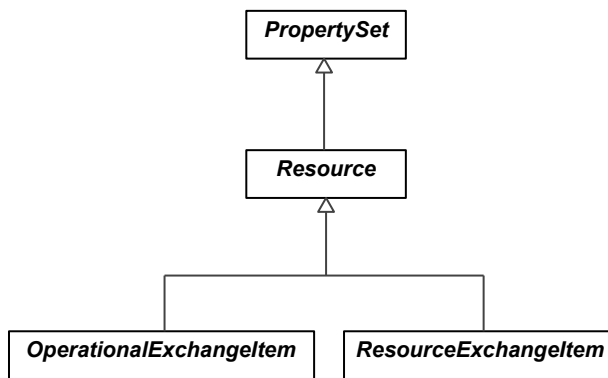


Figure 9:10 - Resource

## Domain MetaModel::Metadata::Processes

### ActivityPerformableUnderCondition

Processes

No

[MeasurableElement](#)

Description

The ActualCondition under which an Activity is performed.

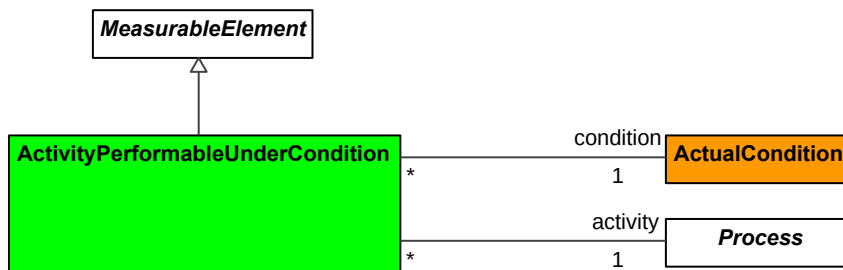


Figure 9:11 - ActivityPerformableUnderCondition

### IsCapableToPerform

Processes

No

[MeasurableElement](#)

Description

A tuple defining the traceability between the structural elements to the Activities that they can perform.

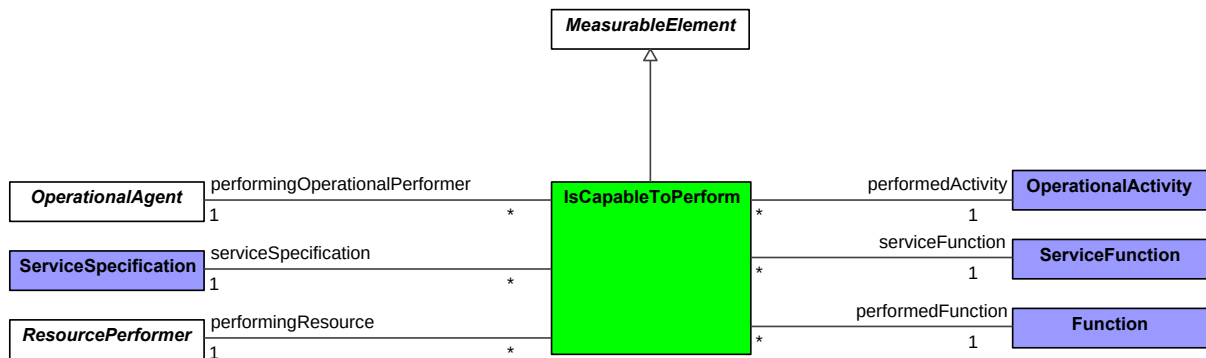


Figure 9:12 - IsCapableToPerform

## PerformsInContext

Processes

No

[MeasurableElement](#)

Description

A tuple that relates an OperationalAction to an OperationalRole, or a FunctionAction to a ResourceRole. It indicates that the action can be carried out by the role when used in a specific context or configuration.

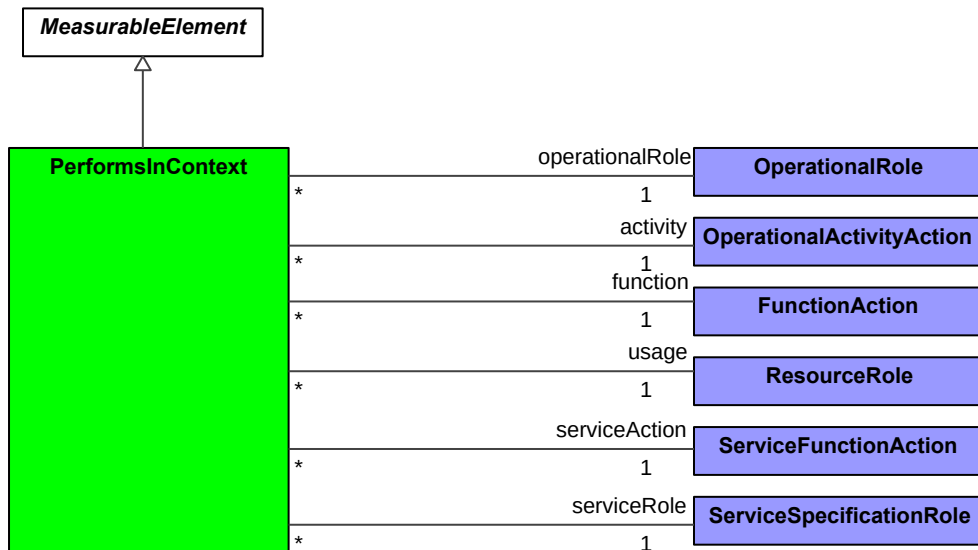


Figure 9:13 - PerformsInContext

## Process

Processes

Yes

[MeasurableElement](#), UML2.5Metamodel::Activity, BPMN2Metamodel::Process

Description

An abstract type that represents a behavior or process (i.e., a Function or OperationalActivity) that can be performed by a Performer.

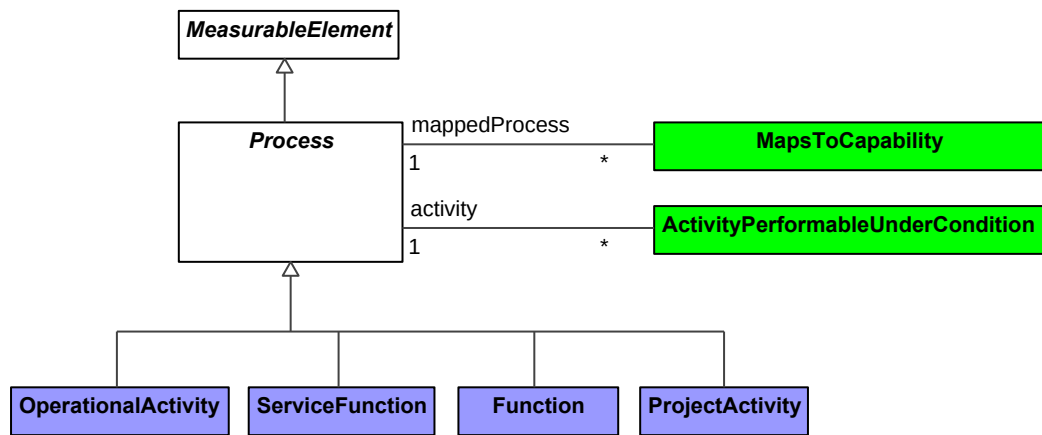


Figure 9:14 - Process

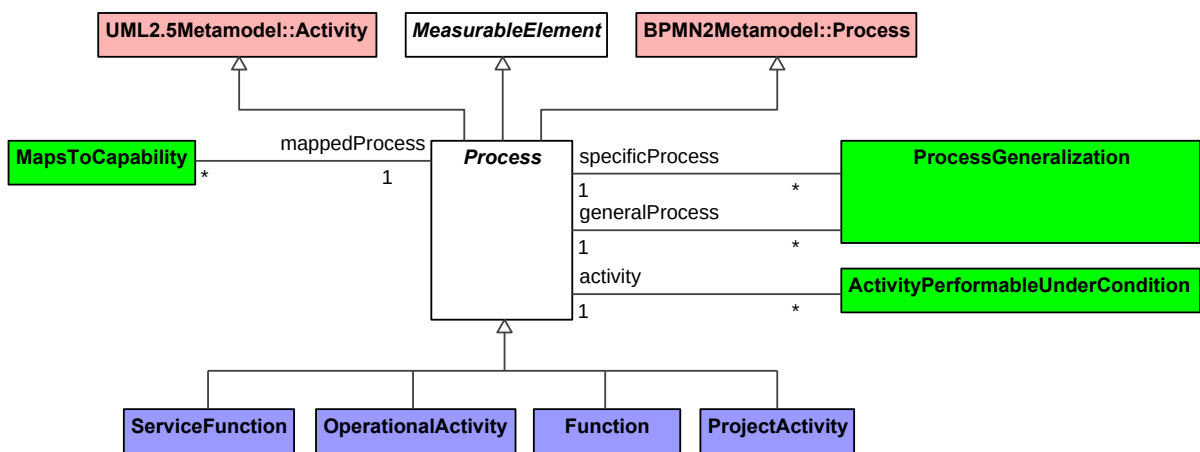


Figure 9:15 - Process

## ProcessEdge

Processes

Yes

[MeasurableElement](#), UML2.5Metamodel::Activity, UML2.5Metamodel::ActivityEdge, BPMN2Metamodel::SequenceFlow

Description

An abstract type that represents a behavior or process (i.e., a Function or OperationalActivity) that can be performed by a Performer.

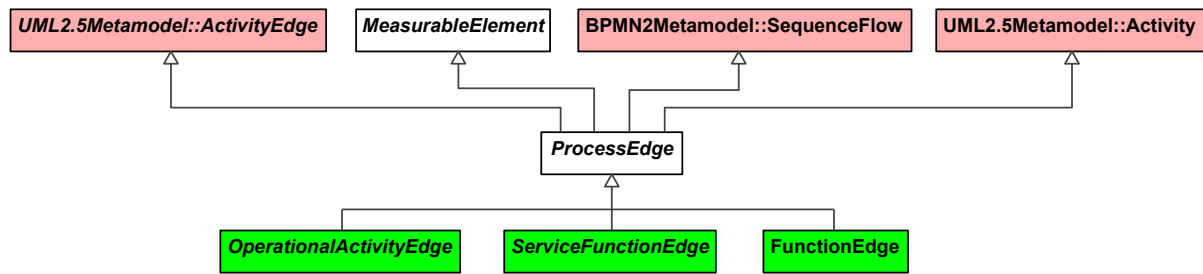


Figure 9:16 - ProcessEdge

## ProcessOperation

Processes

Yes

[MeasurableElement](#), UML2.5Metamodel::Activity, UML2.5Metamodel::Operation

Description

An abstract type that represents a behavior or process (i.e., a Function or OperationalActivity) that can be performed by a Performer.

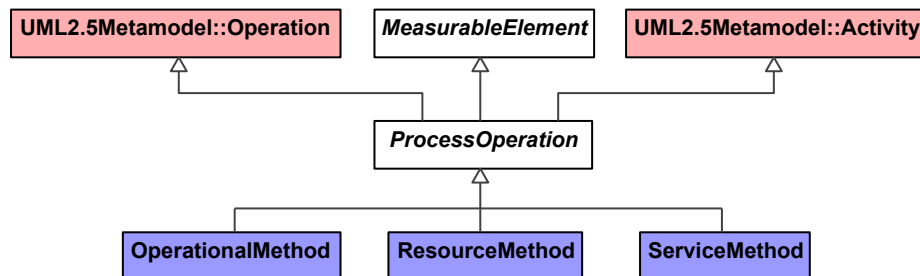


Figure 9:17 - ProcessOperation

## ProcessParameter

Processes

Yes

[MeasurableElement](#), UML2.5Metamodel::Activity, UML2.5Metamodel::CallBehaviorAction, UML2.5Metamodel::Parameter

Description

An abstract type that represents a behavior or process (i.e., a Function or OperationalActivity) that can be performed by a Performer.

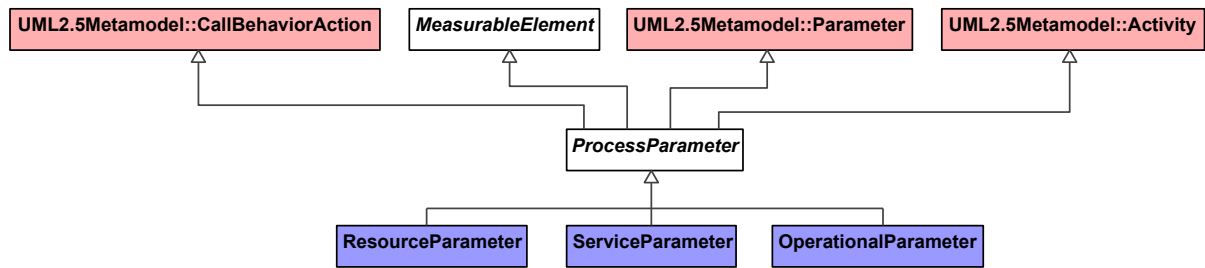


Figure 9:18 - ProcessParameter

## ProcessUsage

Processes

Yes

[MeasurableElement](#), UML2.5Metamodel::Activity, UML2.5Metamodel::CallBehaviorAction, BPMN2Metamodel::CallActivity

Description

An abstract type that represents a behavior or process (i.e., a Function or OperationalActivity) that can be performed by a Performer.

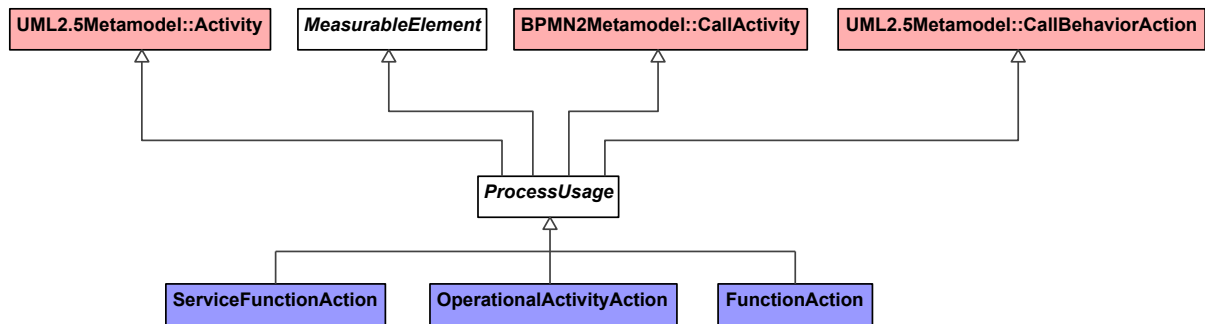


Figure 9:19 - ProcessUsage

## Domain MetaModel::Metadata::States

### StateDescription

States

Yes

UML2.5Metamodel::StateMachine

Description

An abstract type that represents a state machine (i.e., an OperationalStateDescription or ResourceStateDescription), depicting how the Asset responds to various events and the actions.

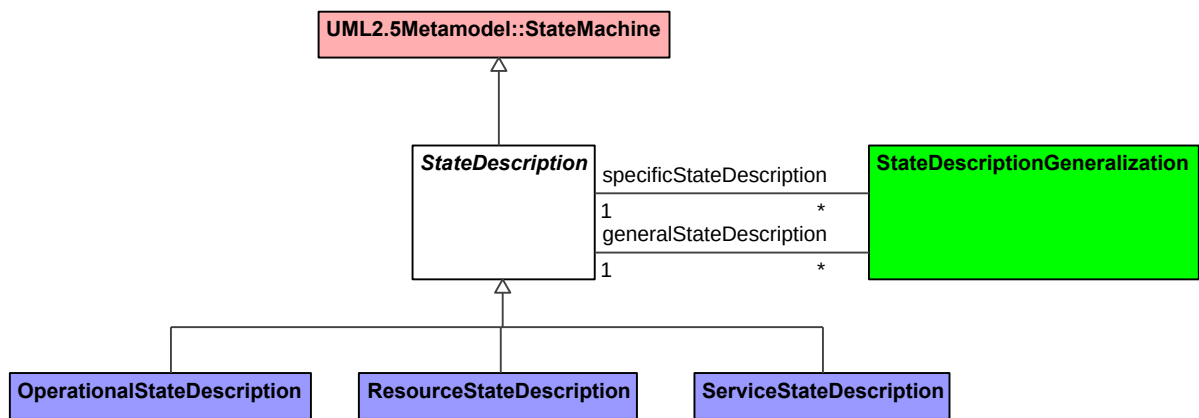


Figure 9:20 - StateDescription

## Domain MetaModel::Metadata::Interaction Scenarios

### InteractionMessage

Interaction Scenarios

Yes

[MeasurableElement](#), UML2.5Metamodel::Activity, BPMN2Metamodel::Process, UML2.5Metamodel::Interaction, UML2.5Metamodel::Message

Description

An abstract type that groups several types of messages used in the InteractionScenario.

### InteractionRole

Interaction Scenarios

Yes

BPMN2Metamodel::ResourceRole

Description

An abstract type that represents an individual participant in the InteractionScenario.

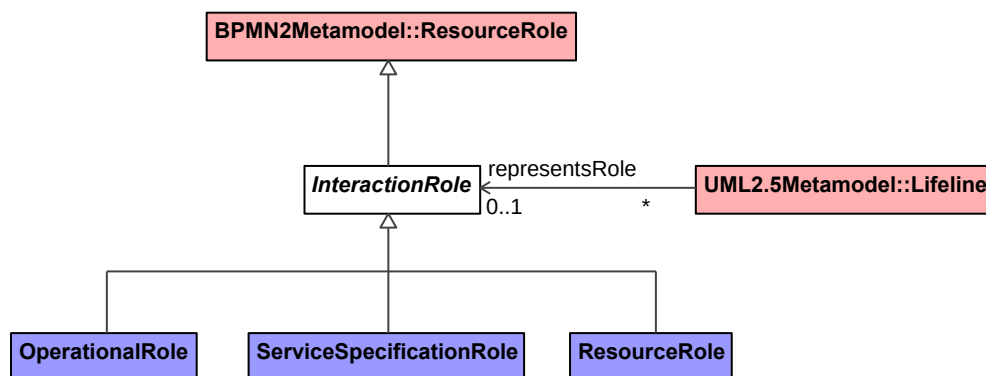


Figure 9:21 - InteractionRole

## InteractionScenario

Interaction Scenarios

Yes

[MeasurableElement](#), UML2.5Metamodel::Activity, BPMN2Metamodel::Process, UML2.5Metamodel::Interaction

Description

An abstract type that specifies interactions between Assets, like ResourcePerformers, and ServiceSpecifications.

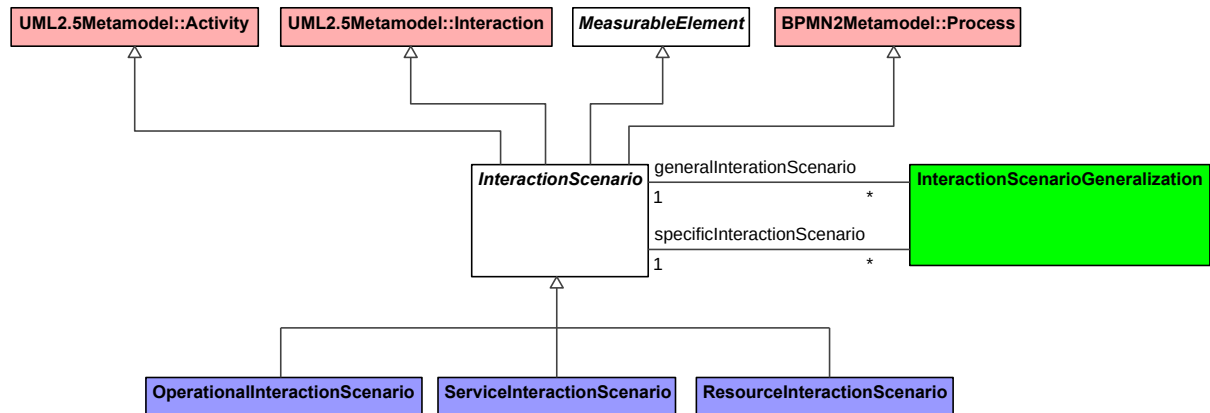


Figure 9:22 - InteractionScenario

## Domain MetaModel::Metadata::Information

### Information

Information

No

[MeasurableElement](#)

Description

A comment that describes the state of an item of interest in any medium or form -- and is communicated or received.

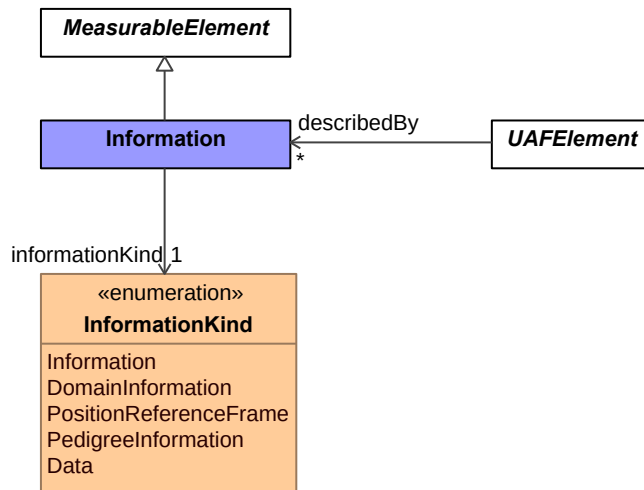


Figure 9:23 - Information

## Domain MetaModel::Metadata::Constraints

### Rule

Constraints

Yes

[MeasurableElement](#)

Description

An abstract type for all types of constraint (i.e., an OperationalConstraint could detail the rules of accountancy best practice).

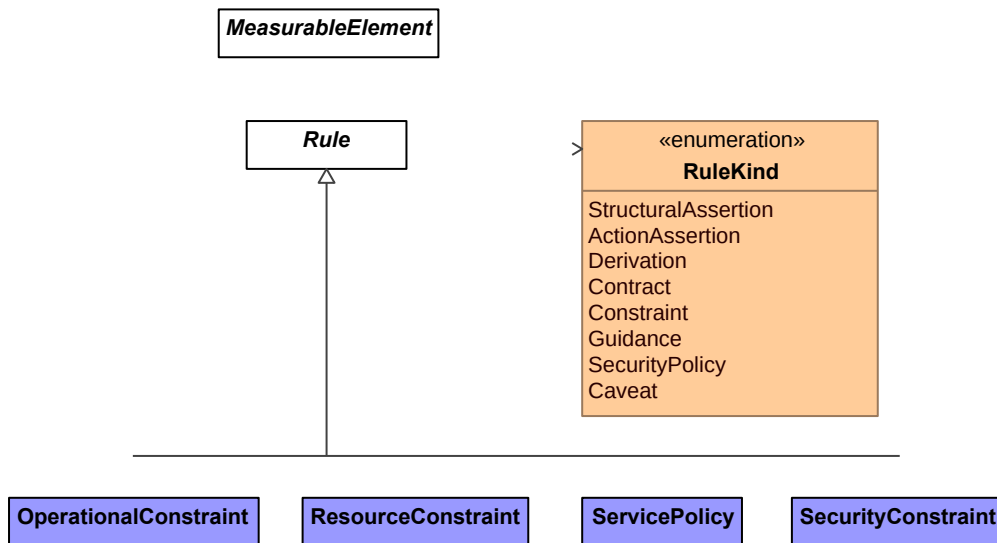


Figure 9:24 - Rule

## Domain MetaModel::Metadata::Traceability

### ArchitecturalReference

Traceability

No

[MeasurableElement](#)

Description

A tuple that specifies that one architectural description refers to another.

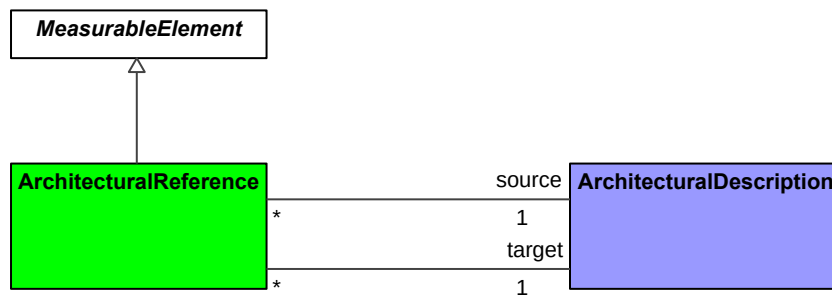


Figure 9:25 - ArchitecturalReference

### Implements

Traceability

No

[MeasurableElement](#)

Description

A tuple that defines how an element in the upper layer of abstraction is implemented by a semantically equivalent element (for example tracing the Functions to the OperationalActivities) in the lower level of abstraction.

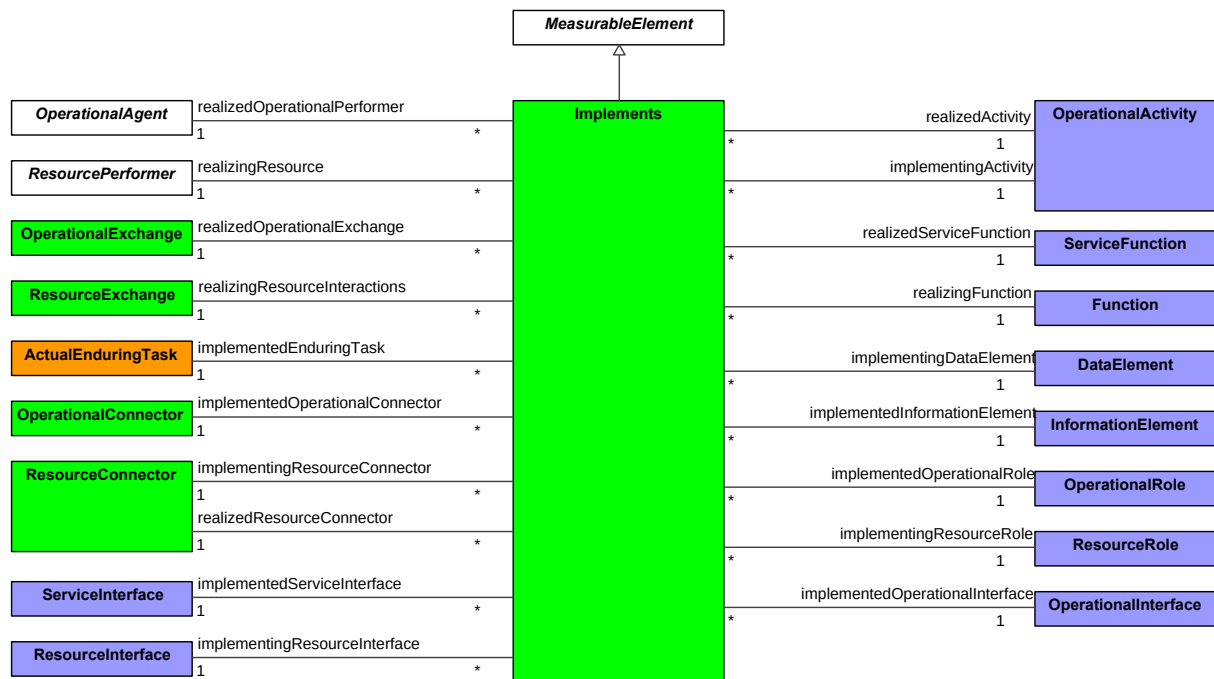


Figure 9:26 - Implements

## 9.1.2 Domain MetaModel::Strategic

### Domain MetaModel::Strategic::Taxonomy

#### Capability

Taxonomy

No

[PropertySet](#), [Desirer](#)

#### Description

A high-level specification of the enterprise's ability to execute a specified course of action.

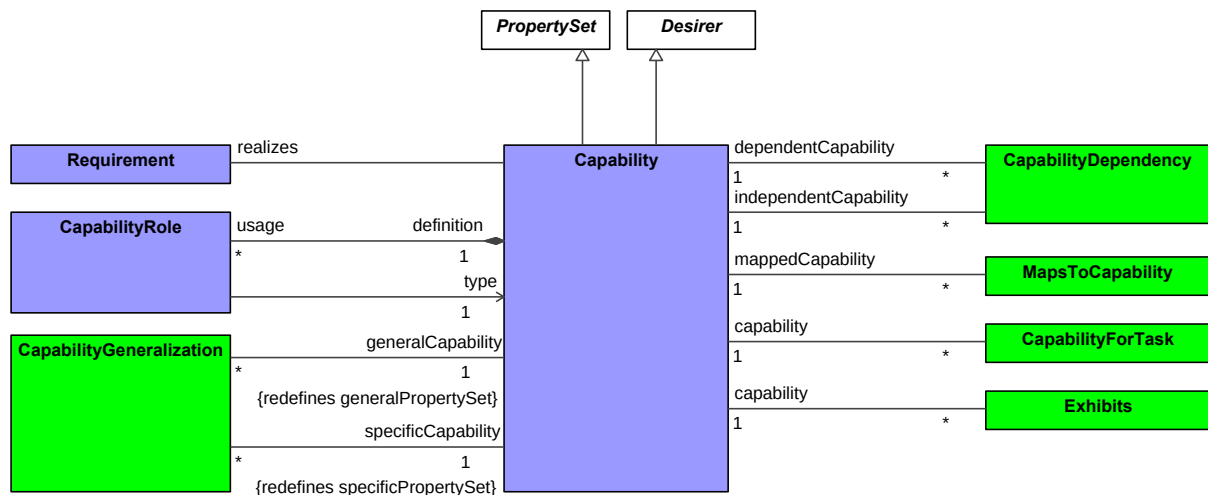


Figure 9:27 - Capability

## CapabilityGeneralization

Taxonomy

No

[PropertySetGeneralization](#)

Description

A CapabilityGeneralization is a taxonomic relationship between a more general Capability and a more specific Capability.



Figure 9:28 - CapabilityGeneralization

## Domain MetaModel::Strategic::Structure

### ActualEnduringTask

Structure

No

[CapableElement](#), [ActualPropertySet](#)

Description

An actual undertaking recognized by an enterprise as being essential to achieving its goals - i.e., a strategic specification of what the enterprise does.

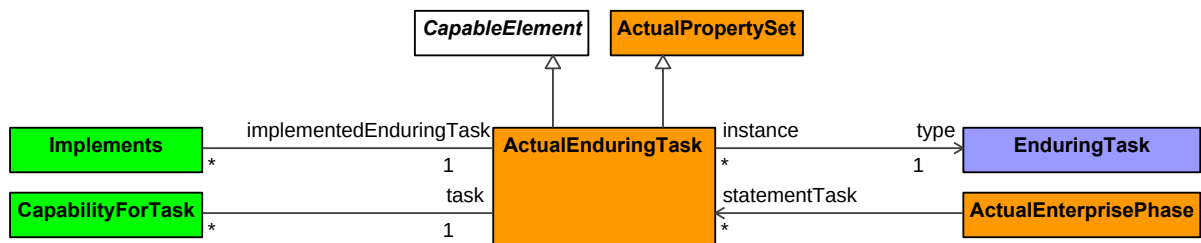


Figure 9:29 - ActualEnduringTask

## ActualEnterprisePhase

Structure

No

[CapableElement](#), [ActualPropertySet](#), [Achiever](#)

Description

An individual that describes the phase of an actual enterprise endeavor.

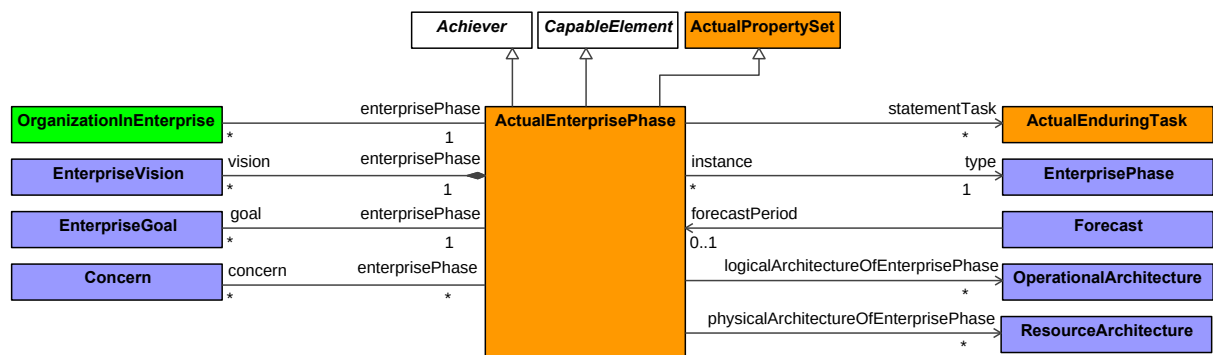


Figure 9:30 - ActualEnterprisePhase

## CapabilityRole

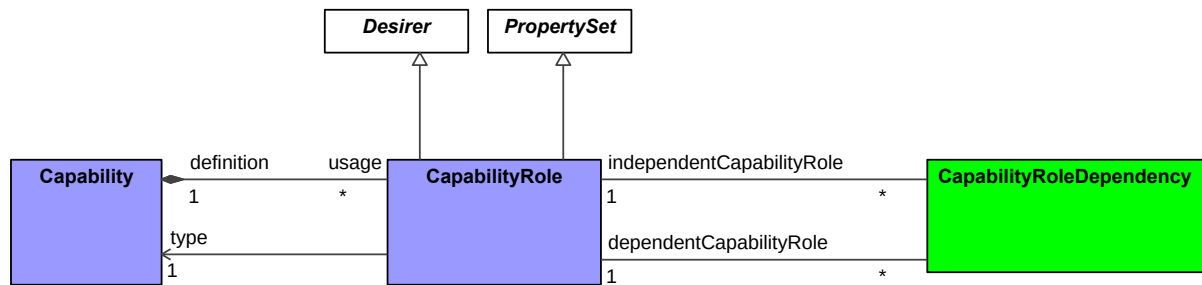
Structure

No

[PropertySet](#), [Desirer](#), [MeasurableElement](#)

Description

A high-level specification of the enterprise's ability to execute a specified course of action.



### Figure 9:31 - CapabilityRole

## EnduringTask

## Structure

No

## PropertySet

### Description

A type of template behavior recognized by an enterprise as being essential to achieving its goals - i.e., a template for a strategic specification of what the enterprise does.



### Figure 9:32 - EnduringTask

## EnterpriseGoal

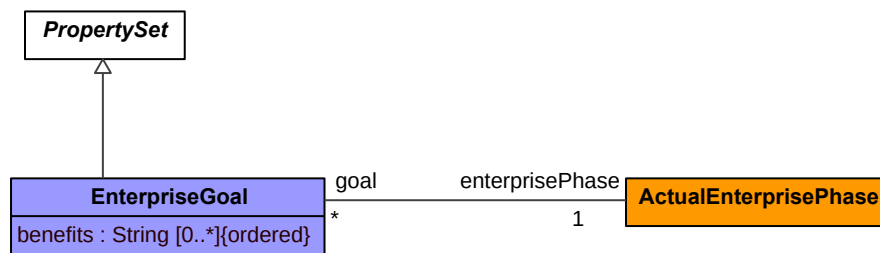
## Structure

No

## PropertySet

### Description

A statement about a state or condition of the enterprise to be brought about or sustained through appropriate Means. An EnterpriseGoal amplifies an EnterpriseVision that is, it indicates what must be satisfied on a continuing basis to effectively attain the EnterpriseVision. BMM: OMG dtc-13-08-24.



**Figure 9:33 - EnterpriseGoal**

## Attributes

benefits : String[0..\*] A description of the usefulness of the Goal in terms of why the state or condition of the Enterprise is worth attaining.

## EnterprisePhase

Structure

No

[PropertySet](#)

## Description

A type of a current or future state of the enterprise.

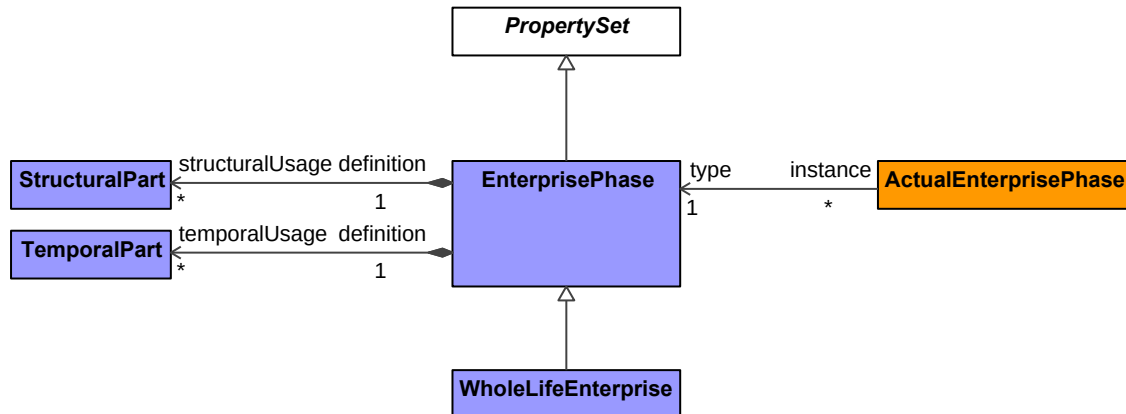


Figure 9:34 - EnterprisePhase

## EnterpriseVision

Structure

No

[PropertySet](#)

## Description

A Vision describes the future state of the enterprise, without regard to how it is to be achieved. BMM: OMG dtc-13-08-24.

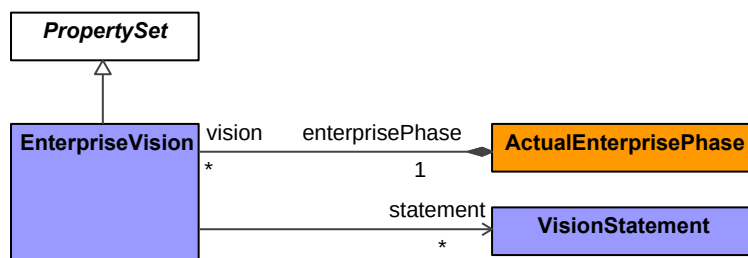


Figure 9:35 - EnterpriseVision

## StructuralPart

Structure

No

## [MeasurableElement](#)

Description

A current or future state of the wholeLifeEnterprise or another EnterprisePhase.

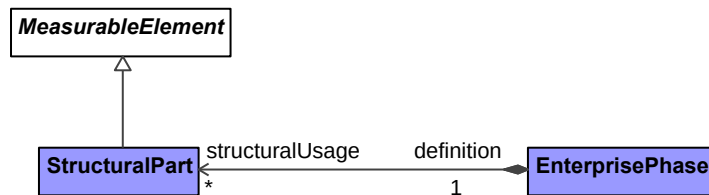


Figure 9:36 - StructuralPart

## TemporalPart

Structure

No

## [MeasurableElement](#)

Description

A current or future state of the wholeLifeEnterprise or another EnterprisePhase.

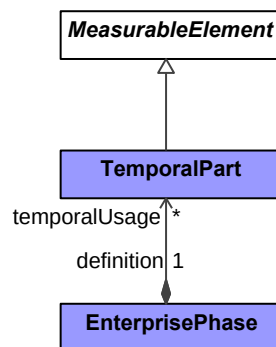


Figure 9:37 - TemporalPart

## VisionStatement

Structure

No

## [MeasurableElement](#)

Description

A type of comment that describes the future state of the enterprise, without regard to how it is to be achieved. BMM: OMG dtc-13-08-24.

**Figure 9:38 - VisionStatement**

## **WholeLifeEnterprise**

Structure

No

[EnterprisePhase](#)

Description

A WholeLifeEnterprise is a purposeful endeavor of any size involving people, organizations and supporting systems. It is made up of TemporalParts and StructuralParts.

9(s)2.70 Tdu -831VisionSomTjEMaC /H4n88C45 (onnec8)Tj33C /H4vC /H48Bis

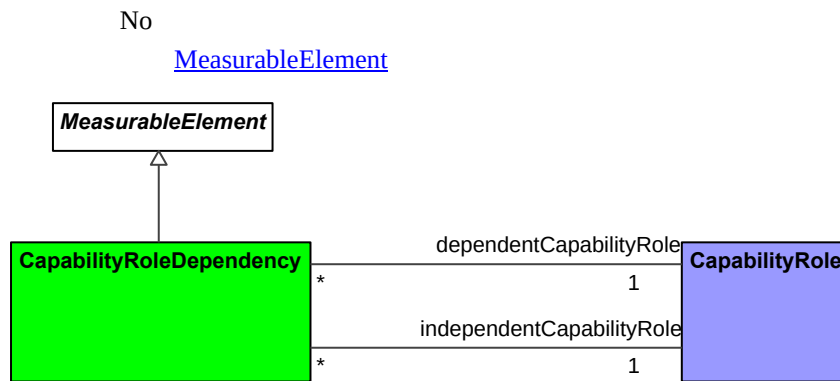


Figure 9:41 - CapabilityRoleDependency

## Domain MetaModel::Strategic::States

### AchievedEffect

States

No

[MeasurableElement](#)

Description

A tuple that exists between an ActualState (e.g., observed/measured during testing) of an element that attempts to achieve a DesiredEffect and an Achiever.

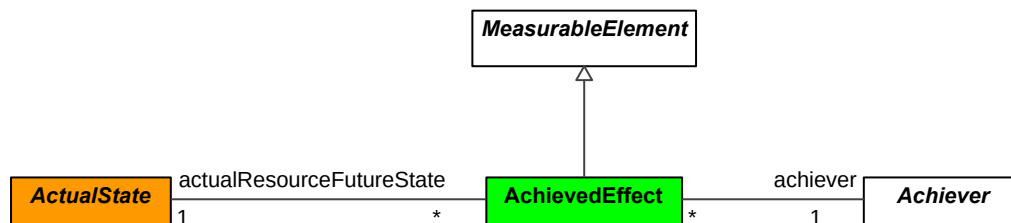


Figure 9:42 - AchievedEffect

### Achiever

States

Yes

[UAFElement](#)

Description

An ActualResource, ActualProject or ActualEnterprisePhase that can deliver a DesiredEffect.

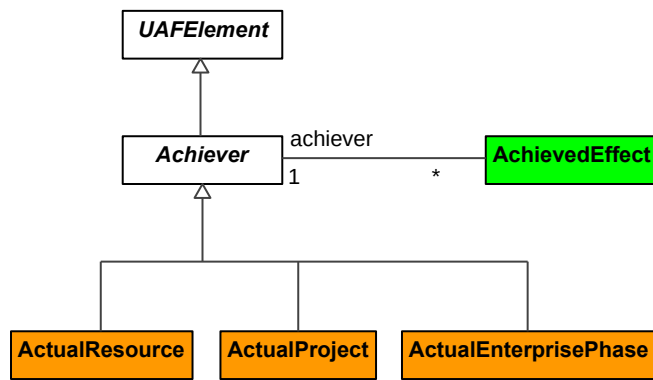


Figure 9:43 - Achiever

## DesiredEffect

States

No

[MeasurableElement](#)

Description

A tuple relating the Desirer (a Capability or OrganizationalResource) to an ActualState.

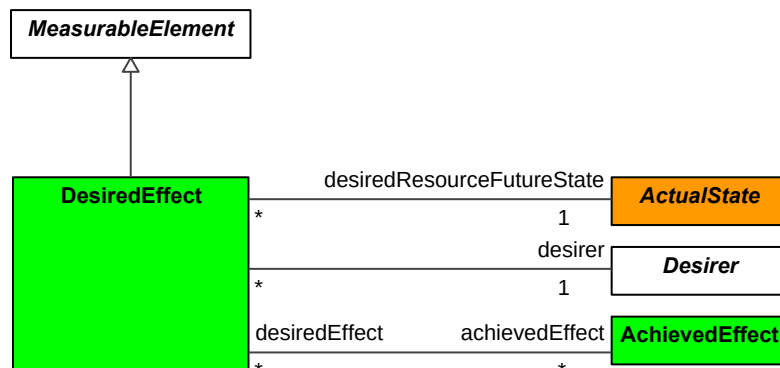


Figure 9:44 - DesiredEffect

## Desirer

States

Yes

[UAFElement](#)

Description

Abstract type used to group architecture elements that might desire a particular effect.

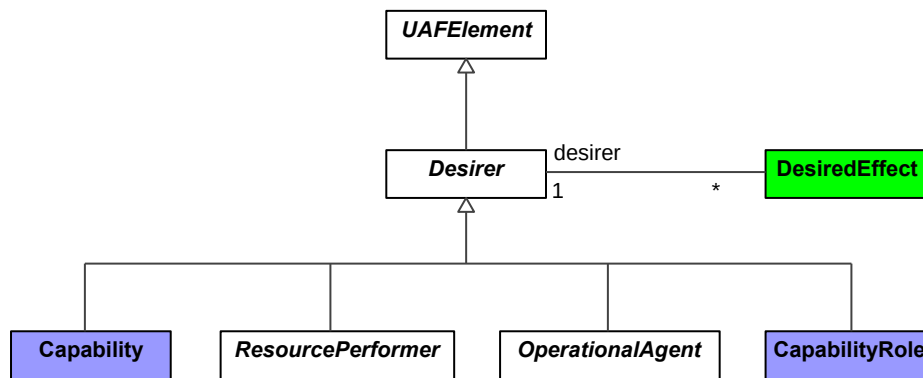


Figure 9:45 - Desirer

## Domain MetaModel::Strategic::Traceability

### CapabilityForTask

Traceability

No

[MeasurableElement](#)

Description

A tuple that asserts that a Capability is required in order for an Enterprise to conduct a phase of an EnduringTask.

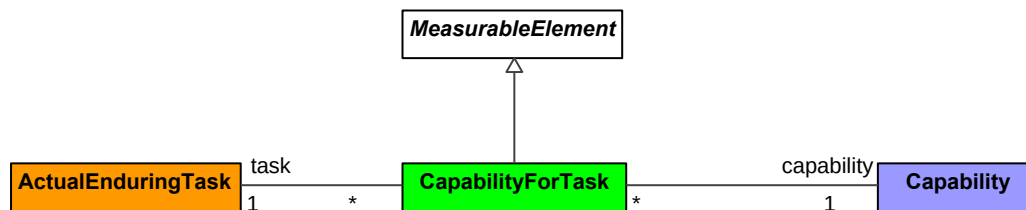


Figure 9:46 - CapabilityForTask

### CapableElement

Traceability

Yes

[UAFEElement](#)

Description

An abstract type that represents a structural element that can exhibit capabilities.

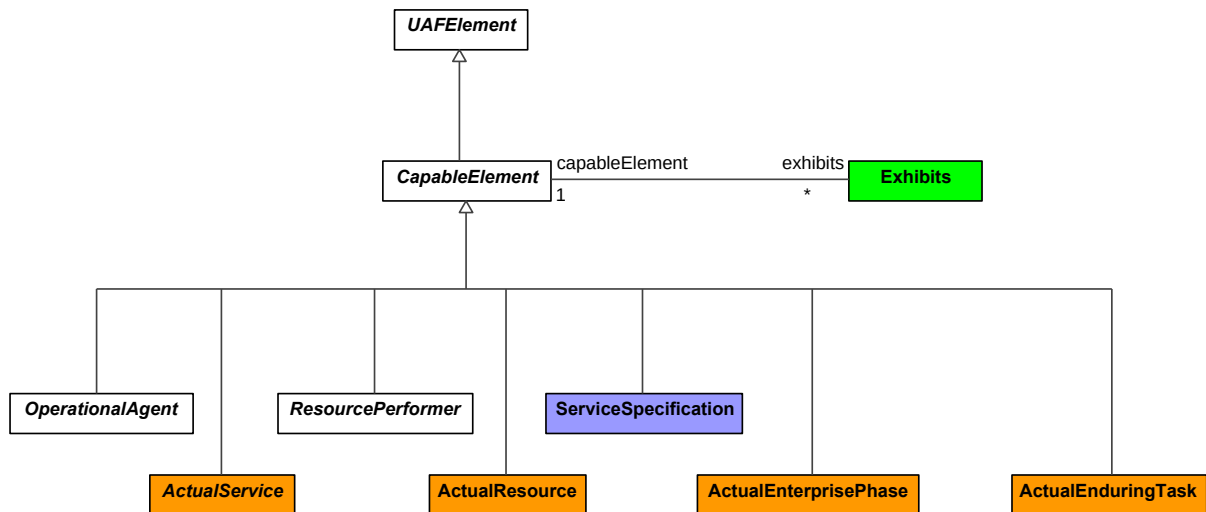


Figure 9:47 - CapableElement

## Exhibits

Traceability

No

[MeasurableElement](#)

Description

A tuple that exists between a CapableElement and a Capability that it meets under specific environmental conditions.

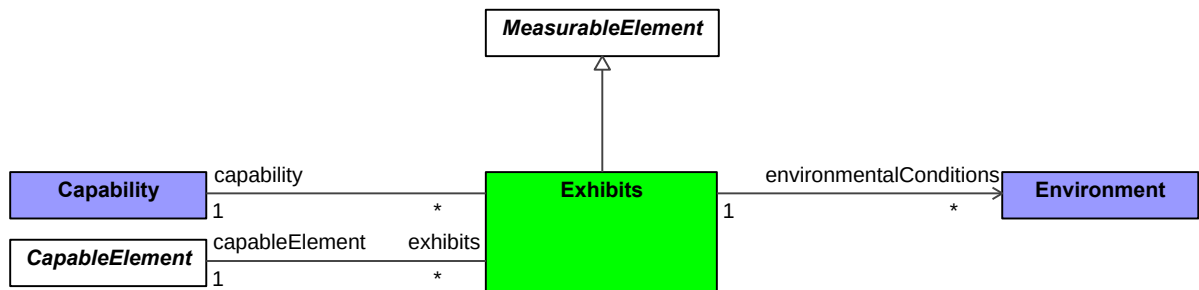


Figure 9:48 - Exhibits

## MapsToCapability

Traceability

No

[MeasurableElement](#)

Description

A tuple denoting that an Activity contributes to providing a Capability.

Figure 9:49 - MapsToCapa iFiM0.Ay(30)JT0 Tc 0 8.55.613 0 Td( )TjEMCH4/P /MC2D 0 BDCc -0.003 Tc 0.001 10.02 -0 20 9 70 M

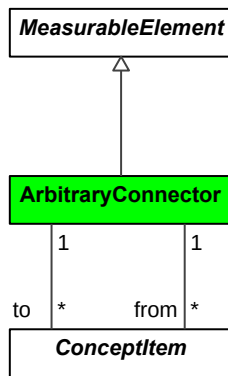


Figure 9:51 - ArbitraryConnector

## ConceptItem

Taxonomy

Yes

[UAFElement](#)

Description

Abstract, an item which may feature in a HighLevelOperationalConcept.

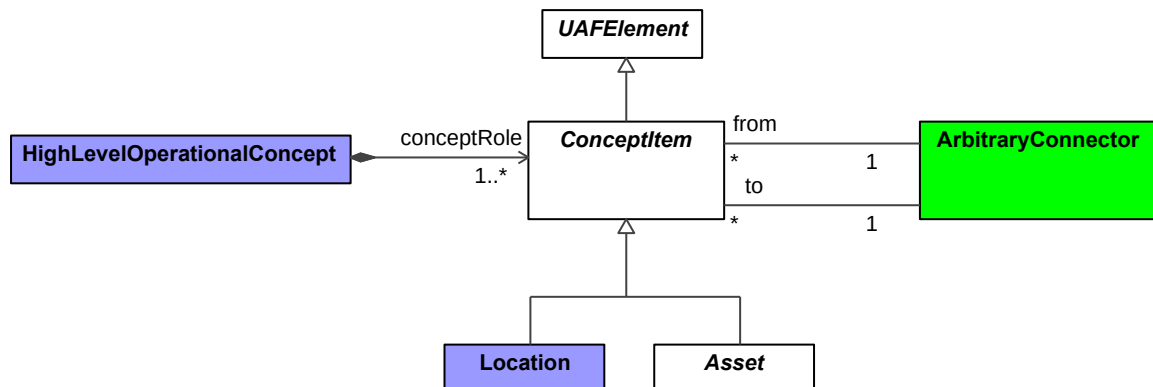


Figure 9:52 – ConceptItem

## HighLevelOperationalConcept

Taxonomy

No

[PropertySet](#)

Description

Describes the Resources and Locations required to meet an operational scenario from an integrated systems point of view. It is used to communicate overall quantitative and qualitative system characteristics to stakeholders.

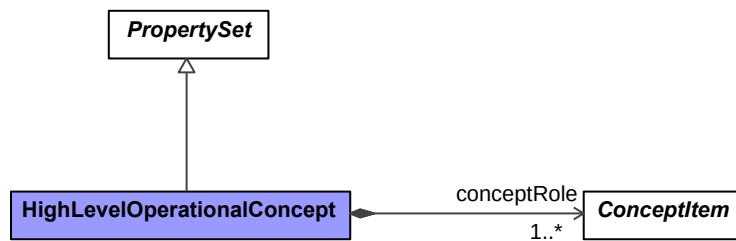


Figure 9:53 - HighLevelOperationalConcept

## Domain MetaModel::Operational::Structure

### KnownResource

Structure  
No

[OperationalPerformer](#), [ResourcePerformer](#)

#### Description

Asserts that a known ResourcePerformer constrains the implementation of the OperationalPerformer that plays the role in the OperationalArchitecture.

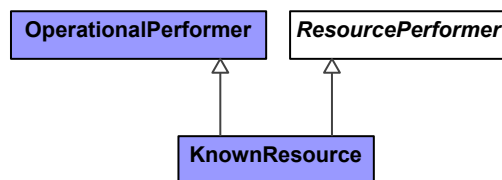


Figure 9:54 - KnownResource

### OperationalAgent

Structure  
Yes

[SubjectOfOperationalConstraint](#), [CapableElement](#), [OperationalAsset](#), [Desirer](#)

#### Description

An abstract type grouping OperationalArchitecture and OperationalPerformer.

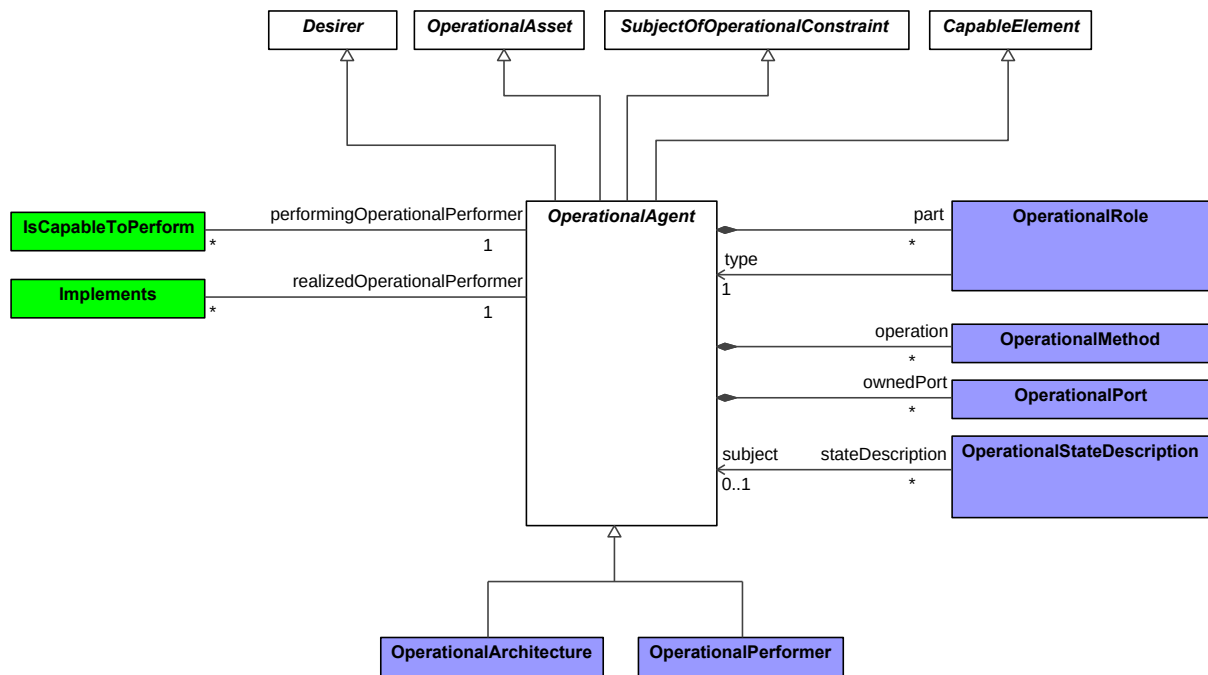


Figure 9:55 - OperationalAgent

## OperationalArchitecture

Structure

No

[OperationalAgent](#), [Architecture](#)

Description

A type used to denote a model of the Architecture, described from the Operational perspective.

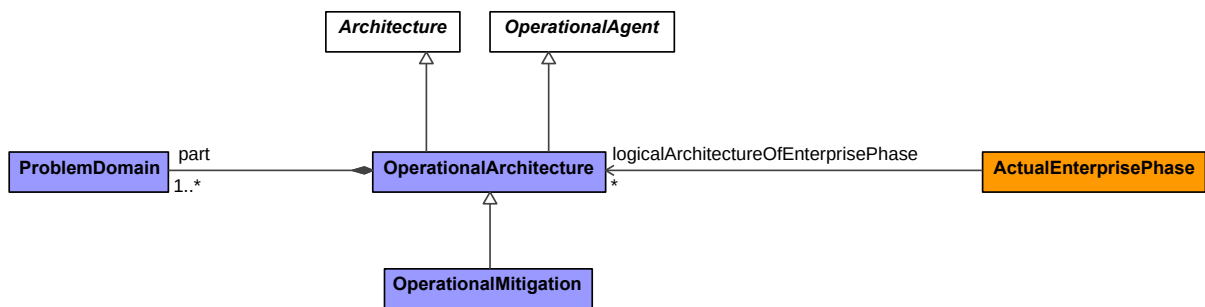


Figure 9:56 - OperationalArchitecture

## OperationalMethod

Structure

No

[ProcessOperation](#)

Description

A behavioral feature of an OperationalAgent whose behavior is specified in an OperationalActivity.

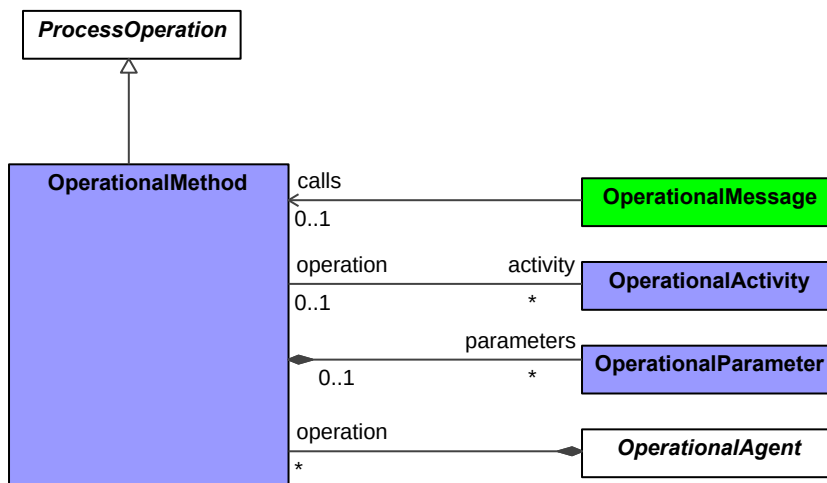


Figure 9:57 - OperationalMethod

## OperationalParameter

Structure

No

[ProcessParameter](#)

Description

A type that represents inputs and outputs of an OperationalActivity. It is typed by an OperationalExchangeItem.

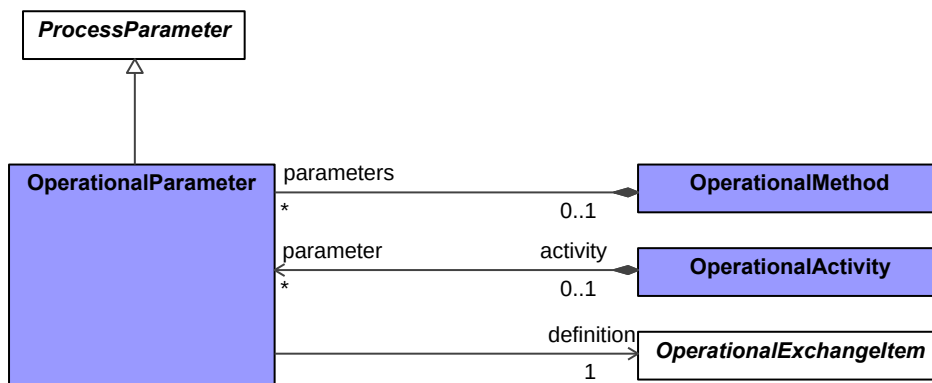


Figure 9:58 - OperationalParameter

## OperationalPerformer

Structure

No

[OperationalAgent](#)

Description

A logical entity that IsCapableToPerform OperationalActivities which produce, consume and process Resources.

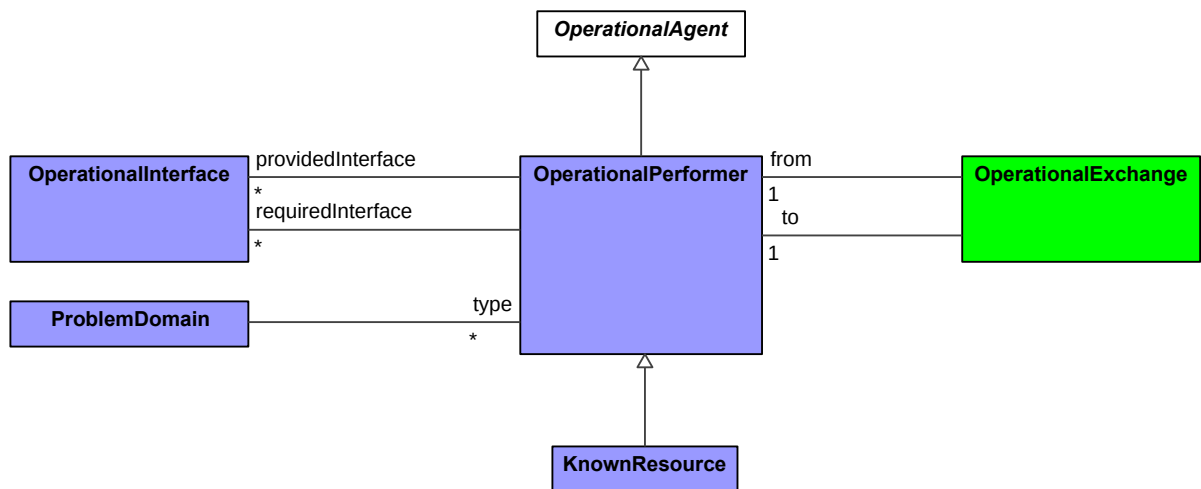


Figure 9:59 - OperationalPerformer

## OperationalRole

Structure

No

[LocationHolder](#), [AssetRole](#), [InteractionRole](#)

Description

Usage of an OperationalPerformer or OperationalArchitecture in the context of another OperationalPerformer or OperationalArchitecture. Creates a whole-part relationship.

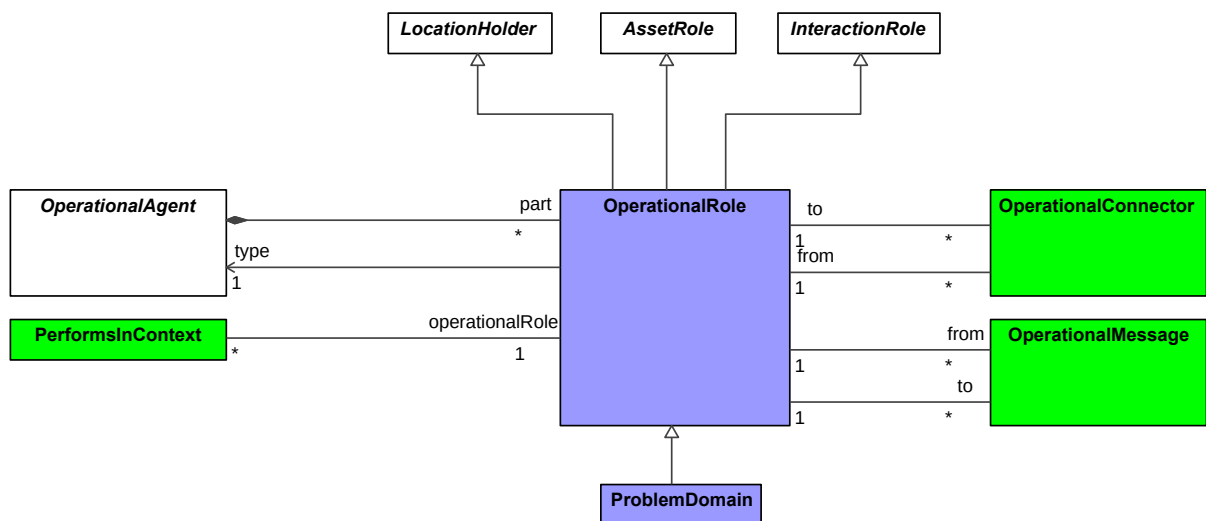


Figure 9:60 - OperationalRole

## ProblemDomain

Structure

No

[OperationalRole](#)

Description

A property associated with an OperationalArchitecture, used to specify the scope of the problem.

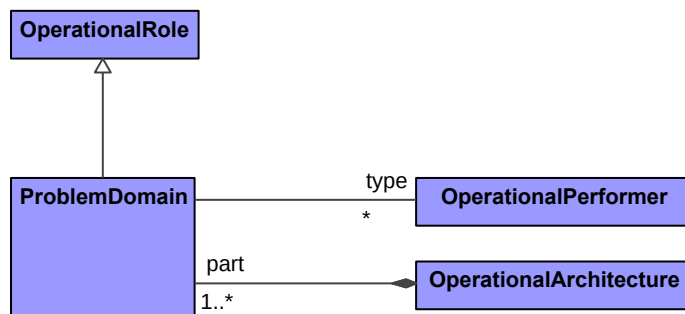


Figure 9:61 - ProblemDomain

## Domain MetaModel::Operational::Connectivity

### OperationalConnector

Connectivity

No

[MeasurableElement](#)

Description

A Connector that goes between OperationalRoles representing a need to exchange Resources. It can carry a number of OperationalExchanges.

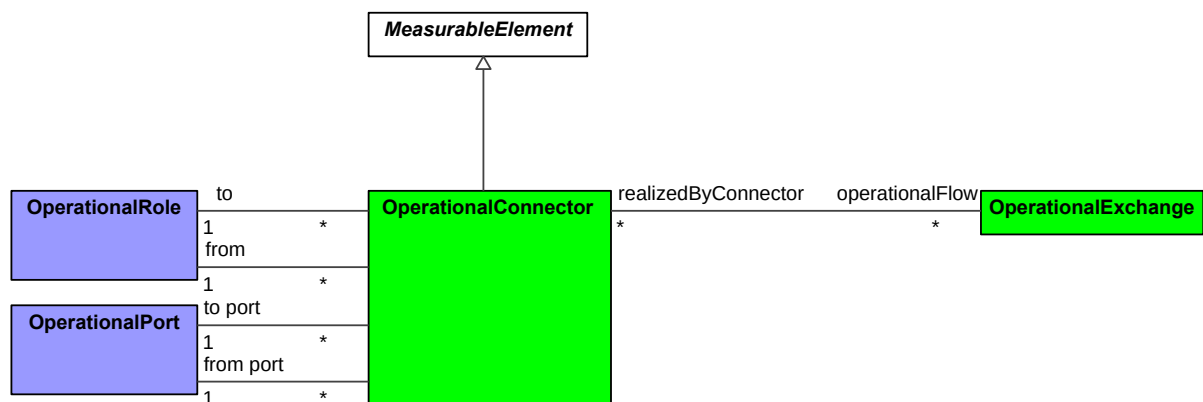


Figure 9:62 - OperationalConnector

### OperationalExchange

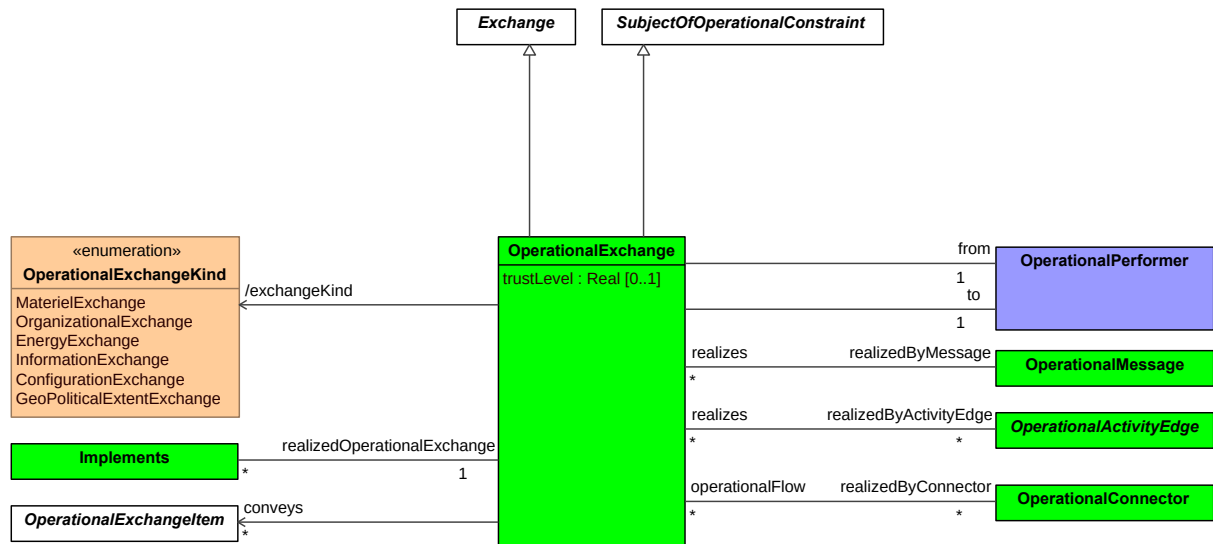
Connectivity

No

[Exchange](#), [SubjectOfOperationalConstraint](#)

Description

Asserts that a flow can exist between OperationalPerformers (i.e., flows of information, people, materiel, or energy).



**Figure 9:63 - OperationalExchange**

Attributes

`trustLevel : Real[0..1]` Captures the directional arbitrary level of trust related to an OperationalExchange between two OperationalPerformers.

## OperationalExchangeItem

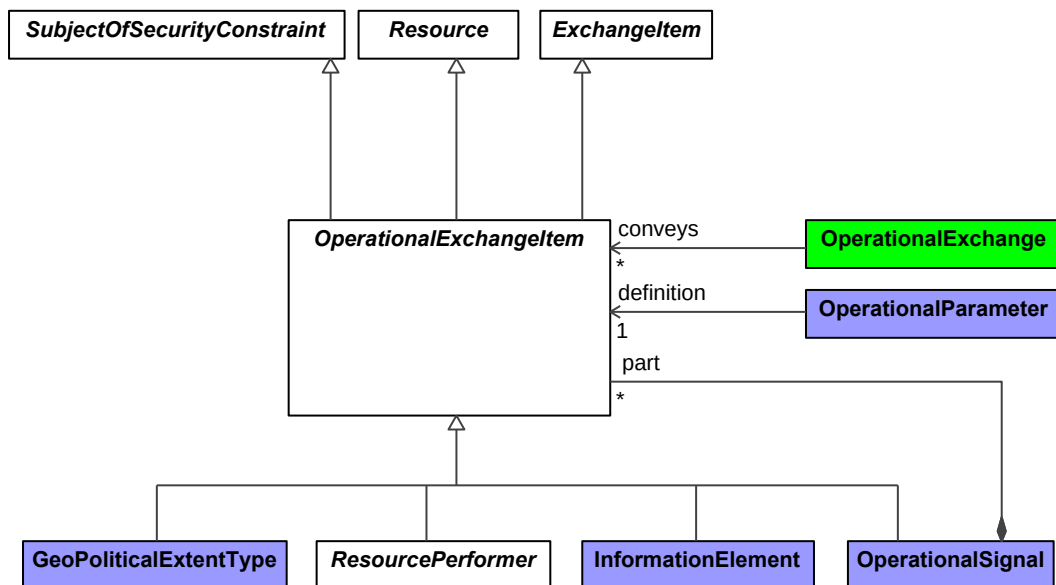
Connectivity

Yes

[Resource](#), [SubjectOfSecurityConstraint](#), [ExchangeItem](#)

Description

An abstract grouping for elements that defines the types of elements that can be exchanged between OperationalPerformers and conveyed by an OperationalExchange.



**Figure 9:64 - OperationalExchangeItem**



## Description

An item of information that flows between OperationalPerformers and is produced and consumed by the OperationalActivities that the OperationalPerformers are capable of performing (see IsCapableToPerform).

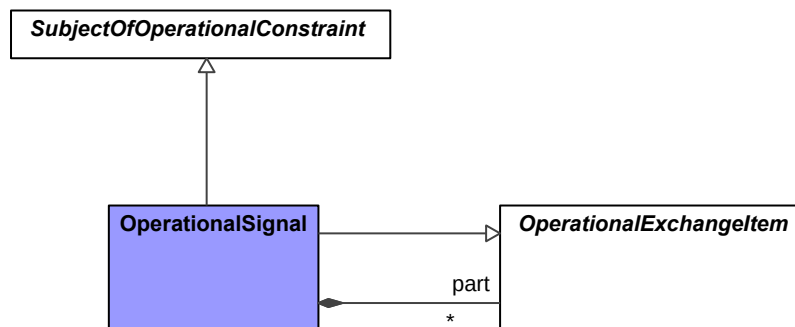


Figure 9:67 - OperationalSignal

## Domain MetaModel::Operational::Processes

### OperationalActivity

Processes

No

[SubjectOfOperationalConstraint](#), [Process](#)

## Description

An Activity that captures a logical process, specified independently of how the process is carried out.

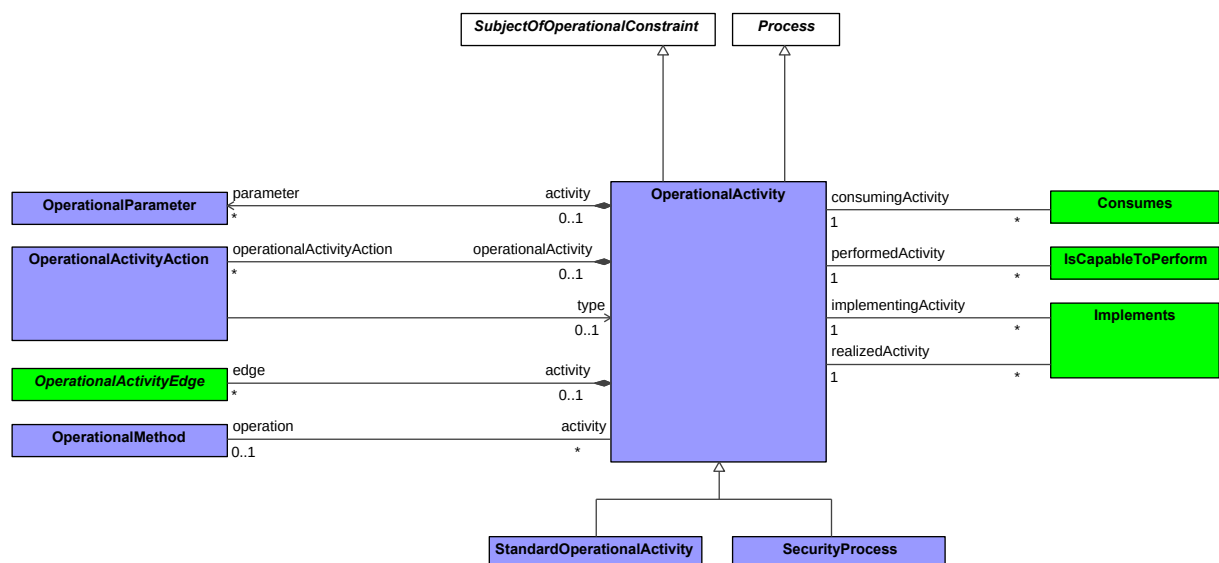


Figure 9:68 - OperationalActivity

### OperationalActivityAction

Processes

No

[ProcessUsage](#)

#### Description

A call of an OperationalActivity in the context of another OperationalActivity.

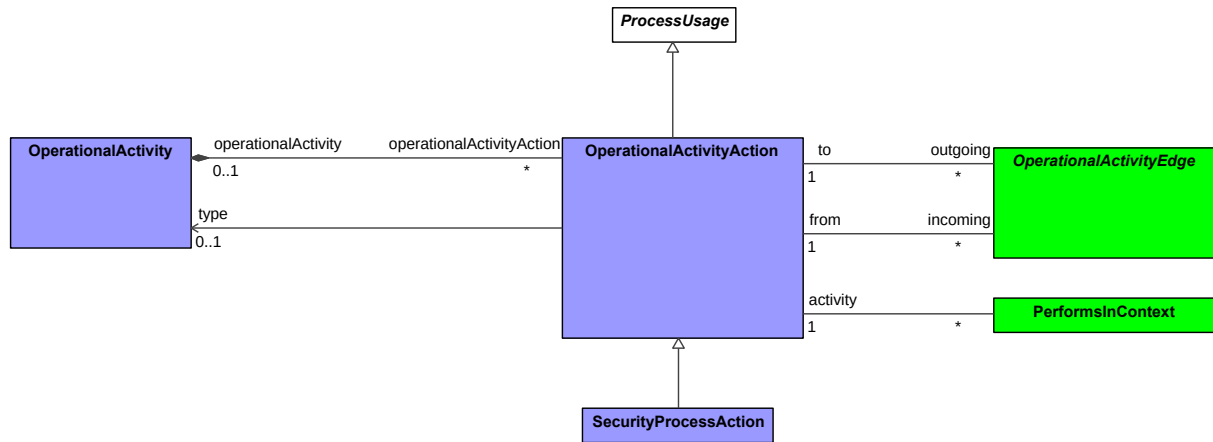


Figure 9:69 - OperationalActivityAction

### OperationalActivityEdge

Processes

Yes

[ProcessEdge](#)

#### Description

A tuple that shows the flow of Resources (objects/information) between OperationalActivityActions.

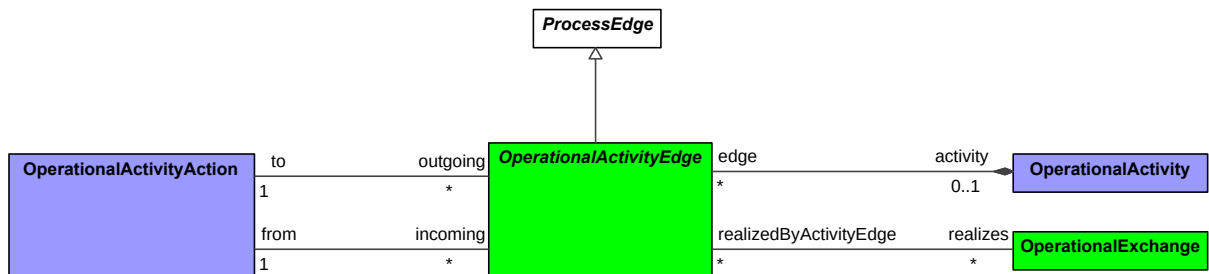


Figure 9:70 - OperationalActivityEdge

### StandardOperationalActivity

Processes

No

[OperationalActivity](#)

#### Description

A sub-type of OperationalActivity that is a standard operating procedure.

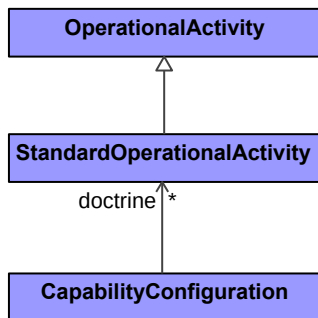


Figure 9:71 - StandardOperationalActivity

## Domain MetaModel::Operational::States

### OperationalStateDescription

States

No

[MeasurableElement](#), [StateDescription](#)

Description

A state machine describing the behavior of an OperationalPerformer, depicting how the OperationalPerformer responds to various events and the actions.

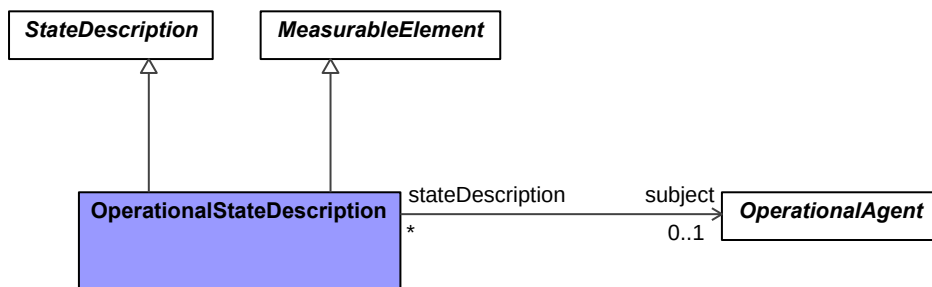


Figure 9:72 - OperationalStateDescription

## Domain MetaModel::Operational::Interaction Scenarios

### OperationalInteractionScenario

Interaction Scenarios

No

[InteractionScenario](#)

Description

A specification of the interactions between OperationalPerformers in an OperationalArchitecture.

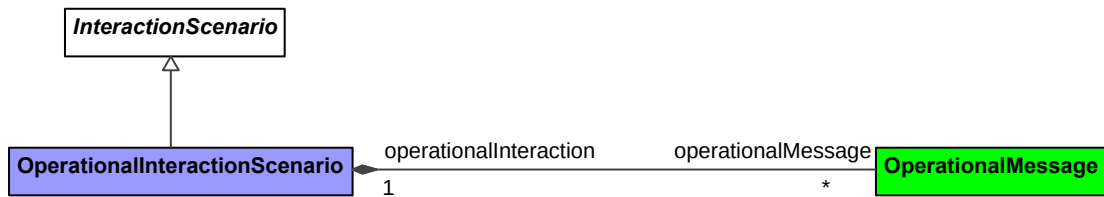


Figure 9:73 - OperationalInteractionScenario

## OperationalMessage

Interaction Scenarios

No

[InteractionMessage](#)

Description

Message for use in an OperationalInteractionScenario which carries any of the subtypes of OperationalExchange.

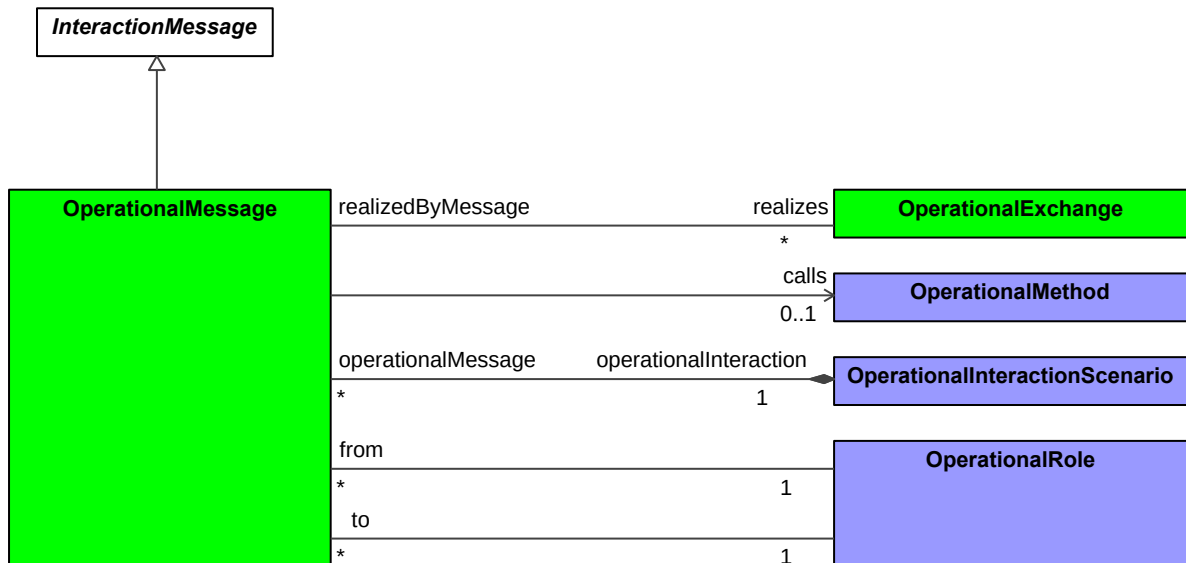


Figure 9:74 - OperationalMessage

## Domain MetaModel::Operational::Information

### InformationElement

Information

No

[SubjectOfOperationalConstraint](#), [OperationalAsset](#), [OperationalExchangeItem](#)

Description

An item of information that flows between OperationalPerformers and is produced and consumed by the OperationalActivities that the OperationalPerformers are capable to perform (see IsCapableToPerform).

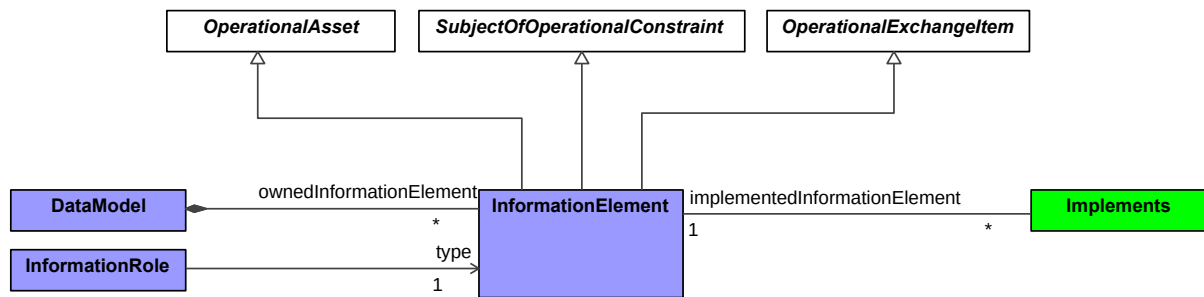


Figure 9:75 - InformationElement

## Domain MetaModel::Operational::Constraints

### OperationalConstraint

Constraints

No

[Rule](#)

Description

A Rule governing an operational architecture element, i.e., OperationalPerformer, OperationalActivity, InformationElement etc.

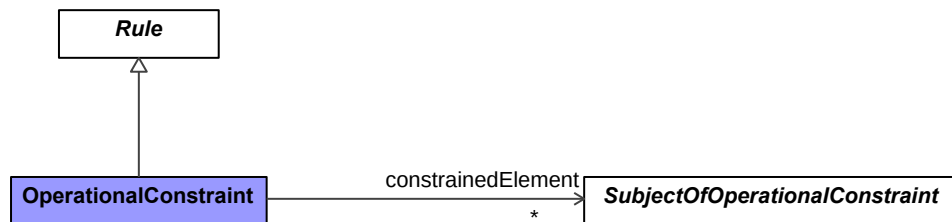


Figure 9:76 - OperationalConstraint

### SubjectOfOperationalConstraint

Constraints

Yes

[UAFElement](#)

Description

An abstract type grouping element that can be the subject of an OperationalConstraint.

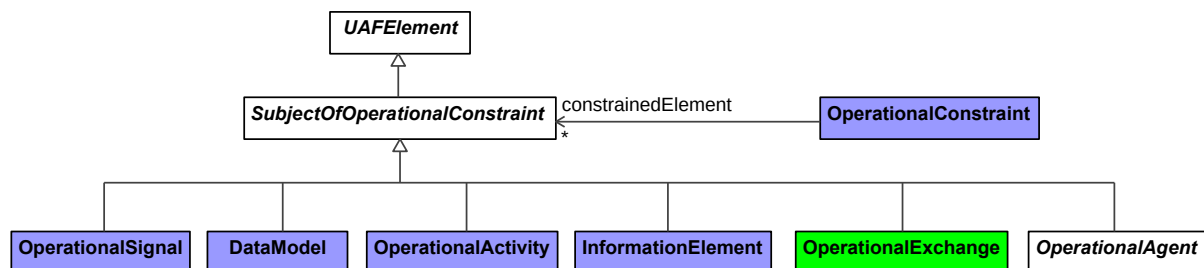


Figure 9:77 - SubjectOfOperationalConstraint

## 9.1.4 Domain MetaModel::Services

Enterprise Architects, Solution Providers, Systems Engineers, Software Architects, Business Architects.  
specifications of services required to exhibit a Capability.

shows Service Specifications and required and provided service levels of these specifications required to exhibit a Capability or to support an Operational Activity.

### Domain MetaModel::Services::Taxonomy

#### ServiceSpecification

Taxonomy

No

[PropertySet](#), [VersionedElement](#), [CapableElement](#)

Description

The specification of a set of functionalities provided by one element for the use of others.

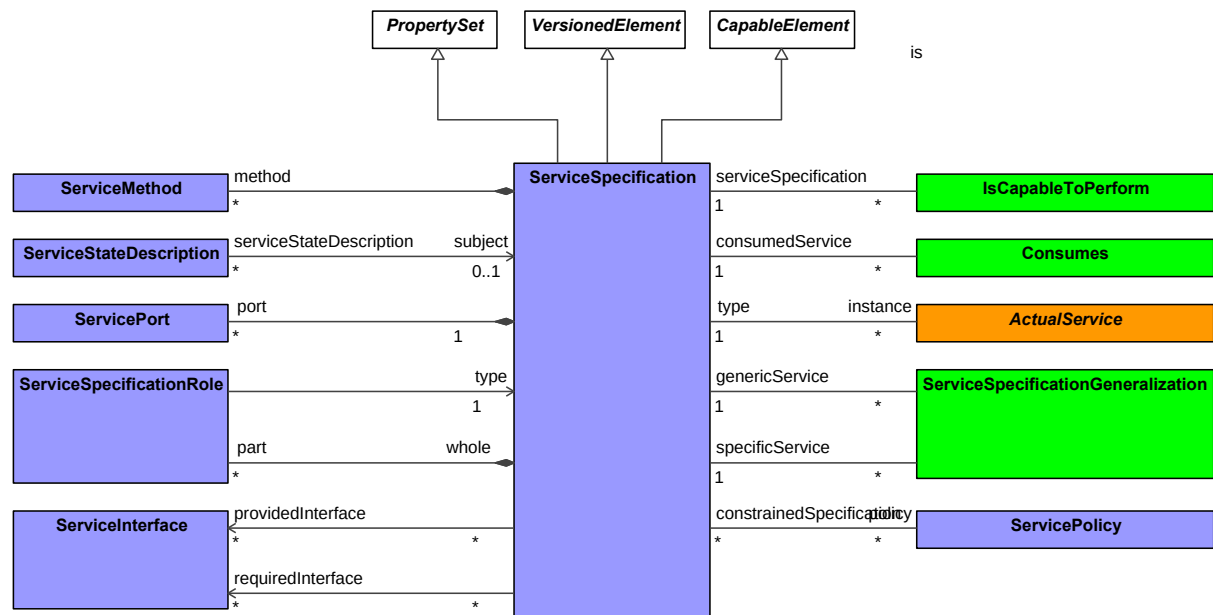


Figure 9:78 - ServiceSpecification

#### ServiceSpecificationGeneralization

Taxonomy

No

[PropertySetGeneralization](#)

Description

A ServiceSpecificationGeneralization is a taxonomic relationship between a more general ServiceSpecification and a more specific ServiceSpecification.

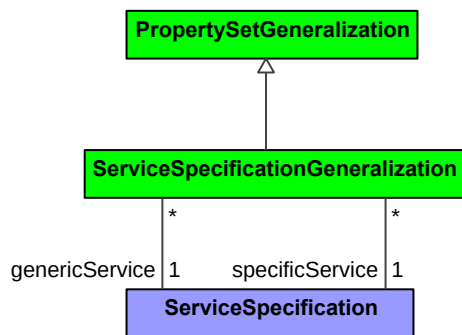


Figure 9:79 - ServiceSpecificationGeneralization

## Domain MetaModel::Services::Structure

### ServiceConnector

Structure

No

[MeasurableElement](#)

Description

A channel for exchange between two ServiceSpecifications. Where one acts as the consumer of the other.

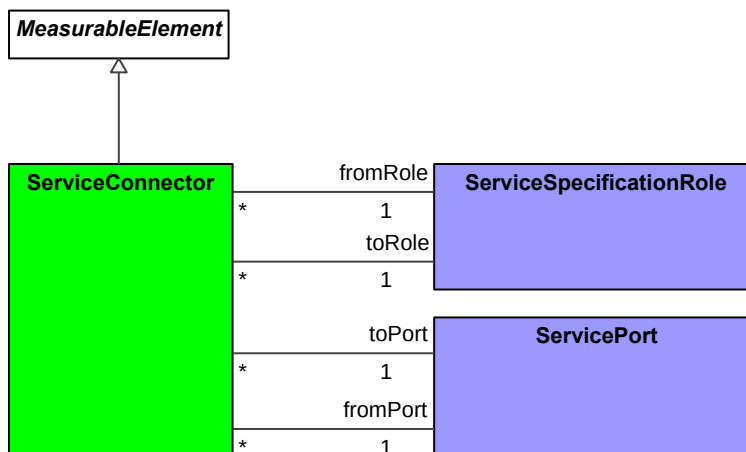


Figure 9:80 - ServiceConnector

### ServiceMethod

Structure

No

[ProcessOperation](#)

Description

A behavioral feature of a ServiceSpecification whose behavior is specified in a ServiceFunction.

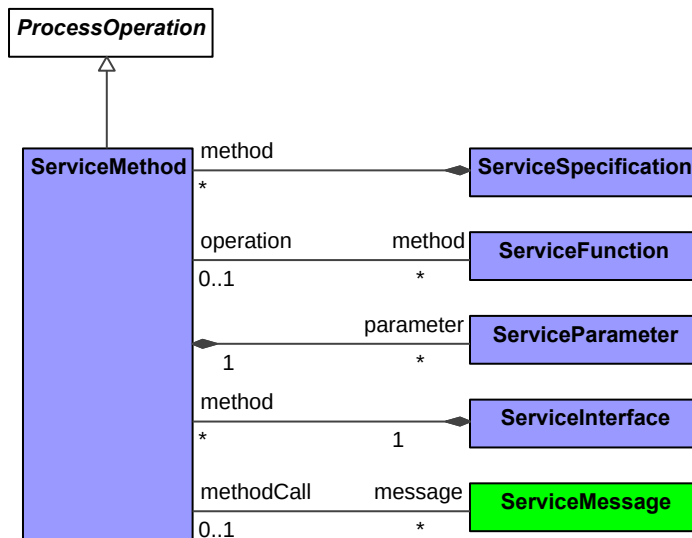


Figure 9:81 - ServiceMethod

## ServiceParameter

Structure

No

[ProcessParameter](#)

Description

A type that represents inputs and outputs of a ServiceFunction, represents inputs and outputs of a ServiceSpecification.

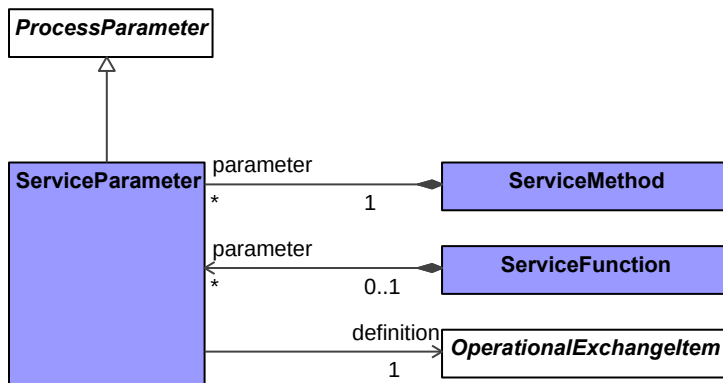


Figure 9:82 - ServiceParameter

## ServiceSpecificationRole

Structure

No

[MeasurableElement](#), [InteractionRole](#)

Description

A behavioral feature of a ServiceSpecification whose behavior is specified in a ServiceFunction.

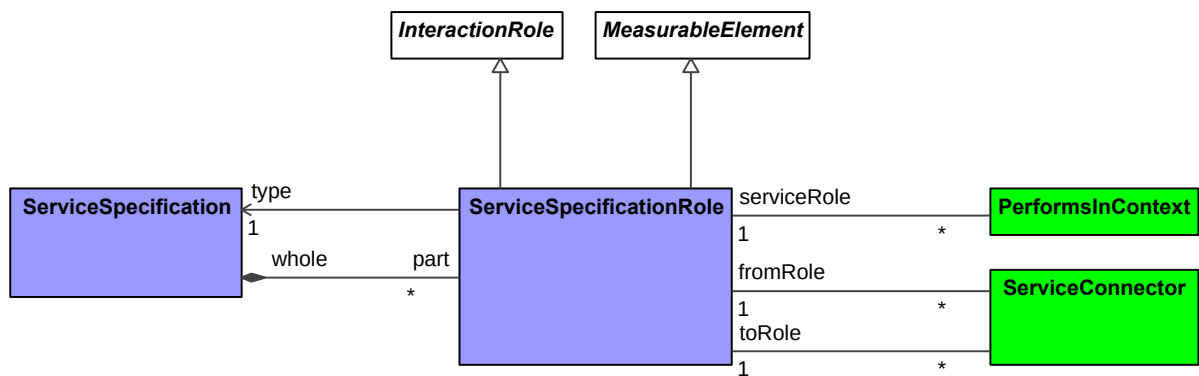


Figure 9:83 - ServiceSpecificationRole

## Domain MetaModel::Services::Connectivity

### ServiceInterface

Connectivity

No

[PropertySet](#)

Description

A contract that defines the ServiceMethods and ServiceMessageHandlers that the ServiceSpecification realizes.

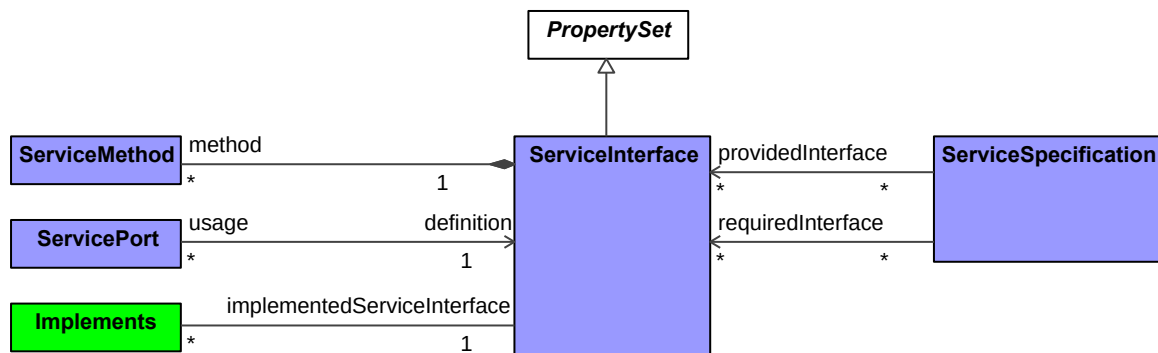


Figure 9:84 - ServiceInterface

### ServicePort

Connectivity

No

[MeasurableElement](#)

Description

An interaction point for a ServiceSpecification through which it can interact with the outside environment and which is defined by a ServiceInterface.

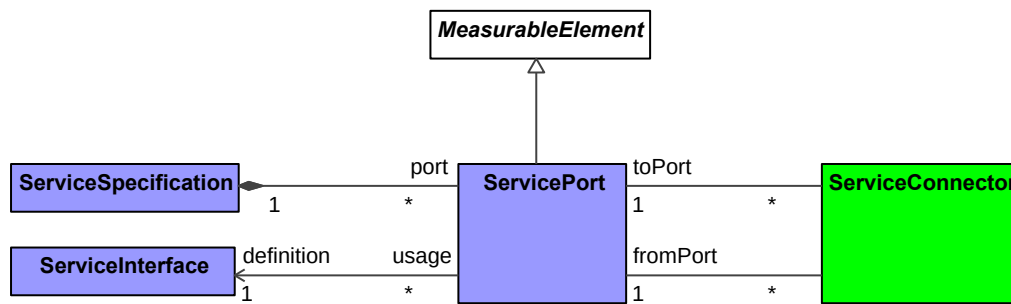


Figure 9:85 - ServicePort

## Domain MetaModel::Services::Processes

### ServiceFunction

Processes

No

[Process](#)

Description

An Activity that describes the abstract behavior of ServiceSpecifications, regardless of the actual implementation.

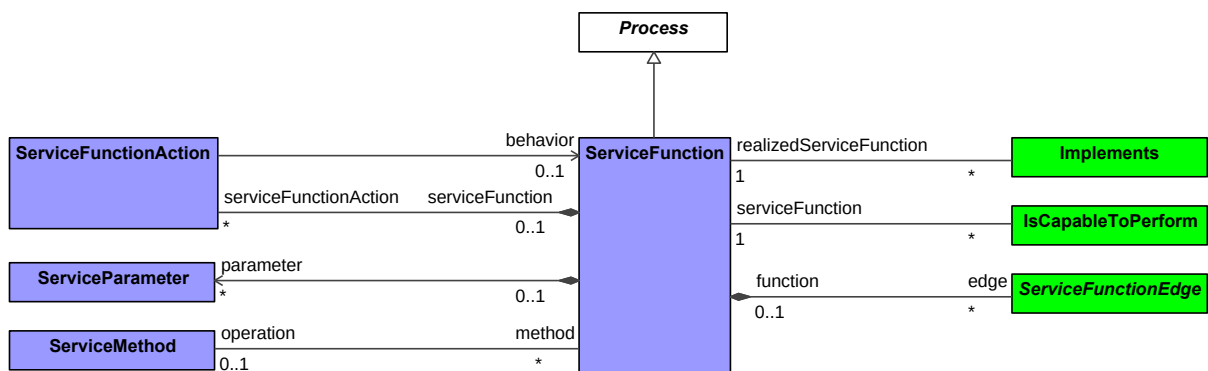


Figure 9:86 - ServiceFunction

### ServiceFunctionAction

Processes

No

[ProcessUsage](#)

Description

A call of a ServiceFunction in the context of another ServiceFunction.

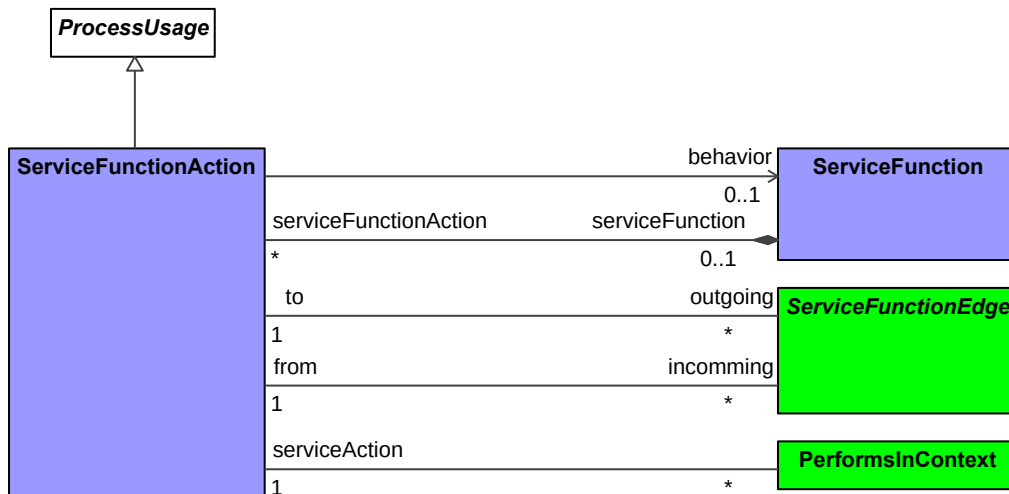


Figure 9:87 - ServiceFunctionAction

## ServiceFunctionEdge

Processes

Yes

[ProcessEdge](#)

Description

A tuple that shows the flow of Resources (objects/information) between OperationalActivityActions.

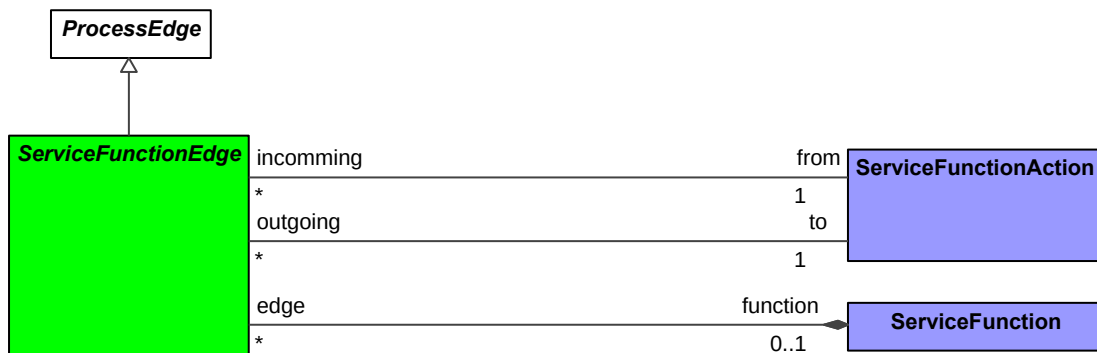


Figure 9:88 - ServiceFunctionEdge

## Domain MetaModel::Services::States

### ServiceStateDescription

States

No

[MeasurableElement](#), [StateDescription](#)

Description

A state machine describing the behavior of a ServiceSpecification, depicting how the ServiceSpecification responds to various events and the actions.

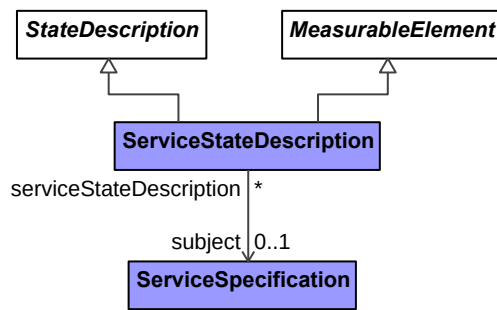


Figure 9:89 - ServiceStateDescription

## Domain MetaModel::Services::Interaction Scenarios

### ServiceInteractionScenario

Interaction Scenarios

No

[InteractionScenario](#)

Description

A specification of the interactions between ServiceSpecifications.

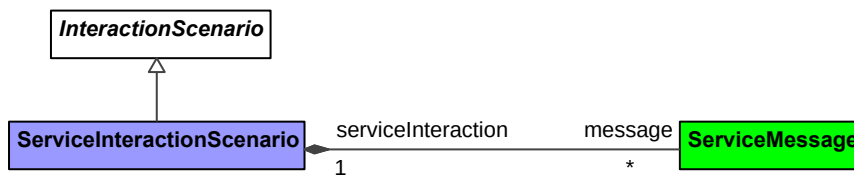


Figure 9:90 - ServiceInteractionScenario

### ServiceMessage

Interaction Scenarios

No

[InteractionMessage](#)

Description

Message for use in a Service Event-Trace.



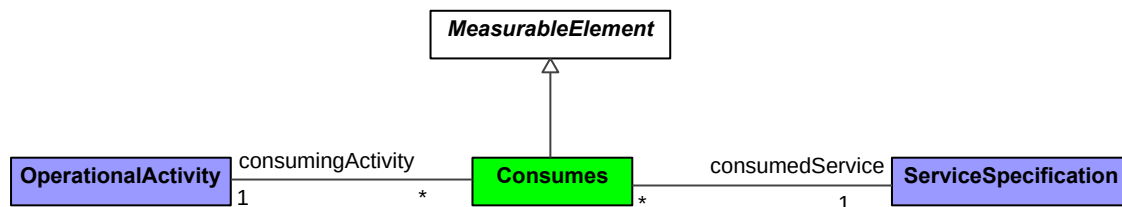


Figure 9:93 - Consumes

## 9.1.5 Domain MetaModel::Personnel

Human resources, Solution Providers, PMs.

human factors.

aims to clarify the role of Human Factors (HF) when creating architectures in order to facilitate both Human Factors Integration (HFI) and systems engineering (SE).

### Domain MetaModel::Personnel::Taxonomy

#### Organization

Taxonomy

No

[OrganizationalResource](#)

Description

A group of OrganizationalResources (Persons, Posts, Organizations and Responsibilities) associated for a particular purpose.

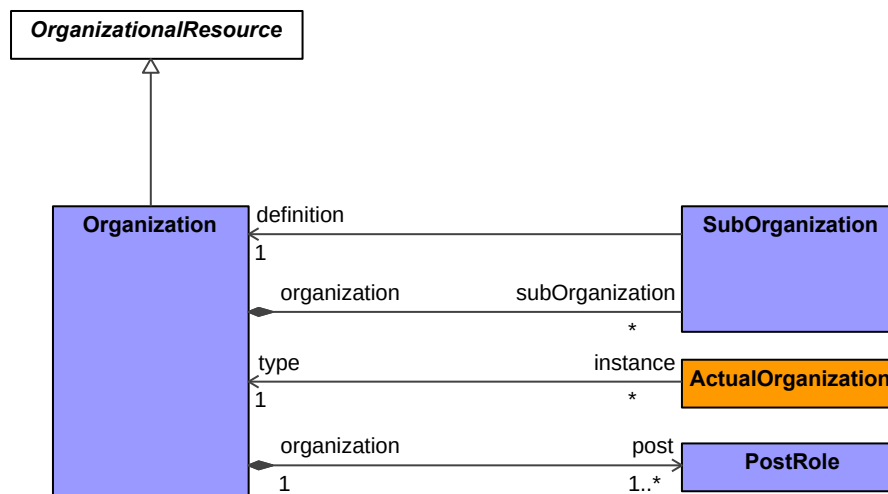


Figure 9:94 - Organization

#### OrganizationalResource

Taxonomy

Yes

[PhysicalResource](#), [Stakeholder](#)

## Description

An abstract type for Organization, Person, Post and Responsibility.

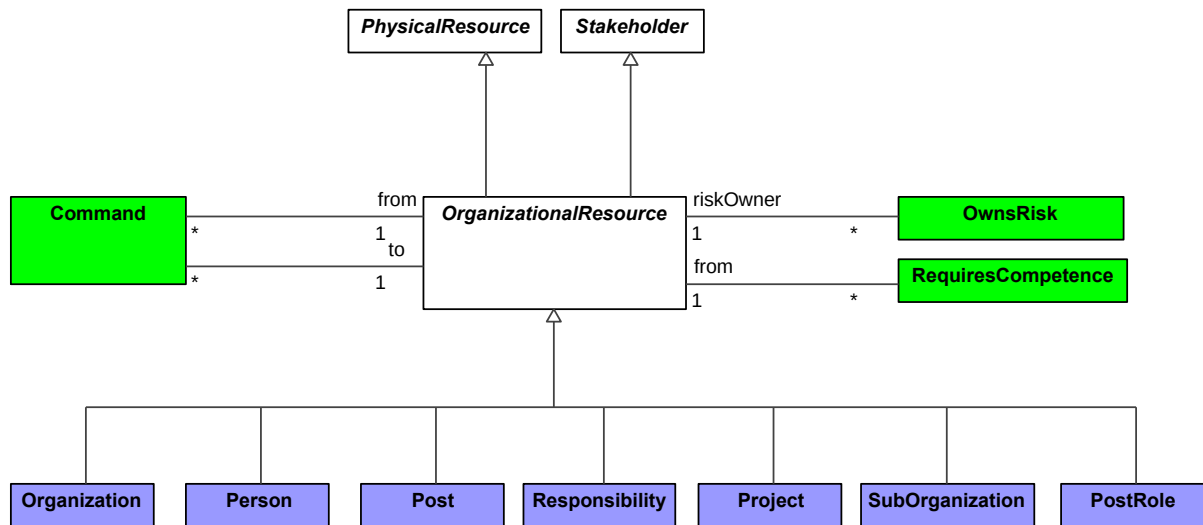


Figure 9:95 - OrganizationalResource

## Person

Taxonomy

No

[OrganizationalResource](#)

## Description

A type of a human being used to define the characteristics that need to be described for ActualPersons (e.g., properties such as address, telephone number, nationality, etc.).

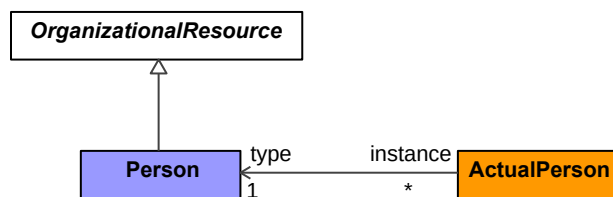


Figure 9:96 - Person

## Post

Taxonomy

No

[OrganizationalResource](#)

## Description

A type of job title or position that a person can fill (e.g., Lawyer, Solution Architect, Machine Operator or Chief Executive Officer).

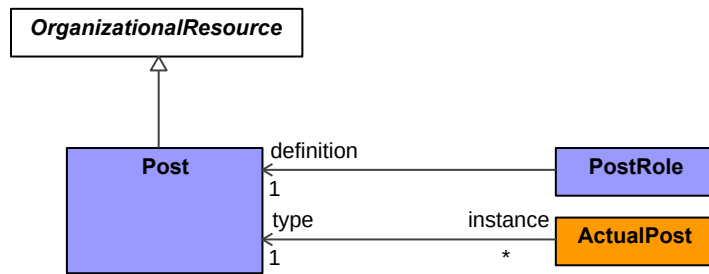


Figure 9:97 - Post

## Responsibility

Taxonomy

No

[OrganizationalResource](#)

Description

The type of duty required of a Person or Organization.

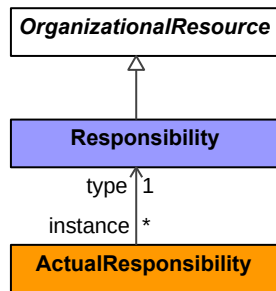


Figure 9:98 - Responsibility

## Domain MetaModel::Personnel::Structure

### PostRole

Structure

No

[OrganizationalResource](#), [ResourceRole](#)

Description

A usage of a post in the context of another OrganizationalResource. Creates a whole-part relationship.

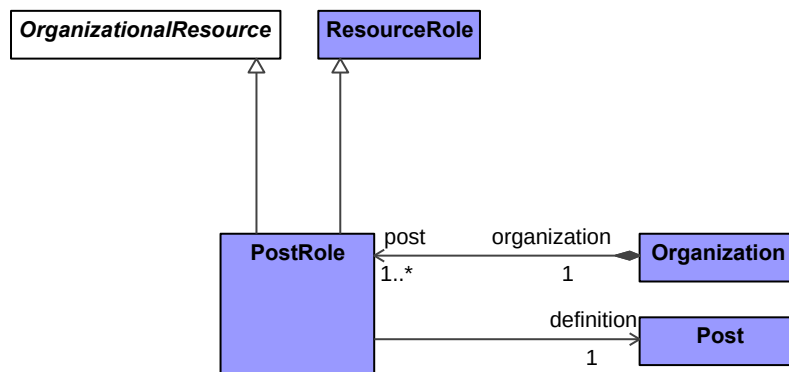


Figure 9:99 - PostRole

## SubOrganization

Structure

No

[OrganizationalResource](#), [ResourceRole](#)

Description

A type of a human being used to define the characteristics that need to be described for ActualPersons (e.g., properties such as address, telephone number, nationality, etc.).

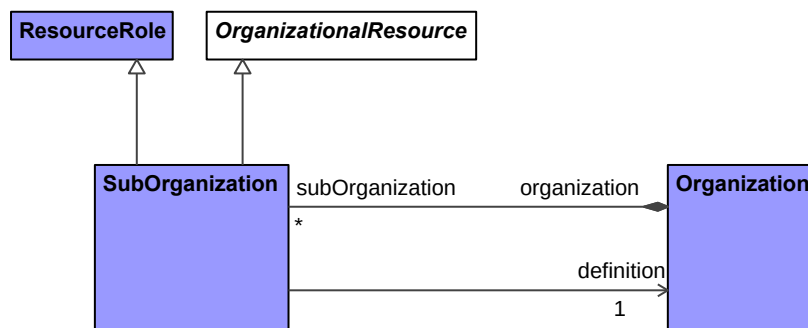


Figure 9:100 - SubOrganization

## Domain MetaModel::Personnel::Connectivity

### Command

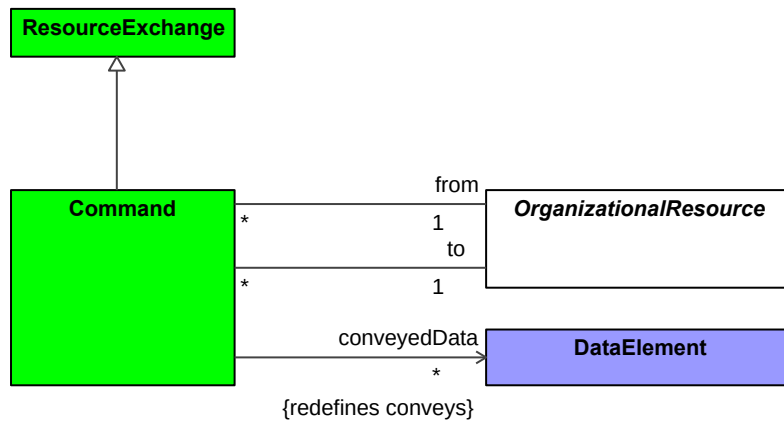
Connectivity

No

[ResourceExchange](#)

Description

A type of ResourceExchange that asserts that one OrganizationalResource commands another.



**Figure 9:101 - Command**

## Control

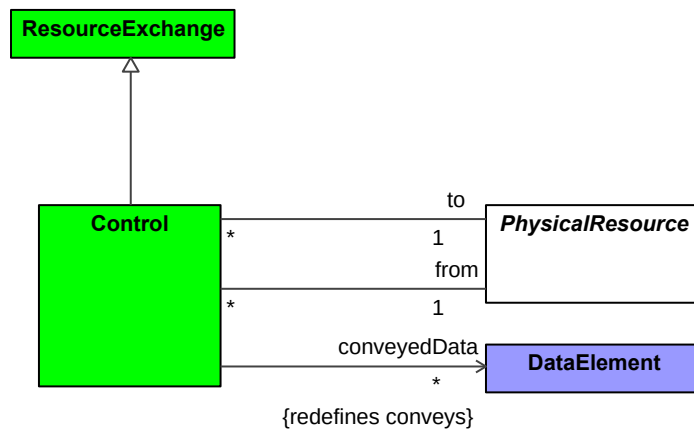
Connectivity

No

[ResourceExchange](#)

Description

A type of ResourceExchange that asserts that one PhysicalResource controls another PhysicalResource (i.e., the driver of a vehicle controlling the vehicle speed or direction).



**Figure 9:102 - Control**

## Domain MetaModel::Personnel::Interaction Scenarios

### ResourceInteractionScenario

Interaction Scenarios

No

[InteractionScenario](#)

Description

A specification of the interactions between ResourcePerformers in a ResourceArchitecture.

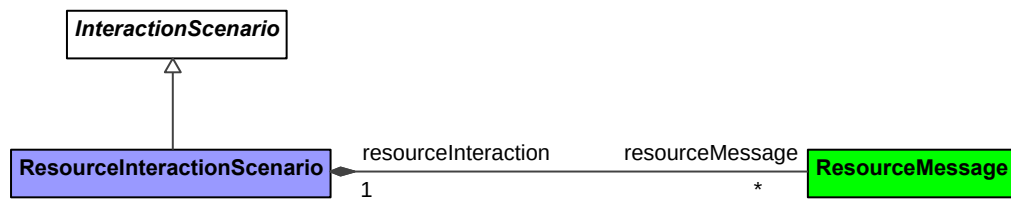


Figure 9:103 - ResourceInteractionScenario

## Domain MetaModel::Personnel::Constraints

### Competence

Constraints

No

[PropertySet](#), [SubjectOfForecast](#)

Description

A specific set of abilities defined by knowledge, skills and aptitude.

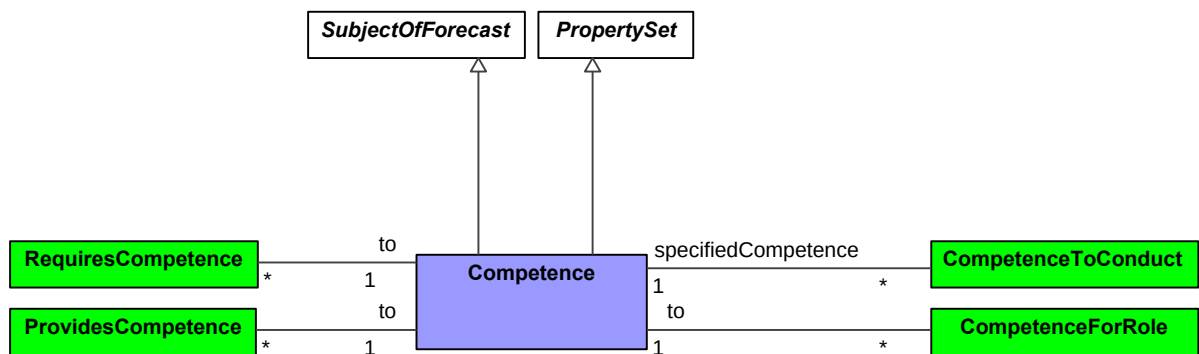


Figure 9:104 - Competence

### CompetenceForRole

Constraints

No

[MeasurableElement](#)

Description

A tuple used to associate an organizational role with a specific set of required competencies.

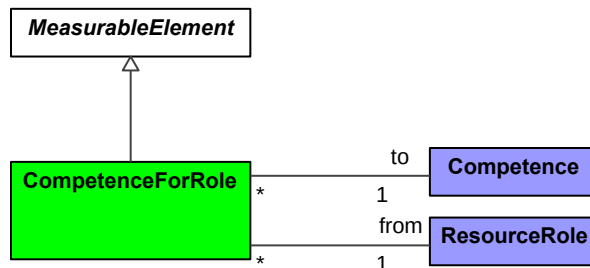


Figure 9:105 - CompetenceForRole

## RequiresCompetence

Constraints

No

[MeasurableElement](#)

Description

A tuple that asserts that an ActualOrganizationalResource is required to have a specific set of Competencies.

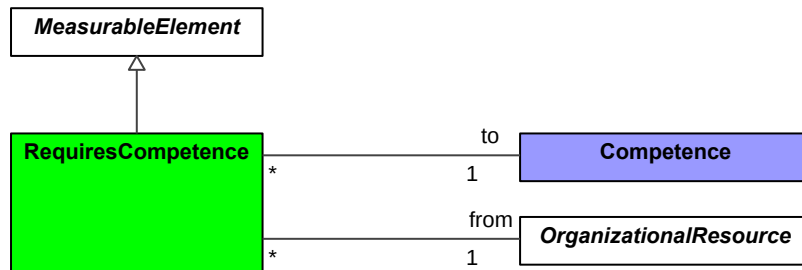


Figure 9:106 - RequiresCompetence

## Domain MetaModel::Personnel::Roadmap

### FillsPost

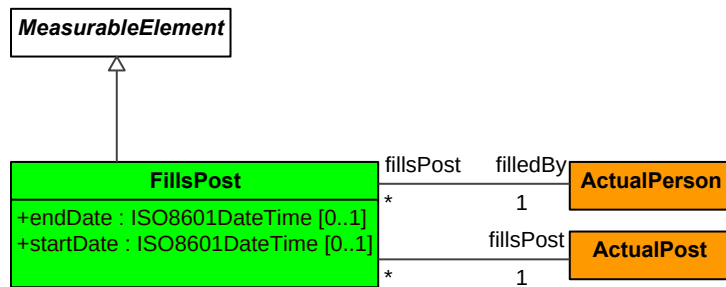
Roadmap

No

[MeasurableElement](#)

Description

A tuple that asserts that an ActualPerson fills an ActualPost.



**Figure 9:107 - FillsPost**

Attributes

endDate : ISO8601DateTime[0..1] End date of an ActualPerson filling an ActualPost.  
 startDate : ISO8601DateTime[0..1] Start date of an ActualPerson filling an ActualPost.

## Domain MetaModel::Personnel::Traceability

### CompetenceToConduct

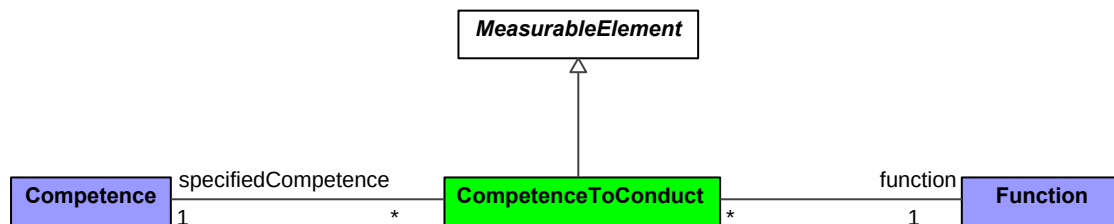
Traceability

No

[MeasurableElement](#)

Description

A tuple used to associate a Function with a specific set of Competencies needed to conduct the Function.



**Figure 9:108 - CompetenceToConduct**

## 9.1.6 Domain MetaModel::Resources

### Domain MetaModel::Resources::Taxonomy

#### CapabilityConfiguration

Taxonomy

No

[ResourceArchitecture](#)

Description

A composite structure representing the physical and human resources (and their interactions) in an enterprise, assembled to meet a capability.

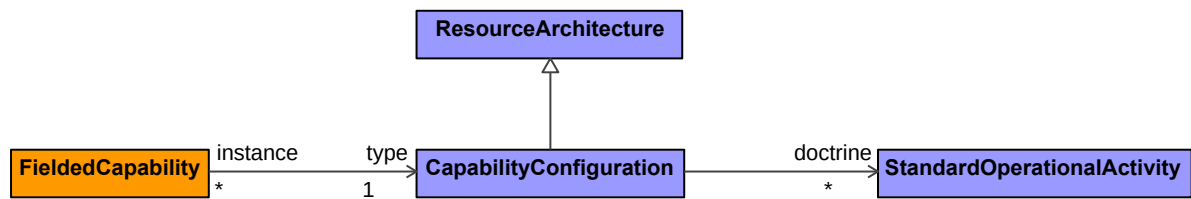


Figure 9:109 - CapabilityConfiguration

## NaturalResource

Taxonomy

No

[PhysicalResource](#)

Description

Type of physical resource that occurs in nature such as oil, water, gas or coal.

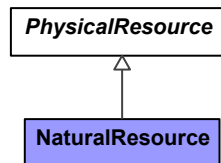


Figure 9:110 - NaturalResource

## PhysicalResource

Taxonomy

Yes

[ResourcePerformer](#)

Description

An abstract type defining physical resources (i.e., OrganizationalResource, ResourceArtifact and NaturalResource).

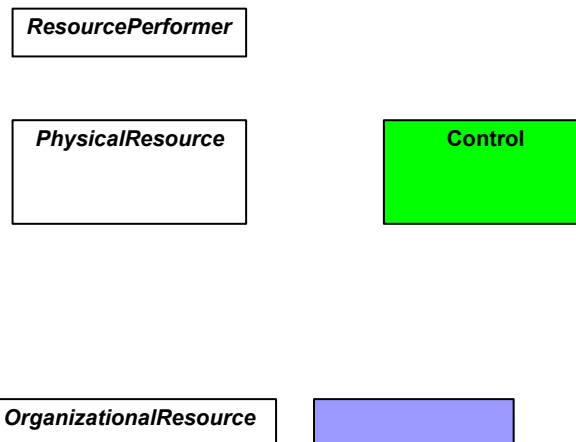


Figure 9:111 - PhysicalResource

## ResourceArchitecture

Taxonomy

No

[ResourcePerformer](#), [Architecture](#)

Description

A type used to denote a model of the Architecture, described from the ResourcePerformer perspective.

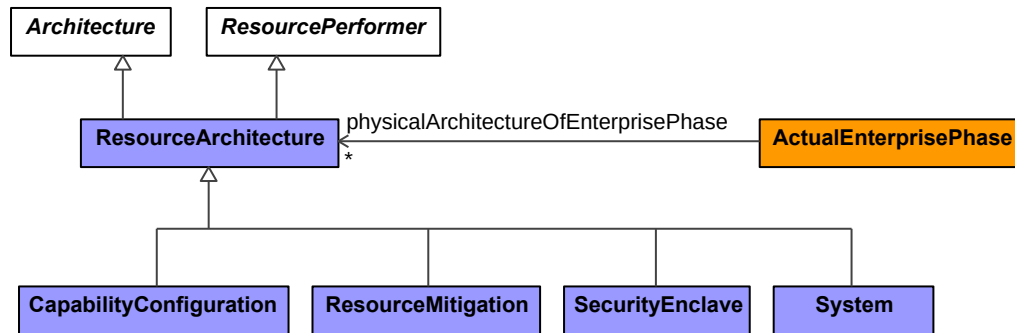


Figure 9:112 - ResourceArchitecture

## ResourceArtifact

Taxonomy

No

[PhysicalResource](#)

Description

A type of man-made object that contains no human beings (i.e., satellite, radio, petrol, gasoline, etc.).

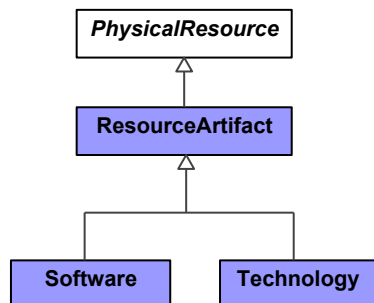


Figure 9:113 - ResourceArtifact

## ResourcePerformer

Taxonomy

Yes

[ResourceExchangeItem](#), [SubjectOfResourceConstraint](#), [OperationalExchangeItem](#), [SubjectOfForecast](#), [CapableElement](#), [Desirer](#), [VersionedElement](#), [ResourceAsset](#)

Description

An abstract grouping of elements that can perform Functions.

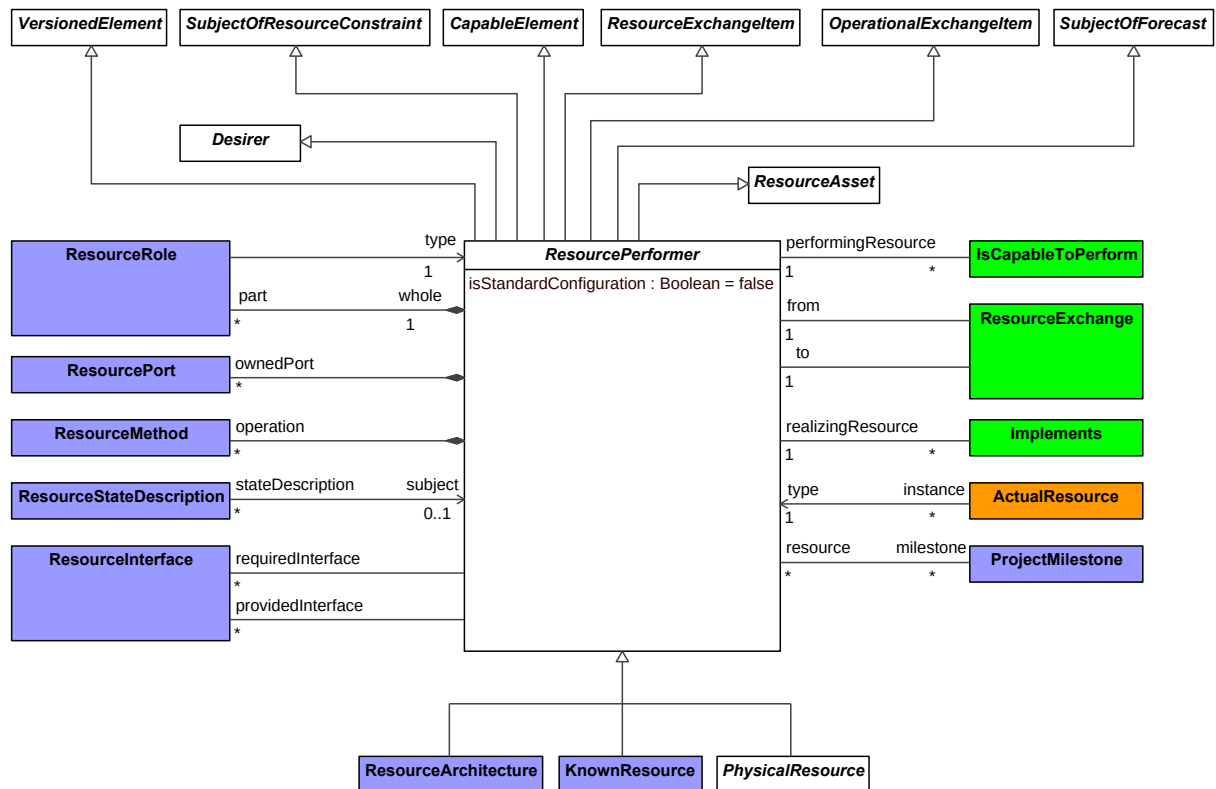


Figure 9:114 – ResourcePerformer

#### Attributes

isStandardConfiguration : Boolean[] Indicates if the ResourcePerformer is StandardConfiguration, default=false.

#### Software

Taxonomy

No

[ResourceArtifact](#)

#### Description

A sub-type of ResourceArtifact that specifies an executable computer program.

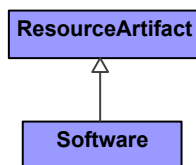


Figure 9:115 - Software

#### System

Taxonomy

No

## [ResourceArchitecture](#)

### Description

An integrated set of elements, subsystems, or assemblies that accomplish a defined objective. These elements include products (hardware, software, firmware), processes, people, information, techniques, facilities, services, and other support elements (INCOSE SE Handbook V4, 2015).

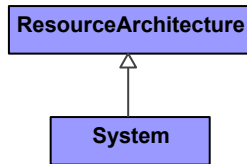


Figure 9:116 - System

## Domain MetaModel::Resources::Structure

### ResourceMethod

Structure

No

## [ProcessOperation](#)

### Description

A behavioral feature of a ResourcePerformer whose behavior is specified in a Function.

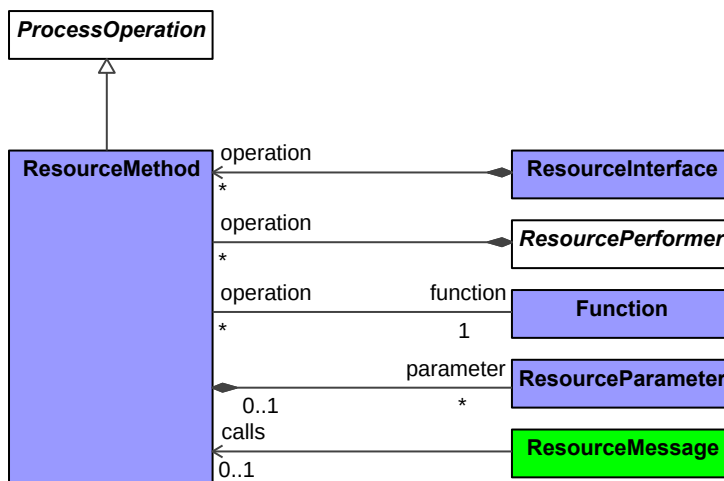


Figure 9:117 - ResourceMethod

### ResourceParameter

Structure

No

## [ProcessParameter](#)

### Description

A type that represents inputs and outputs of a Function. It is typed by a ResourceInteractionItem.

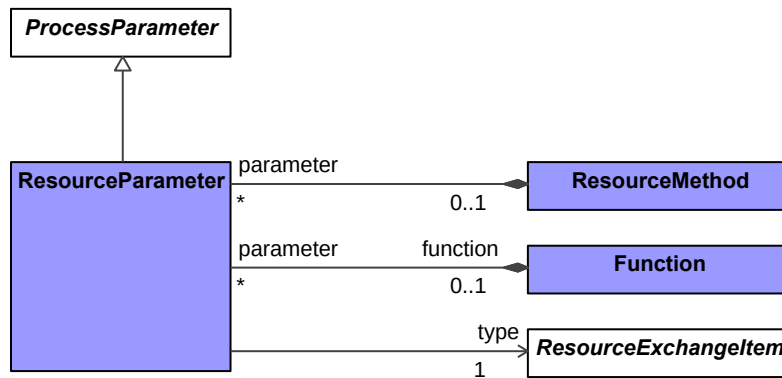


Figure 9:118 - ResourceParameter

## ResourcePort

Structure

No

[ProtocolImplementation](#), [MeasurableElement](#)

Description

An interaction point for a ResourcePerformer through which it can interact with the outside environment and which is defined by a ResourceInterface.

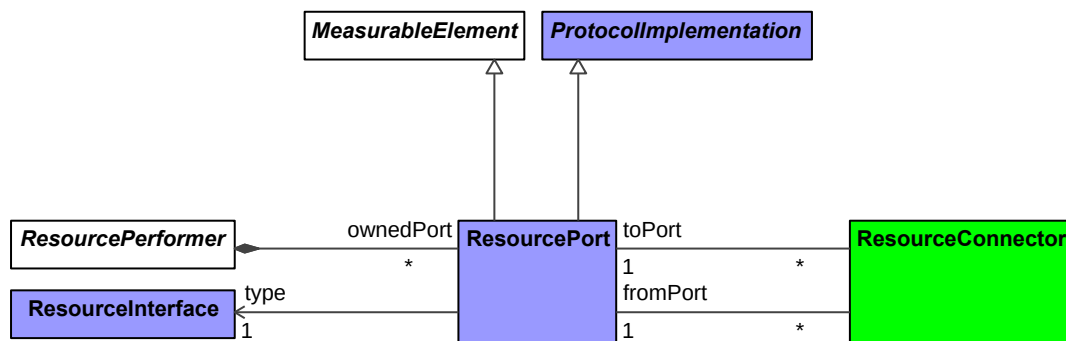


Figure 9:119 - ResourcePort

## ResourceRole

Structure

No

[SubjectOfResourceConstraint](#), [LocationHolder](#), [AssetRole](#), [InteractionRole](#)

Description

Usage of a ResourcePerformer in the context of another ResourcePerformer. Creates a whole-part relationship.

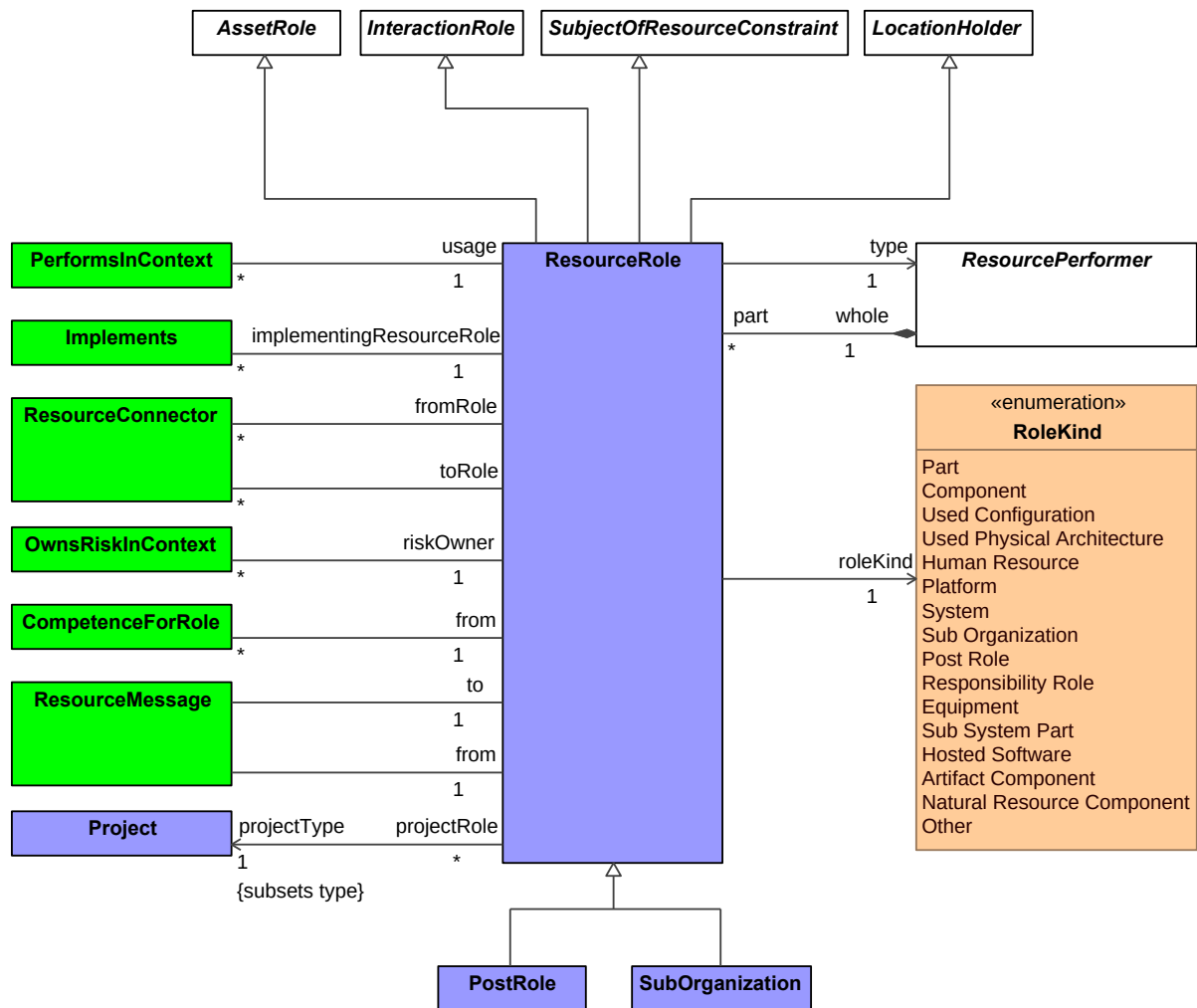


Figure 9:120 - ResourceRole

## Domain MetaModel::Resources::Connectivity

### ResourceConnector

Connectivity

No

[ProtocolImplementation](#), [MeasurableElement](#)

Description

A channel for exchange between two ResourceRoles.

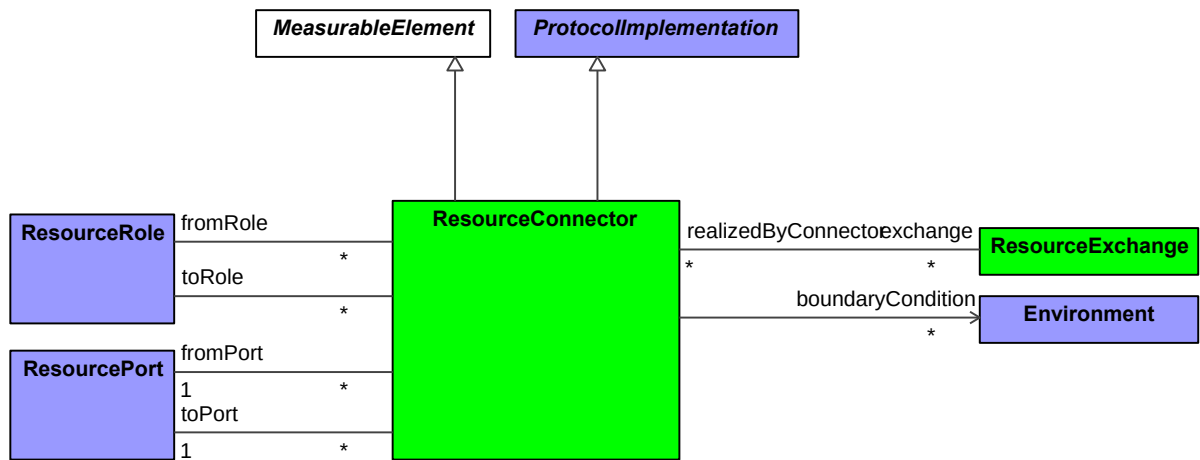


Figure 9:121 - ResourceConnector

## ResourceExchange

Connectivity

No

[Exchange](#)

Description

Asserts that a flow can exist between ResourcePerformers (i.e., flows of data, people, material, or energy).

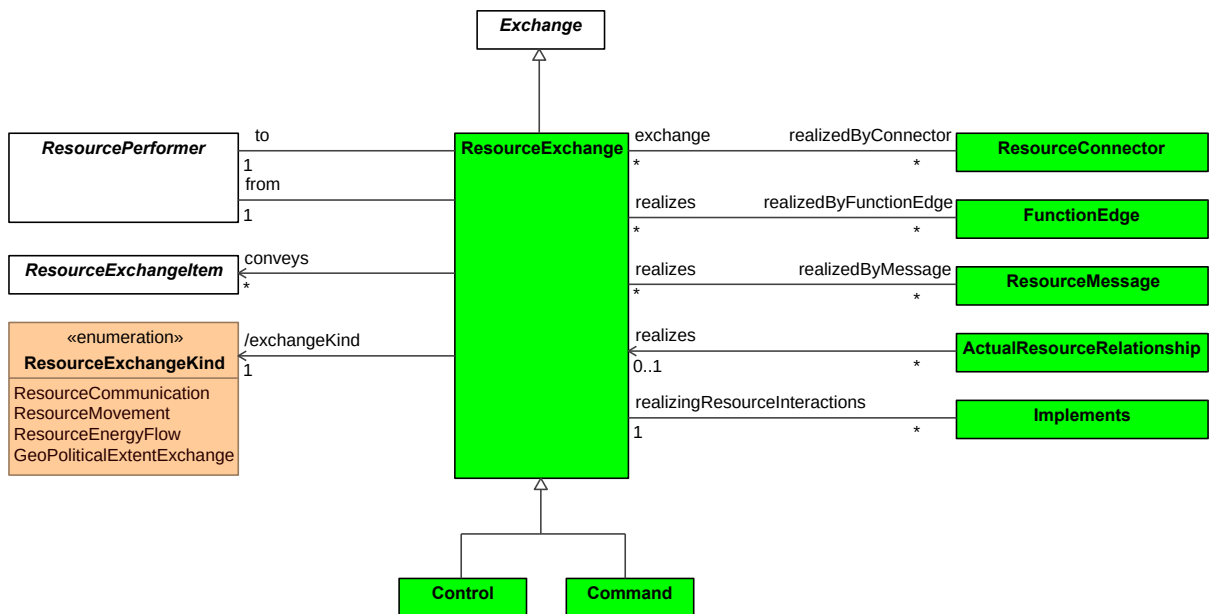


Figure 9:122 - ResourceExchange

## ResourceExchangeItem

Connectivity

Yes

[Resource](#), [SubjectOfSecurityConstraint](#), [ExchangeItem](#)

## Description

An abstract type grouping elements that defines the types of elements that can be exchanged between ResourcePerformers and conveyed by a ResourceExchange.

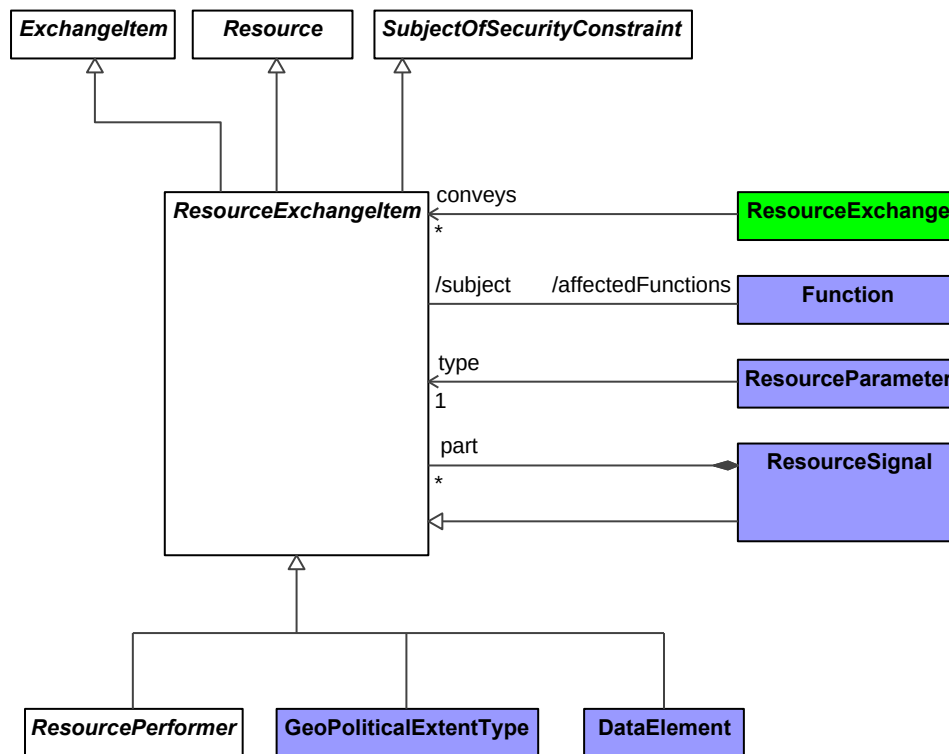


Figure 9:123 - ResourceExchangeltem

## ResourceInterface

Connectivity

No

[PropertySet](#)

## Description

A declaration that specifies a contract between the ResourcePerformers it is related to and any other ResourcePerformers it can interact with. It is also intended to be an implementation of a specification of an Interface in the Business and/or Service layer.

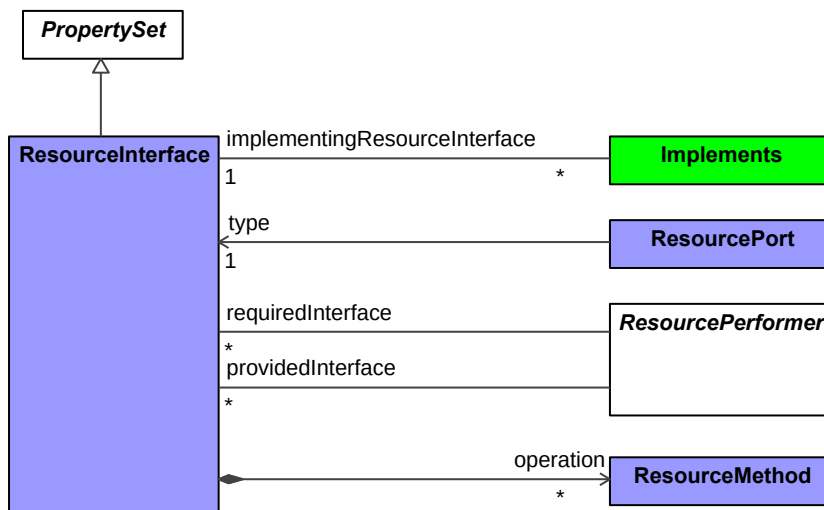


Figure 9:124 - ResourceInterface

## ResourceSignal

Connectivity

No

[ResourceExchangeItem](#)

Description

A property of an element representing something in the physical world, expressed in amounts of a unit of measure.

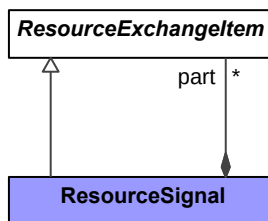


Figure 9:125 - ResourceSignal

## Domain MetaModel::Resources::Processes

### Function

Processes

No

[SubjectOfResourceConstraint](#), [Process](#)

Description

An Activity which is specified in the context to the ResourcePerformer (human or machine) that IsCapableToPerform it.

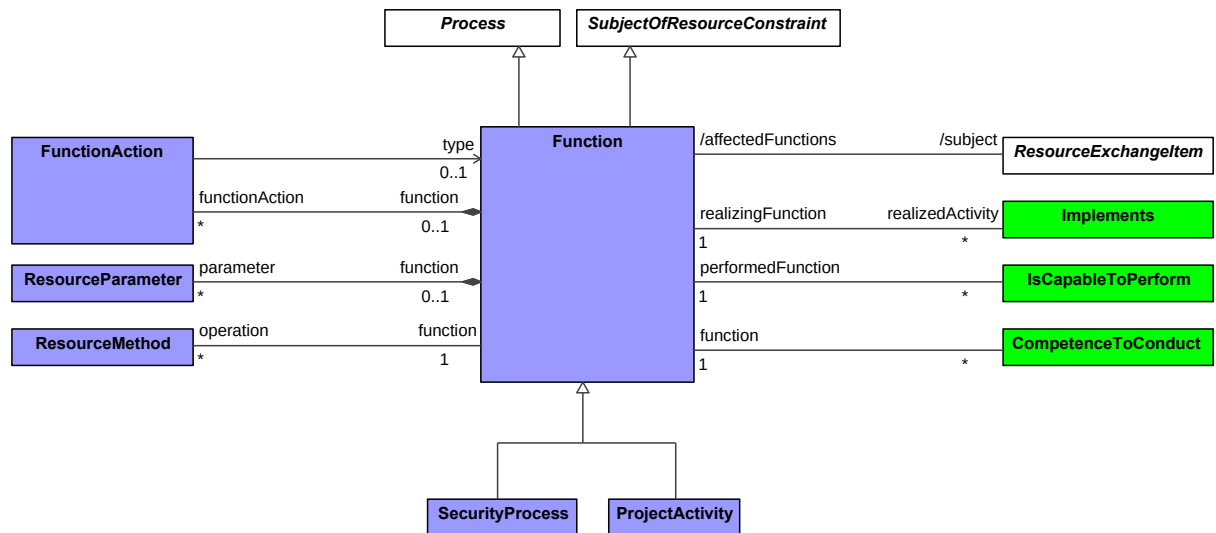


Figure 9:126 - Function

## FunctionAction

Processes

No

[ProcessUsage](#)

Description

A call of a Function indicating that the Function is performed by a ResourceRole in a specific context.

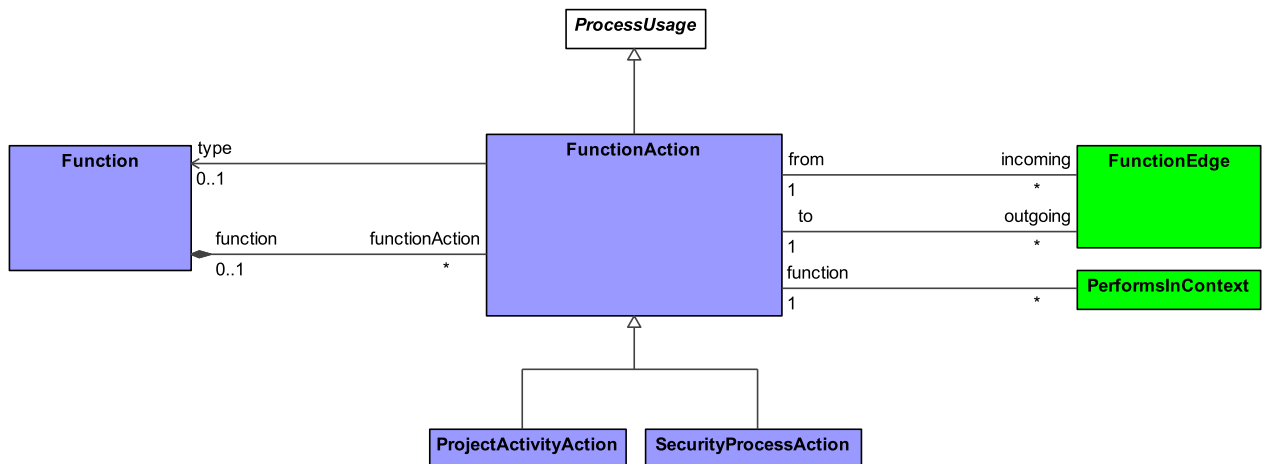


Figure 9:127 - FunctionAction

## FunctionEdge

Processes

No

[ProcessEdge](#)

Description

A tuple that shows the flow of Resources (objects/data) between FunctionActions.

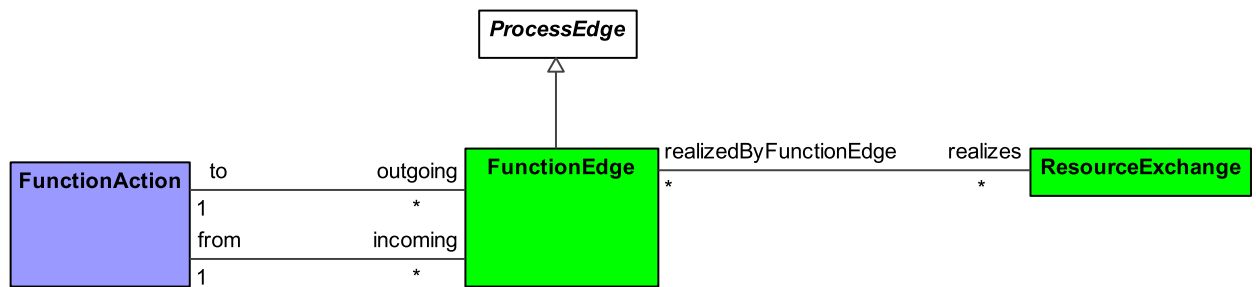


Figure 9:128 - FunctionEdge

## Domain MetaModel::Resources::States

### ResourceStateDescription

States

No

[MeasurableElement](#), [StateDescription](#)

Description

A state machine describing the behavior of a ResourcePerformer, depicting how the ResourcePerformer responds to various events and the actions.

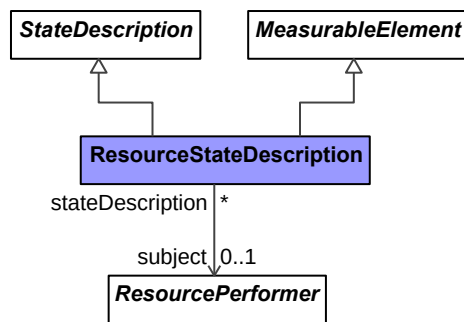


Figure 9:129 - ResourceStateDescription

## Domain MetaModel::Resources::Interaction Scenarios

### ResourceMessage

Interaction Scenarios

No

[InteractionMessage](#)

Description

Message for use in a Resource Event-Trace which carries any of the subtypes of ResourceExchange.

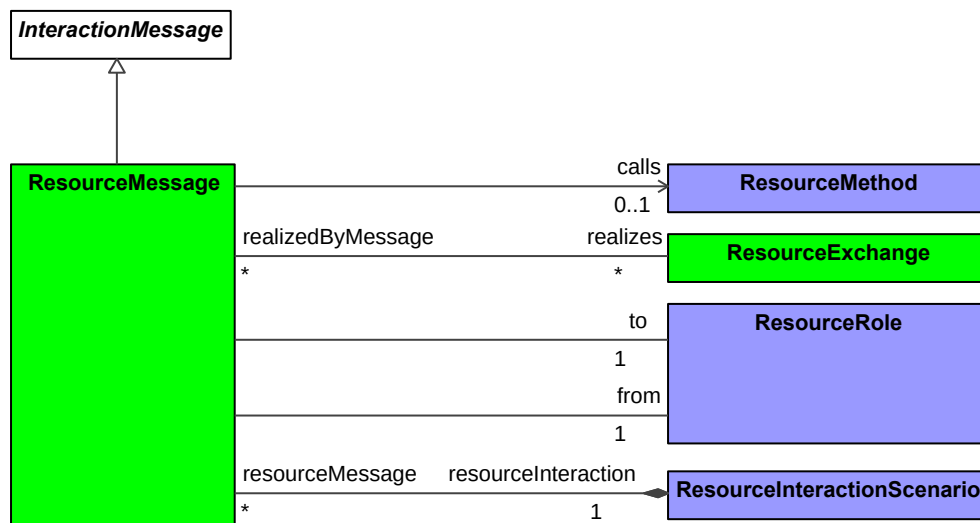


Figure 9:130 - ResourceMessage

## Domain MetaModel::Resources::Information

### DataElement

Information

No

[SubjectOfResourceConstraint](#), [ResourceAsset](#), [ResourceExchangeItem](#)

Description

A formalized representation of data that is managed by or exchanged between resources.

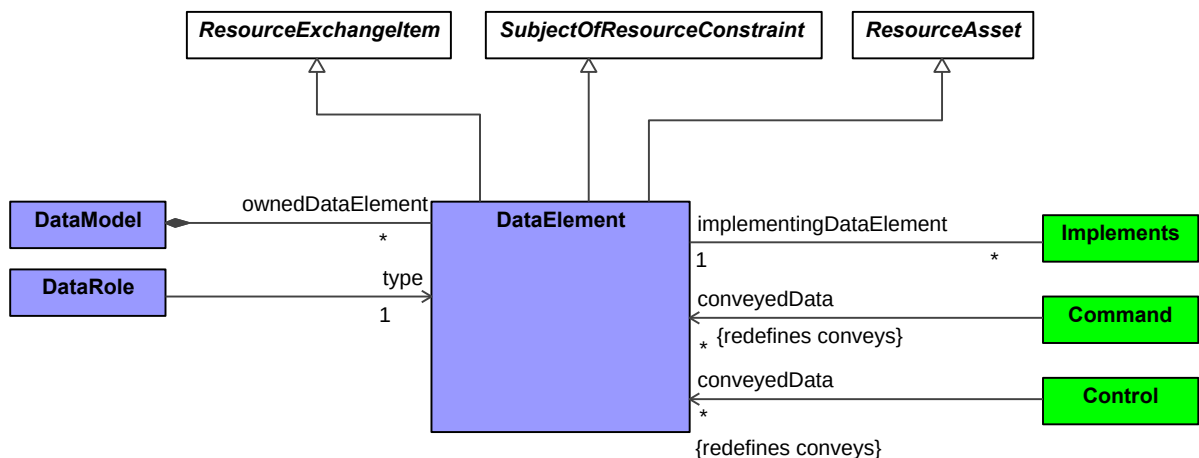


Figure 9:131 - DataElement

### DataRole

Information

No

[AssetRole](#)

## Description

A usage of DataElement that exists in the context of a ResourceAsset. It also allows the representation of the whole-part aggregation of DataElements.

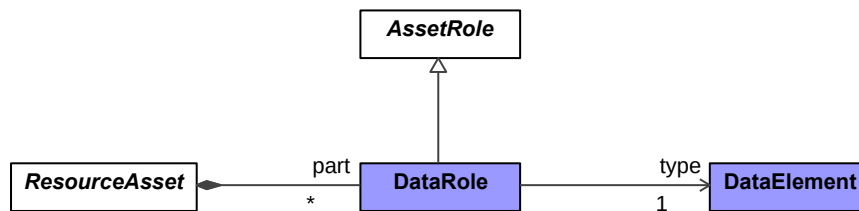


Figure 9:132 - DataRole

## Domain MetaModel::Resources::Constraints

### ResourceConstraint

Constraints

No

[Rule](#)

## Description

A rule governing the structural or functional aspects of an implementation.



Figure 9:133 - ResourceConstraint

### SubjectOfResourceConstraint

Constraints

Yes

[UAFElement](#)

## Description

An abstract type grouping elements that can be the subject of a ResourceConstraint.

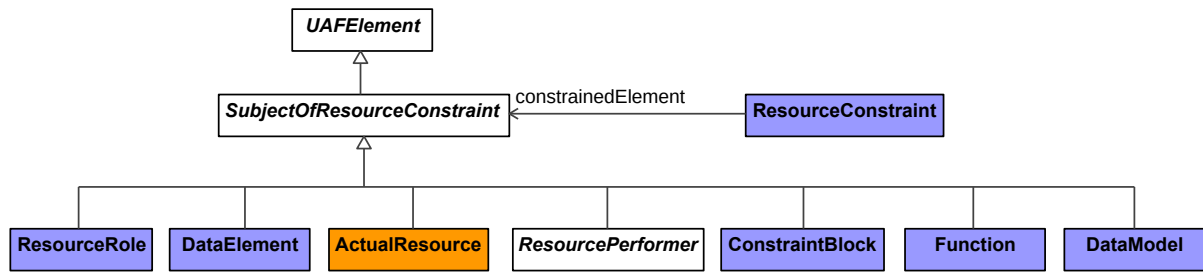


Figure 9:134 - SubjectOfResourceConstraint

## Domain MetaModel::Resources::Roadmap

### Forecast

Roadmap

No

[MeasurableElement](#)

Description

A tuple that specifies a transition from one Asset, Standard, Competence to another future one. It is related to an ActualEnterprisePhase to give it a temporal context.

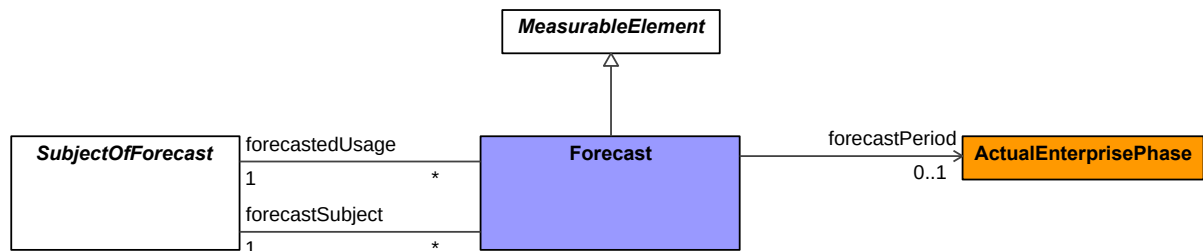


Figure 9:135 - Forecast

### SubjectOfForecast

Roadmap

Yes

[UAFElement](#)

Description

An abstract type grouping elements that can be the subject of a Forecast.

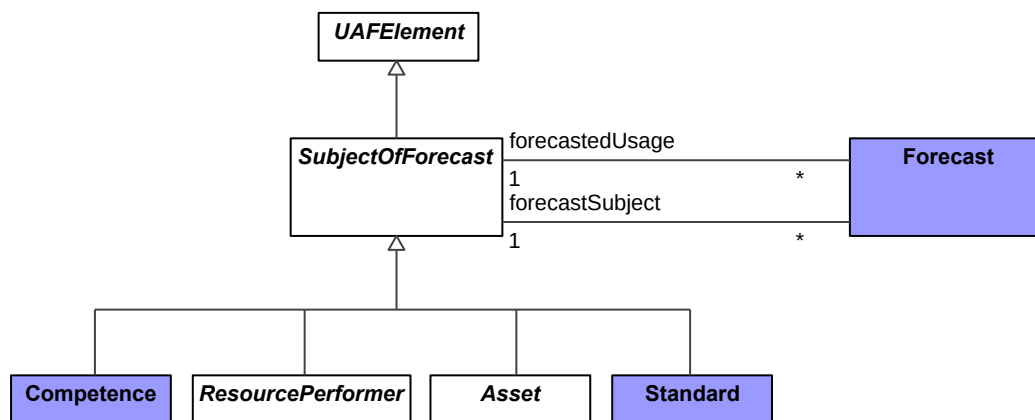


Figure 9:136 - SubjectOfForecast

## Technology

Roadmap

No

[ResourceArtifact](#)

### Description

A sub type of ResourceArtifact that indicates a technology domain, i.e., nuclear, mechanical, electronic, mobile telephony etc.

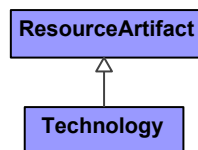


Figure 9:137 - Technology

## VersionedElement

Roadmap

Yes

[UAFElement](#)

### Description

An abstract type grouping ResourcePerformer and ServiceSpecification that allows VersionOfConfiguration to be related to ActualProjectMilestones.

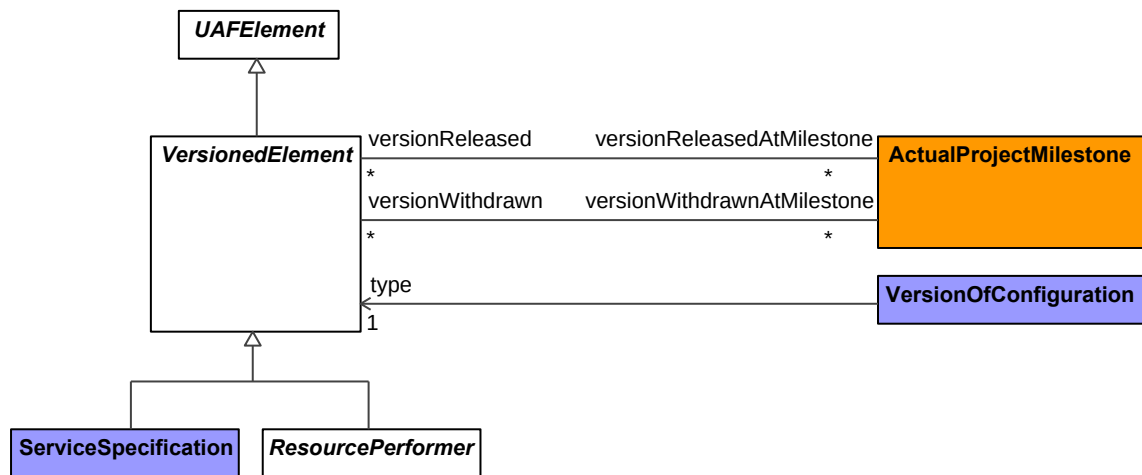


Figure 9:138 - VersionedElement

## VersionOfConfiguration

Roadmap

No

[MeasurableElement](#)

Description

A property of a WholeLifeConfiguration, used in version control of a VersionedElement. It asserts that a VersionedElement is a version of a WholeLifeConfiguration.

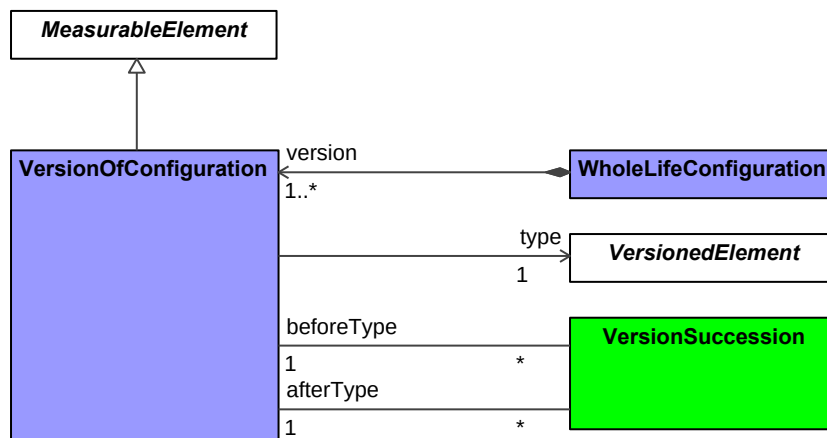


Figure 9:139 - VersionOfConfiguration

## VersionSuccession

Roadmap

No

[MeasurableElement](#)

Description

A tuple between two VersionOfConfigurations that denotes that one VersionOfConfiguration follows from another.

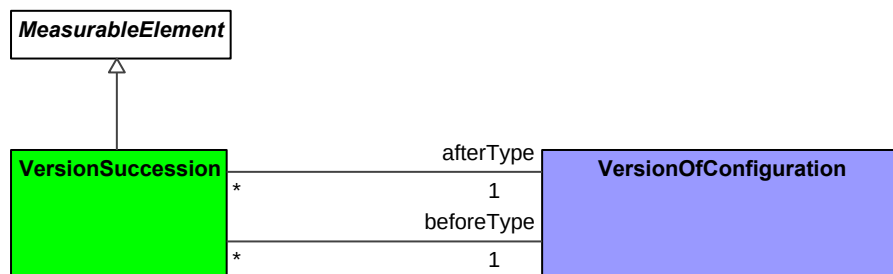


Figure 9:140 - VersionSuccession

## WholeLifeConfiguration

Roadmap

No

[PropertySet](#)

Description

A set of VersionedElements.

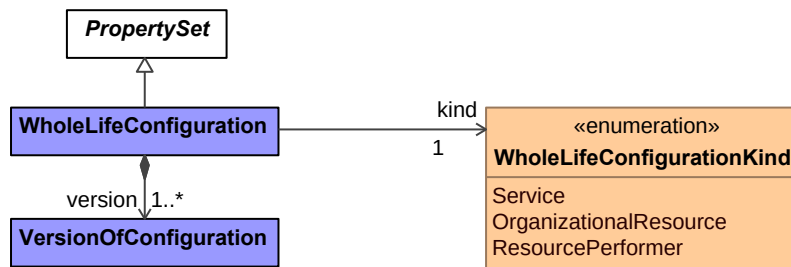


Figure 9:141 - WholeLifeConfiguration

## Domain MetaModel::Resources::Traceability

### ProtocolImplementation

Traceability

Yes

[UAFElement](#)

Description

An abstract type grouping architectural elements that can implement Protocols.

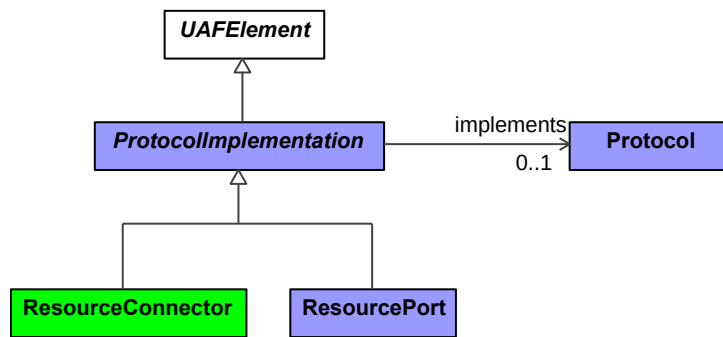


Figure 9:142 - ProtocolImplementation

## 9.1.7 Domain MetaModel::Security

Security Architects, Security Engineers, Systems Engineers, Operational Architects.

addresses the security constraints and information assurance attributes that exist on exchanges between resources and OperationalPerformers

illustrates the security assets, security constraints, security controls, families, and measures required to address specific security concerns.

### Domain MetaModel::Security::Taxonomy

#### Asset

Taxonomy

Yes

[SubjectOfForecast](#), [ConceptItem](#), [LocationHolder](#), [PropertySet](#), [SubjectOfSecurityConstraint](#)

#### Description

Asset as applied to Security views, an abstract type that indicates the types of elements that can be considered as a subject for security analysis.

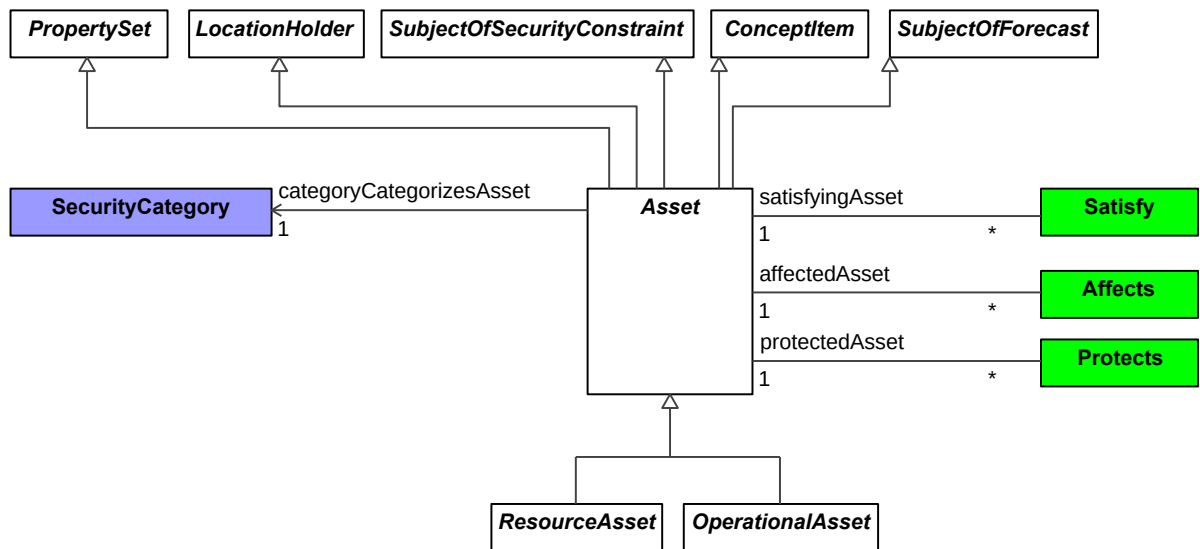


Figure 9:143 - Asset

## OperationalAsset

Taxonomy

Yes

[Asset](#)

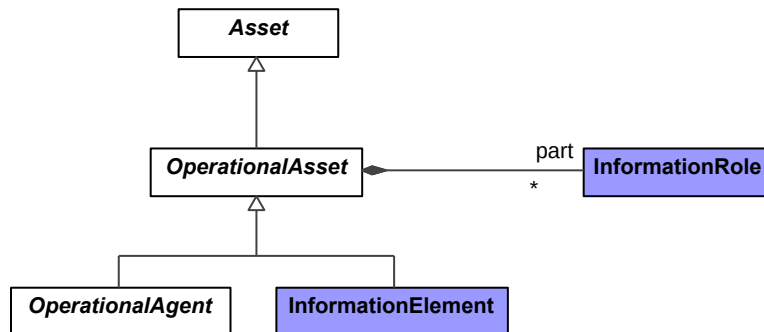


Figure 9:144 - OperationalAsset

## OperationalMitigation

Taxonomy

No

[OperationalArchitecture](#)

Description

A set of **OperationalPerformers** intended to address against specific operational risks.

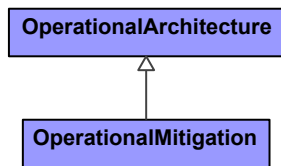


Figure 9:145 - OperationalMitigation

## ResourceAsset

Taxonomy

Yes

[Asset](#)

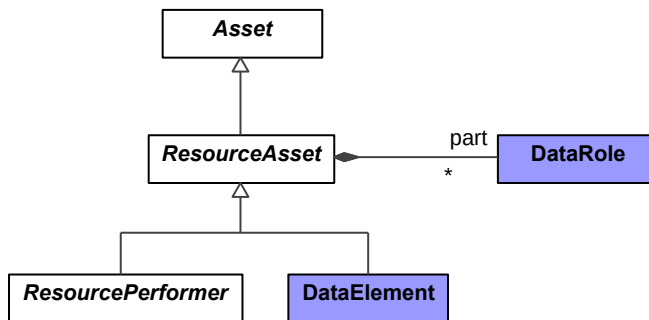


Figure 9:146 - ResourceAsset

## ResourceMitigation

Taxonomy

No

[ResourceArchitecture](#)

Description

A set of ResourcePerformers intended to address against specific risks.

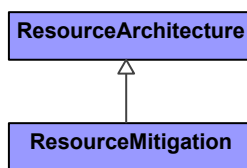


Figure 9:147 - ResourceMitigation

## SecurityEnclave

Taxonomy

No

[ResourceArchitecture](#)

Description

Collection of information systems connected by one or more internal networks under the control of a single authority and security policy. The systems may be structured by physical proximity or by function, independent of location.

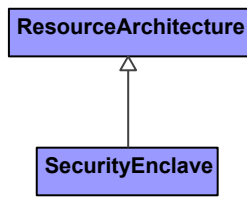


Figure 9:148 - SecurityEnclave

## Domain MetaModel::Security::Structure

### AssetRole

Structure

Yes

BPMN2Metamodel::ResourceRole, [SubjectOfSecurityConstraint](#), [MeasurableElement](#)

Description

AssetRole as applied to Security views, an abstract element that indicates the type of elements that can be considered as a subject for security analysis in the particular context.

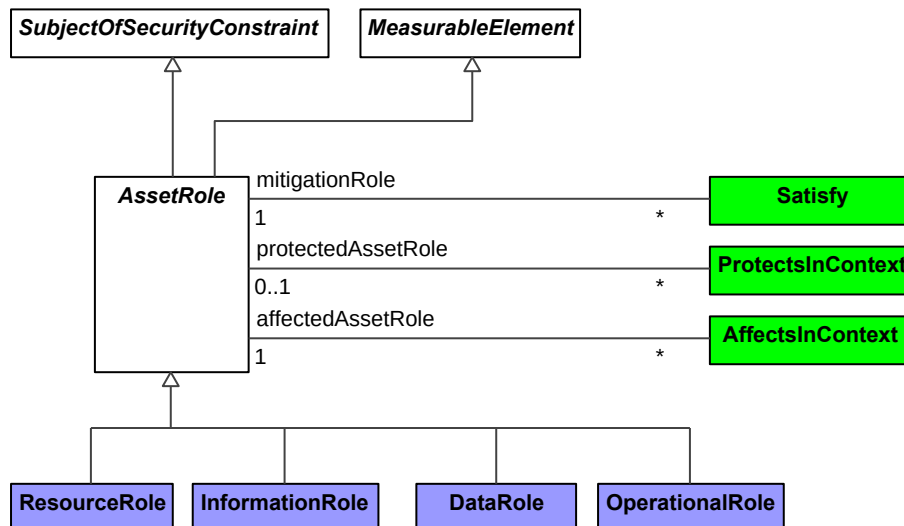


Figure 9:149 - AssetRole

### InformationRole

Structure

No

[AssetRole](#)

Description

A usage of InformationElement that exists in the context of an OperationalAsset. It also allows the representation of the whole-part aggregation of InformationElements.

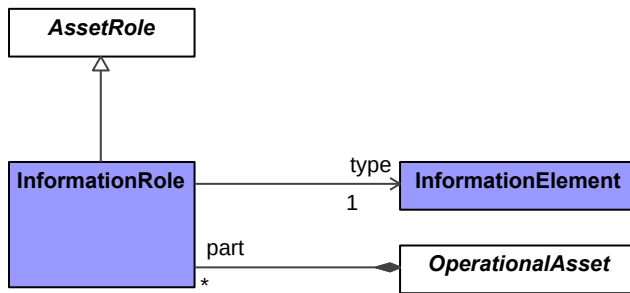


Figure 9:150 - InformationRole

## Domain MetaModel::Security::Processes

### EnhancedSecurityControl

Processes

No

[SecurityControl](#)

Description

Statement of security capability to: (i) build in additional but related, functionality to a basic control; and/or (ii) increase the strength of a basic control.

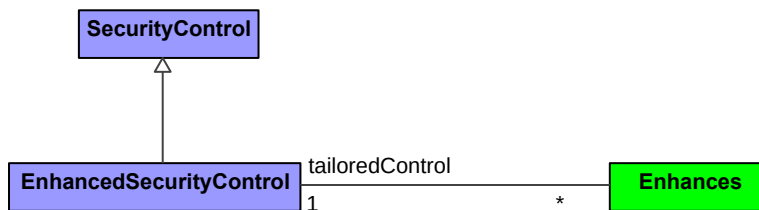


Figure 9:151 - EnhancedSecurityControl

### Enhances

Processes

No

[MeasurableElement](#)

Description

A tuple relating the EnhancedSecurityControl to a SecurityControl.

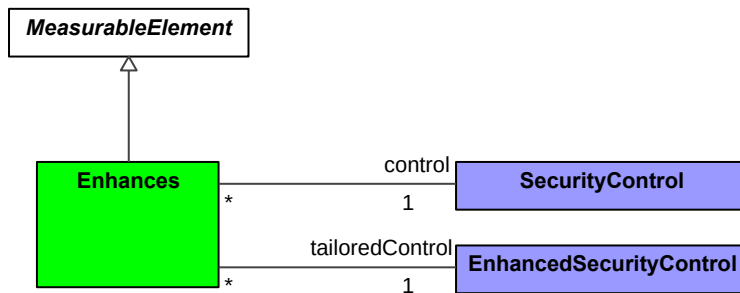


Figure 9:152 - Enhances

## Protects

Processes

No

[MeasurableElement](#)

Description

A tuple that asserts that a SecurityControl is required to protect an Asset.

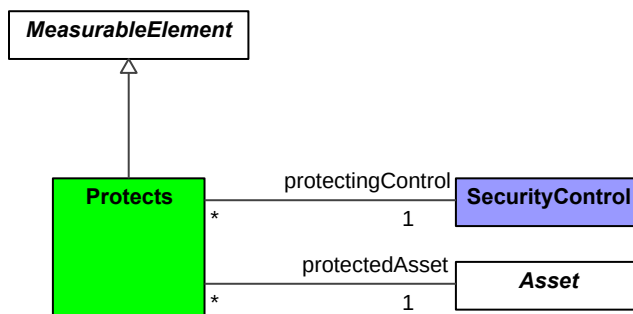


Figure 9:153 - Protects

## ProtectsInContext

Processes

No

[MeasurableElement](#)

Description

A tuple that relates a SecurityControlAction to an OperationalRole, or a ResourceRole. It indicates that SecurityControl is required to protect an Asset in a specific context or configuration.

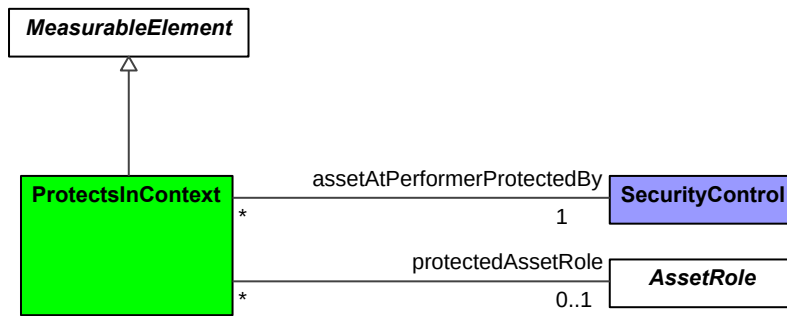


Figure 9:154 - ProtectsInContext

## SecurityProcess

Processes

No

[OperationalActivity](#), [Function](#), [SubjectOfSecurityConstraint](#)

Description

The security-related procedure that satisfies the security control requirement.

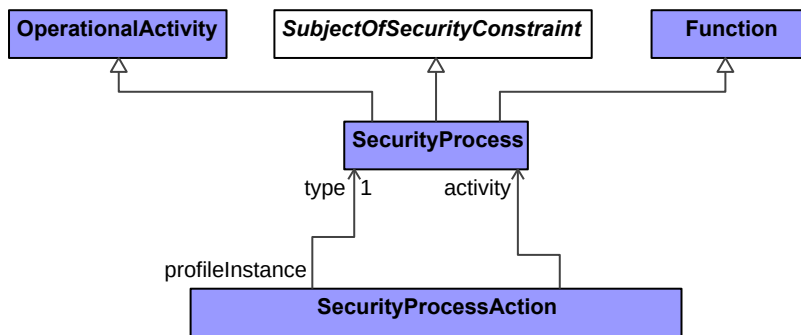


Figure 9:155 - SecurityProcess

## SecurityProcessAction

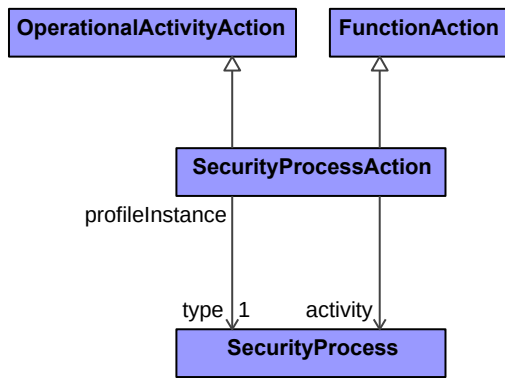
Processes

No

[OperationalActivityAction](#), [FunctionAction](#)

Description

A call of a SecurityProcess in the context of another SecurityProcess.



**Figure 9:156 - SecurityProcessAction**

## Domain MetaModel::Security::Constraints

### ActualRisk

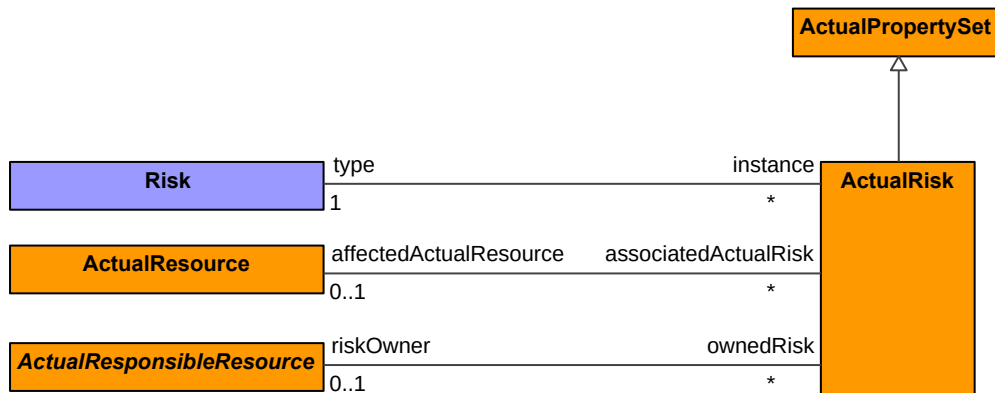
Constraints

No

[ActualPropertySet](#)

Description

An instance of a Risk. A value holder for Risk Measurements.



**Figure 9:157 - ActualRisk**

### Caveat

Constraints

No

[SecurityConstraint](#)

Description

A statement that details alternate conditions under which the rule is not valid.

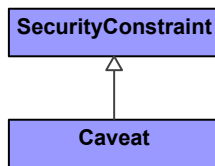


Figure 9:158 - Caveat

## Risk

Constraints

No

[PropertySet](#)

Description

A statement of the impact of an event on Assets. It represents a constraint on an Asset in terms of adverse effects, with an associated measure. The measure is used to capture the extent to which an entity is threatened by a potential circumstance or event. Risk is typically a function of: (i) the adverse impacts that would arise if the circumstance or event occurs; and (ii) the likelihood of occurrence.

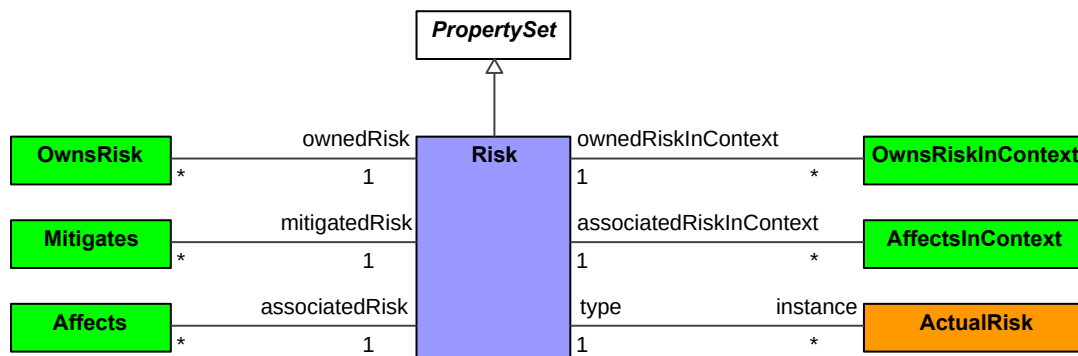


Figure 9:159 - Risk

## SecurityAvailability

Constraints

No

[SecurityMeasurement](#)

Description

Details the potential impact on organization or individuals if the information is not available to those who need to access it.

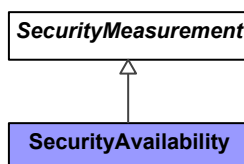


Figure 9:160 - SecurityAvailability

## SecurityCategory

Constraints

No

[MeasurementSet](#)

Description

The security categories that have been determined for each type of information processed, stored, or transmitted by those information systems. The generalized format for expressing the security category (SC) of an information system is:  
SC information system = {(confidentiality, impact), (integrity, impact), (availability, impact)}.

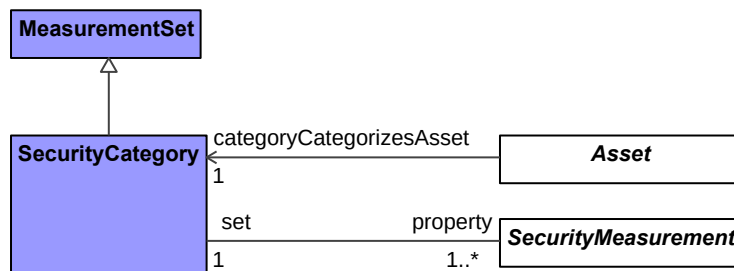


Figure 9:161 - SecurityCategory

## SecurityClassification

Constraints

No

[SecurityMeasurement](#)

Description

Details a classification for the exchange.

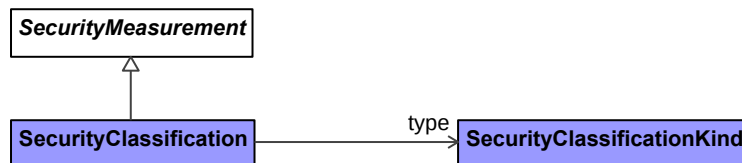


Figure 9:162 - SecurityClassification

## SecurityClassificationKind

Constraints

No

[MeasurableElement](#)

Description

A type that defines acceptable values for the security category (SC) of an information system, where the acceptable values for potential impact are low, moderate, or high.

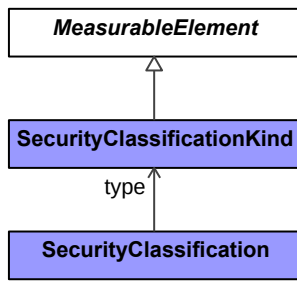


Figure 9:163 - SecurityClassificationKind

## SecurityConstraint

Constraints

No

[Rule](#)

Description

A type of rule that captures a formal statement to define access control policy language.

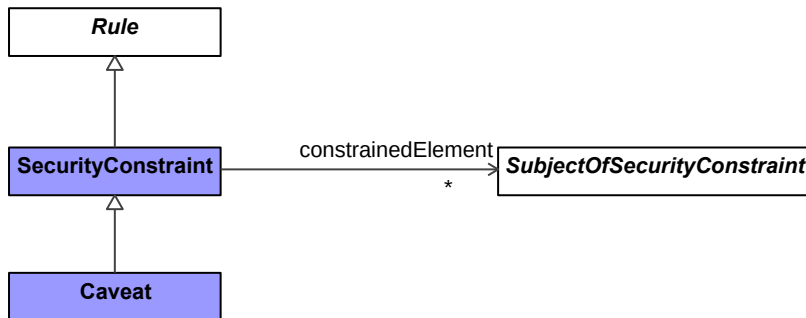


Figure 9:164 - SecurityConstraint

## SecurityControl

Constraints

No

[MeasurableElement](#)

Description

The management, operational, and technical control (i.e., safeguard or countermeasure) prescribed for an information system to protect the confidentiality, integrity, and availability of the system and its information [NIST SP 800-53].



Figure 9:165 - SecurityControl

## SecurityControlFamily

Constraints

No

[SecurityControl](#)

Description

An element that organizes security controls into a family. Each Security Control Family contains security controls related to the general security topic of the family.

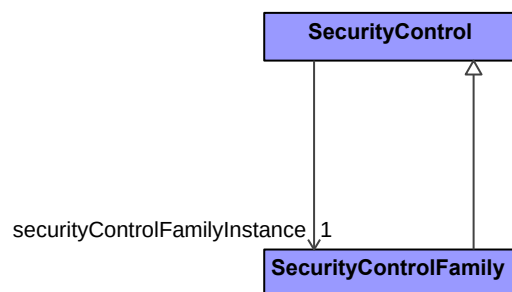


Figure 9:166 – SecurityControlFamily

## SecurityIntegrity

Constraints

No

[SecurityMeasurement](#)

Description

Details the potential impact on organization or individuals due to modification or destruction of information, and includes ensuring information non-repudiation and authenticity.

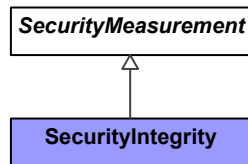


Figure 9:167 - SecurityIntegrity

## SecurityMeasurement

Constraints

Yes

[Measurement](#)

Description

An abstract type grouping all types of security measurements (e.g., SecurityIntegrity, SecurityAvailability).

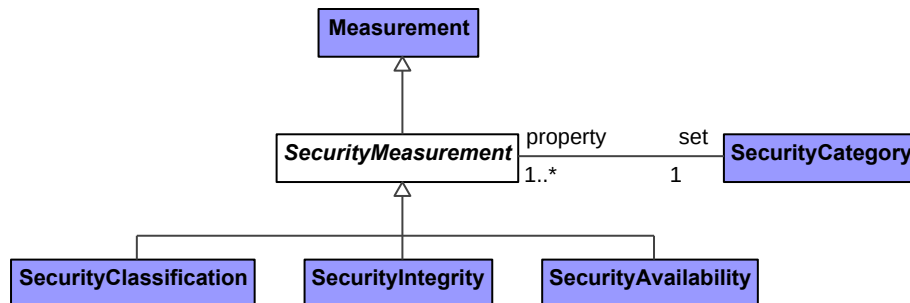


Figure 9:168 - SecurityMeasurement

## SubjectOfSecurityConstraint

Constraints

Yes

[UAFElement](#)

Description

An abstract type grouping elements that can be the subject of a SecurityConstraint.

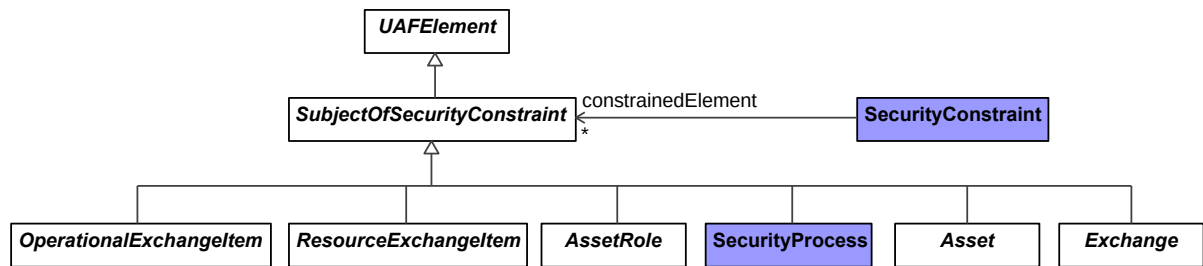


Figure 9:169 - SubjectOfSecurityConstraint

## Domain MetaModel::Security::Traceability

### Affects

Traceability

No

[MeasurableElement](#)

Description

A tuple that asserts that a Risk is applicable to an Asset.

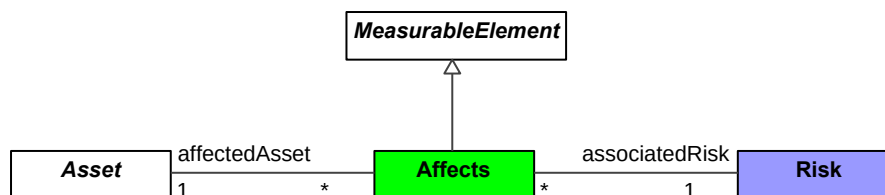


Figure 9:170 - Affects

### AffectsInContext

Traceability

No

[MeasurableElement](#)

Description

A tuple that asserts that a Risk is applicable to an AssetRole in the specific context or configuration.

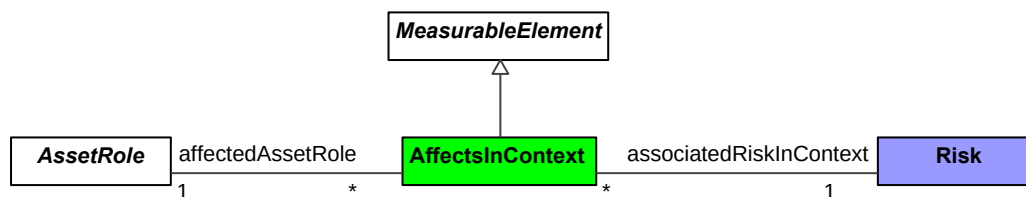


Figure 9:171 – AffectsInContext

## Mitigates

Traceability

No

[MeasurableElement](#)

Description

A tuple relating a Security Control to a Risk. Mitigation is established to manage risk and could be represented as an overall strategy or through techniques (mitigation configurations) and procedures (SecurityProcesses).

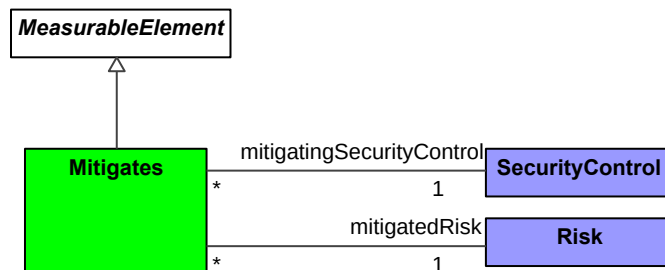


Figure 9:172 - Mitigates

## OwnsRisk

Traceability

No

[MeasurableElement](#)

Description

A tuple relating a Risk to an organizational resource that is responsible for executing the risk mitigation.

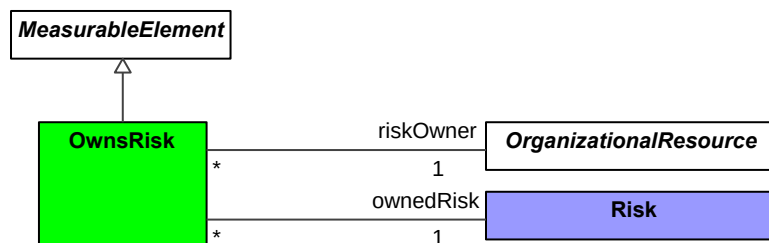


Figure 9:173 - OwnsRisk

## OwnsRiskInContext

Traceability

No

[MeasurableElement](#)

Description

A tuple relating a Risk to an organizational role that is responsible for executing the risk mitigation in the specific context or configuration.

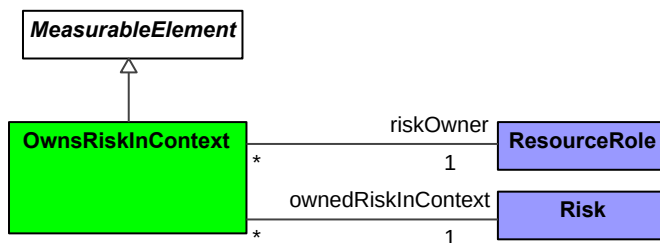


Figure 9:174 - OwnsRiskInContext

## 9.1.8 Domain MetaModel::Projects

### Domain MetaModel::Projects::Taxonomy

#### Project

Taxonomy

No

[OrganizationalResource](#)

Description

A type that describes types of time-limited endeavors that are required to meet one or more Capability needs.

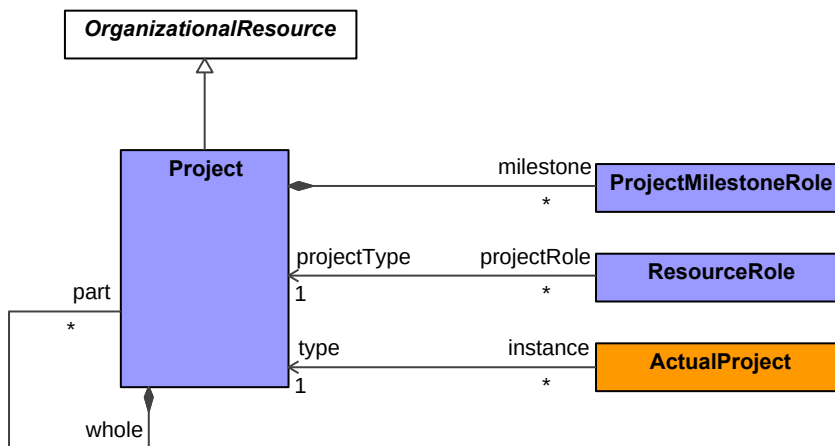


Figure 9:175 - Project

#### ProjectMilestone

Taxonomy

No

[PropertySet](#)

Description

A type of event in a Project by which progress is measured.

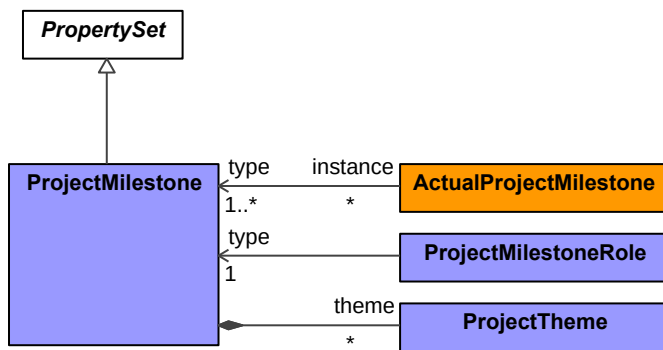


Figure 9:176 - ProjectMilestone

## Domain MetaModel::Projects::Structure

### ActualProjectMilestoneRole

Structure

No

[ActualState](#)

Description

An ActualProjectMilestone that is applied to a ProjectMilestoneRole.



Figure 9:177 - ActualProjectMilestoneRole

### ProjectMilestoneRole

Structure

No

[MeasurableElement](#)

Description

The role played by a ProjectMilestone in the context of a Project.

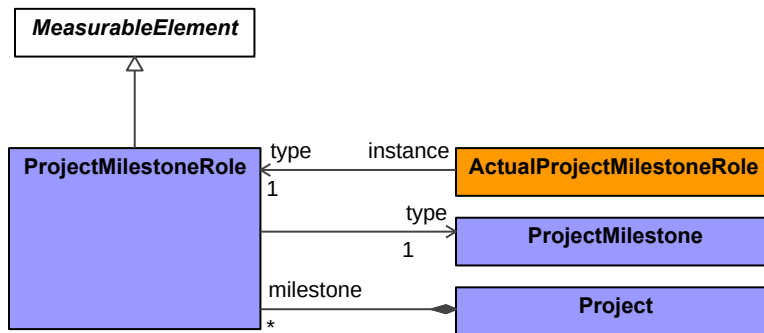


Figure 9:178 - ProjectMilestoneRole

## ProjectStatus

Structure

No

[ActualState](#)

Description

The status (i.e., level of progress) of a ProjectTheme for an ActualProject at the time of the ActualProjectMilestone.

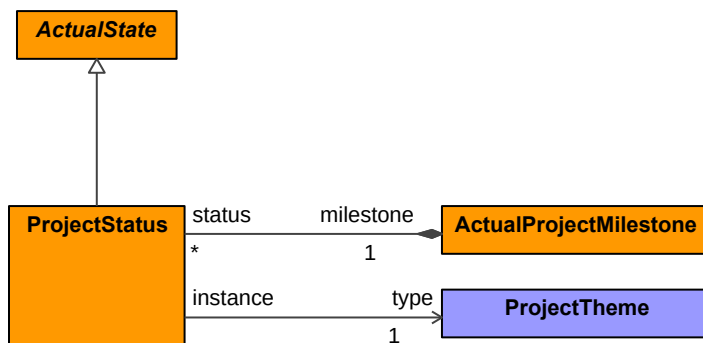


Figure 9:179 - ProjectStatus

## ProjectTheme

Structure

No

[MeasurableElement](#)

Description

A property of a ProjectMilestone that captures an aspect by which the progress of ActualProjects may be measured.

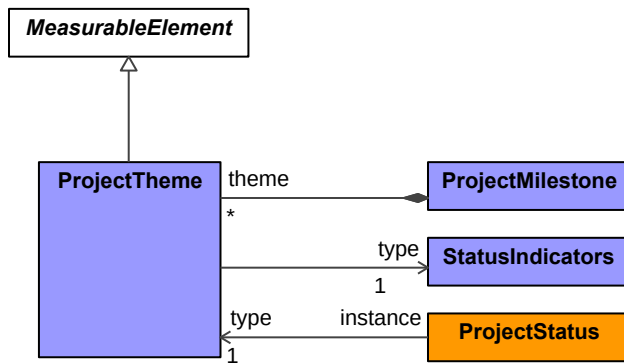


Figure 9:180 - ProjectTheme

## StatusIndicators

Structure

No

[MeasurableElement](#)

Description

An enumerated type that specifies a status for a ProjectTheme.

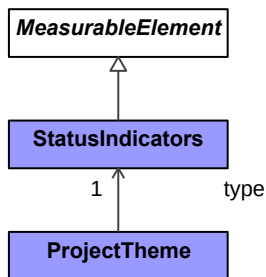


Figure 9:181 - StatusIndicators

## Domain MetaModel::Projects::Connectivity

### MilestoneDependency

Connectivity

No

[MeasurableElement](#)

Description

A tuple between two ActualProjectMilestones that denotes one ActualProjectMilestone follows from another.

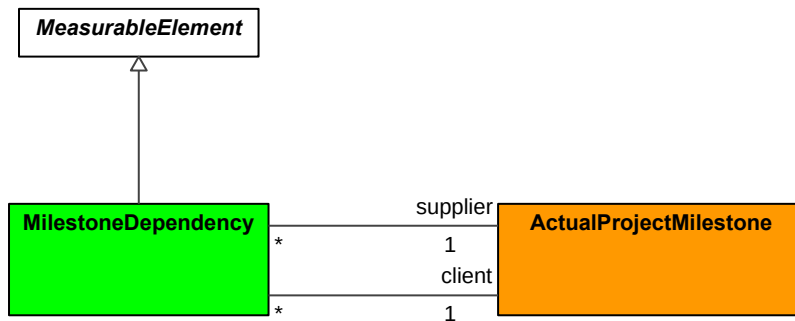


Figure 9:182 - MilestoneDependency

## Domain MetaModel::Projects::Processes

### ProjectActivity

Processes

No

[Function](#), [Process](#)

Description

An activity carried out during a project.

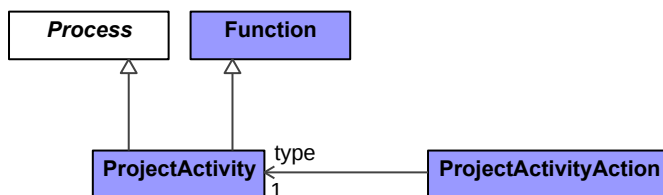


Figure 9:183 - ProjectActivity

### ProjectActivityAction

Processes

No

[FunctionAction](#)

Description

The ProjectActivityAction is defined as a call behavior action that invokes the activity that needs to be performed.

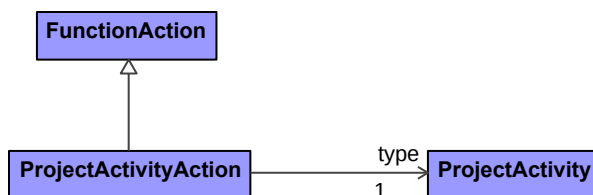


Figure 9:184 - ProjectActivityAction

## Domain MetaModel::Projects::Interaction Scenarios

### ProjectSequence

Interaction Scenarios

isAbstract: No

[MeasurableElement](#)

Description

A tuple between two ActualProjects that denotes one ActualProject cannot start before the previous ActualProject is finished.

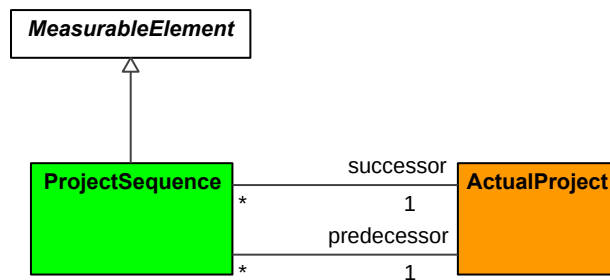


Figure 9:185 - ProjectSequence

## Domain MetaModel::Projects::Roadmap

### ActualProject

Roadmap

No

[ActualOrganizationalResource](#), [Achiever](#)

Description

A time-limited endeavor to provide a specific set of ActualResources that meet specific Capability needs.

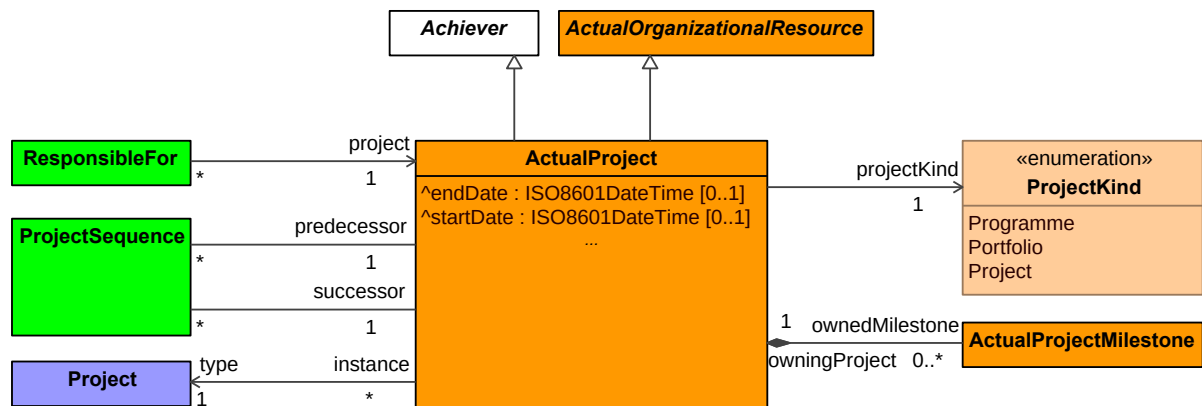


Figure 9:186 – ActualProject

## ActualProjectMilestone

Roadmap

No

[ActualPropertySet](#)

Description

An event with a start date in an ActualProject from which progress is measured.

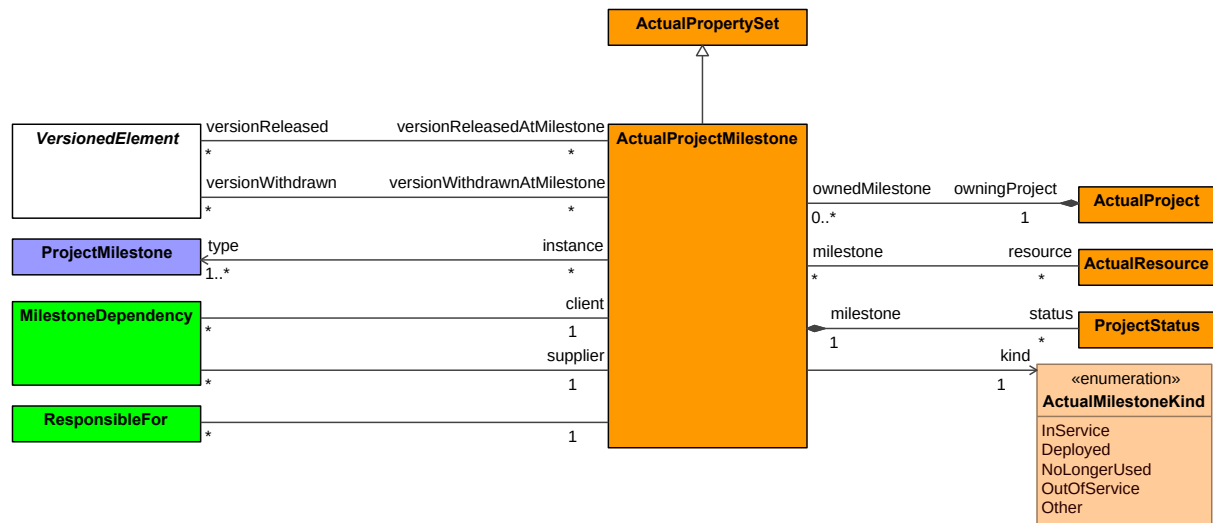


Figure 9:187 - ActualProjectMilestone

Constraints

[1] unnamed1 startTime=endTime

## Domain MetaModel::Projects::Traceability

### ResponsibleFor

Traceability

No

[MeasurableElement](#)

Description

A tuple between an ActualResponsibleResource and an ActualResponsibility or ActualProject. It defines the duties that the ActualResponsibleResource is ResponsibleFor.

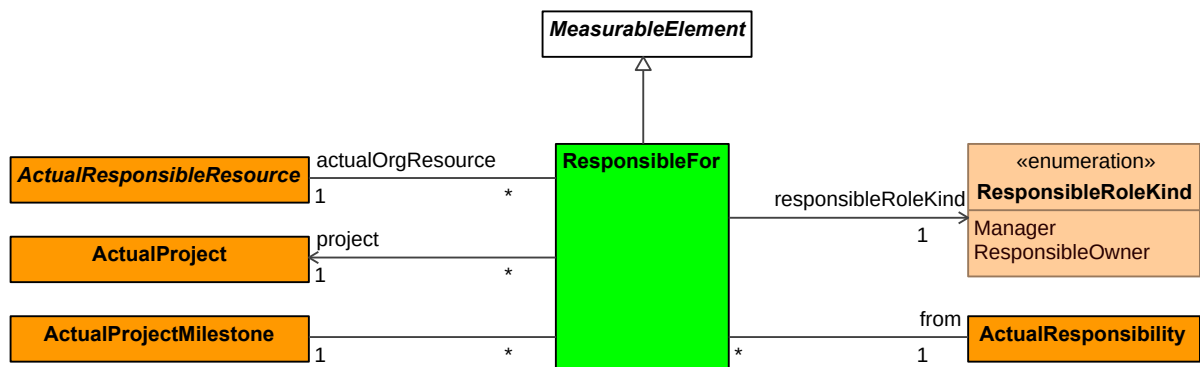


Figure 9:188 – ResponsibleFor

## 9.1.9 Domain MetaModel::Standards

Solution Providers, Systems Engineers, Software Engineers, Systems Architects, Business Architects.  
technical and non-technical Standards applicable to the architecture.

shows the technical, operational, and business Standards applicable to the architecture. Defines the underlying current and expected Standards.

### Domain MetaModel::Standards::Taxonomy

#### Protocol

Taxonomy

No

[Standard](#)

#### Description

A Standard for communication over a network. Protocols may be composite, represented as a ProtocolStack made up of ProtocolLayers.

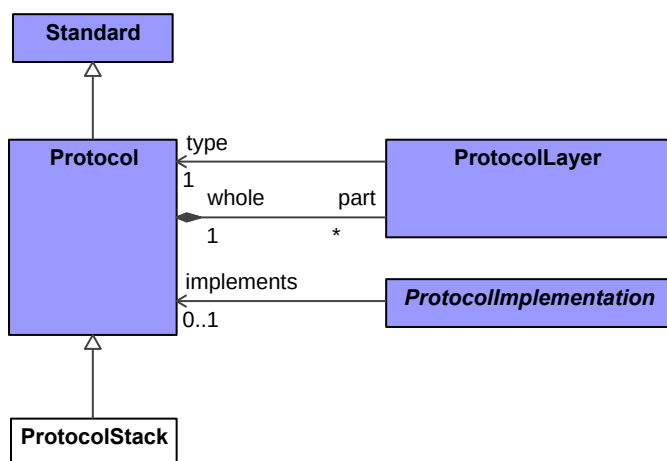


Figure 9:189 - Protocol

## ProtocolStack

Taxonomy

No

[Protocol](#)

Description

A sub type of Protocol that contains the ProtocolLayers, defining a complete stack.

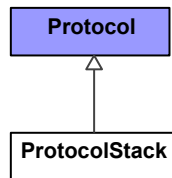


Figure 9:190 - ProtocolStack

## Standard

Taxonomy

No

[SubjectOfForecast](#), [PropertySet](#)

Description

A ratified and peer-reviewed specification that is used to guide or constrain the architecture. A Standard may be applied to any element in the architecture.

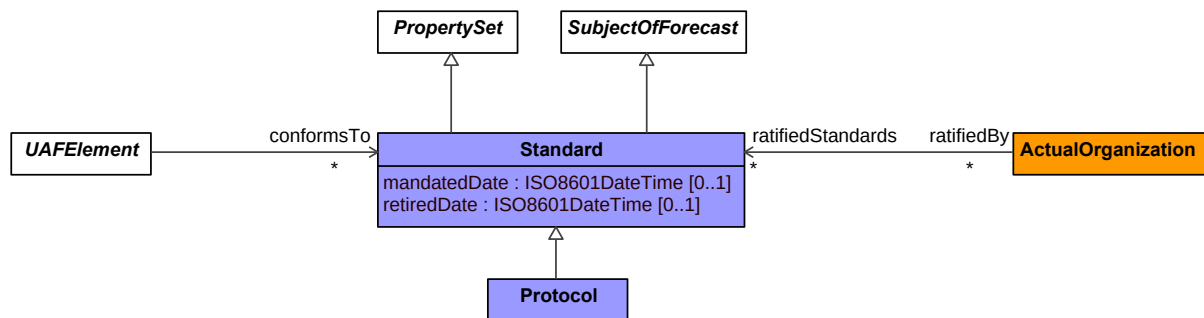


Figure 9:191 - Standard

Attributes

mandatedDate : ISO8601DateTime[0..1] The date when this version of the Standard was published.  
retiredDate : ISO8601DateTime[0..1] The date when this version of the Standard was retired.

## Domain MetaModel::Standards::Structure

### ProtocolLayer

Structure

No

[MeasurableElement](#)

Description

Usage of a Protocol in the context of another Protocol. Creates a whole-part relationship.

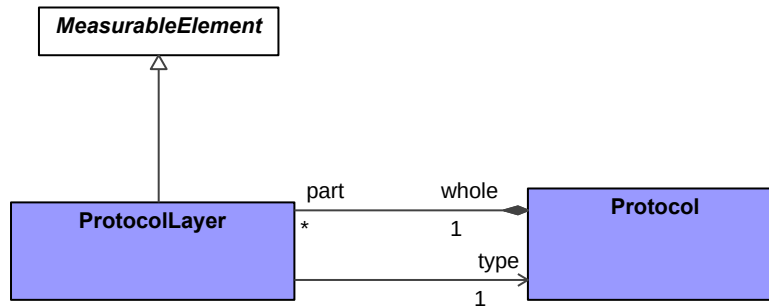


Figure 9:192 – ProtocolLayer

### 9.1.10 Domain MetaModel::Actual Resources

Solution Providers, Systems Engineers, Business Architects, Human Resources.

the analysis.e.g., evaluation of different alternatives, what-if, trade-offs, V&V on the actual resource configurations.

illustrates the expected or achieved actual resource configurations and actual relationships between them.

#### Domain MetaModel::Actual Resources::Taxonomy

##### ActualOrganization

Taxonomy

No

[ActualResponsibleResource](#)

Description

An actual formal or informal organizational unit, e.g., "Driving and Vehicle Licensing Agency", "UAF team Alpha".

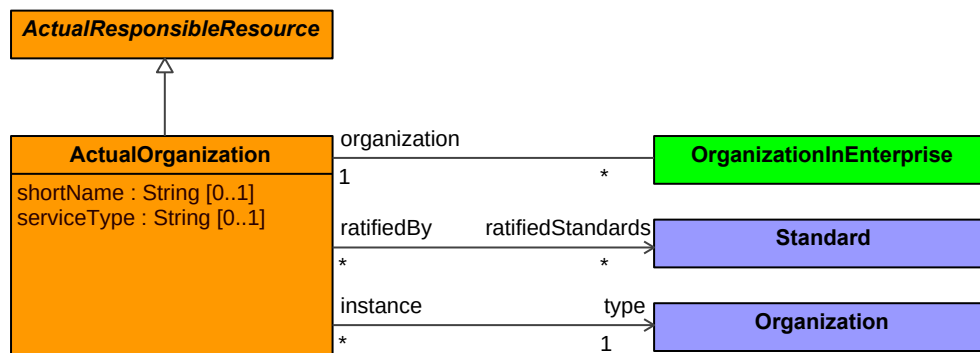


Figure 9:193 - ActualOrganization

Attributes

|                            |                                                                                                                              |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| serviceType : String[0..1] | Service office code or symbol                                                                                                |
| shortName : String[0..1]   | String providing a simplified means of identifying an ActualOrganization, i.e. SoftWareGroup could use SWG as the shortName. |

## **ActualOrganizationalResource**

Taxonomy

Yes

[ActualResource](#), [Stakeholder](#)

Description

Abstract element for an ActualOrganization, ActualPerson or ActualPost.

**Figure 9:194 - ActualOrganizationalResource**

## **ActualPerson**

Taxonomy

No

[ActualResponsibleResource](#)

Description

An individual human being.

**Figure 9:195 – ActualPerson**

## ActualPost

Taxonomy

No

[ActualResponsibleResource](#)

Description

An actual, specific post, an instance of a Post "type" - e.g., "President of the United States of America." where the Post would be president.

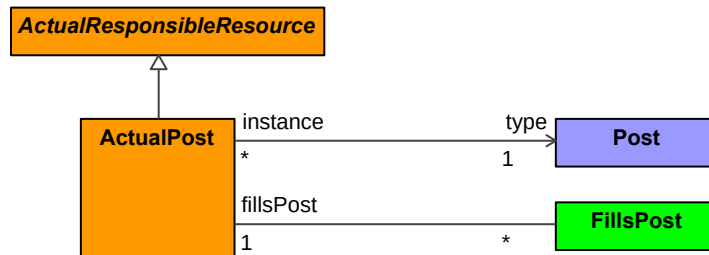


Figure 9:196 - ActualPost

## ActualResource

Taxonomy

No

[ActualPropertySet](#), [SubjectOfResourceConstraint](#), [Achiever](#), [CapableElement](#)

Description

An individual, fully-realized ResourcePerformer.

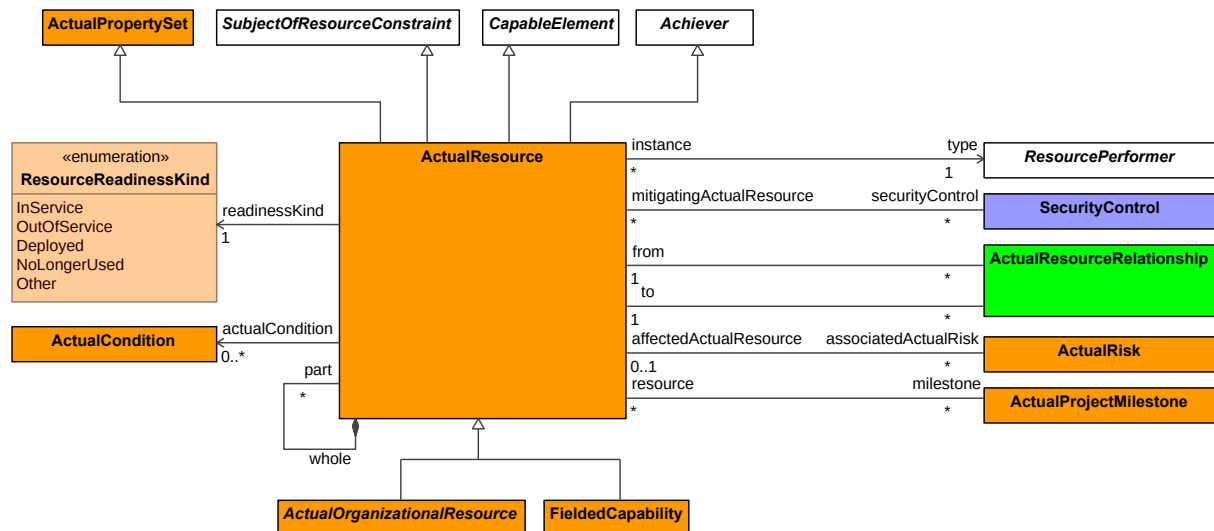


Figure 9:197 – ActualResource

## ActualResourceRelationship

Taxonomy

No

[UAFElement](#)

Description

An actual resource flow existing between ActualResources (i.e., flow of data, people, materiel, or energy).

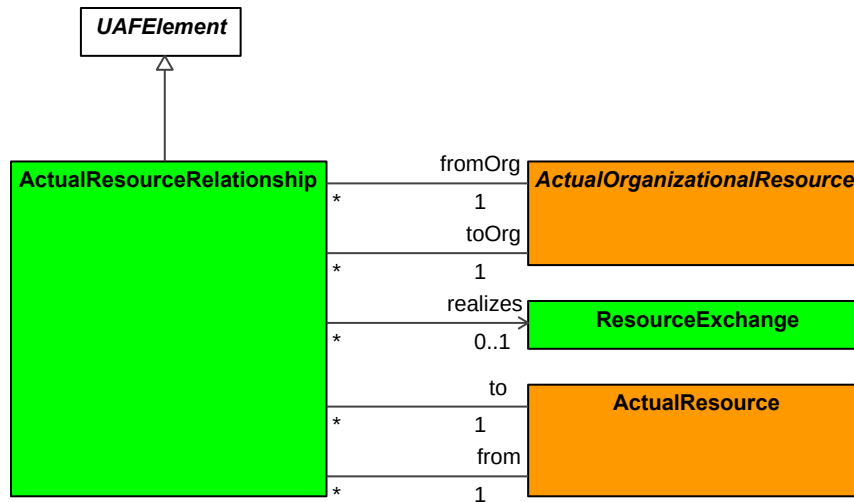


Figure 9:198 - ActualResourceRelationship

## ActualResponsibility

Taxonomy

No

[ActualOrganizationalResource](#)

Description

An actual duty required of a Person or Organization.

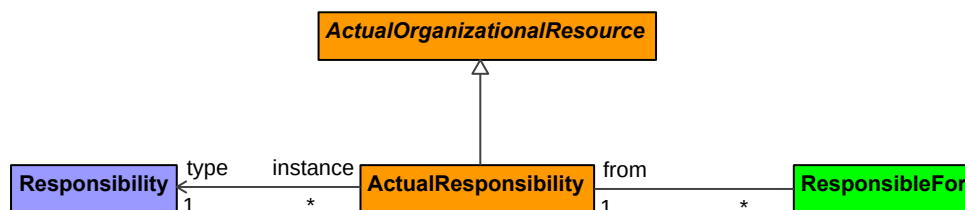


Figure 9:199 – ActualResponsibility

## ActualResponsibleResource

Taxonomy

Yes

[ActualOrganizationalResource](#)

Description

An abstract type grouping responsible OrganizationalResources.

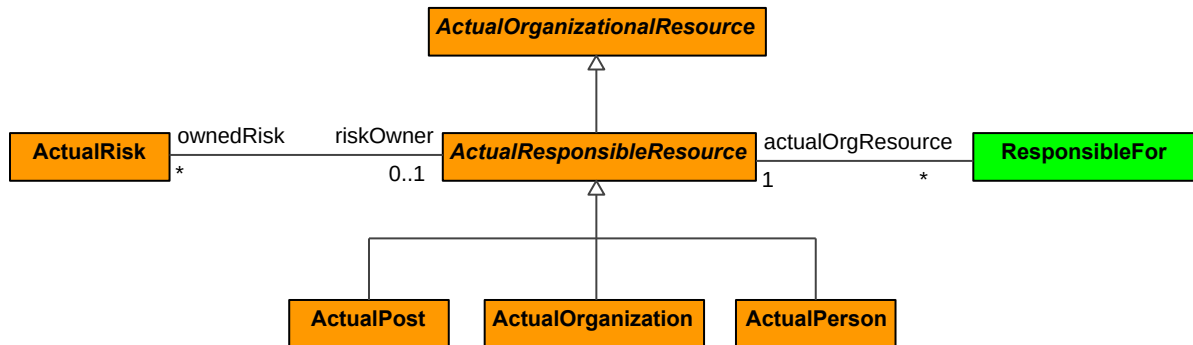


Figure 9:200 - ActualResponsibleResource

## FieldedCapability

Taxonomy

No

[ActualResource](#)

Description

An individual, fully-realized capability.

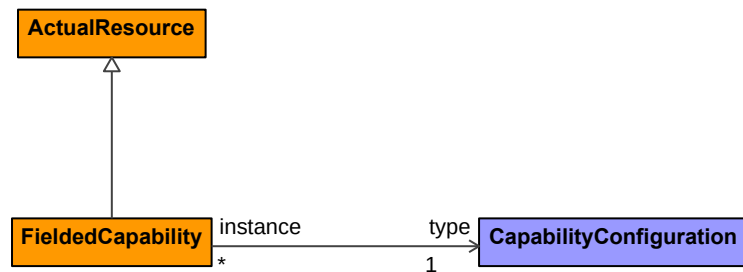


Figure 9:201 - FieldedCapability

## Domain MetaModel::Actual Resources::Constraints

### ActualService

Constraints

Yes

[ActualMeasurementSet](#), [CapableElement](#)

Description

An individual ServiceSpecification.

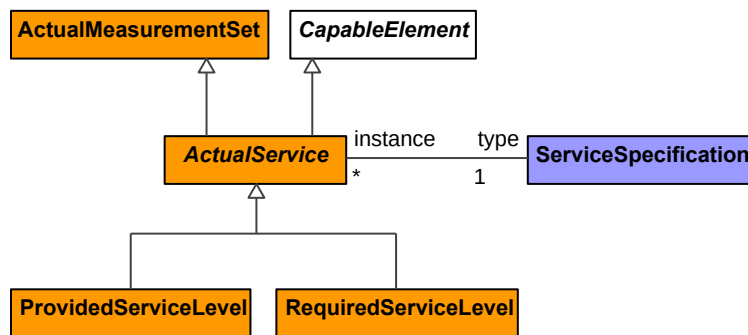


Figure 9:202 - ActualService

## ProvidedServiceLevel

Constraints

No

[ActualService](#)

Description

A sub type of ActualService that details a specific service level delivered by the provider.

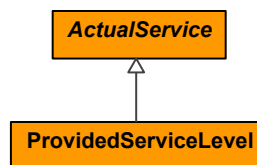


Figure 9:203 - ProvidedServiceLevel

## ProvidesCompetence

Constraints

No

[MeasurableElement](#)

Description

A tuple that asserts that an ActualOrganizationalResource provides a specific set of Competencies.

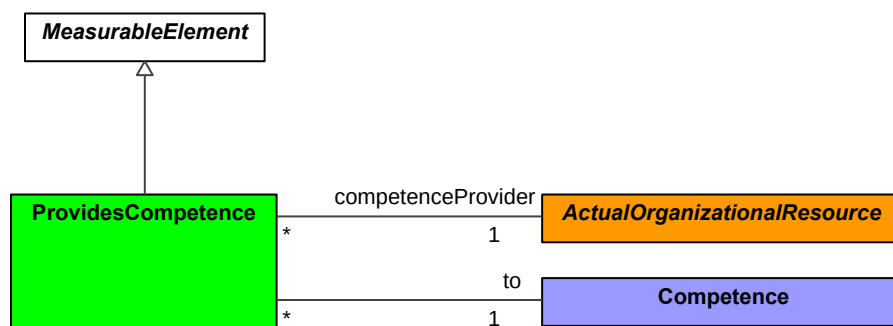


Figure 9:204 - ProvidesCompetence

## RequiredServiceLevel

Constraints

No

[ActualService](#)

Description

A sub type of ActualService that details a specific service level required of the provider.

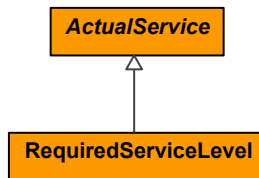


Figure 9:205 - RequiredServiceLevel

### 9.1.11 Domain MetaModel::Dictionary

Architects, users of the architecture, Capability Owners, Systems Engineers, Solution Providers.

Definitions for all the elements in the architecture, libraries of environments and measurements.

Presents all the elements used in an architecture. Can be used specifically to capture:

a. Elements and relationships that are involved in defining the environments applicable to capability, operational concept or set of systems.

b. Measurable properties that can be used to support analysis such as KPIs, MoEs, TPIs etc.

Tabular format, SysML Block Definition Diagram.

## Alias

Dictionary

No

[MeasurableElement](#)

Description

A metamodel Artifact used to define an alternative name for an element.

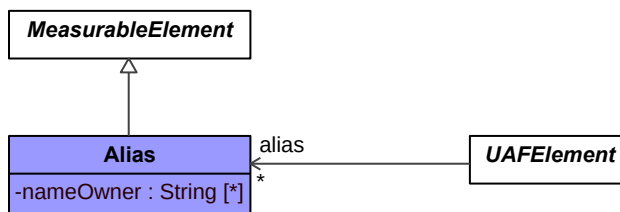


Figure 9:206 – Alias

Attributes

nameOwner : String[\*] Someone or something that uses this alternative name.

## Definition

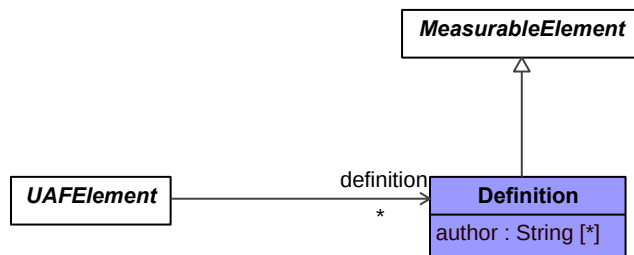
Dictionary

No

[MeasurableElement](#)

Description

A comment containing a description of an element in the architecture.



**Figure 9:207 - Definition**

Attributes

author : String[\*] The original or current person (architect) responsible for the Definition.

## SameAs

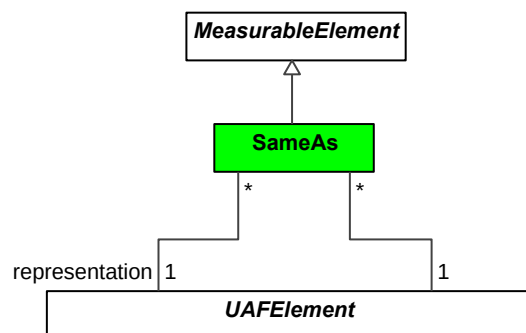
Dictionary

No

[MeasurableElement](#)

Description

A tuple that asserts that two elements refer to the same real-world thing.



**Figure 9:208 – SameAs**

## 9.1.12 Domain MetaModel::Summary & Overview

### ArchitecturalDescription

Summary & Overview

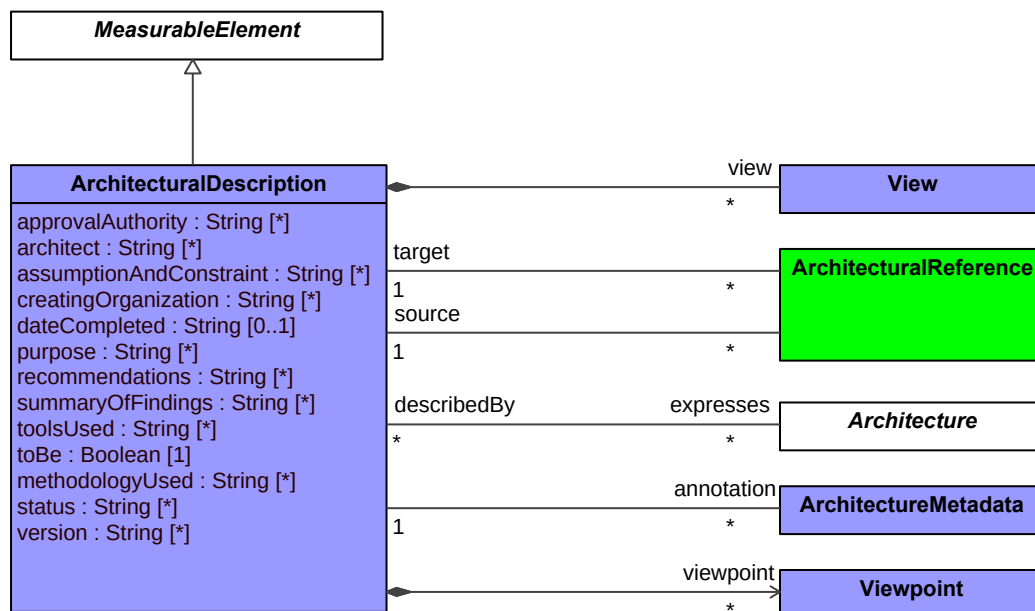
No

[MeasurableElement](#)

Description

An Architecture Description is a work product used to express the Architecture of some System Of Interest.

It provides executive-level summary information about the architecture description in a consistent form to allow quick reference and comparison between architecture descriptions -- It includes assumptions, constraints, and limitations that may affect high-level decisions relating to an architecture-based work program.



**Figure 9:209 - ArchitecturalDescription**

Attributes

|                                                  |                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>approvalAuthority : String[*]</code>       | Someone or something that has the authority to approve the ArchitecturalDescription.                                                                                                                                                                                                                                         |
| <code>architect : String[*]</code>               | Someone responsible for the creation of ArchitecturalDescription.                                                                                                                                                                                                                                                            |
| <code>assumptionAndConstraint : String[*]</code> | Any assumptions, constraints, and limitations contained in the ArchitecturalDescription, including those affecting deployment, communications performance, information assurance environments, etc.                                                                                                                          |
| <code>creatingOrganization : String[*]</code>    | The organization responsible for creating the ArchitecturalDescription.                                                                                                                                                                                                                                                      |
| <code>dateCompleted : String[0..1]</code>        | Date that the ArchitecturalDescription was completed.                                                                                                                                                                                                                                                                        |
| <code>methodologyUsed : String[*]</code>         | The methodology used in developing the architecture.                                                                                                                                                                                                                                                                         |
| <code>purpose : String[*]</code>                 | Explains the need for the Architecture, what it will demonstrate, the types of analyses that will be applied to it, who is expected to perform the analyses, what decisions are expected to be made on the basis of each form of analysis, who is expected to make those decisions, and what actions are expected to result. |

|                               |                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| recommendations : String[*]   | States the recommendations that have been developed based on the architecture effort. Examples include recommended system implementations, and opportunities for technology insertion. |
| status : String[*]            | Approval status of the architecture.                                                                                                                                                   |
| summaryOfFindings : String[*] | Summarizes the findings that have been developed so far. This may be updated several times during the development of the ArchitecturalDescription.                                     |
| toBe : Boolean[1]             | Indicates whether the ArchitecturalDescription represents an Architecture that exists or will exist in the future.                                                                     |
| toolsUsed : String[*]         | Identifies any tools used to develop the ArchitecturalDescription as well as file names and formats if appropriate.                                                                    |
| version : String[*]           | Version number of the architecture.                                                                                                                                                    |

## Architecture

Summary & Overview

Yes

[UAFElement](#)

Description

An abstract type that represents a generic architecture. Subtypes are OperationalArchitecture and PhysicalArchitecture.

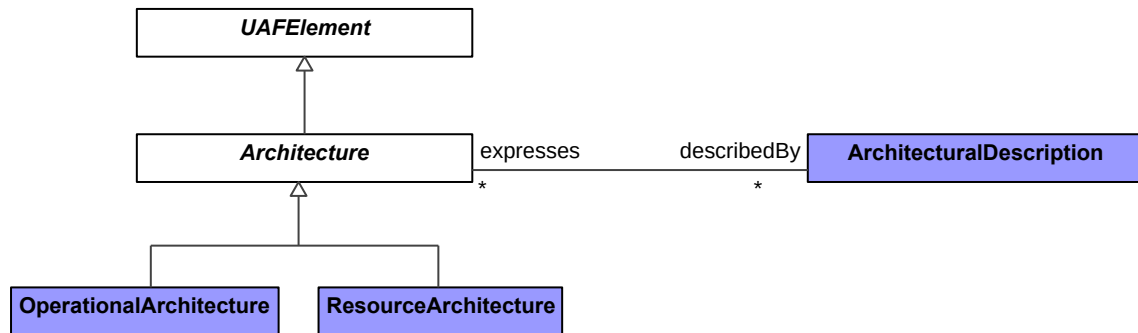


Figure 9:210 - Architecture

## Concern

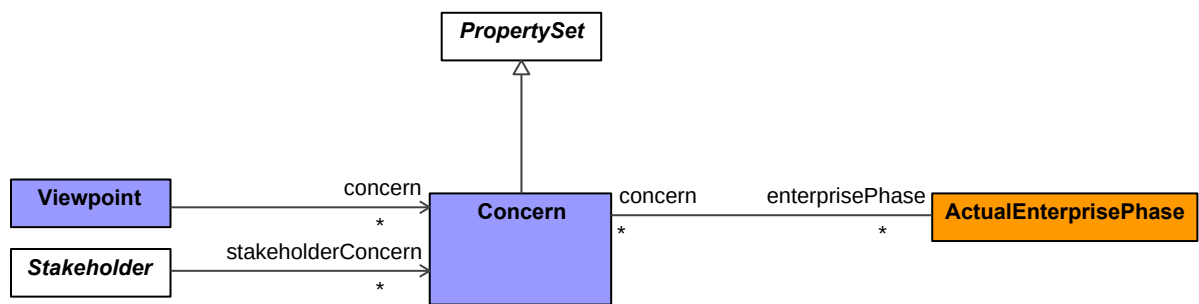
Summary & Overview

No

[PropertySet](#)

Description

Interest in an EnterprisePhase (EnterprisePhase is synonym for System in ISO 42010) relevant to one or more of its stakeholders.



**Figure 9:211 - Concern**

## Stakeholder

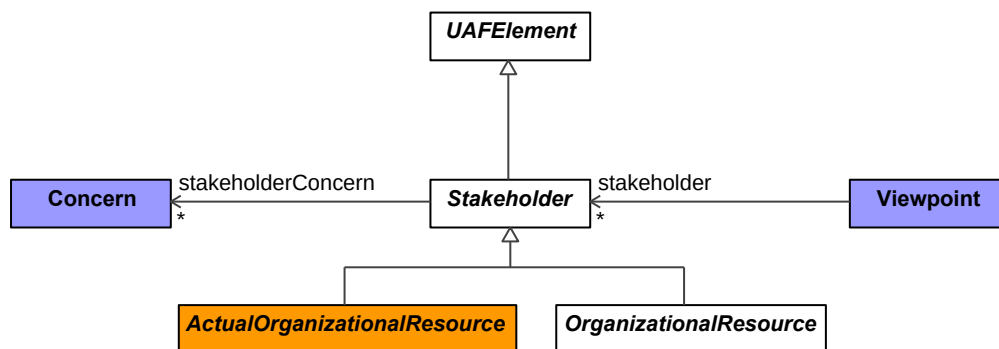
Summary & Overview

Yes

[UAFElement](#)

Description

Individual, team, organization, or classes thereof, having an interest in an EnterprisePhase [ISO/IEC/IEEE 42010:2011].



**Figure 9:212 - Stakeholder**

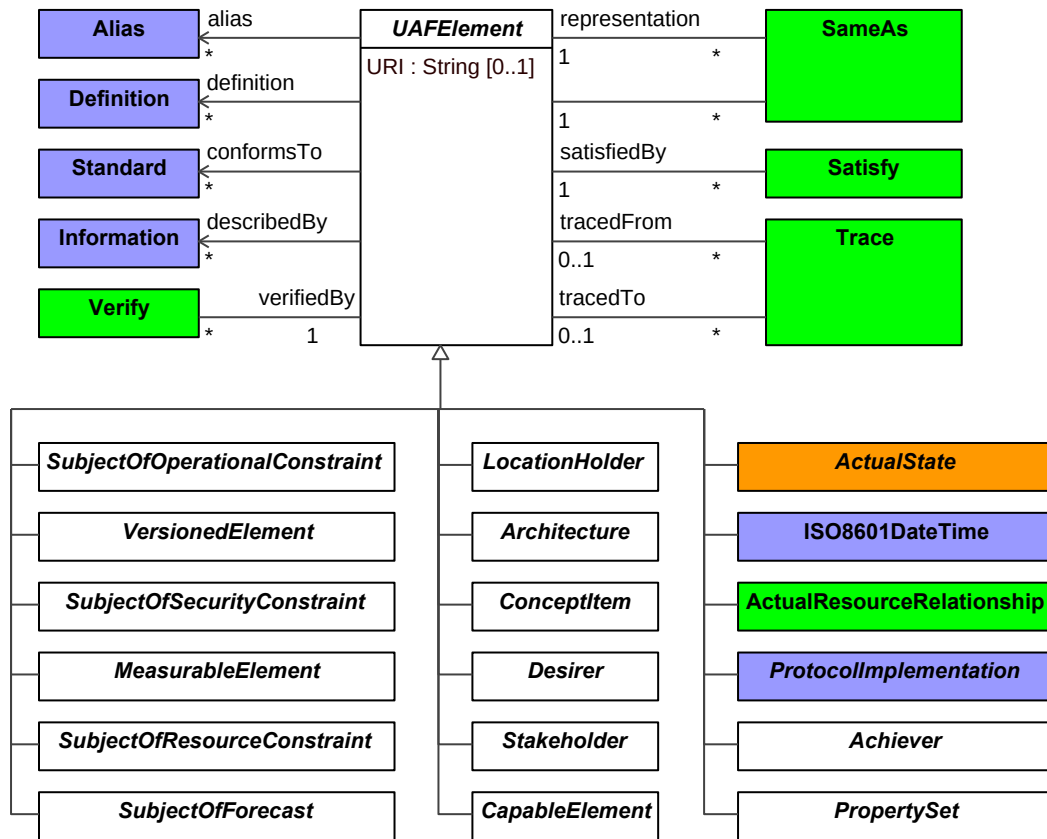
## UAFElement

Summary & Overview

Yes

Description

Abstract super type for all of the UAF elements. It provides a way for all of the UAF elements to have a common set of properties.



**Figure 9:213 - UAFElement**

Attributes

URI : String[0..1] Captures Unique identifier for the element.

## View

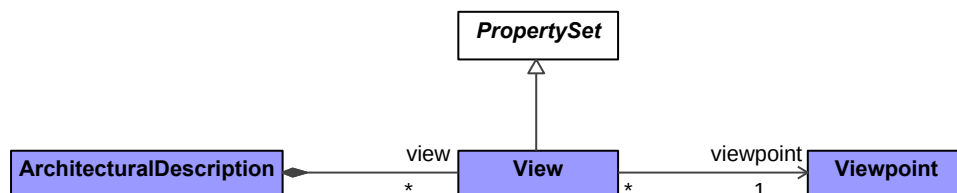
Summary & Overview

No

[PropertySet](#)

Description

An architecture view expresses the architecture of the system-of-interest in accordance with an architecture viewpoint (or simply, viewpoint). [ISO/IEC/IEEE 42010:2011(E)].



**Figure 9:214 - View**

## Viewpoint

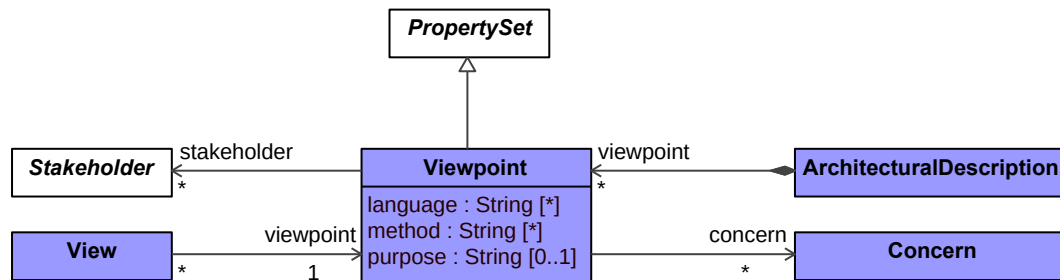
Summary & Overview

No

[PropertySet](#)

Description

An architecture viewpoint frames (to formulate or construct in a particular style or language) one or more concerns. A concern can be framed by more than one viewpoint. [ISO/IEC/IEEE 42010:2011(E)].



**Figure 9:215 - Viewpoint**

Attributes

- language : String[\*] The languages used to express the Viewpoint.
- method : String[\*] The methods employed in the development of the Viewpoint.
- purpose : String[0..1] The purpose of the Viewpoint.

## 9.1.13 Domain MetaModel::Information

### DataModel

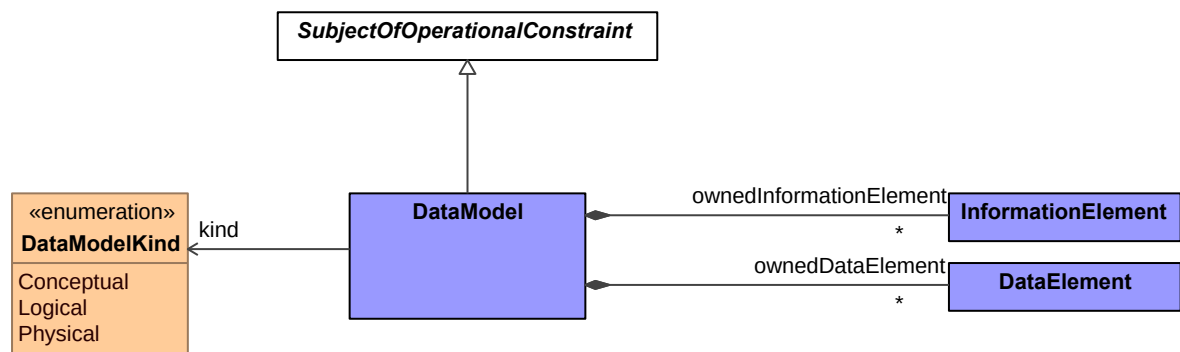
Information

No

[SubjectOfOperationalConstraint](#), [SubjectOfResourceConstraint](#)

Description

A structural specification of data types, showing relationships between them. The type of data captured in the DataModel is described using the enumeration DataModelKind (Conceptual, Logical and Physical).



**Figure 9:216 - DataModel**

## 9.1.14 Domain MetaModel::Parameters

### ActualCondition

Parameters

No

[ActualPropertySet](#)

Description

An individual describing an actual situation with respect to circumstances under which an OperationalActivity, Function or ServiceFunction can be performed.

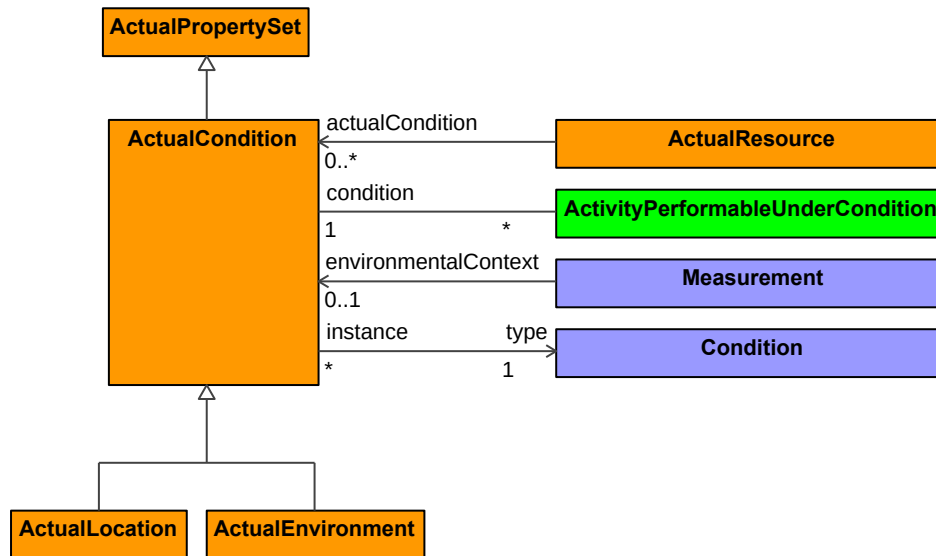


Figure 9:217 - ActualCondition

### ActualEnvironment

Parameters

No

[ActualCondition](#)

Description

An individual that describes the circumstances of an Environment.

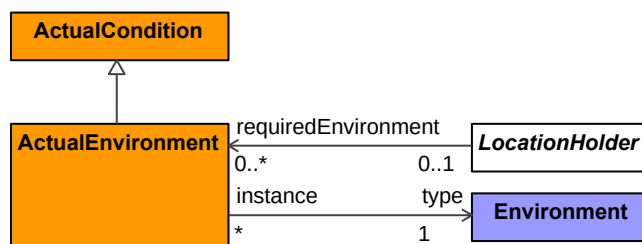


Figure 9:218 - ActualEnvironment

## ActualLocation

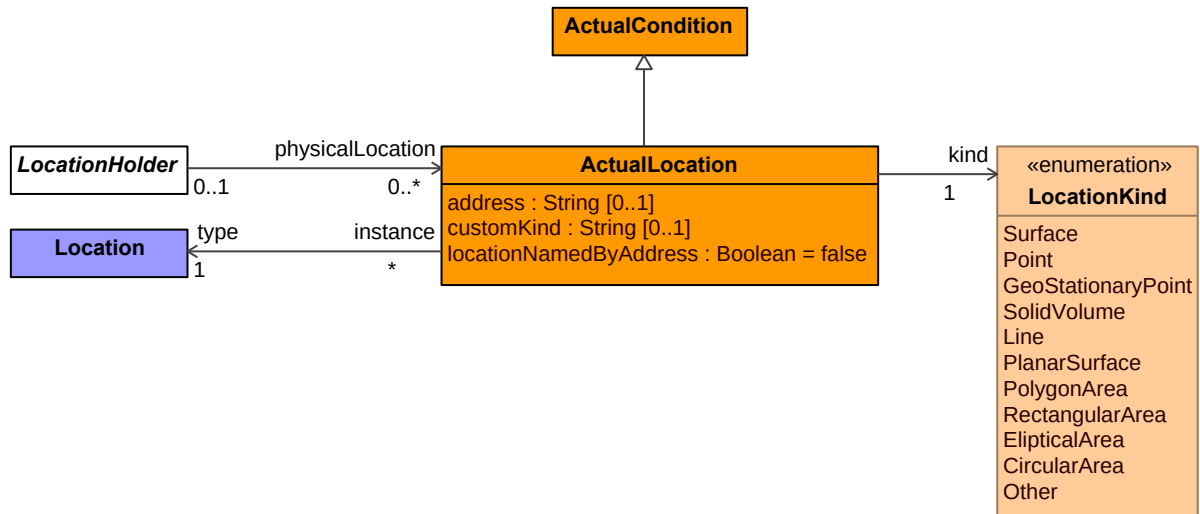
Parameters

No

[ActualCondition](#)

Description

An individual that describes a physical location, for example using text to provide an address, Geo-coordinates, etc.



**Figure 9:219 - ActualLocation**

Attributes

|                                    |                                                                                                                  |
|------------------------------------|------------------------------------------------------------------------------------------------------------------|
| address : String[0..1]             | String describing the address of the ActualLocation, i.e. "1600 Pennsylvania avenue", "The White House"          |
| customKind : String[0..1]          | String describing a location kind that is not in the LocationKind enumerated list                                |
| locationNamedByAddress : Boolean[] | Boolean that indicates if the ActualLocation address is embedded in the ActualLocation name. By default = false. |

## ActualMeasurement

Parameters

No

[ActualState](#)

Description

An actual value that is applied to a Measurement.

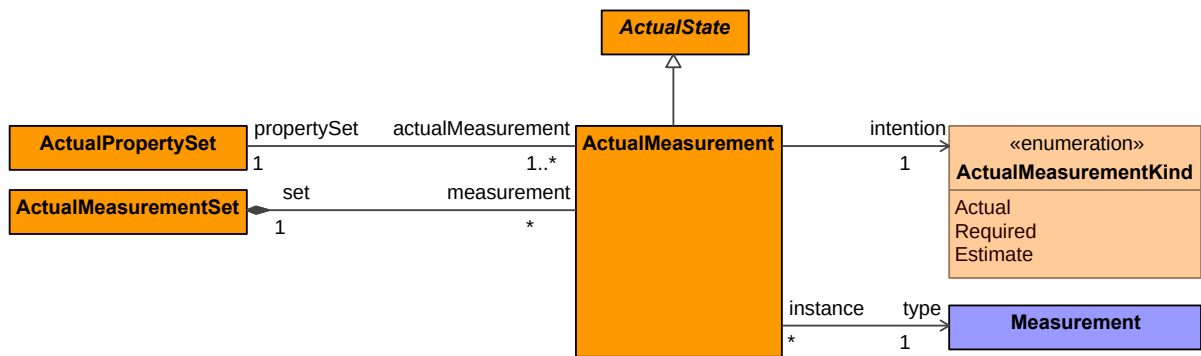


Figure 9:220 - ActualMeasurement

## ActualMeasurementSet

Parameters

No

[ActualPropertySet](#)

Description

A set of ActualMeasurements.

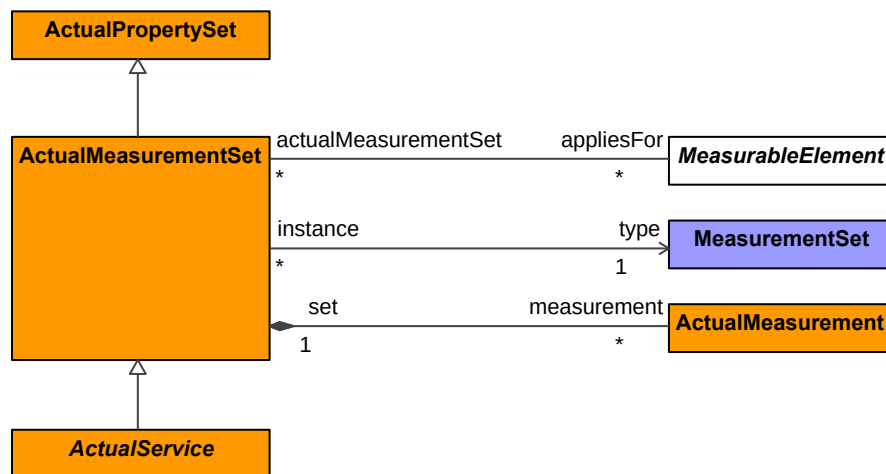


Figure 9:221 - ActualMeasurementSet

## ActualPropertySet

Parameters

No

[ActualState](#)

Description

A set or collection of Actual properties.

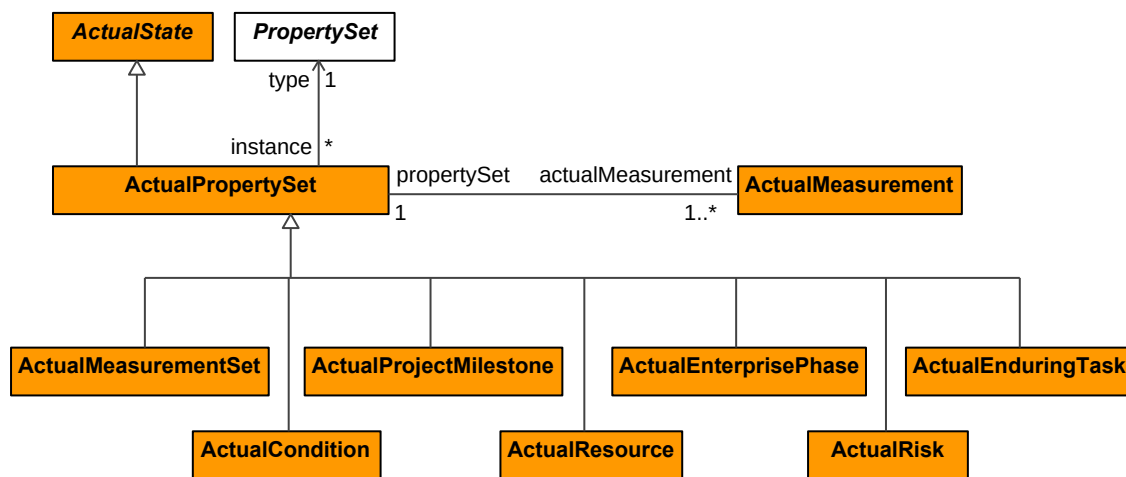


Figure 9:222 - ActualPropertySet

## ActualState

Parameters

Yes

[UAElement](#)

Description

Abstract element that applies temporal extent to a set of elements realized as Instance Specifications.

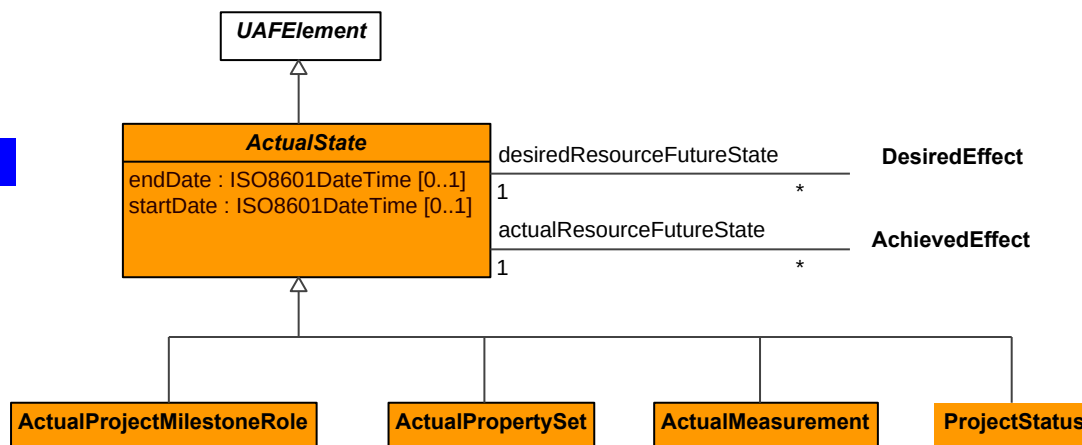


Figure 9:223 - ActualState

Attributes

endDate : ISO8601DateTime[0..1] End time for all individual elements.  
startDate : ISO8601DateTime[0..1] Start time for all individual elements.

## Condition

Parameters

No

[PropertySet](#)

Description

A type that defines the Location, Environment and/or GeoPoliticalExtent.

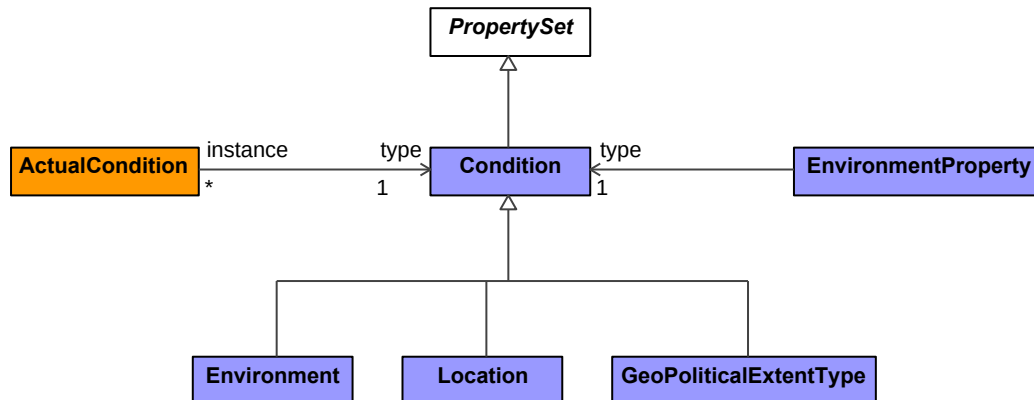


Figure 9:224 - Condition

## Environment

Parameters

No

[Condition](#)

Description

A definition of the environmental factors in which something exists or functions. The definition of an Environment element can be further defined using EnvironmentKind.

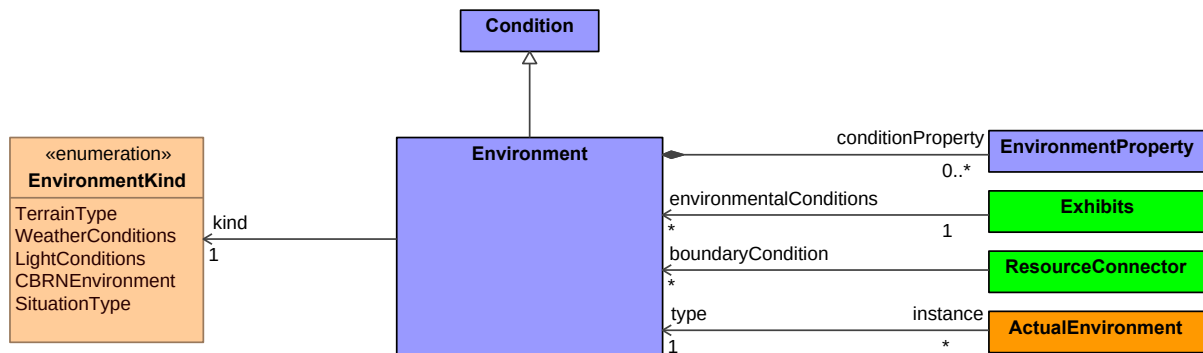


Figure 9:225 - Environment

## GeoPoliticalExtentType

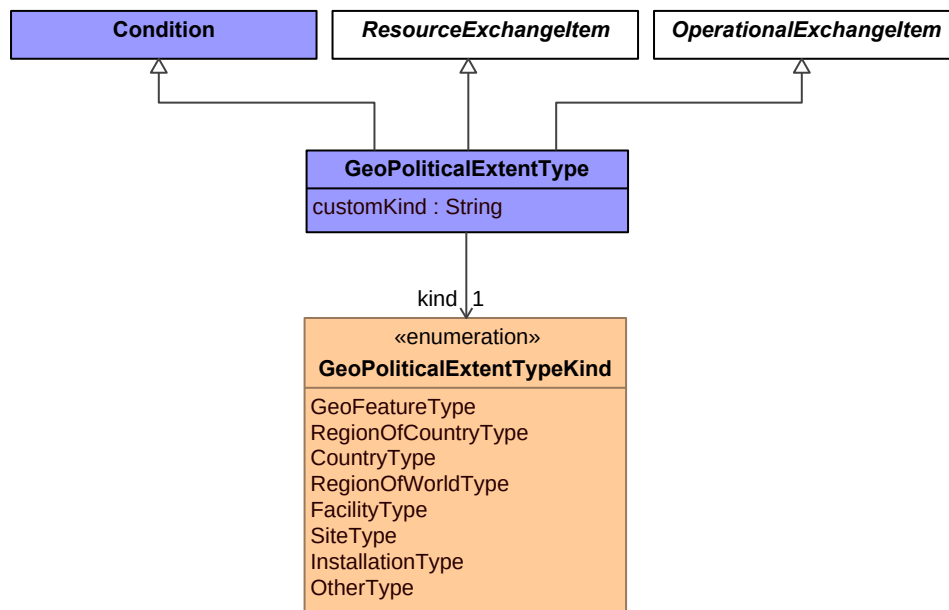
Parameters

No

[Condition](#), [OperationalExchangeItem](#), [ResourceExchangeItem](#)

Description

A type of geospatial extent whose boundaries are defined by declaration or agreement by political parties.



**Figure 9:226 - GeoPoliticalExtentType**

Attributes

customKind : String[] Captures the kind of GeopoliticalExtentType.

## ISO8601DateTime

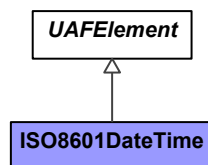
Parameters

No

[UAFElement](#)

Description

A date and time specified in the ISO8601 date-time format including timezone designator (TZD): YYYY-MM-DDThh:mm:ssTZD.



**Figure 9:227 - ISO8601DateTime**

## Location

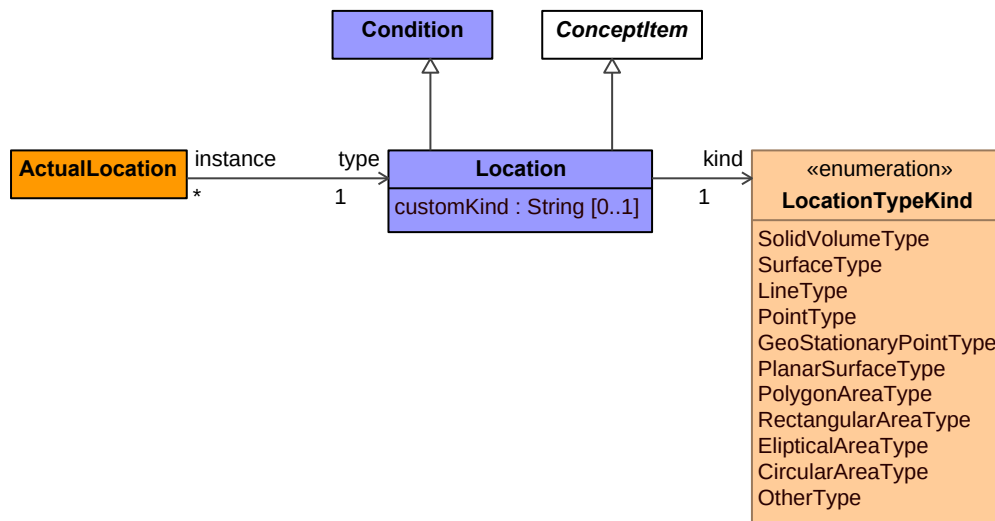
Parameters

No

[ConceptItem](#), [Condition](#)

Description

A specification of the generic area in which a LocationHolder is required to be located.



**Figure 9:228 - Location**

Attributes

customKind : String[0..1] Captures the kind of Location if the LocationTypeKind has been set to "OtherType".

## LocationHolder

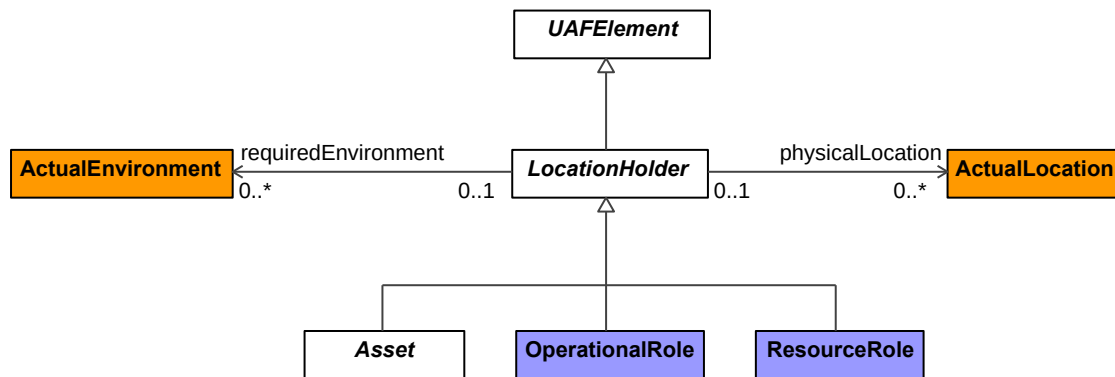
Parameters

Yes

[UAFElement](#)

Description

Abstract type, used to group elements that are allowed to be associated with a Location.



**Figure 9:229 - LocationHolder**

## MeasurableElement

Parameters

Yes

[UAFElement](#)

Description

Abstract type, grouping elements that can be measured by applying MeasurementSets to them.

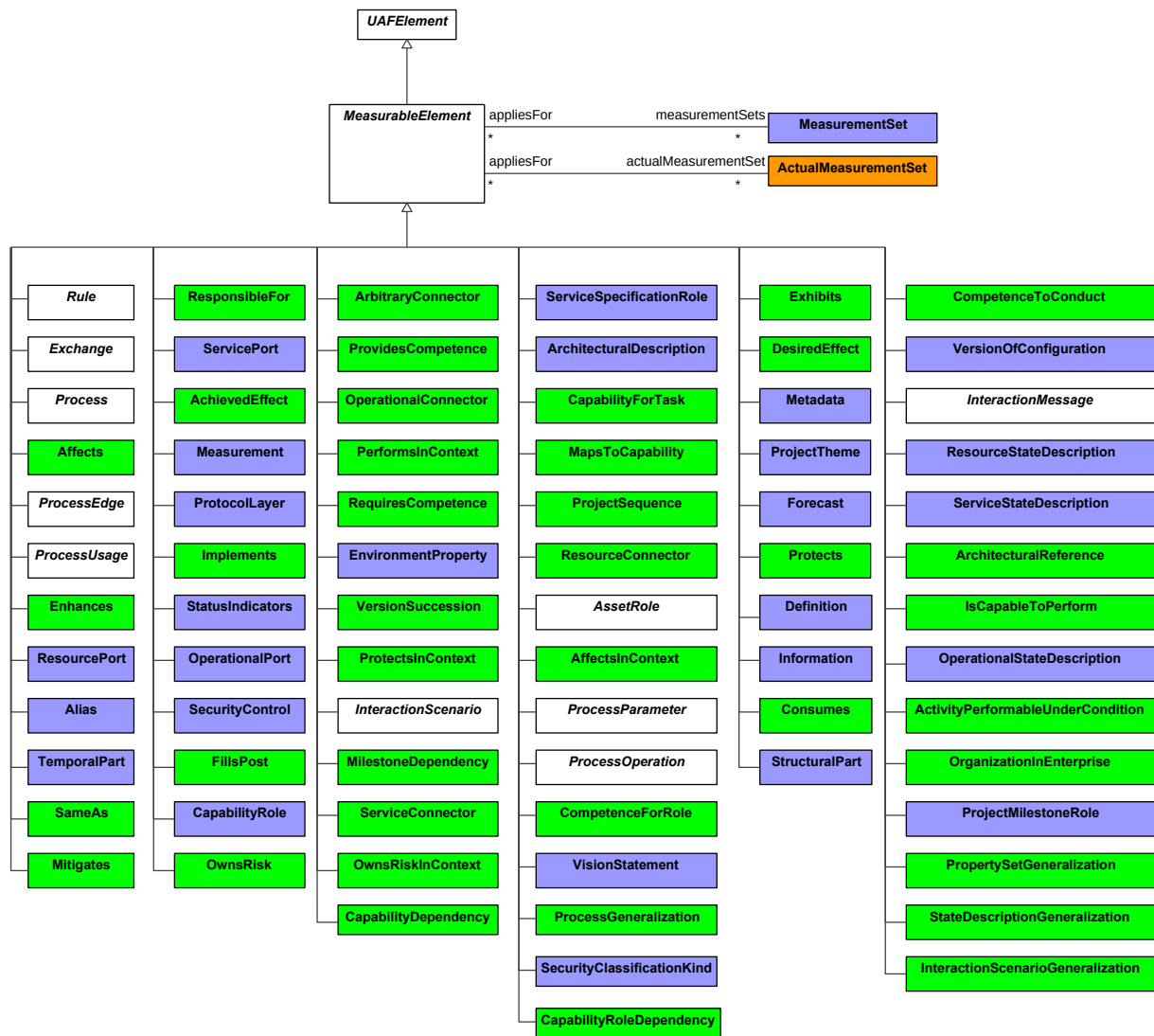


Figure 9:230 - MeasurableElement

## Measurement

Parameters

No

[MeasurableElement](#)

Description

A property of an element representing something in the physical world, expressed in amounts of a unit of measure.

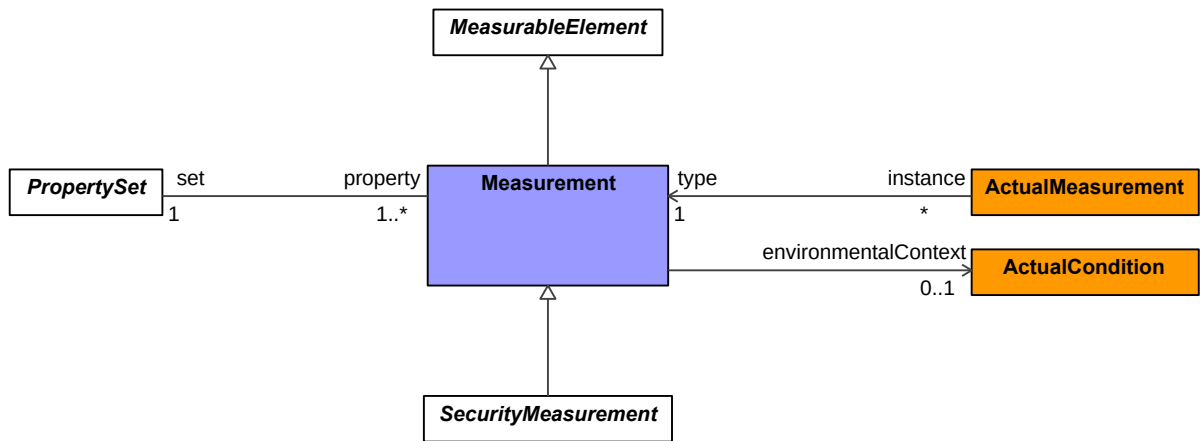


Figure 9:231 - Measurement

## MeasurementSet

Parameters

No

[PropertySet](#)

Description

A collection of Measurements.

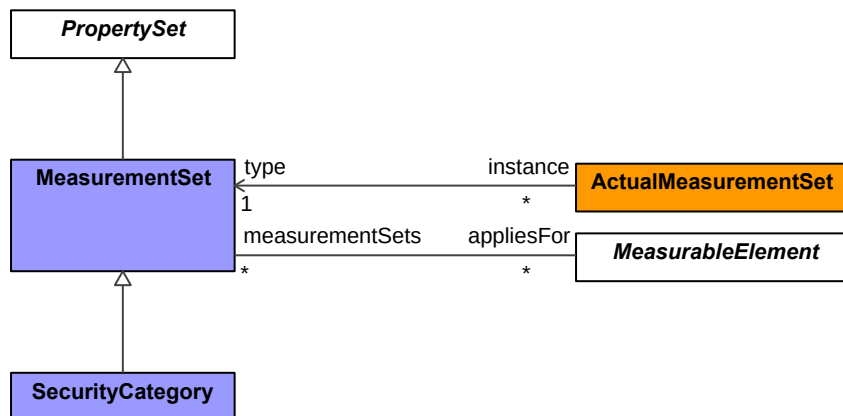


Figure 9:232 - MeasurementSet

## PropertySet

Parameters

Yes

[UAFElement](#)

Description

An abstract type grouping architectural elements that can own Measurements.

